

TABLE 6-6. ΦX174 Genes and Functions

| Gene      | Function                               | Protein | Mass, kDa         |
|-----------|----------------------------------------|---------|-------------------|
|           |                                        | SDS Gel | Sequence          |
| <i>A</i>  | RF replication; viral strand synthesis | 59      | 56 <sup>a</sup>   |
| <i>A*</i> | Shutoff host; DNA synthesis            | 37      |                   |
| <i>B</i>  | Capsid morphogenesis                   | 20      | 13.8 <sup>b</sup> |
| <i>C</i>  | DNA maturation                         | 7       | 9.4 <sup>a</sup>  |
| <i>D</i>  | Capsid morphogenesis                   | 14      | 16.8 <sup>b</sup> |
| <i>E</i>  | Cell lysis                             | 10      | 9.9 <sup>a</sup>  |
| <i>F</i>  | Major coat protein                     | 48      | 48.3 <sup>a</sup> |
| <i>G</i>  | Major spike protein                    | 19      | 19.1 <sup>b</sup> |
| <i>H</i>  | Major spike protein; adsorption        | 37      | 35.8 <sup>a</sup> |
| <i>J</i>  | Core protein; DNA condensation         | 5       | 4.1 <sup>b</sup>  |
| <i>K</i>  | Unknown                                | 6       |                   |

<sup>a</sup>Calculated from DNA sequence  $\left( \frac{\text{number of nucleotides}}{3 \times 0.00915} \right)$ .

<sup>b</sup>Calculated from amino acid sequence.

eclipse stage in which the eclipsed virus has lost infectivity for new cells. Uncoating of the virus is coupled to replication of the viral genome, forming a double-stranded replicative form (RFI). The binding of ΦX174 particles and RFI occurs at areas of adhesion between inner and outer membranes (see Chapter 7) and possibly affords direct access to the host chromosomal replicative apparatus fixed at the inner surface. This  $SS \Rightarrow RFI$  replication is dependent solely on host replication proteins and includes prepriming, priming, elongation, gap filling, ligation, and supercoiling. RFI synthesis commences at the *n'* recognition site in the intervening region between genes *G* and *F* (Fig. 6-23).

The first stage of replication ( $SS \Rightarrow RF$ ) occurs within 1 minute of infection. The second replication stage involves  $RF \Rightarrow RF$  synthesis and occurs up to 20 minutes postinfection. Two additional proteins are required for  $RF \Rightarrow RF$  replication. These are the phage-encoded gene *A* protein and *E. coli rep* product. The gene *A* product nicks supercoiled RFI in the (+) strand between residues 4305 and 4306, forming the open circular form RFII. The gene *A* protein remains attached to the 5' end of the (+) nicked strand by one of two tyrosine residues: Tyr-343 or Tyr-347 (Fig. 6-22). By complexing with RepA (a DNA helicase), the gene *A* protein remains indirectly attached to the (−) strand and is dragged around the genome with the helicase. Leading strand replication then occurs from the 3' end of the open (+) strand, regenerating a new duplex, while the old (+) strand is displaced, extended, and coated with single-stranded DNA-binding proteins. After traversing the entire length of the template, the gene *A* protein cleaves the regenerated origin, again attaching to the 5' phosphoryl end by one of the two tyrosines noted above. The now free 3'-OH end of the (+) strand can attack the original gene *A* protein-5'-phosphoryl bond, generating a single-strand circle coated with SSB. This single strand can be reconverted into a new RFI using host polymerase machinery as before to generate another (−) strand.

Once approximately 35 double-stranded molecules of ΦX174 DNA have accumulated, lagging strand (i.e., minus-strand) synthesis stops and new progeny plus (+)

strands are encapsidated to form progeny phage particles. The switch is brought about by the accumulation of capsomeres and phage protein *C*. Concomitantly, host DNA synthesis is arrested either by the sequestration of a limiting host replication protein to phage DNA replication or by a phage-induced interference protein. This third stage of RF  $\Rightarrow$  SS replication and encapsidation (20–30 minutes postinfection) depends on seven virus-encoded proteins. These proteins complex and encapsidate the (+) strand, making it unavailable for minus-strand synthesis.

$\Phi$ X174 does not encode an endolysin or holin and has no way to attack preexisting bacterial wall structures. So, how does it escape the cell? This phage interferes with bacterial enzymes that make precursors of peptidoglycan. The lack of coordination between peptidoglycan synthesis and bacterial wall construction leaves the wall weak, leading to its ultimate collapse due to osmotic pressures from within.  $\Phi$ X174 protein E, associated with the bacterial membrane, blocks the activity of MraY, a bacterial protein that catalyzes the transfer of peptidoglycan precursors to lipid carriers so that the precursors can be transported through the membrane (see Chapter 7). As the cell disintegrates,  $\Phi$ X174 particles are released to find new, unsuspecting hosts.

## SUMMARY

Most, if not all, free-living microorganisms are susceptible to attack by one or more viruses. Learning how viruses usurp their host cells has provided tremendous insight not only into viruses but into the molecular machinery of the bacterial cell. Take, for example, the N-utilization proteins discovered as a result of investigations with  $\lambda$  phage or the developing model of bacterial DNA replication pieced together from studying the replication of a variety of phage systems. And who could have predicted that a study examining how bacteria attack viral DNA (restriction-modification systems) would lead to the era of recombinant DNA techniques and future gene replacement therapies for human disease (see Chapter 4)? Certainly continued explorations of phage–host interactions promise to yield additional knowledge of the intricacies of life.

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## CHAPTER 7

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# CELL STRUCTURE AND FUNCTION

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As discussed briefly in Chapter 1, prokaryotic cells differ in a number of major ways from eukaryotic cells (Table 7-1). Lower eukaryotes (protozoa, algae, and fungi) lack some of the distinguishing features of metazoan cells. Nevertheless, all eukaryotic cells appear to be similar versions of the same overall plan. In the past, some cytologists considered that bacterial cells were merely smaller replicas of the cells of higher forms and that failure to distinguish comparable subcellular structures in bacteria could be attributed to the limitations of the microscopic and cytological techniques available. This may prove to be the case, since many recent investigations with highly improved techniques and equipment have shown that there may be a closer similarity between eukaryotes and prokaryotes than previously thought. It has also become apparent that the cells of archaeobacteria (*Archaea*) display important structural differences from the cells of either eubacteria or eukaryotes.

### THE EUKARYOTIC NUCLEUS

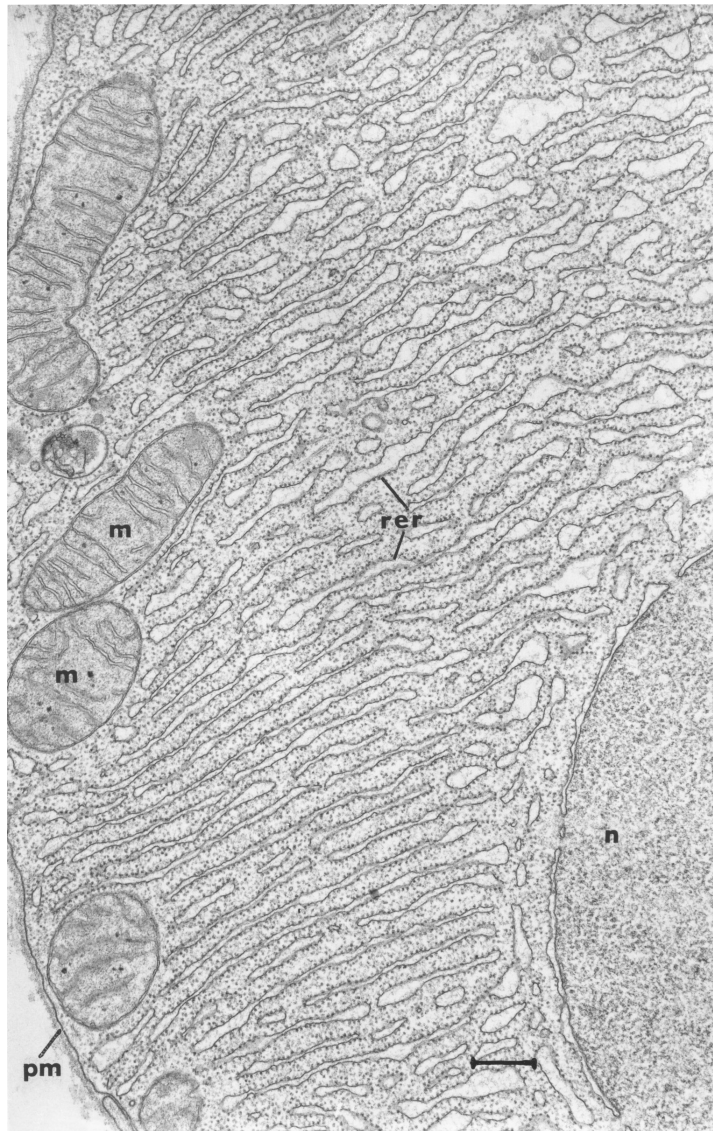
Eukaryotes (*Eukarya*) display a cytologically distinct **nucleus**, the organizational and regulatory center for virtually all of the biochemical and hereditary processes of the cell. With the aid of the electron microscope it has been possible to demonstrate many structural details of the eukaryotic nucleus. Figure 7-1 shows a mammalian cell with a well-defined nuclear membrane composed of at least two distinct layers. The outer surface contains pores with tubules that transcend both membrane layers. Amoebae, protozoa, algae, and fungi (yeasts and molds) also contain a discrete membrane-bound nucleus (Fig. 7-2). The myxomycete *Acyria cinerea* contains a double-layered nuclear membrane that exhibits pores (Fig. 7-3).

TABLE 7-1. Major Components of Cells of Various Classes of Organisms

| Higher Eukaryotes                |                                  | Lower Eukaryotes                 |                                  | Prokaryotes                                 |  |
|----------------------------------|----------------------------------|----------------------------------|----------------------------------|---------------------------------------------|--|
| Metazoan                         | Protozoan                        | Algae                            | Fungi                            | Eukaryotes and Archaeobacteria <sup>a</sup> |  |
| Nucleus                          | Macronucleus                     | Nucleus                          | Nucleus                          | Nucleoid                                    |  |
| Nuclear membrane                 | Nuclear membrane                 | Nuclear membrane                 | Nuclear membrane                 | No nuclear membrane <sup>b</sup>            |  |
| Nucleolus                        | Nuclear elements                 |                                  |                                  |                                             |  |
| Ribosomes                        | Ribosomes                        | Ribosomes                        | Ribosomes                        | Ribosomes                                   |  |
| 40S, 60S/80S                     | 40S, 60S/80S                     | 40S, 60S/80S                     | 40S, 60S/80S                     | 30S, 50S/70S                                |  |
| Cell respiration in mitochondria | Cell respiration in mitochondria | Cell respiration in mitochondria | Cell respiration in mitochondria | Cell respiration in cytoplasmic membrane    |  |
| 70S ribosomes                    | 70S ribosomes                    | 70S ribosome                     | 70S ribosomes                    |                                             |  |
| Endoplasmic reticulum            | Endoplasmic reticulum            | Endoplasmic reticulum            | Cytoplasm                        | Cytoplasm                                   |  |
| Golgi apparatus                  | Dictyosomes                      | Dictyosomes                      |                                  |                                             |  |
| Inclusions                       | Specialized organelles           | Chloroplasts                     | Inclusions                       | Inclusions                                  |  |
| Lysosomes                        |                                  |                                  |                                  |                                             |  |
| Peroxisomes                      |                                  |                                  |                                  |                                             |  |
| Phycobilisomes                   |                                  |                                  |                                  |                                             |  |
| Plasma membrane                  | Plasma membrane                  | Plasma membrane                  | Cytoplasmic membrane             | Cytoplasmic membrane                        |  |
| No cell wall                     | No cell wall                     | Cell wall chitin, glycans        | Cell wall chitin, glycans        | Cell wall peptidoglycan <sup>a</sup>        |  |
| Undulating flagella or cilia     | Undulating flagella or cilia     | Undulating flagella or cilia     | Undulating flagella or cilia     | Rotating flagella                           |  |

<sup>a</sup>Although comparable to eubacteria in many respects, the archaeobacteria (*Archaea*) do not produce a cell wall peptidoglycan comparable to that produced by eubacteria. They also differ from eubacteria in a number of other characteristics not included in this table.

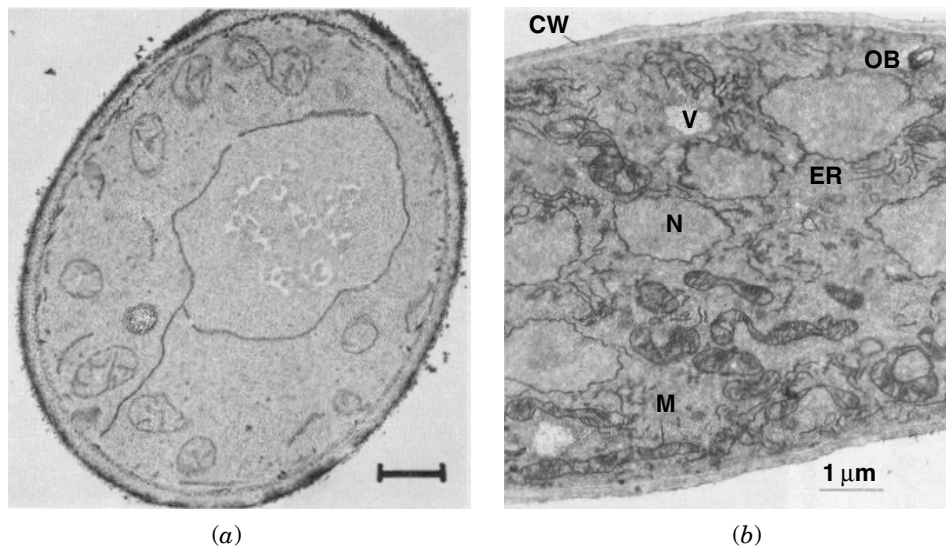
<sup>b</sup>Some species of eubacteria and archaeobacteria have now been shown to have a nuclear membrane.



**Fig. 7-1. Electron micrograph of a portion of a mammalian cell.** The region of this pancreatic exocrine cell between the nucleus (n) on the lower right and the plasmalemma on the lower left is occupied by numerous cisternae of the rough endoplasmic reticulum (rer) and a few mitochondria (m). Prominent pores can be seen in the nuclear membrane. The ribosomes appear as small black dots lining the rer. The nucleolus is not visible in this photograph. Bar equals 1  $\mu$ m. (Source: From Palade, G. E. *Science* **189**:347, 1975.)

## BACTERIAL NUCLEOIDS

In most bacteria the DNA-containing region of the cell (the chromosome) is folded into a cytologically distinct region that does not appear to be bound by a nuclear membrane and is generally termed the **nucleoid** to distinguish it from the eukaryotic nucleus.

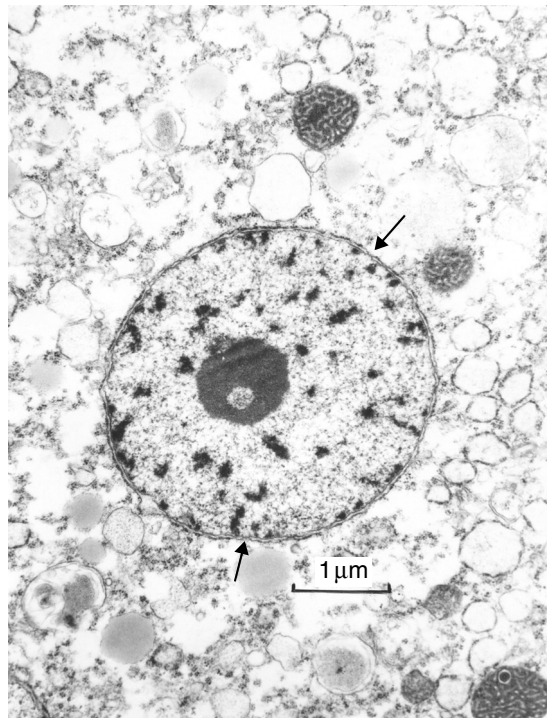


**Fig. 7-2. Fine structure of cells of lower eukaryotes.** (a) General view of a section of a cell of *Saccharomyces cerevisiae*. Prominent features include the cell wall, the nucleus with distinct pores in the nuclear membrane, mitochondria with cristae, and intermembranous structures. Bar equals 100 nm. (Source: From Avers, C. J., et al., *J. Bacteriol.* **100**:1044, 1969.) (b) Fine structure of a hyphal element of the fungus *Mucor genevensis*. Nuclei (N) with pores, vacuoles (V), mitochondria (M), endoplasmic reticulum (ER), cell wall (CW), and dark bodies (DB) are visible. Bar equals 1 μm. (Source: From Clark-Walker, C. D. *J. Bacteriol.* **109**:299, 1972.)

The compaction process is aided by a number of DNA-binding proteins, sometimes called **histone-like proteins**, since they resemble the histones found in the nucleus of eukaryotic cells. Most of the DNA is in the form of supercoils (see Chapter 2) constrained by these binding proteins. Interaction between the DNA supercoils and crowding within the cell is presumed to lead to further compaction and subsequent phase separation between the nucleoid and the cytoplasm.

As shown in Figure 7-4 (a, b, and c), growth in normal or high salt medium and fixation with  $\text{OsO}_4$  results in visualization of a condensed nucleoid, whereas glutaraldehyde fixation (Fig. 7-4d) results in a dispersed appearance of the nucleoid. The development of confocal scanning light microscopy made it possible to observe the shape and substructure of the nucleoid and to compare these images with phase-contrast and electron microscope images. In Figure 7-5, a confocal scanning light microscope image of an  $\text{OsO}_4$ -fixed cell is compared with an electron micrograph and a reconstruction model based on a study of serial sections. Some of the differences observed might reflect the presence of transcription–translation complexes with ribosomes and proteins in the more dense preparations observed with  $\text{OsO}_4$  fixation.

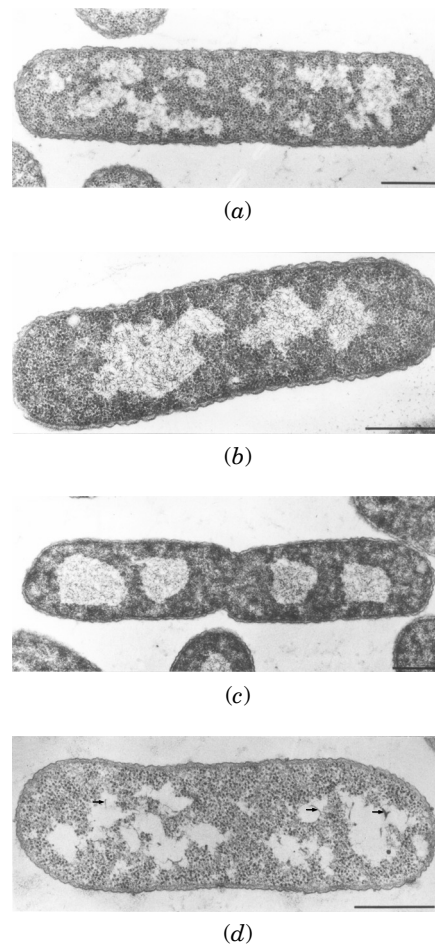
The nucleoid is observed as a coralline (coral-like) shape as shown in Figure 7-6a. The branches of the coralline nucleoid spread far into the cytoplasm and throughout the entire interior of the cell. By viewing serial sections, it has been possible to reconstruct the ribosome-free area of the nucleoid. Figure 7-6b presents a three-dimensional model of how the coralline nucleoid may appear.



**Fig. 7-3. Interphase nucleus of the myxomycete *Arcyria cinerea*.** Prominent pores interrupt the typical nuclear envelope. The nucleolus near the center is well defined, while the chromatin is randomly dispersed throughout the nucleus. (Source: From Mims, C. W. *J. Gen. Microbiol.* **71**:53, 1972.)

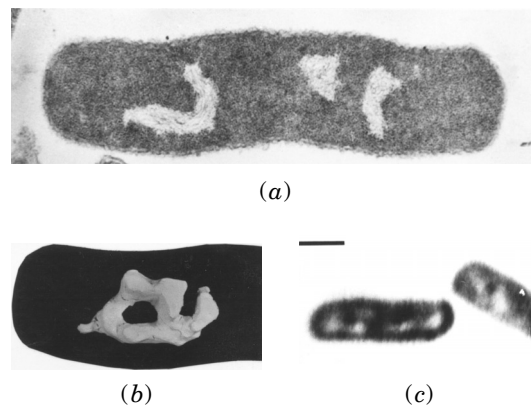
Fast-growing cells harvested and fixed for observation under the electron microscope often display multiple nucleoids. This multinucleate condition occurs because fast-growing cells contain DNA in a state of multifork replication (see Chapter 2). Moreover, it has now become apparent that **multiploidy** (multiple copies of nucleoids or entire chromosomes), a frequent occurrence in eukaryotes, is also common in many bacterial species. The presence of a high copy number of chromosomes (nucleoids) is thought to facilitate rapid cell doubling during fast growth. Separation of the duplicating nucleoid during cell division must occur during the S phase (the period between the initiation of DNA replication and the completion of segregation of the two nucleoids; see Chapter 17).

Fluorescence microscopy studies reveal a mitosis-like movement of immunolabeled nucleoids during replication and cell division in bacteria. Protein components of a chromosomal segregation apparatus have been shown to direct the *oriC* region toward the poles. These proteins (ParB in *Caulobacter crescentus* and SpoOJ in *Bacillus subtilis*) appear to move toward the poles in a mitosis-like manner in concert with *oriC*. In bacteria such as *Escherichia coli* and *Haemophilus influenzae* that lack ParB/SpoOJ homologs, it has been suggested that other systems participate in a mitosis-like mechanism that effects nucleoid segregation. Many of these activities are considered in more detail in relationship to the cell division process discussed in Chapter 17.



**Fig. 7-4. Standard  $\text{OsO}_4$  fixation of *E. coli* by the Ryter-Kellenberger technique of prefixation directly in the culture medium.** (a) Cell was grown in low-salt medium. Bar equals 0.5  $\mu\text{m}$ . (b) Same as (a), except that the cell was grown in normal salt medium (6 g KCl/L). The nucleoid is more confined and localized. Bar equals 0.5  $\mu\text{m}$ . (c) Same as (a), except that the cell was grown in a high-salt medium (21 g KCl/L). The nucleoid is still more confined. (d) Fixation of *E. coli* B with glutaraldehyde embedded in Epon. The cell was grown in high-salt medium. The nucleoid is dispersed into small patches that contain DNA precipitates (arrows). Bar equals 0.5  $\mu\text{m}$ . (Source: From Hobot, J. A., et al. *J. Bacteriol.* **182**:960–971, 1985.)

Isolation of the intact bacterial nucleoid has provided some insight into its properties. Two types of nuclear bodies can be obtained: an envelope-associated nucleoid and an envelope-free nucleoid. Large amounts of RNA, proteins, lipids, and peptidoglycans are found in the envelope-associated nucleoid. The free nucleoid contains lesser amounts of non-DNA components such as protein and RNA. Examination of the isolated nucleoid under the electron microscope (Fig. 7-7) reveals a number of supercoiled loops of DNA extending from the center. The central core appears to contain RNA that can be diminished by washing the grid with RNase.



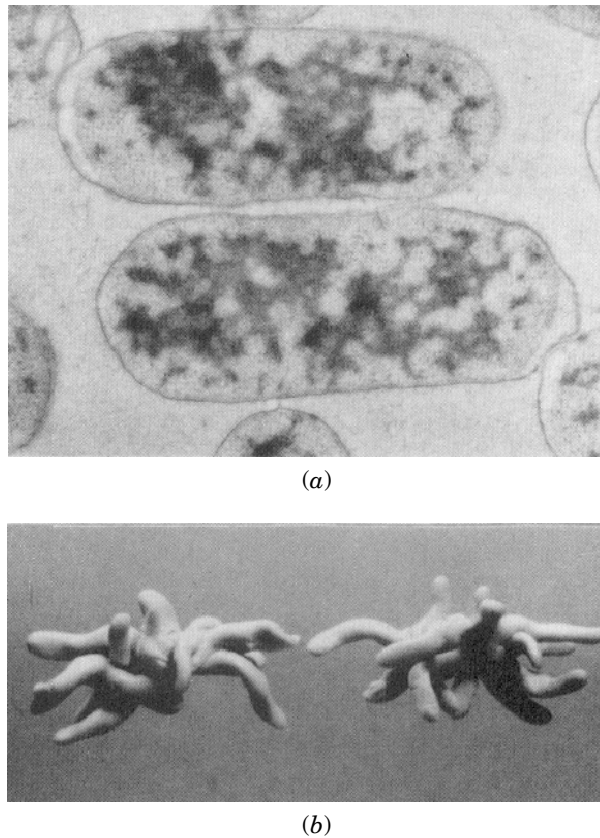
**Fig. 7-5. Fast-growing *E. coli* B/r H266 (generation time, 21 min).** The photographs compare a confocal scanning light micrograph (c) with a reconstruction of an  $\text{OsO}_4$  fixed nucleoid (b), based on serial sections studied with the electron microscope. Only one of these sections is shown here (a). Bar equals 1.0  $\mu\text{m}$ . (Source: From Valkenberg, J. A. C., et al. *J. Bacteriol.* **161**:478–483, 1985.)

## NUCLEOSOMES

Eukaryotic chromosomes are organized into DNA–protein complexes termed **nucleosomes**. The proteins, mainly **histones**, play an important role in the structure of eukaryotic chromosomes by determining the conformation referred to as **chromatin**. The nucleosomes in the repeating unit of DNA organization are frequently seen in a “beads-on-a-string” configuration. Extensive digestion of chromatin with micrococcal nuclease releases the nucleosome core, a small, well-defined particle that has been crystallized. The particle mass is equally distributed between 146 bp of DNA and an octamer formed by two each of four major histone proteins (H2A, H2B, H3, and H4). With the aid of X-ray diffraction it has been possible to develop a three-dimensional model of the structure of the eukaryotic nucleosome core (Fig. 7-8).

The genomic DNA of *E. coli* is compacted into a nucleosome-like structure by interaction with several DNA-binding proteins. As many as 12 species of nucleoid proteins have been identified as shown in Table 7-2. The histone-like protein HU contains two closely related 10 kDa subunits, HU- $\alpha$  and HU- $\beta$ , encoded by *hupA* and *hupB*. Fluorescein-labeled HU taken up by EDTA-treated cells of *E. coli* concentrates in the nucleoid and is uniformly distributed throughout this structure. The proposed role of HU protein is in stabilizing higher-order nucleoprotein structures to confer specificity during DNA interactions. The fact that HU exhibits little sequence specificity suggests that it may be involved in a variety of DNA protein activities requiring coiling of specific DNA sequences. In vitro studies have implicated HU protein in the initiation of DNA replication, binding of repressors, and transposition of bacteriophage Mu. In *E. coli*, *hupA**hupB* double mutants that lack HU protein entirely display severe defects in cell division, DNA folding, and DNA partitioning.

Fis is a small, basic DNA-binding protein that was originally identified by its ability to effect flagellar phase variation in *E. coli* by stimulating DNA inversion. Fis has also been implicated as a participant in rRNA and tRNA transcription. It may play a major role in chromosomal DNA replication as evidenced by its binding to the origin of

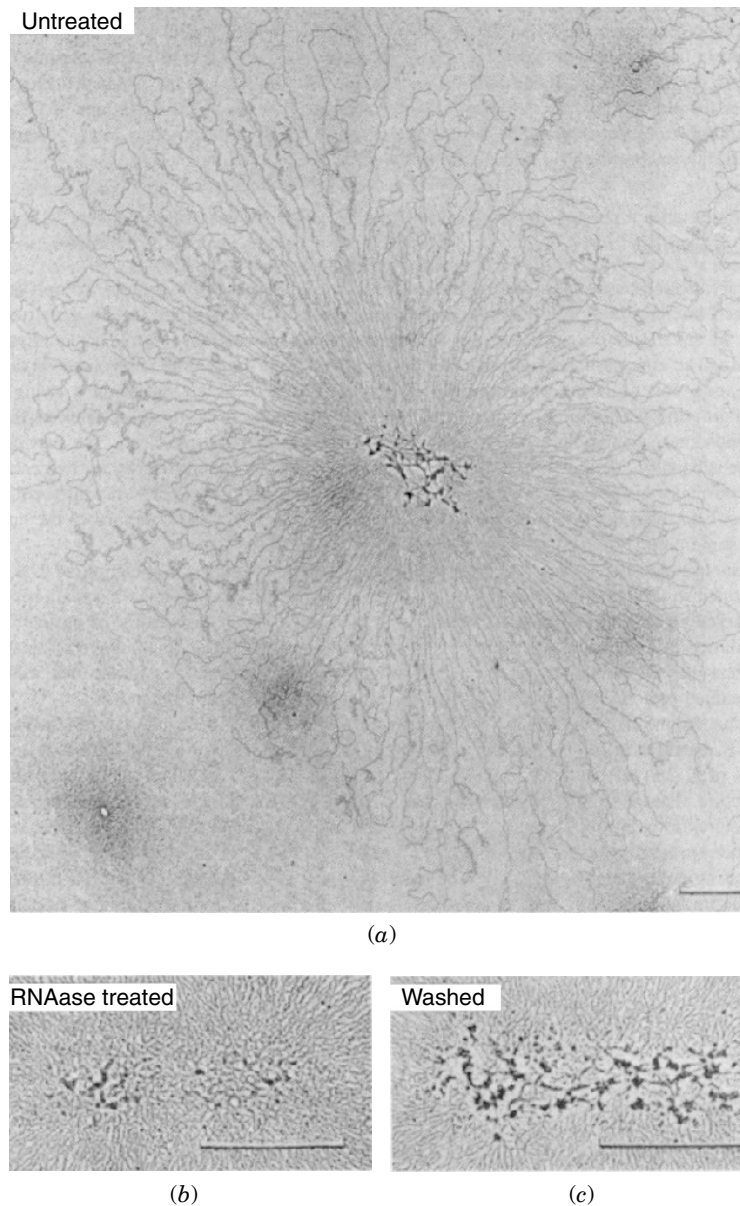


**Fig. 7-6. Coralline shape of the bacterial nucleoid.** (a) Immunostaining of DNA in exponentially growing *E. coli* B using antidouble-stranded DNA mouse IgM, followed by goat IgG antibodies to mouse IgM stained with a mixture of 1 volume of 2%  $\text{KMnO}_4$  and 2 volumes of 2% uranyl acetate. Label is observed as gray-stained patches over DNA-rich area. In this micrograph the upper cell shows two distinct nucleoids. (b) Schematic models of bacterial nucleoids show a pair of nucleoids of growing cells deduced from thin sections of cryofixed, freeze-substituted *E. coli* B. For practical reasons, the models were flattened on the lower side. (Source: From Bohrmann, B., et al. *J. Bacteriol.* **173**:3149–3158, 1991.)

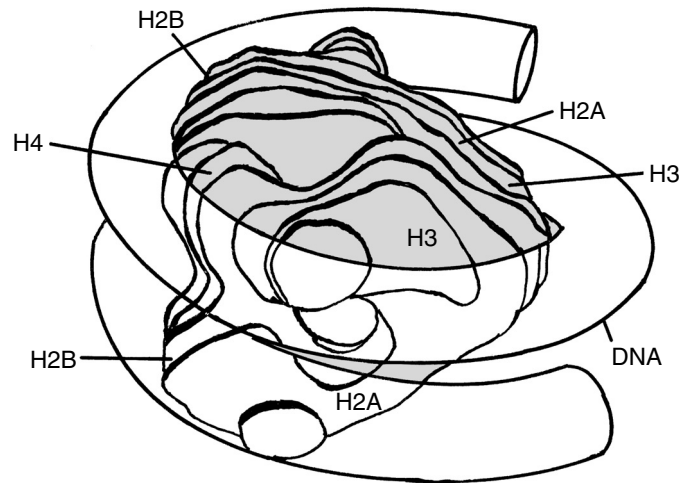
replication (*oriC*) in *E. coli*. Fis levels vary dramatically during the course of cell growth and in response to changing environmental conditions. When stationary-phase cells are subcultured into a rich medium, Fis levels increase from less than 100 to over 50,000 copies per cell prior to the first cell division. As cells commence exponential growth, Fis synthesis is reduced and the intracellular concentration drops as a function of the rate of cell division. Concurrent alteration in *fis* mRNA levels suggests that regulation is at the transcriptional level. Fluctuation in Fis levels may serve as an early signal of a nutritional upshift and may play an important role in governing physiological activities. Fis appears to autoregulate its own expression as shown by the fact that binding to its promoter excludes binding of RNA polymerase.

The bacterial histone-like protein H-NS, encoded by *hns*, inhibits site-specific DNA recombination as evidenced by the dramatic increase in inversion of the *pilA* promoter





**Fig. 7-7. Cell envelope-free nucleoid from *E. coli* 15  $\tau$ -bar.** For preparation of the cell-associated nucleoid, cells are first suspended in 20% sucrose, treated briefly with lysozyme and ethylene-diaminetetraacetic acid (EDTA), and then lysed with a mixture of the nonionic detergents Brij-58 and deoxycholate in the presence of 1 *M* NaCl. Centrifugation at 17,000 *g* for 15 min on a 10–50% sucrose gradient permits isolation of the cell envelope-associated nucleoid. Preparation of the envelope-free nucleoid requires longer lysozyme treatment and use of an ionic detergent such as Sarcosyl. The sedimentation coefficients are 3200S for the envelope-associated nucleoid and 1600S for the envelope-free nucleoid. The hypophase contained 0.4 *M* salt. The chromosome had  $141 \pm 3$  loops and possibly a fork on the loop at about 1 o'clock. In (b), the grid was washed with the control buffer. Bars equals 1.0  $\mu$ m. (Source: From Kavenoff, R. and B. C. Bowen, *Chromosome* 59:89–101, 1976.)



**Fig. 7-8. Model of the nucleosome core.** The model was made by winding a tube simulating the DNA superhelix on a model of the histone octamer, which was built from a three-dimensional map derived from electron micrographs of the histone octamer. The ridges of the periphery of the octamer form a more-or-less continuous helical ramp on which a 146-nucleotide pair length of DNA can be wound. The locations of individual histone molecules (whose boundaries are not defined at this resolution) are proposed here on the basis of chemical cross-linking data. (Source: From Kornberg, R. D., and A. King, *Sci. Am.* **250**:52–64, 1981.)

**TABLE 7-2. Nucleoid (Histone-like) Proteins in Bacteria**

| Protein | Identified Function                                                                |
|---------|------------------------------------------------------------------------------------|
| CbpA    | Curved DNA-binding protein A                                                       |
| CbpB    | Curved DNA-binding protein B<br>[also known as Rob (right origin-binding protein)] |
| DnaA    | DNA-binding protein A                                                              |
| Dps     | DNA-binding protein from starved cells                                             |
| Fis     | Factor for inversion stimulation                                                   |
| Hfq     | Host factor for phage $Q_{\beta}$                                                  |
| H-NS    | Histone-like nucleoid structuring protein                                          |
| HU      | Heat-unstable nucleoid protein                                                     |
| IciA    | Inhibitor of chromosome initiation A                                               |
| IHF     | Integration host factor                                                            |
| Lrp     | Leucine-responsive regulatory protein                                              |
| StpA    | Suppressor of <i>td</i> mutant phenotype A                                         |
| H1      | ?                                                                                  |

region in mutants lacking H-NS (see further discussion in “Pili or Fimbriae” in this chapter).

In the *Archaea*, chromosomal DNA also exists in protein-associated form. Histone-like proteins have been isolated from stable nucleoprotein complexes in *Thermoplasma acidophilum* and *Halobacterium salinarum*. Whole chromosomes of *H. salinarum* consist of regions of both protein-associated DNA and protein-free DNA. In electron

micrographs the protein-associated DNA region appears as nucleosome-like fibrous structures. However, the suggestion that these observations represent a major difference between eubacteria and *Archaea* seems untenable in view of the widespread occurrence of protein-associated DNA and nucleosome-like structures in a variety of bacteria.

## MITOCHONDRIA

Mitochondria are subcellular organelles that carry out oxidative metabolism (respiration) in eukaryotic cells. The highly membranous interior of the mitochondrion contains the cytochrome systems and oxidative phosphorylation processes that yield energy in the form of ATP (see Chapter 9). Lower eukaryotes (yeasts, algae, filamentous fungi, and protozoa) have all been shown to contain mitochondria, although their number and distribution vary widely from one cell type to another. Even within a given species, the number of mitochondria per cell and their intracellular arrangement may vary with the cultural conditions and growth phase.

Mitochondria possess two distinct membranes, **outer** and **inner membranes**, which may be closely associated with each other at certain contact sites. The inner membrane is highly invaginated, forming pleats referred to as **cristae**. Yeasts produce mitochondria actively when grown in an aerobic environment. Under anaerobic conditions, mitochondria are not observed; yet upon shifting back to aerobic conditions, they quickly reappear. Yeast cells grown under strict anaerobiosis contain mitochondria-like particles designated as **promitochondria**. These structures lack most of the characteristic components of the respiratory chain but still contain ATPase, mitochondrial DNA (mtDNA), and various structural proteins. Anaerobic growth apparently induces a dedifferentiation of yeast mitochondria. When yeast cells are grown in the presence of high concentrations of glucose, the respiratory and energy transduction systems are repressed even under highly aerobic conditions. The system responsible for mitochondrial gene replication and expression is completely blocked, and the cell satisfies all its energy requirements by glycolysis.

Mitochondrial DNA (mtDNA) maps as a closed circular molecule, although linear forms may also exist. Despite the presence of its own genetic material, the mitochondrion is not genetically self-sufficient. Most of the mtDNA-specified proteins are components of the complexes that carry out the respiratory functions of the mitochondrion and the RNA components of their protein synthetic apparatus (ribosomal RNAs and tRNAs). In yeast and possibly other lower eukaryotes, the structural gene coding for the ribosome-associated protein VarI is located on mtDNA. Detailed genetic studies indicate that DNA sequences have been transferred from the mitochondrial to the nuclear genome. Nuclear genes code for most other mitochondrial components, including the enzymes of the tricarboxylic acid (TCA) cycle. Transcription of mitochondrial genes appears to be catalyzed by a nucleus-encoded RNA polymerase. One subunit of the yeast RNA polymerase functions as a core enzyme while a second subunit serves as a specificity factor that facilitates recognition of one of the approximately 20 specific mitochondrial transcription initiation signals on mtDNA. Yeast cells lacking functional mtDNA ( $\rho^-$  mutants) do not synthesize mitochondrial proteins. They exhibit irregular shapes, fewer cristae, and aberrant multilamellar configurations. On the other hand, most nuclear mutations that result in respiratory deficiency (*pet*) do not show a marked effect on the morphology of yeast mitochondria.

A considerable body of evidence supports the hypothesis that the mitochondrion originated from a primitive bacterial form ( $\alpha$ -proteobacterium) through development of an endosymbiotic relationship. As discussed in Chapter 2, the primary nucleotide sequences of rRNA have been highly conserved throughout evolution. A comparison of the nucleotide sequence of various classes of mt-rRNA molecules from human beings and other mammals as well as from lower eukaryotes reveals a very close resemblance to bacterial rRNA. Also, it has been shown that a homolog of the bacterial cell division gene *ftsZ* is present in the mitochondrion of the alga *Mallomonas splendens*. Homologs of FtsZ have also been found in chloroplasts. This information, together with the revelation that the mitochondrial genome closely resembles that of bacteria, suggested a prokaryotic origin for mitochondria and chloroplasts. Recent studies with *Saccharomyces cerevisiae* and *Rickettsia prowazekii* suggest that mitochondrial genomes are descended from  $\alpha$ -proteobacterium. However, studies with other organisms provide evidence that challenges the endosymbiosis-based view of the origin of the eukaryotic mitochondrion. These studies have been invoked to support an alternate hypothesis that the mitochondrion may have arisen at the same time that the eukaryotic nucleus originated rather than from some primitive bacterial form.

In eukaryotic cells, programmed cell death (apoptosis) has been shown to involve the mitochondrion. In response to stress (nutrient deprivation or exposure to irradiation), mitochondria rupture or leak and release factors that trigger caspases (protein-splitting enzymes), which disrupt proteins in the cell membranes and the nucleus, resulting in rapid cell death. Two proteins, BAK and BAX, have been shown to be gatekeepers for apoptotic signals that act through the mitochondrion. Activation of a series of reactions leads to release of cytochrome *c* from mitochondria and ultimately to cell death. Only mutants lacking both BAK and BAX proteins fail to undergo apoptosis. The Fas death receptor pathway involves binding of a ligand to the cell surface protein Fas. Activation of Fas results in the production of caspases that ultimately destroy the cell. Recent evidence indicates that the Fas receptor pathway also requires the help of the mitochondrial pathway to trigger cell death.

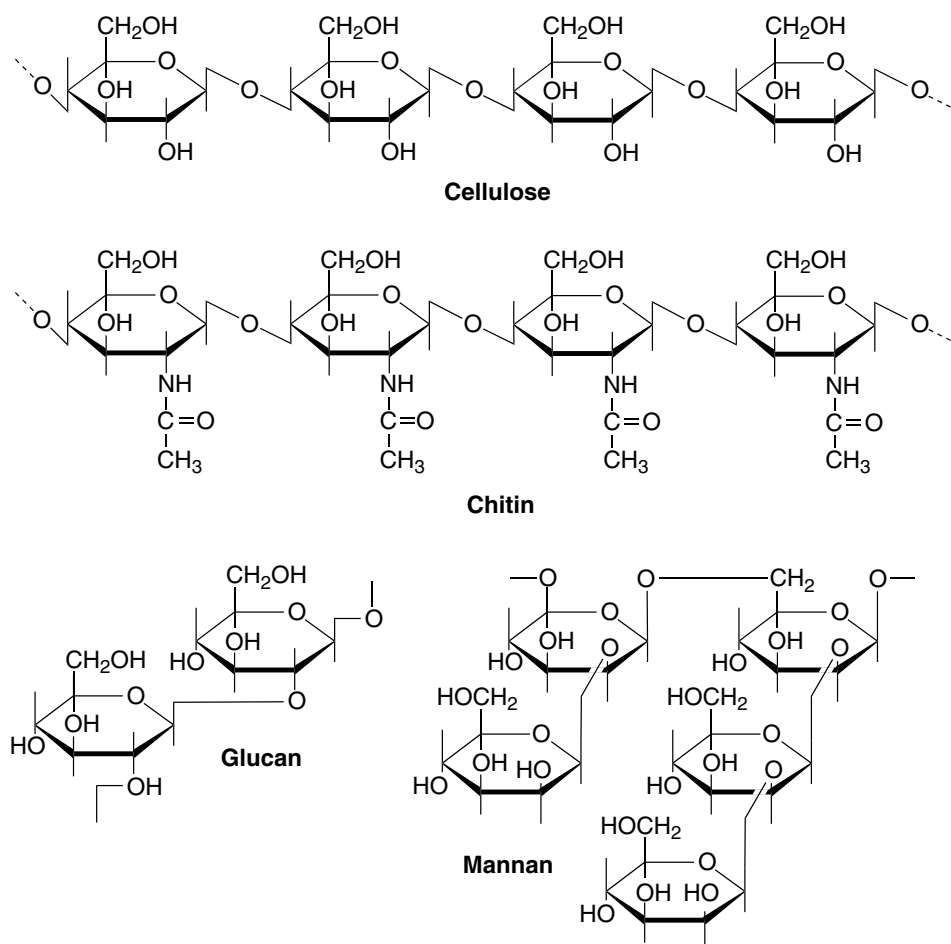
## MICROBIAL CELL SURFACES

### Eukaryotic Cell Surfaces

As discussed briefly in Chapter 1, both eukaryotic and prokaryotic microorganisms are often enclosed within a rigid cell wall. This characteristic is not unique to microorganisms nor is it strictly confined to either the plant or animal kingdom. The cell walls of vascular plants are largely composed of **cellulose**, a  $\beta$ -1,4-linked polymer of glucose (Fig. 7-9). Many algae, as well as higher green plants, contain cellulose as a major cell wall constituent.

The cell walls of some protozoa and many fungi contain **chitin**, a linear polymer of *N*-acetyl glucosamine (Fig. 7-9). Most fungi, algae, and higher plants contain microfibrils of either cellulose or chitin as a prominent skeletal component of their cell walls. Production of these microfibrils occurs at or near the surface of the cell. Among prokaryotes, some species of *Acetobacter* and a few others produce cellulose.

The yeast, *S. cerevisiae*, contains a cell wall that appears as a layered structure when viewed under the electron microscope. The chemical composition of the yeast cell wall is rather uniform over most of the cell surface. Chemical analysis of yeast



**Fig. 7-9. Chemical structures of cell wall polymers.** Cellulose, a  $\beta$ -1,4-linked polymer of glucose; chitin, a linear polymer of *N*-acetyl-D-glucosamine; glucan, a branched polymer of glucose; and mannan, a branched polymer of mannose.

walls reveals the presence of 29%  $\beta$ -glycans (both 1,6- $\beta$ -glycan and 1,3- $\beta$ -glycan are found), 31% mannan, and 13% protein (Fig. 7-9). The mannan-rich outer layer of the wall contains a higher proportion of 1,6- $\beta$ -glycan. Yeast cell walls also contain small percentages of lipids and other materials. Chitin (poly-*N*-acetyl glucosamine) is present in small amounts in vegetative cells of yeast, being confined almost entirely to the ring encircling the septum of budding yeast (bud scar) in mother cells. However, the septum and the chitin ring encircling it show a different chitin : glycan ratio. In the cell walls of all filamentous fungi so far examined, chitin is the only common constituent found.

### Prokaryotic Cell Surfaces

In Chapter 1 it was stated that a high-molecular-weight, sugar-containing, rigid structure called **peptidoglycan** formed the major backbone of the murein sacculus of

the cell wall of both gram-positive and gram-negative bacteria. Indeed, the production of peptidoglycan is considered to be the hallmark of the eubacteria. Archaea produce a **pseudomurein** and an associated surface layer (**S-layer**) composed of protein or glycoprotein. In many archaeobacteria the S-layer may represent the only surface component outside the plasma membrane as shown in Figure 7-10.4. In *B. subtilis* the S-layer is associated with the peptidoglycan-containing sacculus (Fig. 7-10.1). Gram-negative bacteria such as *E. coli* produce an outer membrane (Fig. 7-10.2). The S-layer, if produced, is attached to this outer membrane (Fig. 7-10.3).

### Surface Layers of Bacteria

Bacteria from all of the major phylogenetic groups have been shown to produce a crystalline cell surface layer (S-layer) as the outermost component of the cell (Fig. 7-11). S-layers are composed of identical protein or glycoprotein subunits. They assemble into two-dimensional crystalline arrays showing oblique, square, or hexagonal symmetry in gram-negative or gram-variable bacteria and are associated with the surface of the outer membrane (Fig. 7-10.4). The S-layer may appear quite different in various organisms. In gram-positive bacteria the subunits are linked to the peptidoglycan-containing layer or to secondary cell wall polymers attached to the peptidoglycan layer. In *B. sphaericus* a specific cell wall polymer has been characterized. It contains *N*-acetylglucosamine, *N*-acetylmannosamine, and pyruvic acid and serves as the specific binding site for the N-terminal portion of the S-layer protein. In gram-negative bacteria the S-layer is bound to the lipopolysaccharides of the outer membrane.

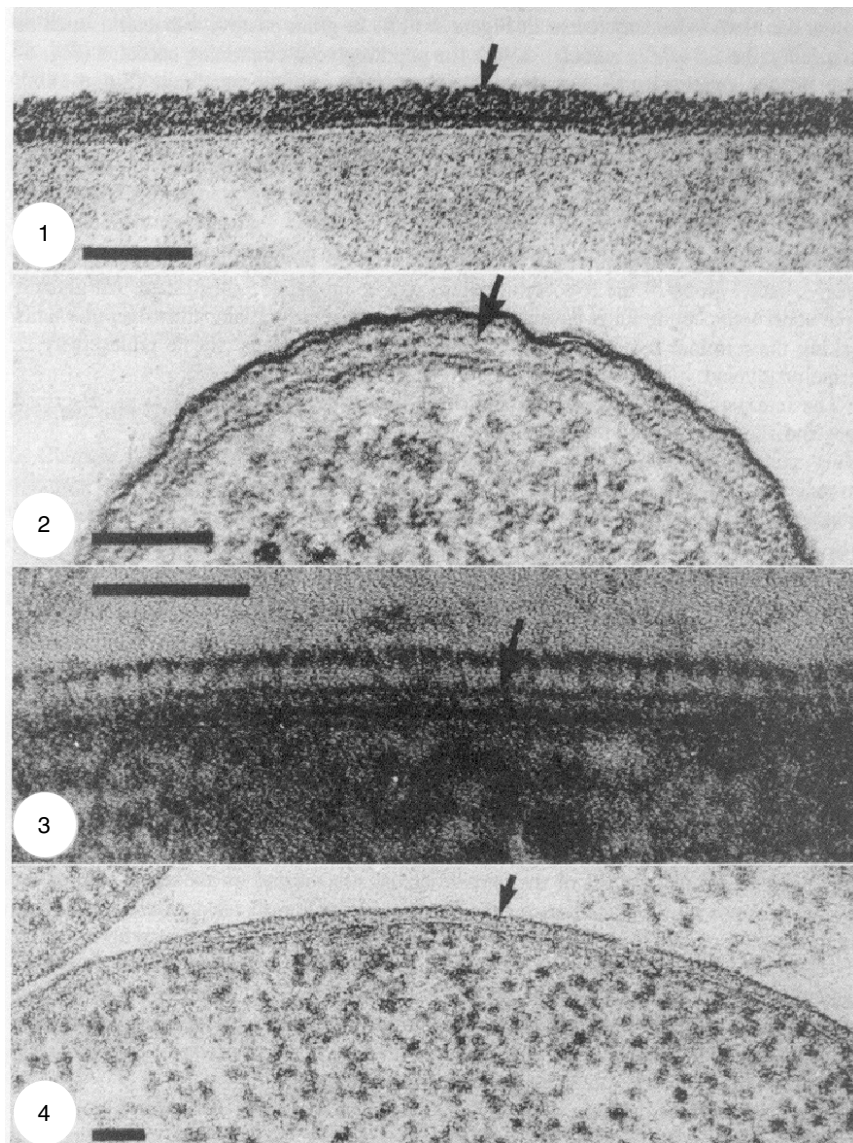
Chemical analysis of purified S-layers shows that they are composed of a single protein or glycoprotein species. In a few organisms, such as *Clostridium difficile* or *B. anthracis*, two types of subunits are observed. S-layers often contain a high proportion of acidic and hydrophobic amino acids. Lysine is the predominant basic amino acid. Arginine, histidine, and methionine are generally low in amount, and the presence of cysteine is rare, being found in only a few S-layer proteins.

Diverse functions have been attributed to the S-layer, including mediation of bacterial attachment to surfaces and to host tissues. In *Campylobacter* and *Aeromonas*, the S-layer serves as a virulence factor. Since the S-layer represents the outermost layer external to the cell membrane in archaea, it must contribute substantially to the shape of the cell (as in *Thermoproteus tenax*). However, there is still a great deal to be learned about the true function(s) of the S-layer in many of the organisms that have been shown to produce this external protein complex.

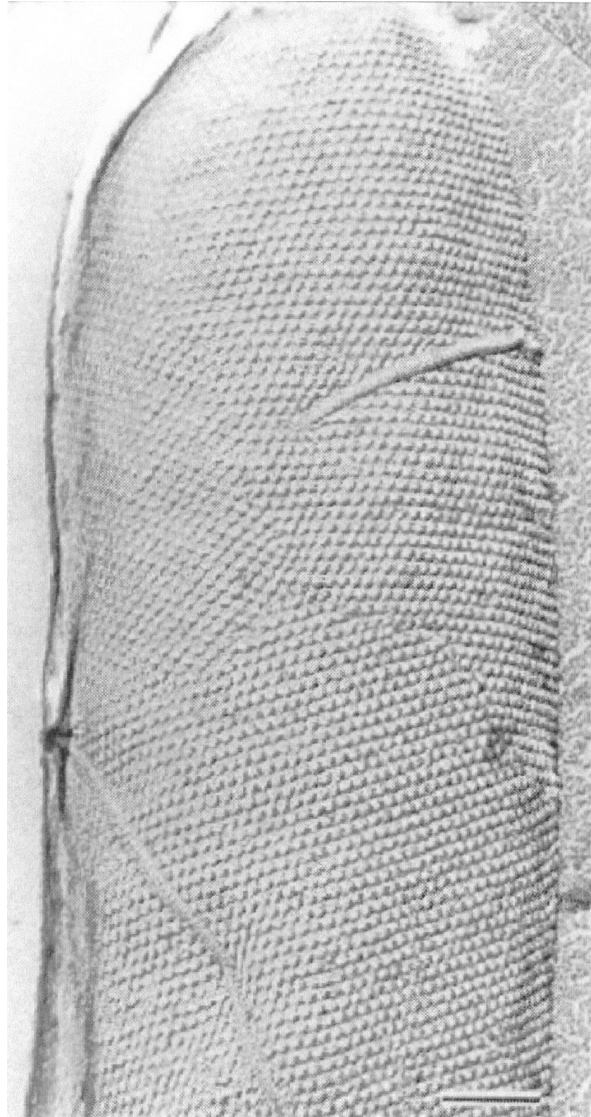
### Peptidoglycans of Bacterial Cell Walls

Peptidoglycan is a heteropolymer of repeating units of  $\beta$ -1,4-*N*-acetylglucosamine (NAG) and *N*-acetyl-muramic acid (NAM). The glycan linkages of peptidoglycan are considered to be uniform in all bacteria with every D-lactyl group of the NAM being peptide substituted. All glycans have short tetrapeptide units terminating with D-alanine or occasionally tripeptide units lacking the terminal D-alanine. The L-alanine at the N terminus can be replaced by glycine.

The interpeptide bridges linking peptidoglycans are of four major types as shown in Figure 7-12 and Table 7-3. Although the proportion of peptide cross-linking varies



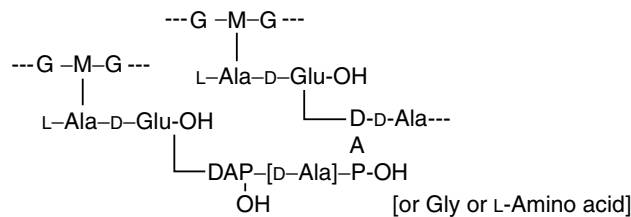
**Fig. 7-10. Envelope profiles of conventionally fixed and embedded bacteria.** Bar equals 50 nm. (1) Amorphous wall fabric of *B. subtilis* 168 (arrow) lies directly above the plasma membrane. (2) *Escherichia coli* possesses a thin peptidoglycan murein layer above the plasma membrane (arrow), over which lays a wavy outer membrane. The region between the outer and plasma membranes is called the periplasm or periplasmic space. The waviness of the outer membrane is believed to be an artifact of the conventional fixation-embedding technique, and much of the periplasm has been leached out. (3) *Clostridium thermosaccharolyticum* is an example of a gram-variable bacterium that has a wall profile intermediate between the gram-positive and gram-negative formats in 1 and 2. The peptidoglycan layer (arrow) is thinner than that of *B. subtilis* but thicker than that of *E. coli*. Above this layer is a proteinaceous S-layer of periodically arrayed subunits. (4) *Methanococcus voltae* is an archaebacterium and possesses only a thin S-layer (arrow) above the plasma membrane as its sole wall layer. (Source: From Beveridge, T. J. and L. L. Graham, *Microbiol. Rev.* **55**:684–705, 1991.)



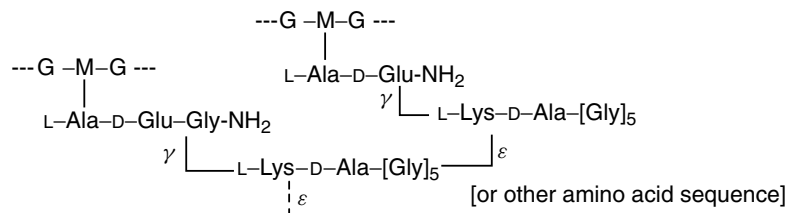
**Fig. 7-11. Electron micrograph of a freeze-etched preparation showing a whole bacterial cell with a hexagonally ordered S-layer lattice.** Bar equals 100 nm. (Source: From Sára, M. and Sleytr, U. B. *J. Bacteriol.* **182**:859–868, 2000.)

considerably from one type to another, in all cases a continuous bag-like sacculus completely surrounds the cell. There is now ample evidence to show that the glycans form helically twisted chains from which the peptide bridges radiate in all directions from the axis of the backbone. In this conformation the peptide cross-links are arranged so that they protrude at  $\sim 90^\circ$  to one another. In a given horizontal plane approximately 50% of the peptide stems will be cross-linked. This orientation permits interpeptide bridging between many different neighboring chains, forming a supramolecular network (mosaic) of peptidoglycan.

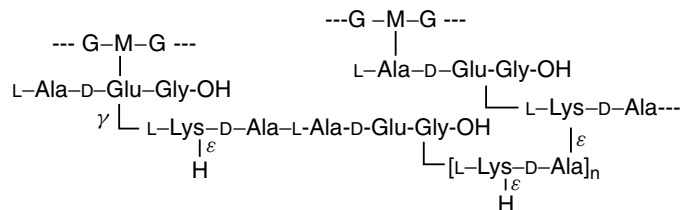




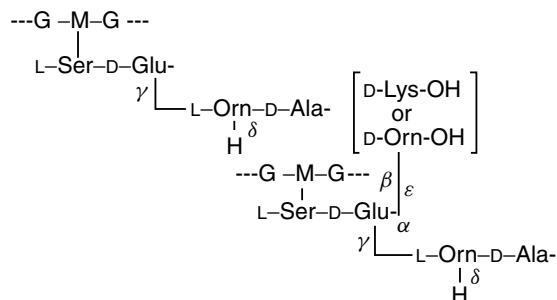
**Type I. Direct D-alanyl-R<sub>3</sub> peptide bond.** Found in *E. coli* and most other gram-negative bacteria; may be found in many bacilli.



**Type II. Pentaglycine or other L- or D-amino acid sequence.**  
This type of interpeptide bridge varies from organism to organism as shown in Table 6-3



**Type III.** Bridge composed of one to several peptides, each having the same amino acid sequence as the peptide unit attached to muramic acid; found in *Micrococcus luteus*.



**Type IV.** Bridge extending between carboxyl groups belonging to either D-alanine or to D-glutamate and a diamino acid residue or a diamino acid-containing short peptide; found in *B. rettgeri*.

**Fig. 7-12. Major types of interpeptide bridges in peptidoglycan.** G, *N*-acetylglucosamine; M, *N*-acetylmuramic acid; DAP, diaminopimelic acid. Other amino acid sequences replacing [Gly]<sub>5</sub> (pentaglycine) in type II bridges are shown in Table 6-3. (Source: Redrawn from Moat, A. G. 1985. Biology of the lactic, acetic, and propionic acid bacteria. In *Biology of Industrial Microorganism*, A. L. Demain and N. A. Solomon (Eds.). Benjamin-Cummings, Menlo Park, CA, pp. 143–188.)

**TABLE 7-3. Types of Interpeptide Bridges in Various Bacteria**

| Amino Acid(s)                               | Organism                             |
|---------------------------------------------|--------------------------------------|
| -[Gly] <sub>5</sub> -                       | <i>Staphylococcus aureus</i>         |
| -[L-Ala] <sub>3</sub> -L-Thr-               | <i>Micrococcus roseus</i>            |
| -[Gly] <sub>5</sub> -[L-Ser] <sub>2</sub> - | <i>Staphylococcus epidermidis</i>    |
| -L-Ser-L-Ala                                | <i>Lactobacillus viridescens</i>     |
| -L-Ala-L-Ala-                               | <i>Streptococcus pyogenes</i>        |
| -L-Ala-                                     | <i>Arthrobacter crystallopoietes</i> |
|                                             | <i>Enterococcus hirae</i>            |
| -Asp-MH <sub>2</sub> -                      | <i>Enterococcus faecium</i>          |
|                                             | <i>Enterococcus hirae</i>            |
|                                             | <i>Lactobacillus casei</i>           |

A fundamentally different concept of the organization of the cell wall has been presented that suggests the glycan strands run perpendicular to the plasma membrane, each strand being cross-linked by peptide bridges with four other strands. This arrangement would permit the formation of a structured matrix pierced with ordered ionophoric channels potentially harboring lipoprotein or teichoic acid in gram-negative organisms or lipoteichoic acid in gram-positive bacteria. Arrangement of the wall components in a perpendicular manner would allow murein synthesis to progress in unison with formation of the associated synthesis machinery and other multicomponent systems. Further studies should resolve the question of whether this altogether different perspective of the structure of the peptidoglycan layer represents a realistic and provable concept.

Peptide cross-bridging between peptidoglycan strands is considered to provide structural rigidity to the cell. This is certainly the case in gram-positive cells in which the multilayered network of cross-linked peptidoglycan ranges between 20 and 40 nm in thickness. In gram-negative cells the peptidoglycan layer is much thinner (2–6 nm), and in most organisms it is probably only a few monolayers in depth. This difference in thickness between gram-positive and gram-negative peptidoglycan layers is generally invoked to explain the fundamental difference in staining with crystal violet (gram stain). In gram-negative bacteria, support from other structures such as the outer membrane with its embedded proteins may provide the additional support required to withstand the turgor pressure from within the cell.

Some organisms possess cell envelopes that are intermediate between these two general types. The cyanobacteria possess envelopes with a combination of these features. Although cyanobacteria have a general structure more closely resembling gram-negative bacteria, their peptidoglycan layers range from 10 to 35 nm, depending on the species examined. The degree of cross-linking is far higher (56–63%) than the usual 20 to 33% cross-linking observed in most other gram-negative organisms. Most cyanobacteria contain the typical *mesodiaminopimelic* acid found in gram-negative bacteria as opposed to the *L*-diaminopimelic acid or *L*-lysine of gram-positive cells. As an exception, *L*-lysine is found in *Anabaena cylindrica*. With the great diversity of bacteria known to exist, it would not be surprising to find other major differences in cell structure and function as more and more organisms are examined more thoroughly.

### Peptidoglycan (Murein) Hydrolases

Almost all bacteria produce peptidoglycan (murein) hydrolases—enzymes that hydrolyze bonds in the peptidoglycan structure. These enzymes are of three basic types:

1. **Glycan-strand hydrolyzing**
  - a. *Endo-N*-acetylmuramidases
  - b. *Endo-N*-acetylglucosaminidases
2. **Endopeptidase hydrolyzing**
  - a. Peptide bonds in the interior of the peptide bridges
  - b. Bonds involving the C-terminal D-alanine residue
3. ***N*-acetylmuramoyl-L-alanine amidase** acting at the junction between the glycan strands and the peptide units

Enlargement of the peptidoglycan sacculus must require an intricate interplay of synthesizing and hydrolyzing enzymes to allow new peptidoglycan subunits to be inserted into the preexisting structure. However, direct proof of the participation of these hydrolases in the formation of the murein of the bacterial cell wall has not been fully verified. Nevertheless, these enzymes appear to play an important role in a number of cellular activities including septum and wall extension during cell growth, cell separation, turnover of wall components, sporulation, competency for transformation, and the excretion of toxins and exoenzymes.

Activation of certain of these hydrolases, particularly under conditions of external stress, may result in cellular autolysis. In fact, it is now considered that triggering of murein hydrolases or other autolysins by environmental stress is responsible for programmed cell death (sometimes referred to as apoptosis).

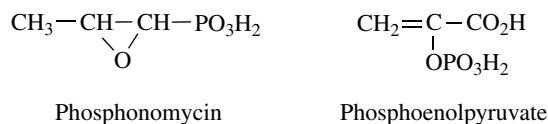
### Peptidoglycan (Murein) Synthesis

Murein biosynthesis involves a number of cytoplasmic, membrane, and periplasmic steps (Fig. 7-13). *N*-acetylglucosamine (GlcNAc) is first coupled with UDP. A portion of the UDP-GlcNAc is converted into UDP-MurNAc (*N*-acetylmuramic acid), and the peptide chain is developed by sequential addition of amino acids. The growing chain is then coupled with undecaprenyl-phosphate, enabling its transfer across the cytoplasmic membrane where it is incorporated into the growing peptidoglycan. At the interface between the growing cell wall and the cell membrane, transglycosidation reactions polymerize the growing chain and transpeptidases introduce cross-linking.

The 2 min region on the chromosome map of *E. coli* contains a large cluster of genes that code for proteins involved in various aspects of peptidoglycan synthesis and cell division. Seven genes mapping in this region (*murC*, *murD*, *murE*, *ddl*, *murF*, *mraY*, *murG*) participate in the pathway of peptidoglycan synthesis from UDP-GlcNAc to the formation of the C<sub>55</sub>-prenol intermediate undecaprenyl-PP-MurNAc-pentapeptide. However, other genes mapping at sites removed from this region also participate in murein synthesis. For example, the *E. coli* enzyme UDP-*N*-acetylglucosamine enolpyruvate transferase (reaction 4 in Fig. 7-13) catalyzes the first committed step in peptidoglycan formation. This enzyme (encoded by *murZ*, *E. coli* map position 69.3 min) is inhibited by the bactericidal antibiotic



phosphomycin (L-*cis*-1,2-epoxypropylphosphonic acid; phosphonomycin), a structural analog of phosphoenolpyruvate:

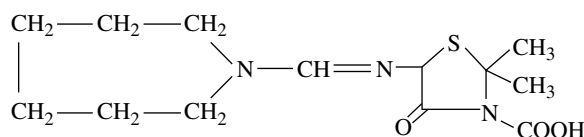


D-glutamic acid, a specific component of peptidoglycan, is added to the UDP-*N*-acetyl-muramyl-L-alanine by the product of the *murD* gene. Another gene, *murI* (*E. coli* map position 90 min), is required for the synthesis of D-glutamate from  $\alpha$ -ketoglutarate.

The *mraY* gene (2 min on the *E. coli* map) encodes the enzyme UDP-*N*-acetylmuramoyl-pentapeptide:undecaprenyl-phosphate. The *MraY* enzyme is involved in the first step of the cycle of reactions leading to the synthesis of C<sub>55</sub>-undecaprenyl intermediates that aid in transport of peptide intermediates across the cell membrane (reactions 12 to 15 in Fig. 7-13). The glycopeptide antibiotics vancomycin and ristocetin bind to the D-Ala-D-Ala terminus of peptidoglycan precursors, preventing cross-linking of adjacent strands. This action prevents the addition of *N*-acetylglucosamine-*N*-acetyl-muramoylpentapeptide-pyrophospholipid to the growing point of the peptidoglycan backbone chain. Bacitracin, a member of a group of low-molecular-weight cyclic peptides, inhibits the dephosphorylation of the lipid-P-P carrier involved in the transfer of precursors into the peptidoglycan structure (reaction 15 in Fig. 7-13). This action prevents the lipid carrier from functioning in the reaction cycle.

Muramyl pentapeptide is an integral structural element of the walls of both gram-positive and gram-negative cells. Peptidases catalyze transpeptidase reactions involving the incorporation of the terminal D-alanyl-D-alanine during the final stages of peptidoglycan biosynthesis (reaction 16 in Fig. 7-13).

Bacterial cells contain a variety of **penicillin-binding proteins (PBPs)**, as shown in Table 7-4. These membrane-bound PBPs interact covalently with penicillin and other  $\beta$ -lactam antibiotics. The five high-molecular-weight PBPs (1A, 1B, 1C, 2, 3) are involved in peptidoglycan biosynthesis. Both PBP1A and PBP1B are associated with cell elongation. PBP2 aids in determination of cell shape, and PBP3 mediates septum formation. The structural similarity between penicillin and the pentapeptide precursor of the bacterial cell wall (Fig. 7-14) allows the  $\beta$ -lactam antibiotics to interact with the active site of a transpeptidase, terminating cross-linking (reaction 16 in Fig. 7-13). Mecillinam, a  $\beta$ -lactam with an amidino side chain, as shown in the structure below, appears to have a different mode of action from that of other penicillins:



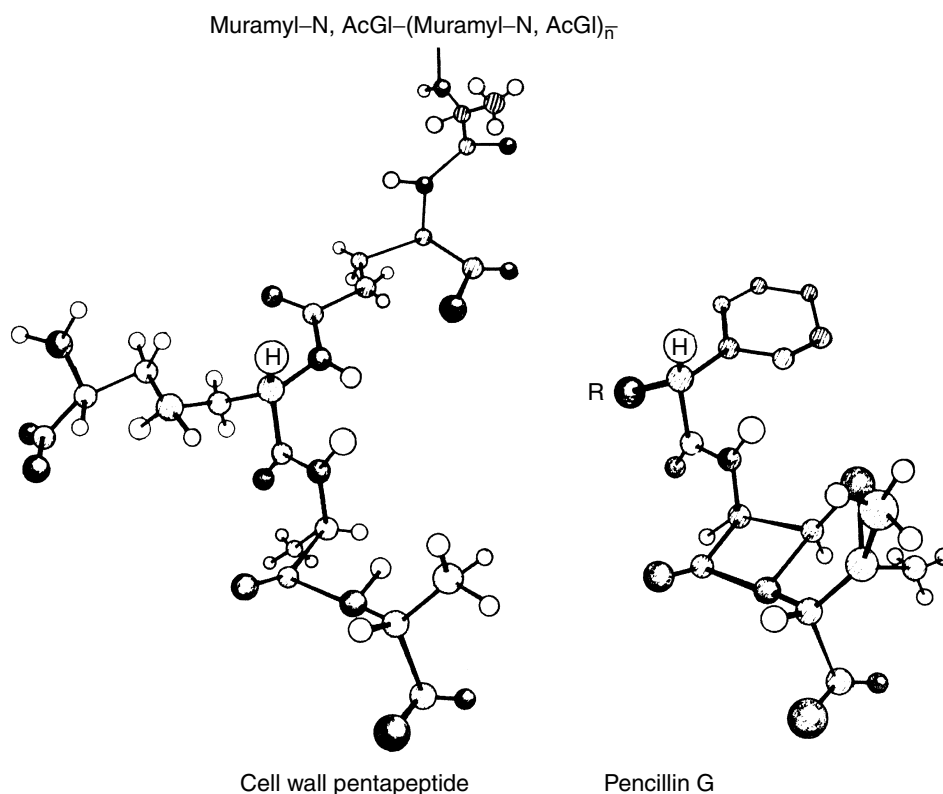
6- $\beta$ -amidino-penicillanic acid  
(Mecillinam or FL-1060)

Gram-negative organisms such as *E. coli* are greater than 100 times more sensitive to this agent than gram-positive organisms such as *S. aureus* or *B. subtilis*. Other

TABLE 7-4. Penicillin-Binding Proteins (Peptidoglycan-Synthesizing Enzymes in *E. coli*)<sup>a</sup>

| PBP No. | Gene<br>Map Position    | Synonyms, Biological Functions, Other Properties                                                                                                                                                                                                                                                                                                                    |
|---------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1A      | <i>mrcA</i><br>75.9 min | ( <i>ponA</i> ) High molecular weight; transpeptidase and transglycosylase in wall elongation; inhibition results in lysis; increase in mecillinam-resistant mutants; deficiency in both PBP1A and PBP1B required for lethality. Inhibition of DNA replication by nalidixic acid prevents lytic blockade of PBP1A and PBP1B by $\beta$ -lactams such as cefsulodin. |
| 1B      | <i>mrcB</i><br>2 min    | ( <i>ponB</i> ) High molecular weight (90,000); transpeptidase and transglycolase activity helps maintain rod shape during wall elongation; lysis triggered by $\beta$ -lactams (cefsulodin) that bind to PBP1A and PBP1B.                                                                                                                                          |
| 1C      |                         | High-molecular-weight bifunctional PBP with transpeptidase and transglycolase activity. Existence revealed by genome analysis.                                                                                                                                                                                                                                      |
| 2       | <i>mrda</i><br>14.3 min | ( <i>pbpA</i> , <i>rodB</i> ) High molecular weight; with RodA, conducts transpeptidation and transglycosylation; maintains rod shape; binds mecillinam specifically; action essential to prevent lysis in PBP1B <sup>-</sup> strains; <i>rodA</i> and <i>pbpA</i> constitute <i>rodA</i> operon.                                                                   |
| 3       | <i>ftsI</i><br>2 min    | ( <i>pbpB</i> , <i>sep</i> ) High molecular weight (63,850). Mediates septum formation; transpeptidation and transglycosylation; interacts with FtsA, FtsQ, FtsW, and RodA; helps maintain rod shape; overproduced in mecillinam-resistant ( <i>mre</i> ) strains; only transpeptidase activity is $\beta$ -lactam sensitive.                                       |
| 4       | <i>dacB</i><br>2 min    | Low molecular weight; DD-carboxypeptidase and endopeptidase; postinsertional modification of new murein.                                                                                                                                                                                                                                                            |
| 5       | <i>dacA</i><br>2 min    | Low molecular weight (41,337); major PB component; DD-carboxypeptidase removes terminal D-Ala from pentapeptide side chains; regulates peptide cross-linkage.                                                                                                                                                                                                       |
| 6       | <i>dacC</i><br>2 min    | Low molecular weight (40,804); D-alanine carboxypeptidase; PBP 5 and PBP 6 account for 85% PB; binds to CM by C-terminal end; bulk of enzyme is in periplasm; regulates peptide cross-linkage; stabilizes PG during stationary phase.                                                                                                                               |
| 6B      | <i>dacD</i><br>44.8 min | D-alanine carboxypeptidase.                                                                                                                                                                                                                                                                                                                                         |
| 7       | <i>pbpG</i><br>47.9 min | PBP8 is a proteolytic artifact of PBP7.                                                                                                                                                                                                                                                                                                                             |
| —       | <i>ampC</i><br>94.3 min | ( <i>ampA</i> ); $\beta$ -lactamase; ampicillin (penicillin) resistance; affects peptidoglycan synthesis; cell morphology.                                                                                                                                                                                                                                          |
| —       | <i>ampH</i><br>8.5 min  | $\beta$ -Lactamase; ampicillin resistance; probable role in peptidoglycan cell wall synthesis; cell morphology.                                                                                                                                                                                                                                                     |

<sup>a</sup>All proteins listed fulfill the classic definition that they bind covalently to a  $\beta$ -lactam agent.



**Fig. 7-14. Structural similarity between penicillin and the pentapeptide precursor of the bacterial cell wall.** (Source: From Butler, K., et al., *J. Inf. Dis.* **122**:Supplement S1, 1970.)

penicillins may bind with more than one PBP, but mecillinam binds specifically to PBP2 of *E. coli*, causing the formation of protoplasts without effect on the activity of other known penicillin-sensitive sites.

The action of different classes of penicillins at different sites suggests that combinations of agents from the two classes of compounds may act synergistically. Synergy has been demonstrated between mecillinam and a variety of other penicillins and cephalosporins both in vitro and in vivo. Similarly, *N*-formimidoyl thienamycin and cephoxitin exhibit synergistic killing action on *E. hirae*, suggesting at least two sites of action of these  $\beta$ -lactams. One block in the cell division cycle, induced by *N*-formimidoyl thienamycin or methicillin, occurs before the completion of chromosome replication. A second block, induced by cephoxitin or cephalothin, takes place later in the cell division cycle. Studies with *E. coli* suggest that on a quantitative basis *N*-formimidoyl thienamycin and methicillin preferentially inhibit PBP3, whereas cephoxitin and cephalothin preferentially inhibit PBP2. The synergistic action of these agents in *E. hirae* suggests a similar mode of action in this organism.

Staphylococci resistant to the action of methicillin and other  $\beta$ -lactam antibiotics produce  $\beta$ -lactamase, encoded by *blaZ*, which hydrolyzes the  $\beta$ -lactam ring. They also produce the penicillin-binding protein PBP2A, a product of the *mecA* gene. The PBP2A protein substitutes for other PBPs, but because of its low affinity for methicillin

and other  $\beta$ -lactams, it does not bind these agents at a clinically effective level. The sensor-transducer proteins BlaR1 and MecR1, and their respective repressors, BlaI and MecI, regulate transcription of *blaZ* and *mecA*. Proteolytic cleavage inactivates the BlaI repressor, resulting in derepression of the *blaZ* gene and increased  $\beta$ -lactamase production. Similarly, induction of MecR1 results in the overexpression of PBP2A. The two events render the organism resistant to  $\beta$ -lactam agents. This appears to be the first demonstration in bacteria of a transmembrane signal transmitted by sequential proteolytic events. It has been proposed that these regulatory pathways may be attractive targets for the development of new drugs to combat resistant strains of staphylococci.

Cycloserine, a compound chemically related to D-alanine, prevents the incorporation of D-alanine into muramyl pentapeptide through its inhibitory action on dipeptide synthetase and alanine racemase (reactions 10 and 11, Fig. 7-13). These combined activities result in the accumulation of mucopeptides lacking the terminal D-alanine.

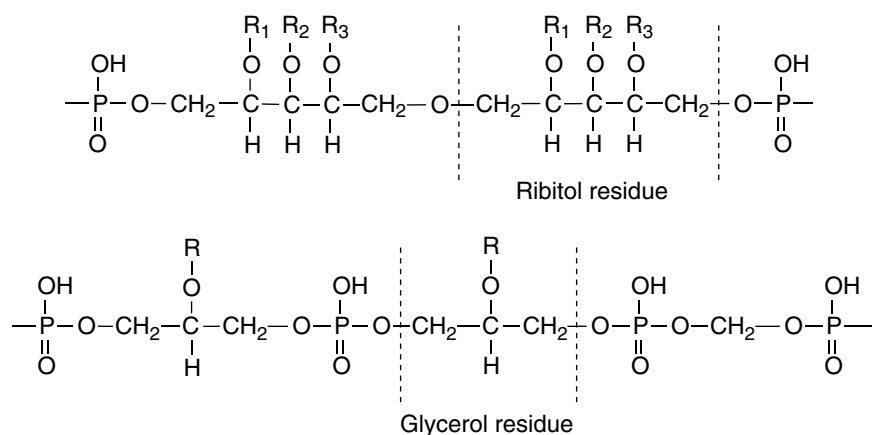
Insertion of new material into the expanding murein layer requires the activity of enzymes that degrade the glycan strand and endopeptidases that degrade the intact murein polymer. *E. coli* produces two soluble lytic transglycosylases (encoded by *slt* and *mlt*) that catalyze the cleavage of the  $\beta$ -1,4-glycosidic bond between *N*-acetylmuramic acid and *N*-acetylglucosamine. At the same time, there is an intramolecular transglycosylation at the *N*-acetylmuramic acid residue, resulting in the formation of a 1,6-anhydro bond. These enzymes apparently play a major role in recycling peptidoglycan components during expansion of the murein layer. For additional discussion regarding the role of genes involved with murein synthesis during the process of growth and cell division, see Chapter 17.

### Teichoic Acids and Lipoteichoic Acids

**Teichoic acids (TAs)** are found in all gram-positive organisms but are absent in gram-negative bacteria. Teichoic acids are polymers of either ribitol phosphate or glycerol phosphate in which the repeating units are joined together through phosphodiester linkages (Fig. 7-15). Sugars, amino sugars, or amino acids may be condensed to the hydroxyl groups of the ribitol or glycerol to provide wide variations in overall structure (Fig. 7-16). By definition, teichoic acids include all polymers containing glycerol phosphate or ribitol phosphate associated with the membrane, cell wall, or capsule. Wall TAs are covalently linked to peptidoglycan through muramic acid and the phosphate group of ribitol or glycerol phosphate. The ribitol TAs of *S. aureus* and some strains of *B. subtilis* are linked to peptidoglycan through a common linkage unit: (glycerol phosphate)<sub>3</sub>-GlcNAc. The glycerol TA of *B. cereus*, several strains of *B. subtilis*, and *B. licheniformis* are joined to peptidoglycan through a common linkage disaccharide, *N*-acetylmannosaminyl-1,4-*N*-acetylglucosamine, irrespective of the structural diversity in the glycosidic branches and backbone chains. Wall TAs are effective antigens and have been identified as the group- or type-specific substances of many organisms.

The wall ribitol TAs of *S. pneumoniae* that serve as the C antigen are more complex than those described above. Choline is present along with glucose, ribitol, phosphorus, galactoseamine, and 2,3,6-trideoxy-2,4-diaminohexose. The walls of *Lactobacillus plantarum* contain a mixture of one polymer of glucosylglycerol phosphate and two polymers of isomeric diglucosylglycerol phosphates. In *L. acidophilus* the TA is a





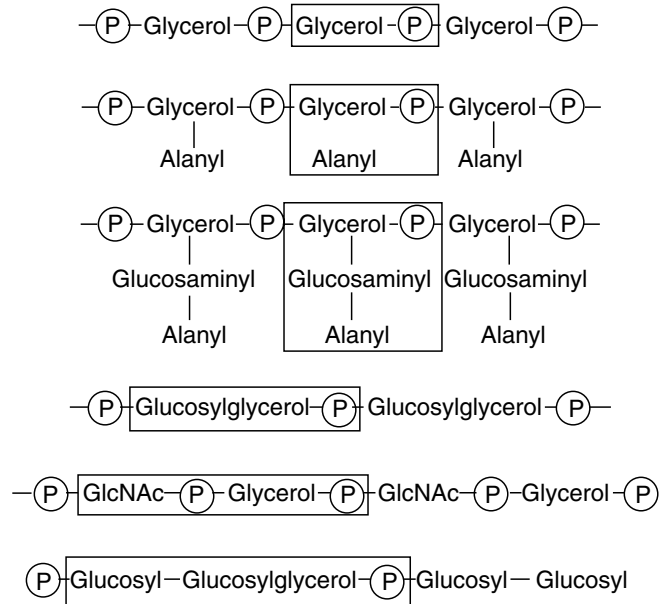
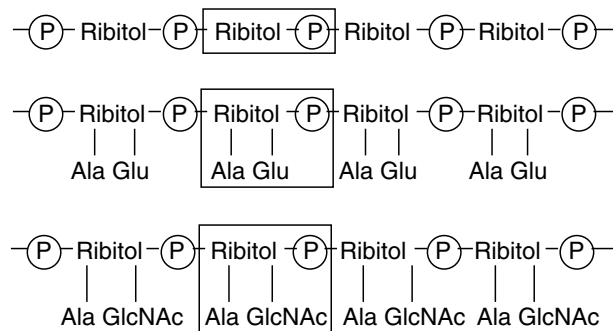
**Fig. 7-15. Teichoic acid structures.** In glycerol teichoic acid, R may be alanine, glucose, or glucosamine. In ribitol teichoic acid, R<sub>1</sub> may be glucose or *N*-acetylglucosamine; R<sub>2</sub> and/or R<sub>3</sub> may be alanine. Some of the variations in the substitutions on teichoic acids are shown in Figure 7-16.

mixture of  $\alpha$ - or  $\beta$ -linked polyglucose polymers with monomeric  $\alpha$ -glycerol phosphate side chains attached at C-2 or C-4.

Other anionic polymers that lack polyol phosphate and may or may not be present in cell walls are not, by definition, teichoic acids. Nevertheless, such components are important. For example, the group-specific polysaccharide of *S. pyogenes* group A contains phosphate, glycerol, rhamnose, and *N*-acetylglucosamine in a molar ratio of 1 : 1 : 2 : 1. Rhamnose and *N*-acetylglucosamine are incorporated from thymidine-5'-diphosphorhamnose and UDP-*N*-acetylglucosamine into the group A polysaccharide of *S. pyogenes*. Assembly of the group A polysaccharide occurs at the cell membrane with the participation of a lipid anchor or acceptor molecule. Immunological specificity of the carbohydrate group antigen of streptococci is based on the relative amounts of *N*-acetylglucosamine or *N*-acetylgalactosamine present.

**Lipoteichoic acids (LTAs)** are membrane-associated polymers characteristic of gram-positive bacteria. The LTAs are linear polymers of 16 to 40 phosphodiester-linked glycerol phosphate residues covalently linked to a membrane anchor (generally a glycolipid or glycerophospholipid). Physiological roles postulated for LTA include regulation of autolysin activity, scavenging of divalent cations such as  $Mg^{2+}$ , electromechanical properties of the cell wall, and interaction of bacteria with cells of infected hosts.

The proposed pathway for the synthesis of LTA occurs in three phases: (1) the glycolipid anchor, (2) the poly(glycerophosphate) component, and (3) the D-alanyl esters linked to poly(glycerophosphate). In *S. aureus* the diglucosyldiacylglycerol moiety of LTA functions as the membrane anchor. The gene (*ypfP*) that encodes diglucosyldiacylglycerol synthase (YpfP) has been cloned from *S. aureus* as well as from *B. subtilis*. The YpfP from *S. aureus* appears to be the only diglucosyldiacylglycerol synthase providing glycolipid for LTA assembly. The YpfP-deficient mutants of *S. aureus* display pleomorphic cells and the glycolipid anchor is replaced by diacylglycerol. When staphylococci are grown under the conditions of phosphate limitation,

Glycerol teichoic acidsRibitol teichoic acids

**Fig. 7-16. Variations in teichoic acid structures.** In each structure the repeating unit is enclosed in a rectangle. Encircled P, interconnecting phosphate group; GlcNAc, *N*-acetylglucosamine. (Source: Redrawn from Moat, A. G. 1985. Biology of the lactic, acetic, and propionic acid bacteria. In *Biology of Industrial Microorganisms*. A. L. Demain and N. A. Solomon (Eds.) Benjamin-Cummings, Menlo Park, CA, pp. 143–188.)

no wall teichoic acid is formed. However, membrane LTAs and teichuronic acids are still produced.

Membrane LTAs of the glycerophosphate polymer type occur in many gram-positive bacteria. These compounds have long polar glycerolphosphate chains linked to a small hydrophobic glycolipid. The LTA of *S. pyogenes* is a polymer-containing glycerophosphate linked to a glycerophosphoryl diglycosyl diglyceride. In *E. hirae* the glycolipid is phosphatidylkojibiosyl diglyceride. The Forssman antigen of *S. pneumoniae* is similar to LTA in that it is found in the cytoplasmic membrane and contains lipids and choline.

Lipoteichoic acids are exposed at the cell surface in many organisms. In *L. plantarum*, specific antiserum to the glycerophosphate sequence shows the label extending from the outer surface through the wall and even outside the boundary of the cell. Spontaneous release of TA and LTA has been described in streptococci and lactobacilli. The LTA of *S. pyogenes* binds via its polyanionic backbone to positively charged surface proteins termed **M proteins**. These proteins have been associated with the virulence of group A,  $\beta$ -hemolytic streptococci. This orientation leaves the lipid end free to interact with fatty acid-binding sites on host cell membranes. In the lactic acid-producing lactobacilli, *L. fermenti*, but not *L. casei*, can be agglutinated by the antisera to LTA. The long polar glycerophosphate chains of LTAs probably extend through the network of the wall to evoke an immune response.

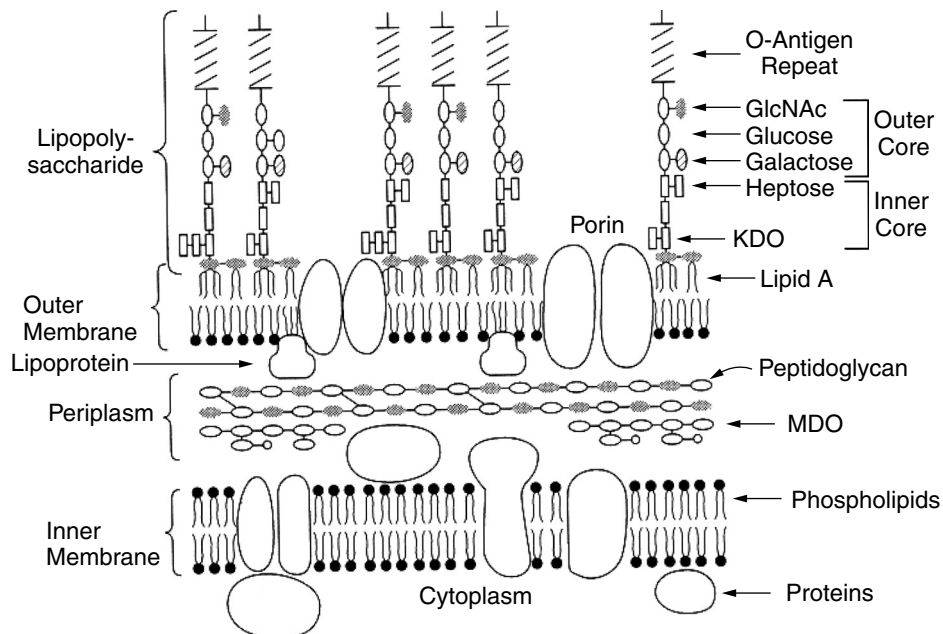
Binding of divalent cations may serve in concentrating  $Mg^{2+}$  or other ions at the cell surface. In *Lactobacillus buchneri*, one  $Mg^{2+}$  ion is bound for every two phosphate groups in the wall TA. Release of TA and LTA may influence the interaction of bacterial pathogens with cells of the invaded host. In addition to their serological reactivity, TAs may activate the alternative pathway of complement and may play a role in the specific adhesion of bacteria to host epithelial surfaces.

The biosynthesis of D-alanyl-LTA has been studied in *L. rhamnosus*, *B. subtilis*, and *S. mutans*. The D-alanyl-carrier protein (Dcp) is required for incorporation of the activated D-alanine into membrane-associated LTA. This activation and ligation of D-alanine to Dcp is catalyzed by D-alanine:Dcp ligase. Genetic analysis reveals that the proteins for D-alanine incorporation are encoded by four genes present in the *dlt* operon: *dltA*, encoding D-alanine:Dcp ligase; *dltC*, encoding Dcp; *dltB*, encoding a putative transmembrane protein involved in the secretion of activated D-alanine; and *dltD*, encoding a membrane-associated thioesterase for mischarged carrier protein. In *S. mutans*, expression of the *dlt* operon is induced in the exponential phase when cells are grown with sugars transported by the phosphoenolpyruvate sugar phosphotransferase system (PTS) but is expressed constitutively when the cells are grown with the non-PTS sugars raffinose and melibiose. Defects in D-alanyl-LTA synthesis in *S. mutans* result in sensitivity to acid. Cells of a *dltC*-defective mutant do not initiate growth below pH 6.5 and cannot survive prolonged (3 hours) exposure in medium buffered at pH 3.5.

A number of gram-positive bacteria are known to lack classical LTAs. In their place, polymers with chemical properties similar to those of LTAs have been identified. These polymers are variously referred to as macroamphiphiles, lipoglycans, or cell surface glycolipids.

### Outer Membranes of Gram-Negative Bacteria

Gram-negative bacteria display a prominent outer membrane that is peripheral to the periplasmic region and the peptidoglycan sacculus (Fig. 7-17). This outer membrane is covalently attached to the peptidoglycan layer through lipoprotein and serves to reinforce the shape of the cell and to provide a protective barrier against the external environment. Under the electron microscope the outer and inner membranes present a similar appearance as double-track layers. Chemical analysis shows that they are similar in lipid content. Some components of the outer membrane are qualitatively similar to those of the cytoplasmic membrane; however, the outer membrane appears as an asymmetric bilayer, with the external layer being composed primarily of LPS



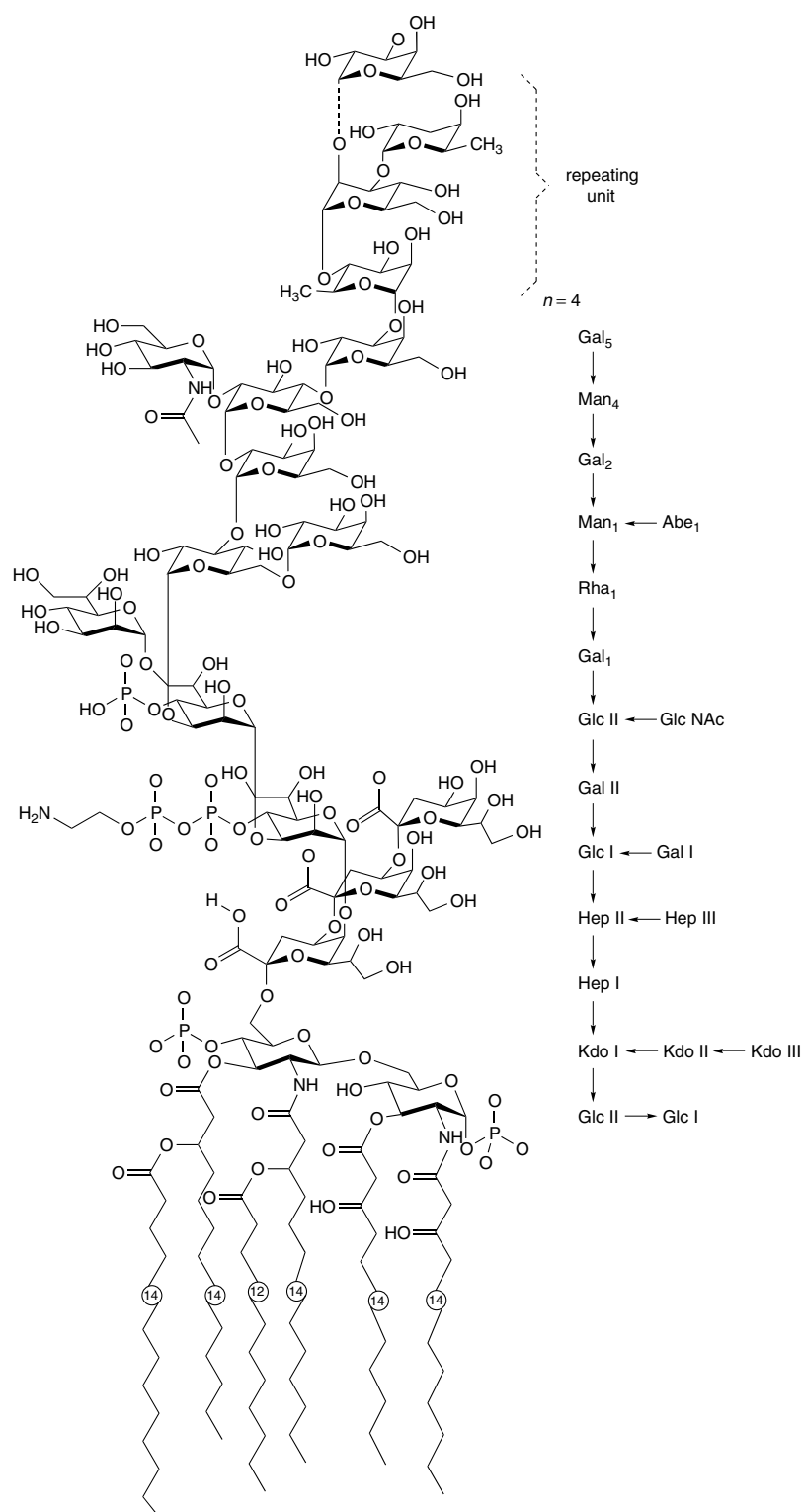
**Fig. 7-17. Diagram of gram-negative bacterial cell surface.** Ovals and rectangles represent sugar residues; circles depict the polar head groups of glycerophospholipids (phosphatidylethanolamine and phosphatidylglycerol). MDO = membrane-derived oligosaccharides. The core region of LPS is that of *E. coli* K-12, a strain that does not normally contain an O-antigen repeat unless transformed with an appropriate plasmid. (Source: From Raetz, C. R. H. *J. Bacteriol.* **175**:5745–5753, 1993.)

and the inner layer containing primarily phospholipids (Fig. 7-17). **Outer membrane proteins (OMPs)**, called porins, form large water-filled pores with diameters of 1 to 2 nm that traverse the membrane and regulate the access of hydrophilic solutes to the cytoplasmic membrane.

**Lipopolysaccharides.** Lipopolysaccharides consist of three basic components or regions, as shown in Figures 7-17 and 7-18. **Region I**, the outermost portion, contains repeating carbohydrate units that represent the “O” antigen. Alteration in the sugar composition of the O antigen results in a change in the immunological specificity. The sugars found in the O-antigen region can occur in a wide variety of combinations, accounting for tremendous antigenic diversity and many hundreds of chemical types or serotypes of *Salmonella*, *Shigella*, and other *Enterobacteriaceae*.

The **core region (region II)** consists of an outer and an inner core. The outer core shows high-to-moderate structural variability, whereas the inner core shows very low

**Fig. 7-18. Covalent structure of a model LPS.** The O-specific chain (region I) consists of four repeating units and a terminal Gal residue. A hexaacyl lipid A component is shown, and the two main chain heptoses are substituted in position 4 by phosphate and 2-amino-ethylpyrophosphate, respectively. On the right the designations of each of the saccharide residues are given. (Source: From Kastowsky, M., et al., *J. Bacteriol.* **174**:4798–4806, 1992.)



structural variability, particularly within a very closely related group of organisms, for example, the *Salmonella*. The oligosaccharide subunits of the core region of *E. coli* and *Shigella* differ only slightly from those of *Salmonella*. In other gram-negative bacteria less closely related to the enteric bacteria, a greater diversity in the structure of the core region may be encountered. However, the unique octose sugar, 2-keto-3-deoxyoctulosonic acid (KDO), appears to be a common component of the core region of most gram-negative organisms.

**Lipid A (region III)**, embedded in the outer membrane, has been extensively studied in *Salmonella*. In this organism, the chemical composition of lipid A has been shown to consist of a chain of D-glucosamine disaccharide units with all of the hydroxyl groups substituted. The substituents are the core polysaccharide units on the one hand and long-chain fatty acids on the other. The most commonly observed fatty acid is  $\beta$ -hydroxymyristic acid (3-hydroxy-tetradecanoic acid), a C<sub>14</sub> saturated fatty acid that is substituted on the amino groups.

The hydroxyl groups are also esterified with other long-chain fatty acids such as lauric, myristic, and palmitic acids. The structure shown in Figure 7-18 is a monomeric unit of LPS. An average of three such subunits are linked together through pyrophosphate bridges between the lipid A molecules. The exact manner in which the LPS is linked to other surface structures is uncertain. However, LPS is very tightly associated with OMPs embedded in the outer membrane, particularly OmpA. The OmpA contributes to the stability of the outer membrane since it spans the membrane and is cross-linked to the underlying peptidoglycan layer. OmpA is exposed at the surface, where it also serves as a receptor for T-even phage and plays a role in conjugation and the action of colicins K and L.

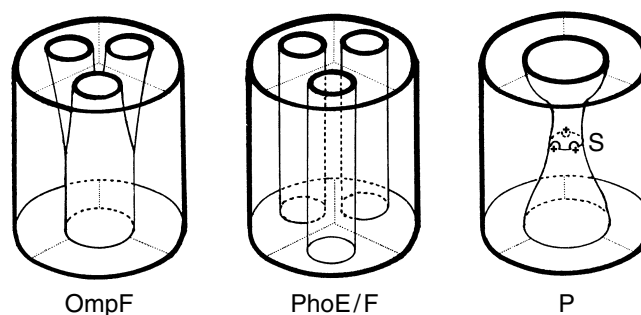
**Porins.** A variety of other OMPs are found in the *Enterobacteriaceae* and other gram-negative bacteria. As shown in Table 7-5, these OMPs play a variety of overlapping roles in the physiology of the cell. Some of these proteins facilitate the entry of specific metabolites such as vitamin B<sub>12</sub>, iron, or maltose, while others, such as OmpC and OmpF, constitute components of general porins that allow hydrophilic solutes of a molecular weight less than 700 to traverse the outer membrane. Both OmpF and OmpC have similar functional and structural properties, but expression of their structural genes, *ompF* and *ompC*, is regulated in opposite directions by the osmolarity of the growth medium.

As shown in Figure 7-19, the channel-forming porin trimers of *E. coli* span the outer membrane. Electron microscopy and image reconstruction techniques show that the three channels of OmpF on the outer surface merge into a single channel at the periplasmic face. By comparison, the PhoE porin of *E. coli* forms three separate channels that traverse the width of the membrane. *Pseudomonas aeruginosa*, on the other hand, forms a single, small, highly anion-selective channel (OmpP) in which the permeability is related to the presence of a selectivity filter (S) containing three positively charged lysine molecules. A study of 12 different porins from *E. coli*, *P. aeruginosa*, and *Yersinia pestis* revealed that most porins appear to be cation selective. Only 3 of the 12 showed anion selectivity.

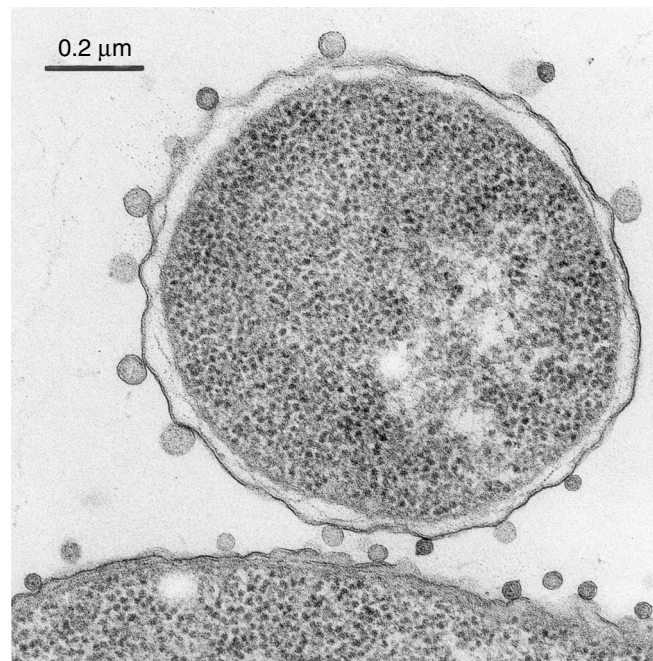
In *E. coli*, a major OMP, LamB, serves as a receptor for bacteriophage  $\lambda$  (Chapter 6). Production of this protein is induced by growth on maltose and is involved in the passage of maltose and maltose-containing oligosaccharides through the outer membrane. Immunoelectron microscopy of newly induced LamB at the surface of *E. coli* reveals that LamB is inserted homogeneously over the entire surface of the cell.

TABLE 7-5. Outer Membrane Proteins of Gram-Negative Bacteria

| Protein                                     | Functions                                                                                                                                            |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| OmpA                                        | Stabilization of outer membrane and mating aggregates in F-dependent conjugation; receptor for phage TuII                                            |
| Murein lipoprotein<br>(Braun's lipoprotein) | Most abundant surface protein in <i>E. coli</i> and <i>S. enterica</i> ; major structural protein; in conjunction with OmpA, stabilizes cell surface |
| OmpB (porin)                                | Diffusion channel for various metabolites including maltose                                                                                          |
| LamB (malto porin)                          | Specific porin for maltose and maltodextrin; receptor for bacteriophage $\lambda$                                                                    |
| OmpC (porin)                                | Diffusion channel for small molecules; receptor for phages TuIb and T4                                                                               |
| OmpF (porin)                                | Diffusion channel for small molecules; receptor for phages Tula and T2                                                                               |
| OmpT                                        | Protease                                                                                                                                             |
| PhoE (protein E)                            | Anion-selective diffusion channels induced under phosphate limitation                                                                                |
| Protein P                                   | Anion-selective diffusion channel in <i>Pseudomonas aeruginosa</i> ; induced under phosphate limitation                                              |
| TolA                                        | Maintenance of OM integrity; activity of group A colicins                                                                                            |
| TonA                                        | Ferrichrome siderophore uptake; receptor for phages T1, T5, 80, and colicin M                                                                        |
| TonB                                        | Siderophore-mediated iron transport; B <sub>12</sub> transport                                                                                       |
| Tsx                                         | Nucleoside-specific channel; receptor for T-even phages and colicin K                                                                                |



**Fig. 7-19. Schematic three-dimensional representations of the structures of three different porin proteins.** The proteins are oriented such that the top of the figure is the portion exposed to the external environment, whereas the bottom is the portion extending into the periplasmic space. The channels traverse the width of the outer membrane. The OmpF porin of *E. coli* forms coalescing channels. The PhoE porin of *E. coli* and the protein F porin of *Pseudomonas aeruginosa* form three separate and distinct channels. Protein P of *P. aeruginosa* forms a single anion-specific channel containing a selectivity filter (S) consisting of three charged lysine molecules (+). (Source: From Hancock, R. E. W., *J. Bacteriol.* **169**:929–933, 1987.)



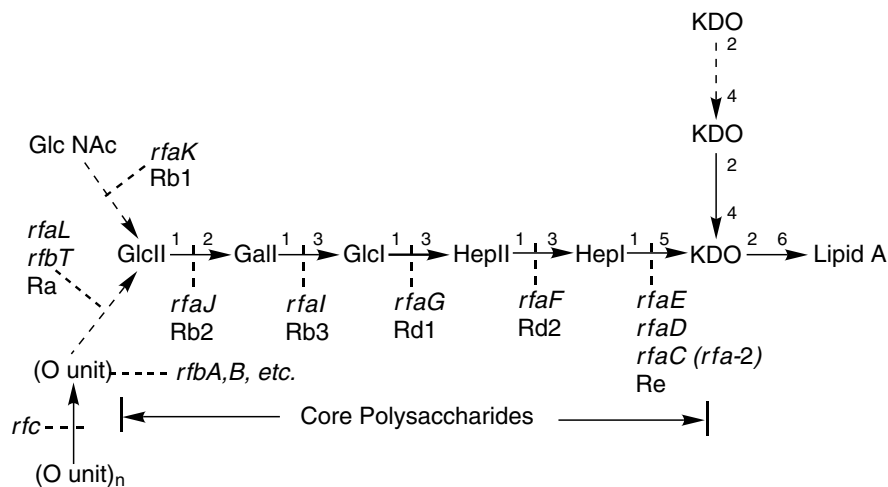
**Fig. 7-20.** Effect of loss of outer membrane lipoprotein (Lpp) and OmpA on the cell envelope and shape of *E. coli*. (Source: From Sonntag, I., et al. *J. Bacteriol.* **136**:280, 1978.)

**Murein lipoprotein** is a major OMP present in large quantities in *E. coli*. Lipoprotein molecules serve to anchor the outer membrane to the peptidoglycan layer. Mutants lacking *lpp*, the structural gene for lipoprotein, and *ompA*, the structural gene for OmpA, display spherical morphology and abundant blebbing of the outer membrane (Fig. 7-20). In these mutants the murein layer is no longer associated with the outer membrane. These mutants also display an increased growth requirement for  $Mg^{2+}$  and  $Ca^{2+}$  and are sensitive to hydrophobic antibiotics such as novobiocin, or to detergents, suggesting a protective function of the outer membrane. For additional discussion of the role of porins in nutrient transport, see Chapter 9.

### Lipopolysaccharide Biosynthesis

The three components of LPS (**O-antigen**, **core oligosaccharide**, and **lipid A**) are synthesized independently of each other and later ligated in or on the inner membrane. After assembly, the intact LPS is translocated to the outer membrane. The LPS mutant strains of *S. enterica* and of *E. coli* K-12 have been isolated and their LPS structures defined in terms of polysaccharide composition as shown in Figure 7-21. Most of the genetic determinants responsible for oligosaccharide core biosynthesis in *E. coli* and *S. enterica* have been shown to reside in the *rfa* gene cluster. The genes coding for the five transferases (*rfaKJIGB*) required for assembly of the outer core region of *E. coli* LPS have been identified as shown in Figure 7-21. Similarly, genes coding for enzymes responsible for inner core biosynthesis (*rfaC*, *rfaD*) have been identified. The *rfaD* gene codes for ADP-L-glycero-D-mannoheptose-6-epimerase. Mutation in the





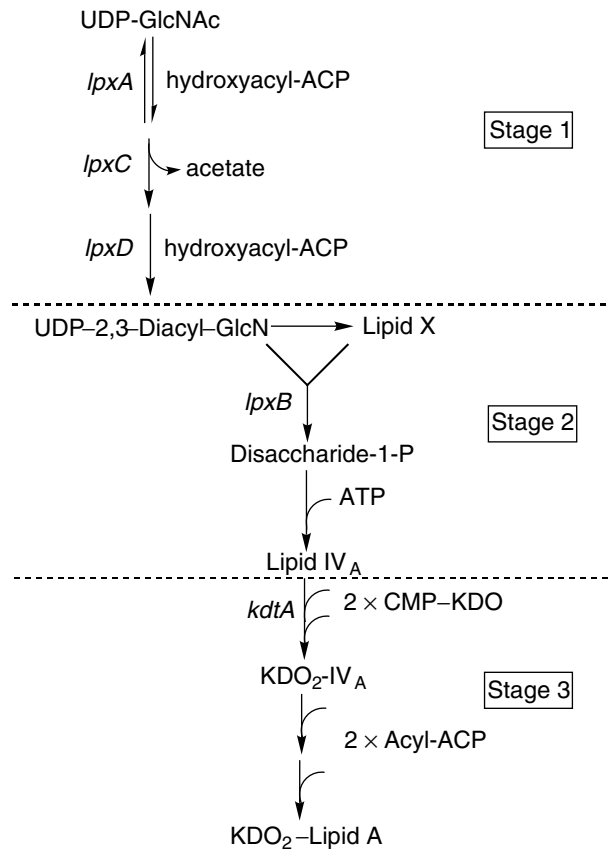
**Fig. 7-21. Schematic representation of the genetic determinants of the LPS of *S. enterica* serovar Typhimurium.** The genes (*rfaBCDEFGHIJK*) and the LPS structures (chemotypes Re, Rd2, Rd1, Rc, Rb3, Rb2, and Ra) of mutants blocked at various stages of LPS biosynthesis are indicated. The dotted lines indicate the defective LPS termination points. KDO, 2-keto-3-deoxyoctulosonic acid; Hep, L-glycero-D-mannoheptose; Glc, glucose; Gal, galactose; GlcNAc, *N*-acetylglucosamine; (O unit)<sub>n</sub>, number of O-antigen side chains. The structural genes presumed to be responsible for LPS core biosynthesis are as follows: *rfaE*, specific function unknown; *rfaC*, ADP-heptose:LPS heptosyltransferase 1; *rfaD*, UDP-glucose:LPS glycosyl-transferase 1; *rfaB*, UDP-galactose:LPS- $\alpha$ -1,3-galactosyltransferase; *rfaJ*, UDP-galactose:LPS glucosyltransferase 1; *rfaK*, UDP-*N*-acetylglucosamine:LPS glucosaminyltransferase. (Source: From Chen, L., and W. G. Coleman, Jr. *J. Bacteriol.* **175**: 2534–2540, 1993.)

*rfaC* gene results in production of a heptoseless LPS structure referred to as chemotype Re. These mutants display increased permeability to both hydrophobic and hydrophilic agents. In *S. enterica*, the *rfaL* gene encodes a component of the O-antigen ligase and *rfaK* encodes the *N*-acetylglucosamine transferase. The order of genes within the *rfa* cluster at 79 units on the *S. enterica* linkage map has been shown to be *cysE-rfaDFCLKZYJIBG-pyrE*.

The major pathway leading to lipid A biosynthesis takes place in three stages: stage 1, UDP-GlcNAc acylation; stage 2, disaccharide formation and 4' kinase action; and stage 3, KDO transfer and late acylation. The genes and gene products involved in this pathway are shown along with the intermediates in Figure 7-21. UDP-2,3-diacylglycosamine plays a key role in lipid A biosynthesis, as shown in Figures 7-22 and 7-23.

### Enterobacterial Common Antigen

Enterobacterial common antigen (ECA) is a cell surface glycolipid synthesized by all members of the *Enterobacteriaceae*. It is a heteropolymer containing *N*-acetyl-D-glucosamine (GlcNAc), *N*-acetyl-D-mannosaminuronic acid (ManNAcA) and 4-acetamido-4,6-dideoxy-D-galactose (Fuc4NAc) linked together to form chains of trisaccharide repeat units. Enzymes encoded by genes in the *rfe* and *rfb* gene clusters,

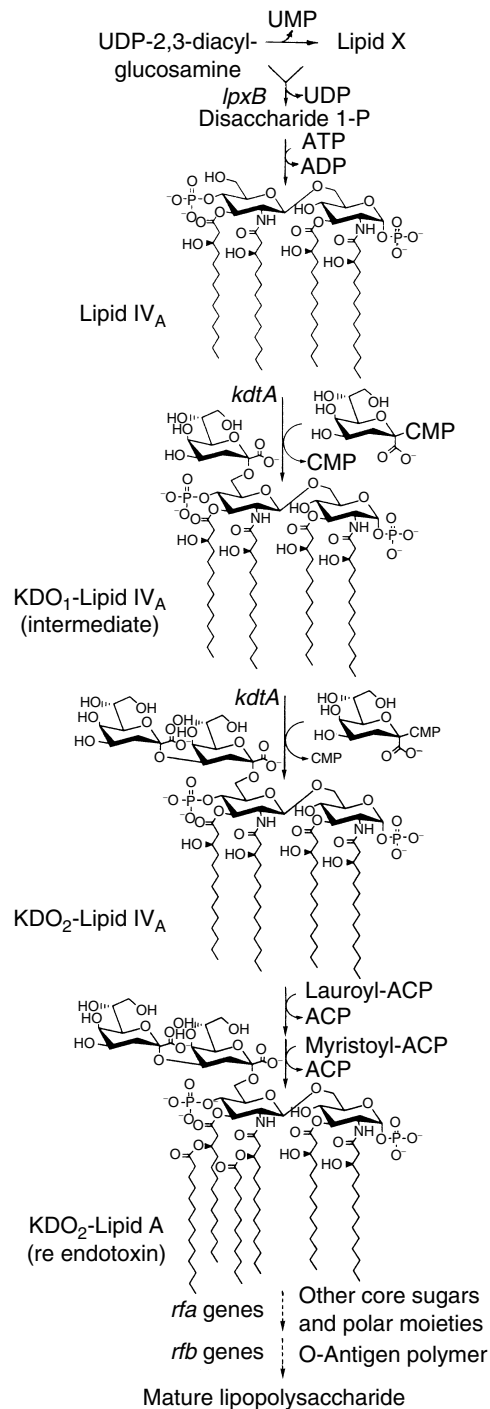


**Fig. 7-22. Genes and enzymes involved in lipid A biosynthesis in *E. coli*.** The three stages of the pathway are stage 1, UDP-GlcNAc acylation; stage 2, disaccharide formation and 4' kinase action; and stage 3, KDO transfer and late acylation. (Source: Redrawn from Raetz, C. R. H. *J. Bacteriol.* **175**:5745–5753, 1993.)

as well as the *rff* genes, catalyze the biosynthesis of ECA. The *rfe* and *rfb* gene clusters are also involved in LPS biosynthesis. This anomaly is explained by the fact that common intermediates are required for both LPS and ECA synthesis.

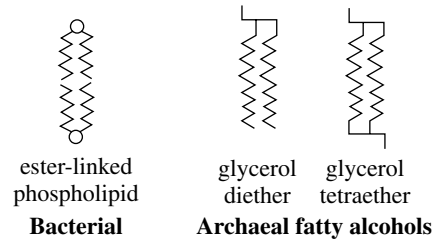
## CYTOPLASMIC MEMBRANES

The cytoplasmic membrane of bacterial cells is a bimolecular lipid leaflet of phospholipid molecules aligned at their hydrophobic ends. The polar phospholipids are hydrophilic and face the external or outside of the membrane and the internal or cytoplasmic side of the membrane. Various proteins and other components of the cell may be partially or wholly embedded in the membrane layer (Fig. 7-17). The cell membrane serves as a permeability barrier, preventing most solutes from gaining entrance to the cytoplasm except through specific transport proteins present in the membrane layer.



**Fig. 7-23. Pathway of lipid A biosynthesis.** The *lpxB* and *kdtA* genes code for the lipid A disaccharide synthase and KDO transferase. In wild-type cells, the level of the intermediates is very low (<1000 molecules per cell). (Source: From Raetz, A. *J. Bacteriol.* **175**:5745–5753, 1993.)

The phospholipids of bacteria contain ester-linked phospholipids whereas the archaea contain fatty alcohols linked to glycerol via either diether or tetraether bonds:



Further discussion of the structure of ester-linked phospholipids and the diether- or tetraether-linked fatty alcohols can be found in Chapter 13.

Removal of the outer membrane and the peptidoglycan layer by the action of various chemical and enzymatic agents does not alter the integrity of the cell with respect to retention of its internal contents. Thus, the inner membrane of gram-negative cells serves the same function as the cytoplasmic membrane of gram-positive cells. The cytoplasmic membrane of bacteria and extensions of it assume many of the functions attributable to specialized organelles in eukaryotic cells. Systems governing the active transport of metabolites into and out of the cell, oxidative phosphorylation, cell wall biosynthesis, phospholipid biosynthesis, and the secretion of various extracellular enzymes and other proteins have all been associated with the cell membrane. In addition, anchoring of DNA to the membrane to aid its distribution to the daughter cells during cell division has also been described. In gram-positive cells, lipoteichoic acids are anchored hydrophobically in the cytoplasmic membrane. A discussion of the synthesis of the lipid and phospholipid components of the cell membrane can be found in Chapter 13.

Using any of the enzymes that selectively degrade the peptidoglycan and digest away the wall structure allows isolation of the cell membrane of gram-positive bacteria. If cells are treated with lysins while suspended in a solute to which the cell is impermeable (e.g., sucrose) at a concentration that approximately balances the high internal osmotic pressure, then an osmotically fragile body (**protoplast**) is formed. The protoplast can then be lysed osmotically and the membrane isolated. The cytoplasmic membrane contains numerous enzymes that perform a myriad of functions. Very few enzymes are located in the outer membrane of gram-negative bacteria. Terminal electron transport enzymes and metabolite transport functions are found in the cytoplasmic membrane. Only those compounds of a molecular weight less than 700 are aided in their passage through the outer membrane by porins.

Initially, it was not an easy task to identify the cell wall and cell membrane as distinct entities. The ability of cells to remain intact as protoplasts in an osmotically stable environment after the cell wall was removed by treatment with lytic agents provided the most convincing evidence that the wall and membrane were separate structures. Protoplasts are capable of conducting most, if not all, of the various activities attributable to intact cells (e.g., enzyme synthesis, various metabolic activities, bacteriophage synthesis, and spore formation) and are even capable of multiplication in a properly controlled osmotic environment. Some organisms, such as the *Mycoplasma*, exist as stable forms in nature without an outer cell wall. Sterols serve to stabilize the membrane of these organisms. All of the available evidence indicates that the cell

membrane is the structure responsible for retention of the intracellular contents and controls the entrance and exit of metabolites.

Electron micrographs of gram-negative cells reveal two unit membranes: an outer membrane composed of phospholipids, protein, and LPS and an inner membrane composed of phospholipids, glycolipids, and phosphatidylglycolipids. Embedded in the lipid bilayer are various proteins and lipoproteins that function in metabolite transport and the synthesis of macromolecular constituents of the cell wall and the outer membrane. Certain proteins serve to maintain the integrity of the wall–membrane complex, while others provide protection against phagocytosis or act as attachment sites for bacteriophage or chemical agents such as antibiotics.

### Permeability and Transport

Control mechanisms that regulate the transport of substances into and out of the cell have been thoroughly investigated and we now understand the permeability properties of the membrane and its components reasonably well. At one time, cell physiologists regarded the cytoplasmic membrane as an osmotic barrier equivalent to a collodion membrane. Metabolites were considered to enter or leave the cell primarily on the basis of the concentration gradient of metabolites from one side of the membrane to the other. Membranes were considered to have pores of finite size that permitted penetration of molecules into the cell. Size of the molecule, the relative internal and external concentrations, the charge distribution of the membrane and solute, and the pH and ionic strength of the internal and external environments were considered to be the primary factors governing the entrance of compounds into the cell. While all of these factors have their influences on the passage of materials into and out of the cell, it is now well established that cells have highly specialized transport proteins either inserted in the cytoplasmic membrane or occasionally found in the periplasmic region of gram-negative bacteria. The function of these transport systems in the movement of metabolites into and out of the cell is considered in detail in Chapter 9.

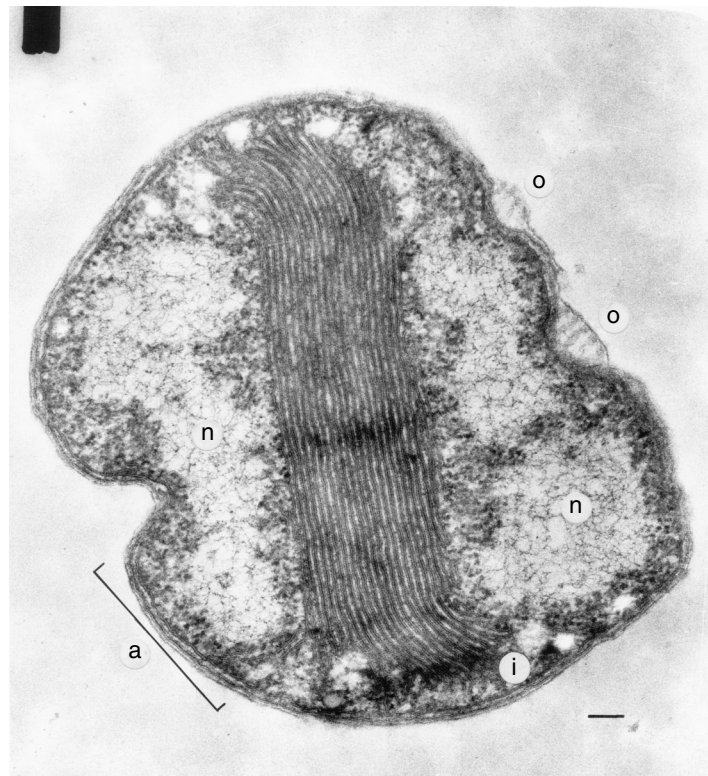
### Periplasm

In gram-negative bacteria a region that includes the peptidoglycan layer separates the inner and outer membranes. This region is the periplasmic space. Certain types of cytological evidence suggest that there is no truly empty space between the outer and inner membranes. Peptidoglycan in a hydrated state represents a gel that fills the entire space between the two membranes. The degree of cross-linking of the peptidoglycan probably diminishes toward the cytoplasmic membrane, resulting in the formation of a **periplasmic gel** in which proteins are freely diffusible. This periplasmic region has been estimated to comprise approximately 10% of the total cell volume. Compression of this gel by a sudden reduction of the external osmotic pressure (osmotic shock) results in the abrupt release of enzymes and other periplasmic constituents. Periplasmic proteins include binding proteins for amino acids, carbohydrates, and vitamins; a variety of enzymes; and a number of other proteins with diverse functions. In addition to their function in transport, periplasmic-binding proteins are involved in the chemotactic response to attractants and repellants in the environment (see further discussion in the section on chemotaxis later in this chapter).

### Other Membranous Organelles

Under the electron microscope certain organisms have been shown to contain membranous structures other than the cytoplasmic membrane. Membranous structures that appear to be sac-like invaginations of the cytoplasmic membrane of bacteria, termed **mesosomes**, have been observed with considerable regularity. Originally, a variety of physiological functions were attributed to mesosomes. However, these structures have been shown to vary in size and appearance depending on the fixation procedures used in preparing cells for electron microscopy. As a result, some consider these structures to be artifacts that can be manipulated by changes in fixatives, and much of the earlier work on mesosomes has been discredited. Nevertheless, these invaginated structures may represent functionally different membranous entities that will require further study before they are dismissed as artifacts of chemical fixation procedures.

The marine bacterium, *Nitrocystis oceanus*, possesses laminar membranous organelles (Fig. 7-24). These membranous organelles consist of approximately 20 vesicles so flattened that the lumen is only 100 Å thick. The outer surfaces are in contact



**Fig. 7-24. Cross-section of a marine bacterium, *Nitrocystis oceanus*.** The electron micrograph shows the cell wall and plasma membrane (area a), wall organelles (o), nuclear material (n), and the surrounding ribosome-packed cytoplasm. The lamellae of the membranous organelle traverse the entire cell. Other inclusions (i) may be seen within the cytoplasm. Bar equals 0.1  $\mu$ m. (Source: From Murray, R. G. E. and S. W. Watson, *J. Bacteriol.* **89**:1594, 1974.)

and form a triplet structure with an accentuated center line; these lamellae almost traverse the cell, displace the cytoplasm and the nuclear material, and form the most prominent cytological feature of the cell. *Nitrosomonas europaea* and other nitrifying organisms also produce distinctive membrane systems that differ significantly from one another. The presence of these lamellar organelles in nitrifying organisms (organisms that convert  $\text{NH}_4^+$  to  $\text{NO}_2^-$  and  $\text{NO}_3^-$ ) must somehow be related to the specialized mechanisms for acquiring energy from this process. However, the association of the intricate membrane systems observed in these organisms with the actual oxidation process is largely assumptive. It has been considered that these lamellar membranous organelles may be equivalent to mesosomes, plasmalemmasomes, or chondroids that have been studied in other kinds of bacteria, but their distinctive appearance and high degree of organization suggest that they are unique structures that may ultimately be directly associated with the oxidative processes of the autotrophic nitrifying bacteria.

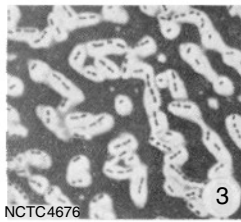
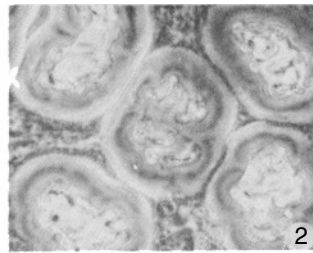
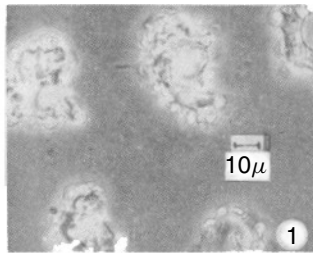
## CAPSULES

Many microorganisms produce an external layer of mucoid polysaccharide or polypeptide material that adheres to the cell with sufficient tenacity to be observed by simple techniques such as negative staining with India ink (Fig. 7-25). When this outer layer of material reaches sufficient size to be readily visible, it is referred to as a **capsule**. Several examples of well-defined capsules are shown in Figure 7-25. Many microorganisms produce a variety of extracellular materials that may be too sparse or too soluble in the medium to be observed by microscopic techniques. These materials are often referred to as a **slime layer**. The production of capsules or slime layers may impart the ability to form biofilms that enhance their ability to attach to surfaces.

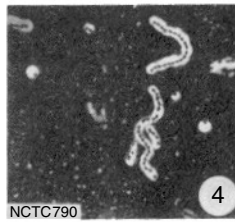
The extracellular polysaccharide colanic acid or M antigen produced by *E. coli* strains and other enteric bacteria (Fig. 7-26) is considered to be a slime layer. Most of these surface components can be separated from the underlying cell wall without impairing the function or structural integrity of the cell or its viability. This is not to say that the properties of the cell may not be altered in some manner by removal of the material. For example, strains of *Streptococcus pneumoniae* that fail to produce the polysaccharide capsular material are essentially avirulent. Similarly, the D-polypeptide capsule markedly increases the virulence of the anthrax bacillus (*Bacillus anthracis*).

From time to time, other materials have been shown to be associated with the outer surface of the cell. However, these materials are usually considered to be transient metabolic excretions or chance adherents acquired from the environment and do not represent true capsular material. True capsules are generally composed of high-molecular-weight components, are highly viscous materials, and stain poorly, if at all, with the usual stains. The best staining procedures employ mordants that cause the precipitation of the capsular material by metal ions, alcohol, acetic acid, and so on. Reaction of the capsule with specific antiserum is frequently employed to enlarge the capsular area so that it may be more readily visualized under the microscope (Figs. 7-25.6 and 7-25.7). Techniques have also been developed for visualization of the capsule under the electron microscope (Figs. 7-25.8 and 7-25.9).

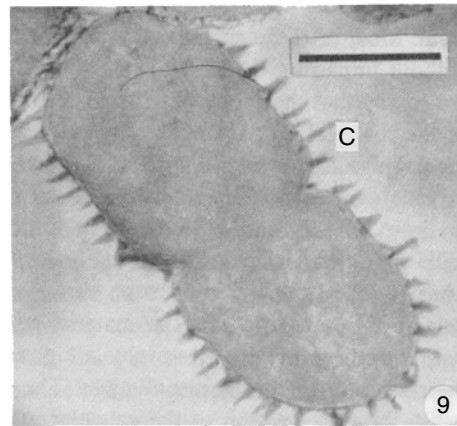
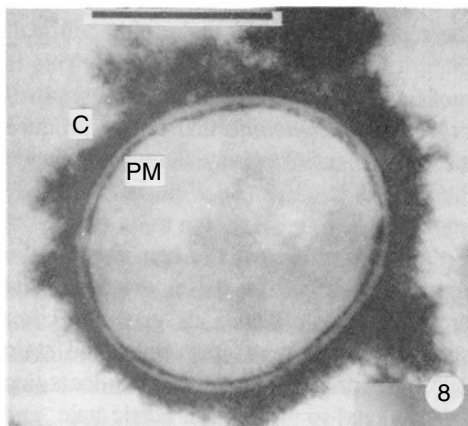
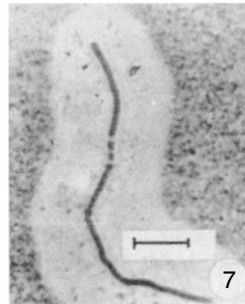
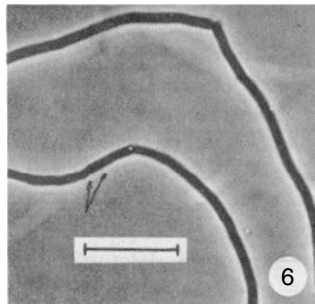
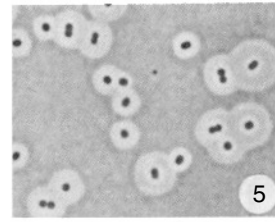
The polysaccharide capsular substances produced by *S. pneumoniae* confer immunological specificity and are also associated with virulence. Ninety distinct capsular serotypes, each differing in sugar composition and/or linkages, have been recognized



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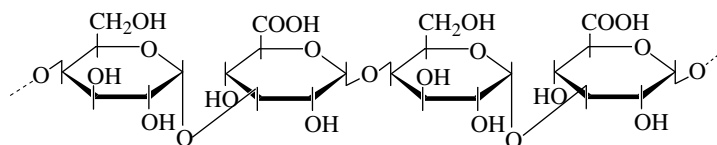


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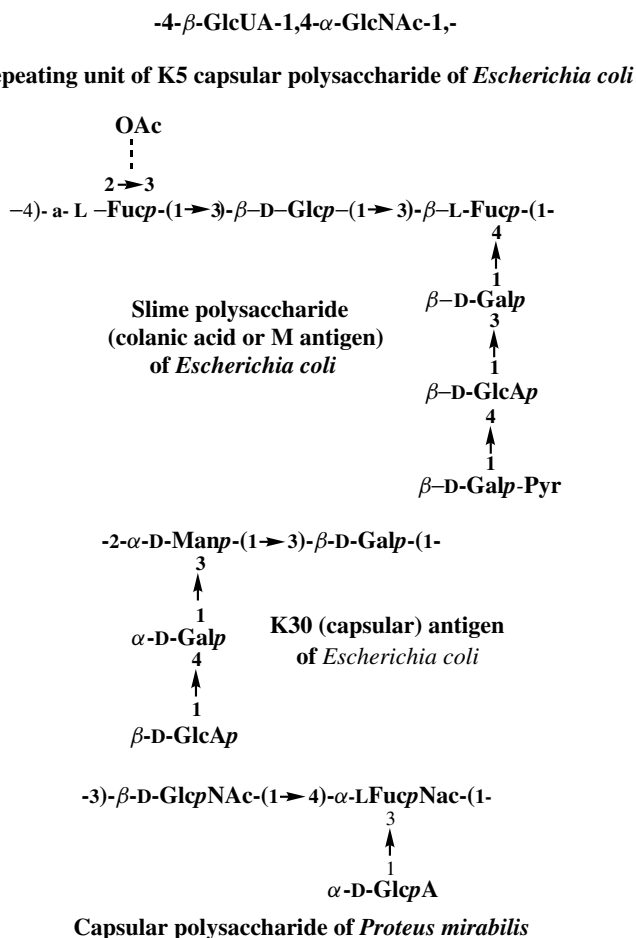
and the structures of a large number of them have been determined. Type 3 pneumococcal polysaccharide is composed of glucopyranose and glucuronic acid in alternating  $\beta$ -1,3- and  $\beta$ -1,4-linkages:



A given *S. pneumoniae* isolate expresses only one capsular polysaccharide, and the genetic basis of this unique attribute lies in the set of genes encoding specific biosynthetic reactions for specific polysaccharides. Genetic exchange of the genes from a donor to a recipient strain results in the replacement of the recipient's serotype-specific genes with those of the donor and subsequent expression of the donor capsular polysaccharide.

Synthesis of the type 3 capsular polysaccharide of *S. pneumoniae* requires UDP-glucose (UDP-Glc) and UDP-glucuronic acid (UDP-GlcUA) for production of the  $[3\text{-}\beta\text{-D-GlcUA-(1,4)-}\beta\text{-D-Glc-(1}\rightarrow\text{)}_n]$  polymer. The generation of UDP-Glc proceeds by conversion of Glc-6-P to Glc-1-P to UDP-Glc and is mediated by a phosphoglucomutase (PGM) and a Glc-1-P uridylyltransferase, respectively. Genes encoding both a Glc-1-P uridylyltransferase (*cps3U*) and a PGM homolog (*cps3M*) are present in the type 3 capsule locus, but these genes are not essential for capsule production. Spontaneous mutation in a distant gene (*pgm*, encoding a second PGM homolog) resulted in a 4-fold reduction in the amount of capsular material produced as compared with the

**Fig. 7-25. Representative examples of encapsulated microorganisms.** (1 and 2) Dark-phase-contrast photomicrographs of a pond alga. In (2) skimmed milk run under the coverslip permits visualization of the capsule as a broad hyaline periphery that is virtually unseen in (1). Bar equals 1  $\mu\text{m}$  in (1) and (2). (Source: From Dondero, N., *J. Bacteriol.* **85**:1171, 1963.) (3 and 4) Hyaluronic acid containing capsules of group C streptococci. The dense capsules are visualized on plates of moist 20% horse serum in 1% glucose agar with India ink background stain. The capsules are absent from smears of hyaluronidase cultures. (Source: From MacLennan, A. P. *J. Gen. Microbiol.* **15**:485, 1956.) (5) Phase-contrast photomicrograph of a gram-negative, capsule-forming coccus. These capsules were formed after the organisms had been stripped of their original capsules. The presence of oxygen and an oxidizable carbon source were the only requisites for resynthesis of the capsular polysaccharide. (Source: From Juni, E., and G. A. Heym, *J. Bacteriol.* **87**:461, 1964.) (6 and 7) Phase-contrast photomicrographs of *B. anthracis* encapsulated cells. Bars in (6) and (7) equal 10  $\mu\text{m}$ . (6) Membrane-like outline of capsule in untreated cells. (7) Same cells after treatment with antiserum. Negative staining with India ink. Note the marked swelling of the capsule. (Source: From Avakyan, A. A., et al., *J. Bacteriol.* **90**:1082, 1965.) (8 and 9) Ultrastructure of the capsules of *Streptococcus pneumoniae* type II and *Klebsiella pneumoniae* type I as seen in the electron microscope using ruthenium red in combination with osmium tetroxide. (8) A mat-like capsule (C) surrounds the cell. Ruthenium red penetration is evident at the plasma membrane (PM) and into the cytoplasm. Bar equals 0.5  $\mu\text{m}$ . (9) *Klebsiella pneumoniae* capsular fibrils (C) are seen at regular intervals along the wall. Bar equals 0.5  $\mu\text{m}$ . (Source: From Springer, E. L. and I. L. Roth, *J. Gen. Microbiol.* **74**:21, 1973.)

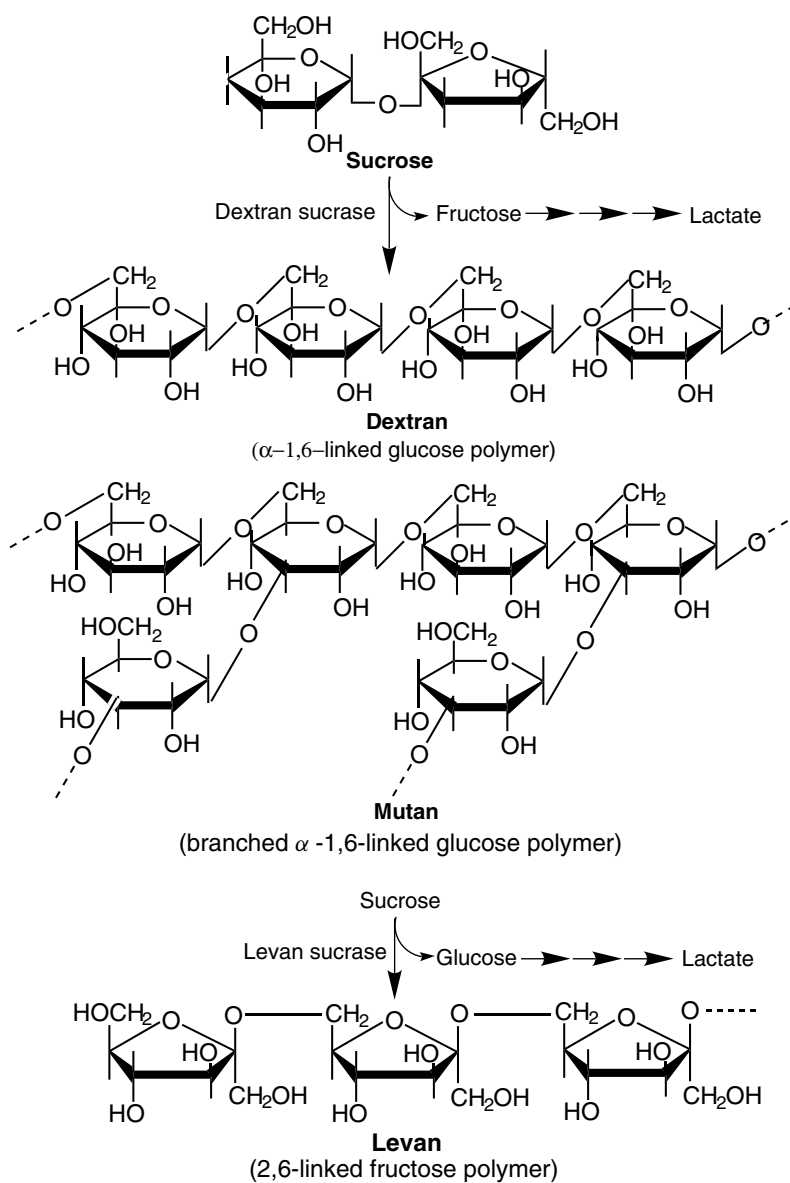


**Fig. 7-26. Chemical structures of capsules or slime layers of gram-negative bacteria.** GlcpNAc, *N*-acetylglucosamine (italicized *p* indicates pyranose form of the sugar); Fuc, L-fucose; Gal, D-galactose; Glc, D-glucose; Man, D-mannose; FucpNAc, *N*-acetylfucosamine.

wild-type parental strain. Thus, it appears that most of the PGM activity required for type 3 capsule biosynthesis is derived from the cellular PGM.

In other *S. pneumoniae* serotypes and in other streptococci, the enzymes expected to be essential for capsule production are often not encoded by genes in the capsule locus. For example, the type 14 locus lacks genes necessary for the synthesis of UDP-glucose, UDP-galactose, and UDP-*N*-acetylglucosamine, precursors of the type 14 polysaccharide. In type 14 and other strains of *S. pneumoniae* or *S. pyogenes* the sugar or other intermediate is an important component of teichoic acid or peptidoglycan, and the cell uses existing cellular pools of these precursors for synthesis of the capsular polysaccharide.

In each individual case, alteration in the immunological specificity results from the incorporation of monosaccharides other than glucose and glucuronic acid. For example, galacturonic acid, rhamnose, *N*-acetyl-amino sugars, and uronic acid may be added in various combinations. Enhancement of virulence by the presence of a capsule results



**Fig. 7-27. Production of dextran and levan by bacteria.** The enzyme dextran sucrose splits sucrose to form fructose and dextran. The fructose is fermented to lactic acid. Levansucrose splits sucrose to form glucose and levan. The glucose is fermented to lactic acid. The structure of mutan, a branched dextran polymer, is also shown.

from inhibition of phagocytosis. Although physical size of the capsule correlates with the degree of virulence of the pneumococcus, the precise mechanism whereby capsules interfere with phagocytosis is not known.

Certain members of the streptococci produce both soluble and insoluble forms of dextrans and levans (Fig. 7-27). *S. salivarius* synthesizes levan, a polymer of fructose,

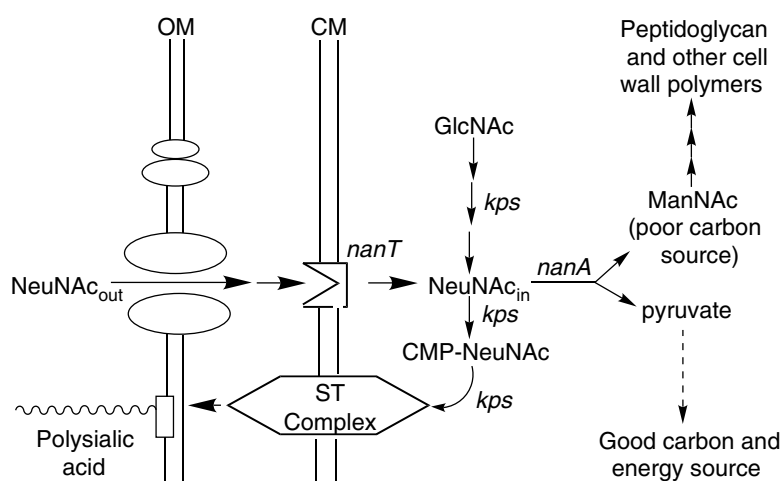
from sucrose using levan sucrase (fructosyltransferase). Other streptococci produce both soluble and insoluble forms of dextrans (polymers of glucose) by the action of dextran sucrase. Mutans are branched glucose polymers with linear  $\alpha$ -1,3-linked glucose units (Fig. 7-27). *S. mutans* produces FruA, an extracellular fructan hydrolase encoded by *fruA*, which releases fructose from levan, inulin, and raffinose, and cleaves sucrose into glucose and fructose. Expression of *fruA* is regulated by induction and carbon catabolite repression. The ability to synthesize fructan polymers is thought to aid in the formation of oral biofilms (bacterial aggregates referred to as plaque) that aid in the adherence of *S. mutans* and other oral bacteria to the enamel surfaces of teeth.

$\beta$ -Hemolytic streptococci can be classified serologically on the basis of differences in their Lancefield group antigens. The capsular substances of Lancefield groups A and C streptococci are polysaccharides containing glucosamine and glucuronic acid units, the components of hyaluronic acid. This substance provides protection from the destructive effects of atmospheric oxygen by virtue of its ability to aid in the formation of cell aggregates. Disruption of the aggregates with hyaluronidase results in an increased oxygen uptake and the production of toxic levels of hydrogen peroxide. Unencapsulated variants are sensitive to oxygen.

A number of strains of *E. coli* produce capsular polysaccharides commonly referred to as K antigens. The K antigens are structurally diverse and give rise to serological specificity. There are over 70 recognized K antigens in *E. coli*. The K antigens are placed in either capsular Group I (heat-stable) or capsular group II (temperature regulated, expressed at 37 °C but not at  $\leq 20$  °C). For example, the K30 antigen of *E. coli* is a member of Group I. The K5 antigen is in Group II. The Group II K antigens are also characterized by acidic components, such as 2-keto-3-deoxy-D-mannooctulonic acid, *N*-acetylneuraminic acid (sialic acid), and *N*-acetylmannosaminouronic acid (Fig. 7-26). The K5 repeating unit, shown in Figure 7-26, is identical to that of the first polymeric intermediate in the biosynthesis of heparin. As a consequence, the K5 polysaccharide is practically nonimmunogenic and strains of *E. coli* expressing this capsule are quite virulent since host cells do not mount an active immune response to it. A correlation has been drawn between activity and active production of K5 polysaccharide. The capsular polysaccharide 17-kb multi-gene cluster encodes the proteins required for the synthesis, activation, and assembly of the K antigens of *E. coli*. Three distinct regions have been identified in this gene cluster. The central region (Region 2) is composed of *neu* genes concerned with polysaccharide synthesis and polymerization. The *kps* genes in Regions 1 and 3 are involved in polymer assembly and transport and govern the same functions in all *E. coli* –producing group II capsules.

A model for polysialic (poly-*N*-acetylneuraminic) acid synthesis and translocation is presented in Figure 7-28. Two genes important in *N*-acetylneuraminic acid (neuAc) synthesis, *neuA* and *neuC*, are located at the 2.7 kb *Eco* R1-*Hind*III fragment of the *E. coli* K-1 gene cluster. The *neuA* gene encodes CMP-neuAc synthetase, and *neuC* encodes a 45,000-molecular-weight protein (P7) required for the synthesis of neuAc. Region 3 of the *kps* gene cluster contains two genes, *kpsM* and *kpsT*, which encode protein constituents of a system for the transport of polysialic acid.

Under appropriate growth conditions, *E. coli* K-12 and other enteric bacteria produce a slime polysaccharide, colanic acid, termed the M antigen (Fig. 7-26). The *E. coli* *rcaA*<sub>K12</sub> and *rcaA*<sub>K30</sub> genes are transcriptional activators involved in the expression of colanic acid synthesis. There is evidence that the RcsA protein may be a relatively



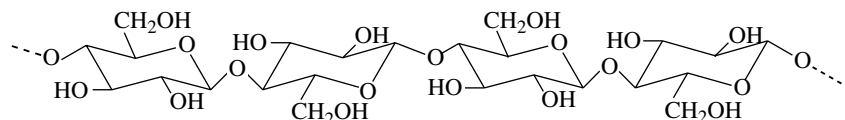
**Fig. 7-28. Model for synthesis and translocation of polysialic acid (poly-*N*-acetylneuraminic acid) in *E. coli* K-1.** External *N*-acetylneuraminic acid (NeuNAc<sub>out</sub>) diffuses into the periplasm through outer membrane pores and is transported by the sialic acid permease encoded by *nanT*. Internal sialic acid (NeuNAc<sub>in</sub>), transported or produced endogenously by *kps* gene products, is either degraded by the aldolase encoded by *nanA* or activated to the sugar nucleotide precursor CMP-NeuNAc. Sialic acid is polymerized and translocated to the outer membrane as PSA by the sialyltransferase (ST) complex. The aldolase cleavage products either enter intermediary metabolic pathways or are disseminated into surface polysaccharides. (Source: From Vimr, E. R. *J. Bacteriol.* **174**:6191–6197, 1991.)

widespread regulatory component for the synthesis of enterobacterial extracellular polysaccharides.

*Proteus mirabilis* produces an acidic capsular polysaccharide that is a high-molecular-weight polymer of branched trisaccharide units composed of 2-acetamido-2-deoxy-D-glucose (*N*-acetyl-D-glucosamine), 2-acetamido-2,6-dideoxy-L-galactose (*N*-acetyl-L-fucosamine), and D-glucuronic acid (Fig. 7-26). *Proteus mirabilis* 2573 also produces an O:6 serotype LPS in which the O-chain component has the same structure as the homologous capsular polysaccharide. The acidic nature of this polysaccharide may play a role in urinary calculi (stone) formation in a manner comparable to the roles of acid proteins and glycoproteins in mineralization and control of crystal formation in biological tissues. This process is initiated by a minute focus of solid matter that grows continually larger as chemical crystals that precipitate from the urine adhering to it.

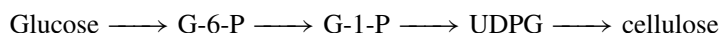
High-molecular-weight peptide capsules are produced by *Yersinia pestis* and *Bacillus anthracis*. The *B. anthracis* capsule is a polypeptide of  $\gamma$ -D-glutamyl subunits originally thought to be produced only in vivo during infection but later shown to be produced in culture if an excess of carbon dioxide or bicarbonate is present. Other species of bacilli (e.g., *B. subtilis*) produce a polypeptide capsule containing a mixture of D- and L-glutamic acid subunits. The ratio of the two isomers can be controlled by culture conditions. It is not known specifically what governs the ratio of D : L isomers, but aeration tends to favor an approximately 1 : 1 ratio of D : L subunits. In addition to carbon dioxide, the presence of DL-isoleucine, DL-phenylalanine, and glutamic acid affect *B. anthracis* capsule production. This capsule is of special interest because of its association with virulence of the organism through its interference with phagocytosis.

*Acetobacter xylinum* synthesizes cellulose, a polymer of  $\beta$ -1,4-linked glucose units:



Cellulose structure showing 180° flip of each glucose unit with respect to its neighbor

Cellulose accumulates as an extracellular aggregate of crystalline microfibrils. Intertwined ribbons from large numbers of cells form a tough pellicle on the surface of the culture medium. Although cellulose synthesis has been studied extensively in higher plants as well as in bacteria, the intermediary steps in the pathway have not been entirely elucidated. Resting cells and particulate membrane-bound preparations of *A. xylinum* incorporate [1- $^{14}$ C]-glucose into glucose-6-phosphate, glucose-1-phosphate, uridine glucose-5'-phosphate (UDPG), and cellulose. Labeling studies and demonstration of enzyme activities in cell-free extracts indicate that the sequence of reactions leading to cellulose synthesis is



Lipid-linked and protein-linked cellodextrins (partially polymerized glucose units) may function as intermediates between UDPG and cellulose. A multienzyme complex orchestrates intracellular cellulose synthesis. Extrusion through pores in the LPS layer results in the aggregation of bundles of cellulose that undergo crystallization by self-assembly at the cell surface.

### Microbial Biofilms

Many organisms are known to produce layers of material external to the usual cell surface layers. These external layers may contain carbohydrates, peptides, proteins, lipids, or combinations of these substances, which contribute to the adherence of the organism to solid surfaces to produce **biofilms**—complex communities of microorganisms attached to surfaces. Studies of organisms grown in artificial media in the laboratory may bear very little relationship to what occurs in the natural environment. In many cases, organisms grown in the laboratory lose their ability to produce the protective outer layers observed during growth in nature, apparently because these materials provide no selective advantage in laboratory cultures. It is now widely recognized that most bacteria found in natural, clinical, and industrial settings persist in association with surfaces. These microbial communities may often involve multiple species that interact with each other as well as with the environment. Recent inventive experimental approaches and improved methodologies have permitted exploration of the metabolic interactions, phylogenetic relationships, and competition in these complex communities. Following microbial adhesion to solid surfaces, biofilms can develop on almost any solid surface that comes in contact with aqueous liquids.

A number of factors play a role in adherence of bacteria to solid surfaces. When encountering a solid surface, bacteria experience a totally different set of conditions than in an aqueous environment. Under these circumstances, cells often undergo structural and functional adaptations. For example, in *V. parahaemolyticus* flagellar

synthesis is triggered, while in *P. aeruginosa* polysaccharide production is increased. In *E. coli* the adhesion of type 1 fimbriated cells to abiotic surfaces leads to altered composition of the outer membrane proteins. It has been suggested that physical interactions between type 1 fimbriae and the surface are part of a surface-sensing mechanism in which protein turnover may contribute to the observed change in composition of outer membrane proteins. The resultant alteration of the cell envelope may influence adhesion. Further discussion of the physiology of organisms forming biofilms can be found in “Pili or Fimbriae” and also in Chapter 15.

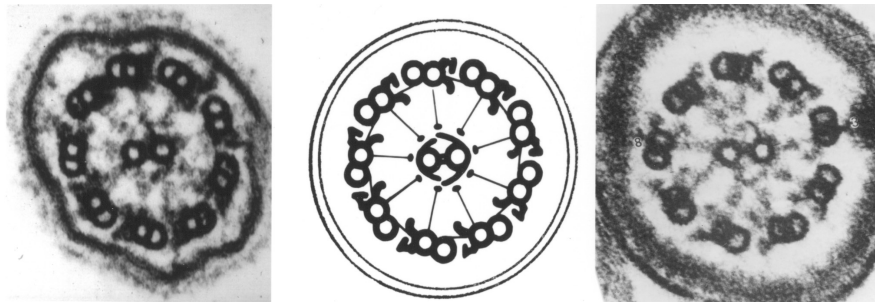
## ORGANS OF LOCOMOTION

Many microorganisms are motile—that is, they are able to move about in a concerted manner in an aqueous environment. Motile bacteria and protozoa produce flagella or cilia. Some organisms employ pseudopodal or ameboid movement. Spirochetes swim by virtue of a screw-like movement that is effective even in a viscous (semisolid) medium. Motile organisms are placed at an advantage in seeking food, in avoiding toxic chemicals or predators, and in colonizing favorable ecological niches. As a result, other forms of motility have evolved: swarming, which involves the development of specialized flagella; gliding, which involves cooperation between groups of cells (rafts); and twitching, a pilus-dependent means of translocation over solid surfaces. This last type of movement is discussed in “Pili or Fimbriae.” All motile organisms have a common feature: energy is required for locomotion. Eukaryotes use ATP to activate the contractile movement of complex cilia or flagella. In prokaryotes, proton motive force (PMF) activates a motor that turns the flagellum like the propeller of a boat. In some of the other forms of motility, the detailed mechanisms have yet to be elucidated.

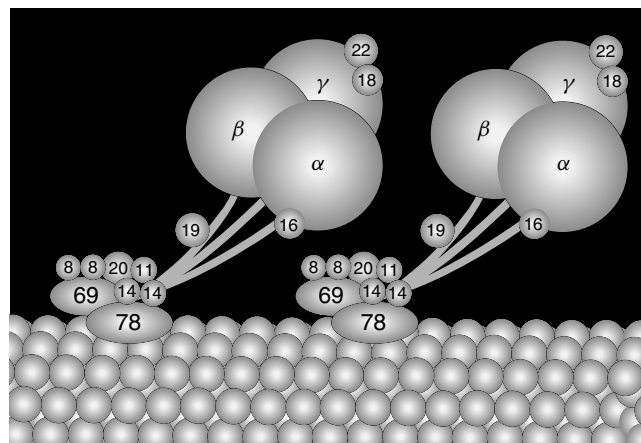
### Cilia and Flagella of Eukaryotes

Cilia and flagella of virtually all eukaryotic microorganisms have the same basic arrangement. A cross-sectional view under the electron microscope reveals paired tubules arranged in a typical  $9 + 2$  pattern that traverse the length of the flagellum of *Chlamydomonas reinhardtii* (Fig. 7-29). The nine peripheral pairs and one central pair of tubules are arranged in a bundle termed the **axoneme**. The individual doublet microtubules are cross-linked by nexin, a protein linker, and connected to the central core by radial spokes (Fig. 7-30). As a result of this rigid structure, when the dynein molecules in one doublet exert a force on the neighboring doublet, the entire structure bends instead of the microtubules sliding against each other. In normally beating flagellar axonemes, the ATPase-containing dynein arms generate longitudinal shear forces between the outer doublets, causing a localized sliding opposed by resistive components that convert it into a transverse bending movement.

Axonemal dynein contains ATPase and functions as a motor enzyme that provides the energy for ciliary and flagellar bending. This motor enzyme complex contains several proteins. Three heavy chains of approximately 500 kDa each contain the motor and ATPase sites. Two intermediate chains of 70 to 80 kDa mediate cargo attachment. The ATP cyclic cross-bridging of the dynein arms appears to involve the interaction of the projections on the globular heads with tubulin subunits in the tubule of the adjacent doublet (Fig. 7-30). The regulatory mechanism responsible for the control of the interdoublet sliding is not thoroughly characterized, but it is considered that



**Fig. 7-29. Electron micrographs of cross-sections through eukaryotic flagella and diagram of axonemal components.** The electron micrographs show cross sections of a *Chlamydomonas* flagellum (left) and a ram sperm flagellum (right). The ram sperm flagellum has a fibrous sheath between the axoneme and membrane. Otherwise, these flagella from two widely separated genera are remarkably similar, indicating that axoneme structure and composition has been highly conserved throughout evolution. (Photos and diagram courtesy of Dr. G. B. Witman, Department of Cell Biology, University of Massachusetts Medical School, Worcester, MA.)



**Fig. 7-30. Diagrammatic view of two *Chlamydomonas* dynein arms attached to an A-tubule.** Each dynein heavy chain ( $\alpha$ ,  $\beta$ , and  $\gamma$ ) has a head that contains the ATP hydrolytic site and interacts with the B tubule of the adjacent doublet (not shown) to generate interdoublet sliding, and a fibrous stem that extends to the base of the dynein. At the base there is an intermediate chain/light-chain complex containing the two dynein intermediate chains (IC69 and IC78) and several light chains. Additional dynein light chains are associated with the heads and stems. Numbers indicate apparent molecular weights of subunits. (Courtesy of Dr. G. B. Witman, Department of Cell Biology, University of Massachusetts Medical School, Worcester, MA.)

interactions between the radial spokes attached to each doublet tubule and the central complex of the axoneme are involved in some manner. It has been shown that the tubulin in the microtubules of *C. reinhardtii* is acetylated and partially detyrosinated. Detyrosination is largely confined to the outer doublets and has not been detected in the microtubules of the central pair. These posttranslational modifications of the tubulin structure may have some functional role at the molecular level. Also, there are several kinases and phosphorylases in the axoneme that operate to control dynein activity.



### Bacterial (Prokaryotic) Flagella

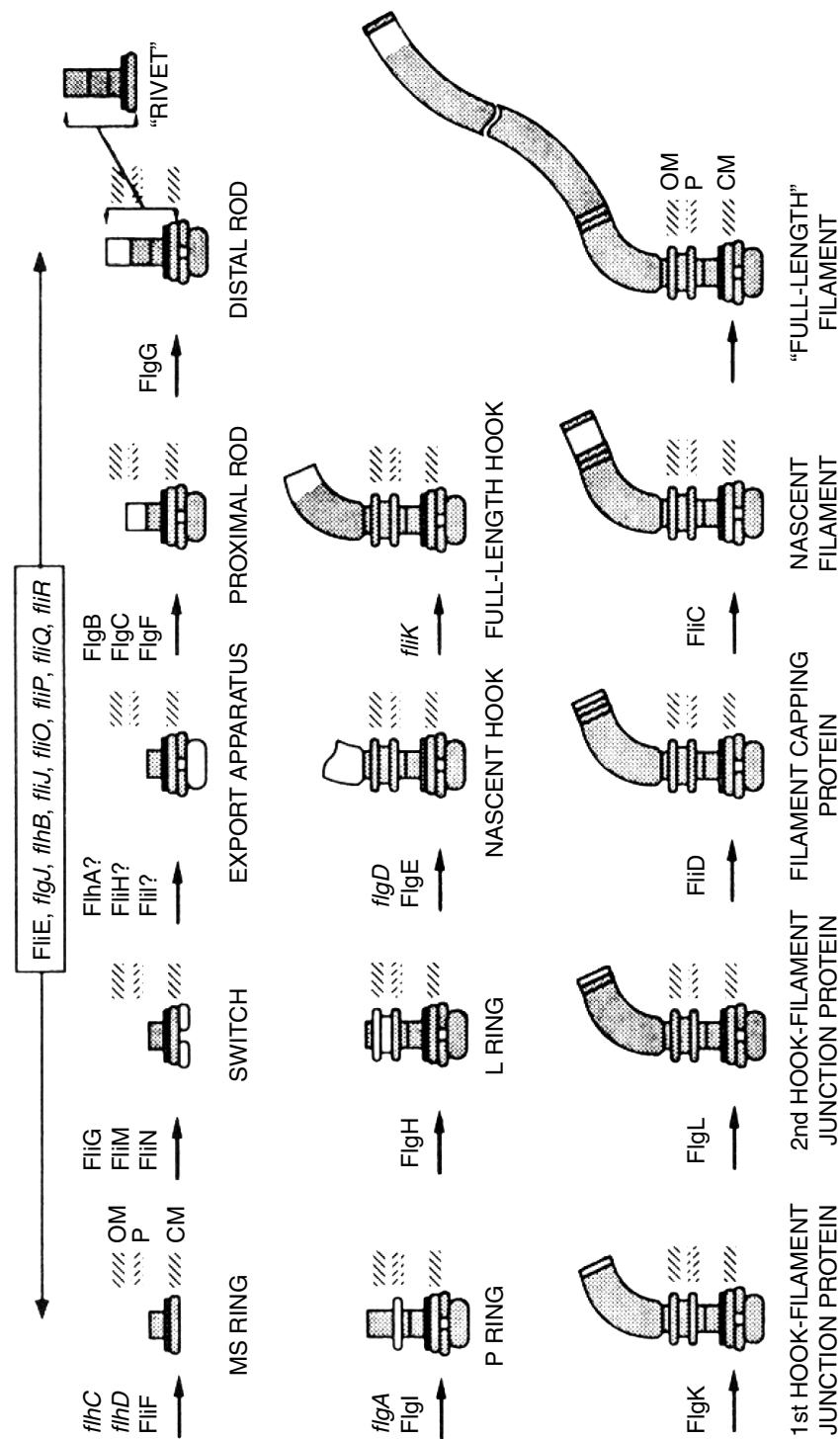
Bacterial flagella consist of three major components: a **basal body**, a **proximal hook**, and a long **helical filament** (Fig. 7-31). The filament and hook are hollow cylindrical structures that extend from the cell surface. The basal body is embedded in the cell surface and contains a motor that turns the helical flagellum to drive the cell through a liquid environment much as a propeller drives a boat. The motor is reversible and is under the control of a chemotactic system that enables the cell to move toward favorable or away from unfavorable environmental conditions.

The genetic system governing the synthesis and function of flagella in *E. coli* is composed of 14 operons arranged in a regular cascade of three classes. The class 1 operon encodes the transcriptional activator of class 2 operons. Class 2 genes include structural components of the rotary motor and hook structure as well as the transcriptional activator for class 3 operons. The genes in class 3 include flagellar filament structural genes and the chemotaxis signal transduction system that directs cellular motion. A checkpoint mechanism ensures that class 3 genes are not transcribed before the basic structural assembly has been completed.

Flagella assembly and function requires over 40 genes (Table 7-6). The biosynthetic processing in *E. coli* and *S. enterica* occurs in a sequence that starts with the assembly of the M and S rings under the influence of regulatory proteins FlhC and FlhD (Fig. 7-31). This step is followed by the addition of switch proteins FliG, FliM, and FliN; assembly of the export apparatus containing FlhA, FliH, and FliI; and formation of the proximal and distal rods containing FlgB, FlgC, FlgF, and FlgG. Mutants defective in any of the genes coding for these proteins result in recovery of the MS ring only. A series of light chains of 8 to 22 kDa are involved in assembly and has been interpreted to mean that intermediary stages of rod assembly are relatively unstable, and, as a result, incomplete rod stages have not been found.

The next simplest isolated structure is the “rivet” portion of the distal rod that contains FliF, FliE, and four rod proteins (FlgB, FlgC, FlgF, and FlgG). The P and L rings (outer cylinder) contain FlgI and FlgH. These proteins appear to be the only flagellar components exported by the primary cellular export pathway, as shown by the presence of signal peptides. The next stages involve formation of the hook under the influence of *flgD* and addition of the hook proteins FlgE and FliK. Addition of the first hook–filament junction protein (FlgK) and the second hook–filament junction protein (FlgL) followed by the filament capping protein (FliD) completes the formation of components necessary for filament assembly. The addition of FliC (filament protein or **flagellin**) completes the biogenic process of flagellar assembly.

The filament portion of common bacterial flagella is composed of 100% protein (flagellin). Electron density maps show that the filament is composed of densely packed subunits (Fig. 7-32). The inner part of the filament forms a dense core of 30 to 50 Å radius. Flagellar growth occurs at the distal end of the filament. The hook portion is composed of at least three types of protein—that is, **hook-associated proteins (HAPS)** HAP1, HAP2, and HAP3. HAP2 forms a cap structure that presumably serves to prevent arriving flagellin subunits from diffusing away before polymerization occurs. Electron density studies show well-defined subunit packing in the core region and a central hole of approximately 60 Å diameter, which is large enough to accommodate the folded flagellin. A flagellum-specific pathway exports most of the flagellar components. Only the P and L proteins of the outer ring seem to be transported by the signal peptide-dependent pathway.



**Fig. 7-31. Biogenesis of the bacterial flagellum.** Succeeding stages of increasingly complex structure are shown along with the genes needed for each stage. Each incremental feature is shown in white with all preceding structures shown with stippling. The structure known as the rivet is, after the MS ring, the simplest substructure that has been detected by electron microscopy, and has lost the switch and export complex and perhaps other structures during the isolation procedure. The filament does not have a well-defined mature length, "full-length" simply implies that the filament is long enough to function in propulsion. Where the gene product is known to incorporate into structure, its symbol is given in Roman letters; where this is not known, the gene symbol is given in italics. The gene product and genes indicated in the box are needed at approximately the stages shown and certainly prior to the assembly of the distal rod. OM, outer membrane; P, periplasmic space and peptidoglycan layer; CM, cell membrane. (*Source*: From Macnab, R. M. *Annu. Rev. Genet.* **26**:131–158, 1992.)

Certain flagellar components, for example, the Mot proteins (MotA and MotB), are associated with the cytoplasmic membrane and surround the MS-ring component (Fig. 7-33). MotA and MotB are components of torque generators that enable motor rotation. The MotA protein is involved in conducting protons across the cytoplasmic membrane. Although MotB is associated with the cytoplasmic membrane, most of the protein protrudes into the periplasmic space. MotB is considered to be a linker that

**TABLE 7-6. Flagellar, Motility, and Chemotaxis Gene Products of *S. enterica* and *E. coli* and Their Known or Suspected Functions**

| Gene Product                                         | Function/Location                                                                                                                        |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Regulatory Proteins</i>                           |                                                                                                                                          |
| FlhC, FlhD                                           | Master regulators of the flagellar region acting in class 2 operons; transcription initiation ( $\sigma$ ) factors?                      |
| FliA                                                 | Transcription initiation ( $\sigma$ ) factor for classes 3a and 3b operons.                                                              |
| FlgM flagellar                                       | Anti-FliA (anti- $\sigma$ ) factor. Also known as RflB. Active only when assembly has not proceeded through completion of the hook.      |
| FliS, FliT, FliD?                                    | Repressor of classes 3a and 3b operons (RflA activity).                                                                                  |
| FljA                                                 | Repressor of <i>fliC</i> operon.                                                                                                         |
| Hin                                                  | Site-specific recombinases, affecting <i>fljB</i> promoter.                                                                              |
| <i>Proteins Involved in the Assembly Process</i>     |                                                                                                                                          |
| FlhA, FliH, FliI factors                             | Export of flagellar proteins? FlhA is homolog of various virulence factors.<br>FliI is homolog of catalytic subunit of $F_0 F_1$ ATPase. |
| FlgA                                                 | Assembly of basal body periplasmic P ring.                                                                                               |
| FlgD                                                 | Initiation of hook assembly.                                                                                                             |
| FliK                                                 | Control of hook length.                                                                                                                  |
| FliB                                                 | Methylation of lysine residues on the filament protein, flagellin; functions of this modification unknown.                               |
| <i>Flagellar Structural Components</i>               |                                                                                                                                          |
| FliG, FliM, FliN                                     | Components of flagellar switch, enabling rotation and determining its direction (CCW vs. CW).                                            |
| MotA, MotB                                           | Enable motor rotation. No effect on switching.                                                                                           |
| FliF                                                 | Basal body MS (membrane and supramembrane) ring and collar FliE.<br>Basal body component, possibly at MS ring-rod junction.              |
| FlgB, FlgC, FlgF                                     | Cell-proximal portion of basal body rod.                                                                                                 |
| FlgG                                                 | Cell-distal portion of basal body rod.                                                                                                   |
| FlgI                                                 | Basal body periplasmic P ring.                                                                                                           |
| FlgH                                                 | Basal body outer membrane L (LPS layer) ring.                                                                                            |
| FlgE                                                 | Hook.                                                                                                                                    |
| FlgK, FlgL                                           | Hook-filament junction.                                                                                                                  |
| FliC, FljB                                           | Filament (flagellin protein), FljB (found in <i>S. enterica</i> only) is an alternative, serotypically distinct, flagellin.              |
| FliD                                                 | Filament cap, enabling filament assembly.                                                                                                |
| <i>Flagellar Proteins of Unknown Function</i>        |                                                                                                                                          |
| FlgJ, FlhB, FlhE, FliJ, FliL, FliO, FliP, FliQ, FliR |                                                                                                                                          |

TABLE 7-6. (continued)

| Gene Product                                                                                                                              | Function/Location                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <i>Sensory Transduction Components</i>                                                                                                    |                                                           |
| CheA                                                                                                                                      | CheY and CheB kinase.                                     |
| CheZ                                                                                                                                      | CheY phosphatase. Antagonist of CheY as switch regulator. |
| CheW                                                                                                                                      | Positive regulator of CheA activity.                      |
| CheY                                                                                                                                      | Switch regulator, placing it in CW state.                 |
| CheR                                                                                                                                      | Methylation of receptors. Sensory adaptation.             |
| CheB                                                                                                                                      | Demethylation of receptors. Sensory adaptation.           |
| <i>Chemoreceptors</i>                                                                                                                     |                                                           |
| Tar (aspartate); Tap (peptide, <i>E. coli</i> only); Trg (ribose); Tsr (serine); sugar receptors of the phosphotransferase uptake system. |                                                           |

Source: From Macnab, 1992.

fastens MotA and other components of the torque-generating machinery to the cell wall, keeping the motor components stationary with respect to the rest of the cell. The MotA and MotB proteins and the flagellar switch proteins FliG, FliM, and FliN constitute the flagellar motor. The FliG, FliM, and FliN proteins affect motor rotation and the switch from CW to CCW rotation. Additional aspects of the switch are discussed later in conjunction with chemotaxis. A change in expression from one flagellin gene to another (phase variation) is discussed in some detail in Chapter 5.

### Chemotaxis

Many microorganisms, including *E. coli* and *Salmonella*, exhibit a wide range of metabolic and physiological versatility, enabling them to grow and survive in diverse environments. This versatility is employed to greatest advantage when a motile organism is able to direct its movement toward more favorable or away from unfavorable environments. This capability is called **chemotaxis**. But how can bacteria knowingly change direction? The trick is to control the gyration of their flagella. The rotation of the flagellar rotor is reversible, capable of turning in clockwise (CW) or counterclockwise (CCW) directions. When all the motors in a single cell rotate in a CCW direction, the flagella sweep around the cell in a common axis, forming a concerted bundle (Fig. 7-34a). The result is a smooth swimming motion of the cell. However, if the motors reverse and turn in a CW direction, the flagella disperse (Fig. 7-34b) and the cell undergoes a tumbling motion. This run-and-tumble strategy does not hold for all bacteria. For example, in *Halobacterium* spp., CCW rotation causes movement in one direction and CW rotation causes movement in the opposite direction.

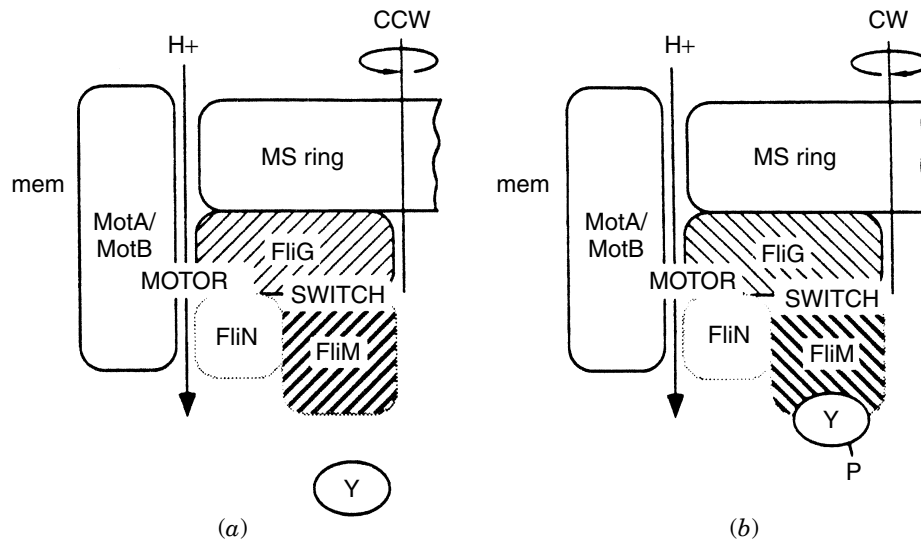
Chemotaxis is the result of regulating the switch between CW and CCW rotation. A cell placed in a medium containing a unidirectional gradient of attractant, such as glucose, will suppress the onset of CW rotation if it is moving toward the attractant but will tumble more often to reorientate its direction if it is moving away from the attractant (Fig. 7-35).

The critical question here is to explain how bacteria interface with their environment to sense various attractants and repellants. A first step toward understanding the

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**Fig. 7-32. Three-dimensional density map of the cap filament complex and its central section.** (a) End-on view from the top, showing a pentagonal shape of the plate domain of the cap. The five vertices of the pentagon are labeled with Greek letters,  $\alpha$  to  $\epsilon$ , which are used to guide the orientation of the pentagon in the side views in (c). (b) Side view showing a regular helical array of flagellin subunits and the plate domain of the cap. Scale bar, filament diameter of 230 Å. (c) Five side views, showing each of the five gaps between the plate and the filament end. The viewing directions are as indicated by arrows in (a) labeled with corresponding numbers. Greek letters labeling the vertices as in (a) also indicate the orientations of the pentagonal plate. (Source: From Yonekura, K., et al. *Science* **290**:2148–2152, 2000.)

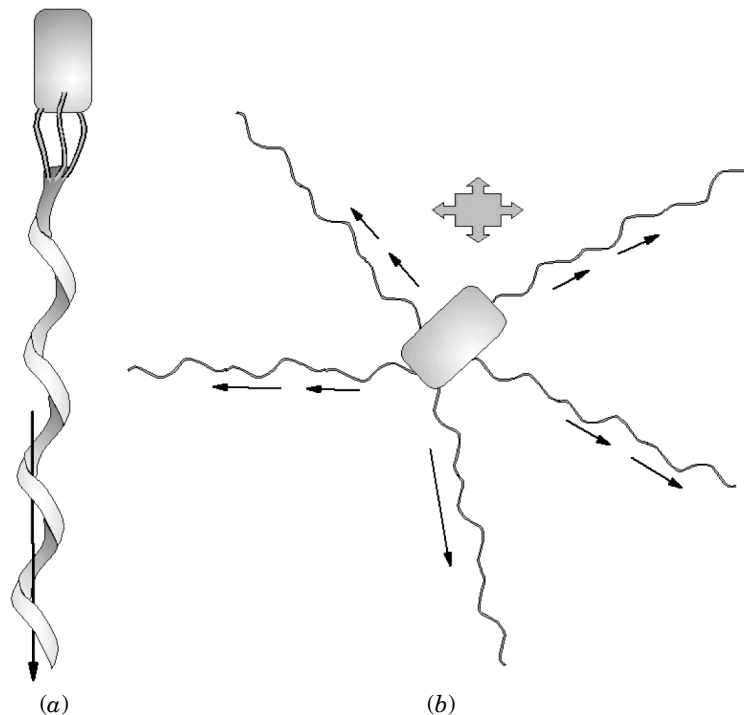
chemotactic response is to view the time-course (kinetic) response of a bacterium to the pulse addition of an attractant. As illustrated in Figure 7-36, there is a latency phase (0.2 s) followed by an excitation phase in which CCW flagellar rotation (smooth swimming) is favored. A gradual decrease in the relative probability of CCW versus CW rotation marks the adaptation phase. The adaptation phase may last from seconds to minutes. Basically, the cell recognizes the lack of further change in the concentration of attractant, and it now requires a higher concentration of attractant to trigger another excitation phase.



**Fig. 7-33. Model for the structure of the flagellar switch and its function.** The FliG is located in the cytoplasmic face of the MS ring. Both MotA and MotB are integral membrane (mem) proteins that surround the MS ring. The FliM and FliN are considered to interact with FliG; the locations shown for them are speculative but are consistent with available evidence. The FliM is postulated to be the target for CheY (Y), based on the large number of positions within its sequence that affect the counterclockwise (CCW) versus clockwise (CW) state of the switch. Both FliM and FliG are postulated to constitute the switching function, while FliG and FliN (perhaps with MotA and MotB) constitute the motor function. (a) When CheY is not bound to FliM, both FliM and FliG are in their CCW states and the motor rotates CCW. (b) When CheY is phosphorylated (Y-P) and binds to FliM, it places FliM in its CW state; this in turn causes FliG to change to the CW state, and so the motor rotates CW. (Source: From Irikura, V. M., et al. *J. Bacteriol.* **175**:802–810, 1993.)

In *E. coli*, the mechanism regulating the CW/CCW switch involves a family of membrane receptors and six cytoplasmic proteins called Che proteins that transduce information from the receptors to the motor. The membrane receptors (also called methyl-accepting chemotaxis proteins) are homodimers that span the membrane. The sensory domain lies in the periplasm and the signaling domain resides in the cytoplasm. There are five chemoreceptors (Trg, Tap, Tar, Tsr, Aer; think of them as the five senses of the bacterium), each of which senses a different subset of attractants and repellants. The signal-transducing domains of chemoreceptors interact with CheW and CheA as depicted in Figure 7-37. The CheA protein has an autokinase activity that is stimulated 100-fold when associated with the receptor.

Amazingly, the vast majority of receptor signaling complexes are clustered within a single patch generally localized to one of the poles of *E. coli*. These complexes are huge—1.4 million molecular weight—with a subunit ratio of approximately 28 receptors to 6 CheW and 4 CheA molecules. This chemotaxis complex is not homogeneous with respect to receptors; each complex contains representatives of all five sensory receptors. In fact, the receptors Trg, Tap, and Aer will not function in the absence of both Tar and Tsr. Thus, the minor receptors function in conjunction with the major receptors within a single structure composed of thousands of receptor

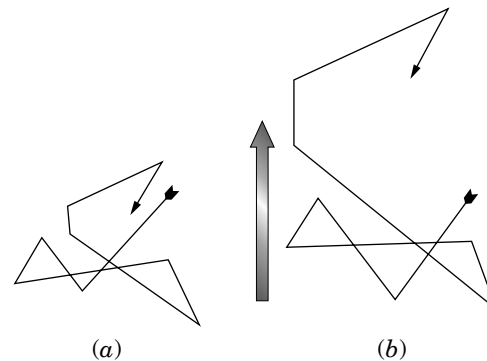


**Fig. 7-34. Behavior of flagella of *E. coli* and *S. typhimurium* during swimming and tumbling.** (a) Swimming: with the motors in CCW rotation, the flagellar filaments form a propulsive bundle, with wave propagation proceeding from proximal to distal. (b) Tumbling: CW rotation of the motors causes the normal left-handed form of the filament to undergo a polymorphic transition to the curly right-handed form. While the filaments are undergoing such transitions, the bundle is dispersed and the cell body moves chaotically, end over end—that is, the cell tumbles and reorients randomly, ready for the next swimming interval.

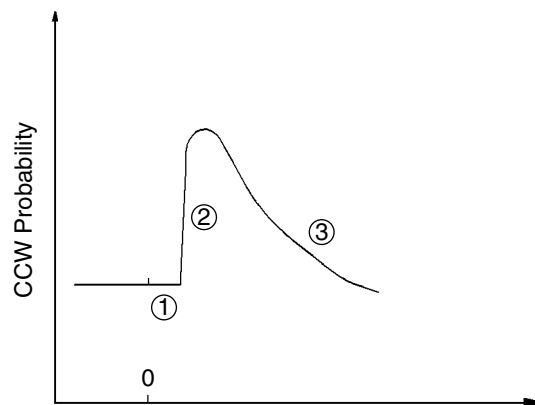
monomers organized within an ordered array by the association of CheW and CheA with their common C-terminal extensions. The role of CheW in this complex appears to be architectural.

The natural tendency of the flagellar motor is toward CCW rotation and, thus, smooth swimming. When transphosphorylated by CheA, a protein called CheY can interact with the FlhM component of the motor and trigger CW rotation (tumbling). The goal of the chemotactic machinery, then, is to modulate the level of CheY-PO<sub>4</sub> and, thus, the frequency of tumbling. When a receptor interacts with an attractant, it introduces an asymmetry in the receptor structure that is thought to alter its ability to pack into an array compatible with formation of a complex between the receptor signaling domain, CheW, and CheA. As a result, CheA autophosphorylation stops, as does transphosphorylation of CheY. The CheZ protein is a CheY phosphatase that lowers the level of CheY-PO<sub>4</sub> (Fig. 7-37). Since unphosphorylated CheY does not affect the motor switch, the result is suppression of tumble frequency, an effect that increases the length of smooth swimming.

How does the cell know that it should continue to move toward higher concentrations of attractant? It adapts or desensitizes itself to the most recently encountered



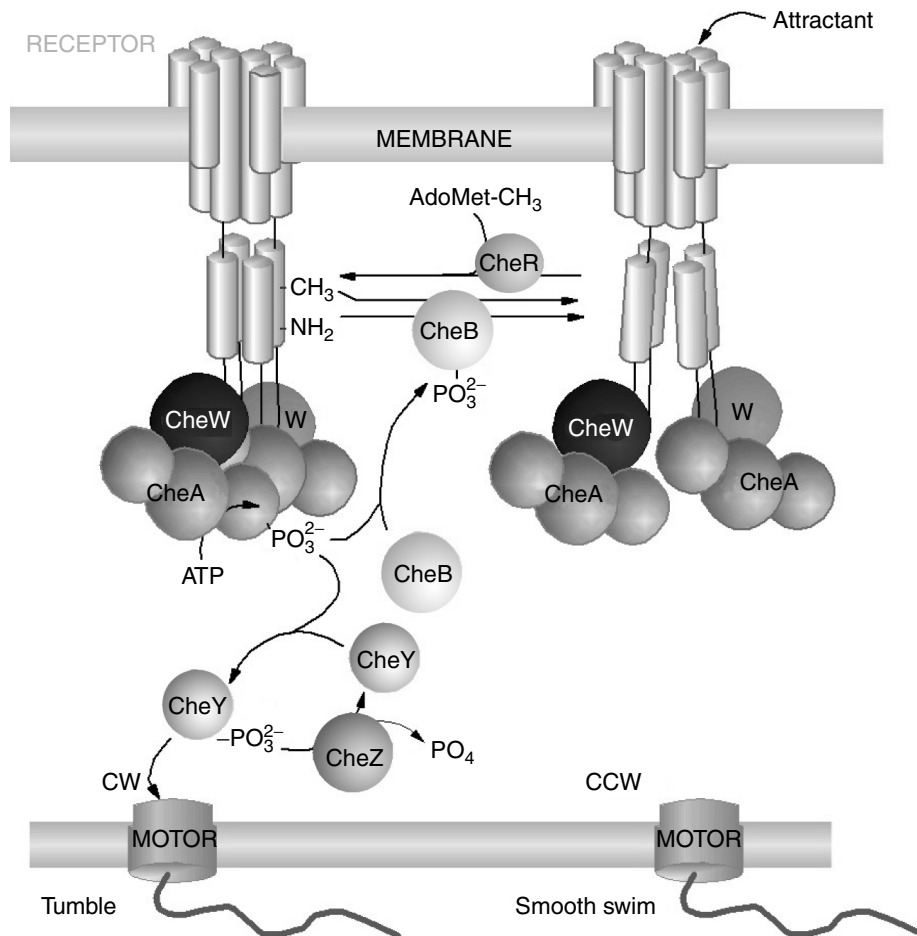
**Fig. 7-35. Idealized trajectory of a swimming cell of *E. coli* or *S. typhimurium*** (a) in an isotropic medium and (b) in a unidirectional gradient of attractant (open arrow). In either case, the cell swims in a straight line, randomizes its direction by tumbling, swims again, tumbles, and so on, yielding a three-dimensional random walk. The effect of an attractant gradient is to extend (by tumbling suppression) the mean duration of swimming segments in an up-gradient direction (heavy lines); segments in the down-gradient direction (fine lines) are not appreciably shortened. To illustrate this, the cells (a) and (b) start at the same point (arrow tail) and execute the same number of tumbles with the same resulting directional changes, but the cell (b) in the gradient has up-gradient segments extended. Therefore, it displays migration in the gradient direction by executing a time-based random walk.



**Fig. 7-36. Time course of the response of a bacterium to temporal stimulation by the sudden addition of an attractant at time zero.** After an initial latency phase (1) of about 0.2 s, the excitation phase (2) manifests itself as a rapid increase in the probability that the flagellar motors will be in CCW versus CW rotational state. The degree of excitation progressively decreases during the adaptation phase (3), which can last from seconds to minutes, depending on the magnitude of the stimulus.

concentration of attractant. Chemotactic “memory” involves methylation of the membrane receptor signal domain following excitation; hence the designation of these proteins as methyl-accepting chemotaxis proteins (MCPs). Methylation of glutamate residues in the signaling domains of an MCP dimer (after excitation) is thought to neutralize negative charges on those domains. This would reduce the repulsive





**Fig. 7-37. The chemotaxis switch in *E. coli*. Tumble mode.** Symmetrical assembly of CheA and CheW with the membrane receptors stimulates CheA kinase. Transfer of the phosphate from CheA to CheY and subsequent interaction of CheY-PO<sub>4</sub> with the flagellar motor causes clockwise rotation and tumble. *Smooth swimming mode.* Binding of attractant to the periplasmic domain of the membrane receptor causes asymmetry in the receptor structure and decreased kinase activity. Dephosphorylated CheY (catalyzed by CheZ phosphatase) will not interact with the motor, thereby allowing CCW rotation and smooth swimming. *Resetting the switch.* Following excitation, CheR methylates regions on the MCP that allow reassociation of the MCP C-terminal struts and reactivation of CheA kinase, reinstituting tumble. Concomitant phosphorylation of CheB will demethylate and deaminate residues on the MCP struts, once again disrupting the assembly structure downregulating CheA kinase.

electrostatic effects, forcing separation of the two C-terminal struts, allowing them to move closer to each other and reestablish an active CheW–CheA-receptor complex (i.e., restoration of CheA kinase activity). This increases the CheA-PO<sub>4</sub> level, and tumbling occurs even though attractant is still present.

Two proteins are involved with adding and removing methyl groups on the MCP receptors. The methyltransferase CheR uses SAM as a methyl donor to methylate the

glutamate residues while the methylesterase CheB can remove those methyl groups. CheB can also deamidate glutamine residues in the receptor, which will increase the net negative charge on the receptor as well as provide more substrate for CheR methylase. The CheB methylesterase is really part of a feedback loop that “resets” the receptor for excitation just in case the organism encounters a higher concentration of attractant or moves toward a lower concentration. The amino-terminal end of CheB is homologous to CheY and will compete with CheY for CheA-PO<sub>4</sub>. The phosphorylation of CheB then activates the esterase activity (Fig. 7-37).

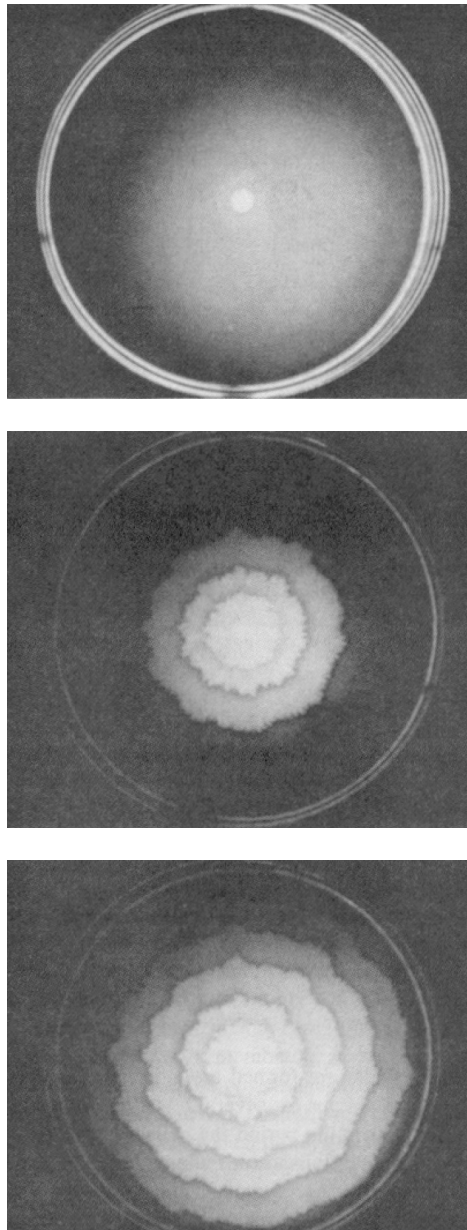
Consequently, the increased kinase activity that occurs following activation of the receptor-CheW-CheA complex not only elevates CheA-PO<sub>4</sub>, which increases tumble frequency, but increases CheB esterase-dependent removal of MCP methyl groups. The resulting increase in negative charges intensifies the electrostatic force, repelling the MCP struts from each other. Movement of the struts then deactivates the CheA kinase complex, leading to lower CheA-PO<sub>4</sub> levels and smooth swimming. In the absence of attractant, this switch occurs at a relatively fixed frequency. However, an encounter with attractant (or with a higher concentration of attractant after an adaptation phase) hastens inactivation of the CheW-CheA-receptor kinase activity and extends the smooth swim toward the attractant.

### Swarming Motility

A number of motile gram-negative bacteria that display characteristic swimming and chemotactic behavior when grown in liquid media undergo a differentiation into a swarming type of motility when grown on agar of appropriate consistency (usually 0.5 to 0.8% agar). *Proteus mirabilis*, *Proteus vulgaris*, *Vibrio parahaemolyticus*, *Serratia marcescens*, and some strains of *Bacillus* and *Clostridium* have been shown to display this phenomenon—that is, radially spreading colonies on semisolid agar medium (Fig. 7-38). Swarmer cells undergo group translocation on an agar surface within a slime layer containing a complex mixture of polysaccharides, surfactants, proteins, and peptides. Swarming has been associated with the formation of biofilms and may contribute to bacterial virulence.

When grown in liquid medium, *V. parahaemolyticus* swims by means of a single sheathed polar flagellum that is structurally and functionally similar to the flagella of other gram-negative bacteria. However, on solid surfaces the cell develops into a swarmer cell that is highly elongated and produces extensive lateral flagella (Laf). The polar flagellar system (Fla) is expressed constitutively, whereas the Laf system is expressed only when the organism is grown on a solid surface. Lateral flagella are unsheathed and display a different subunit composition from polar flagella.

The gene systems controlling the two distinct motility systems are large, each composed of 40 or more genes without overlap except for shared chemotaxis components. Genes belonging to the Laf system have been identified and ordered in a hierarchical scheme of gene control. These genes encode structural components of the flagellum and the motor and a  $\sigma$  factor of RNA polymerase important for directing swarmer cell development. One gene (*lafA*) codes for lateral flagella. Other genes required for swarming but not for swimming have been identified by gene replacement mutagenesis. Reversible motors embedded in the cytoplasmic membrane drive both types of flagella. However, as compared with the systems of other gram-negative bacteria, the polar flagellar motor of *V. parahaemolyticus* is powered by sodium-motive

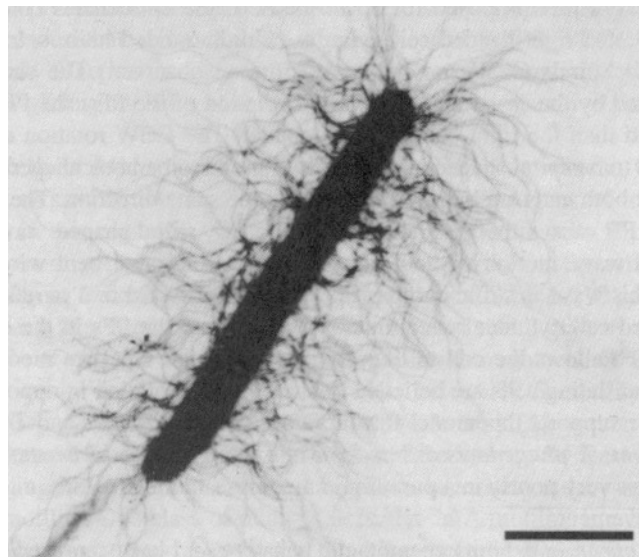


**Fig. 7-38. Development of the *P. mirabilis* swarming colony.** Multicellular swarming behavior and swarming colony formation were observed by inoculating cells at the center of an agar plate and incubating them at 37°C. The swarming colony was then photographed at 6 (upper), 24 (middle), and 48 (bottom) hours postinoculation. Consolidation zones are seen as the dark areas separating lighter regions of swarming motility. (Source: From Belas, R. *ASM News* **58**:115–22, 1992.)

force. The two independent flagellar organelles appear to be directed by a common chemosensory control system.

*Proteus mirabilis* is a motile gram-negative bacterium that produces characteristic peritrichous flagella when grown in liquid media. When transferred to solid media, the cells begin to elongate, the normal septation mechanism is inhibited, and giant swarmer cells are produced. The swarmer cells undergo extensive production of new flagella (Fig. 7-39). This alteration enables the swarmer cell to move over solid medium (swarming motility). By definition, individual swarmer cells cannot “swarm.” Swarming is the result of the coordinated effort of groups of differentiated swarmer cells that move about in rafts. The swarming process is cyclic or periodic. During swarming, concentric zones of bacterial growth (swarm bands) are formed around a central colony. Swarming occurs for a period of 1 to 2 hours and then stops for several hours before the appearance of a second swarm. At the end of the swarming period, the cells slow their activity and divide into short, sparsely flagellated cells. Transposon mutagenesis techniques have been used to study the behavior of mutant phenotypes of *P. mirabilis* with altered swarmer cell differentiation. These mutants appear to lack the appropriate intercellular signals required to regulate the cyclic swarming process.

The relatively recent demonstration of swarming motility in *Escherichia coli* and *Salmonella enterica* serovar Typhimurium (hereinafter referred to as *S. enterica*) provided the opportunity to study this phenomenon in genetically well-characterized organisms. Mutants obtained by transposon mutagenesis as well as existing mutants have been used to identify genes involved in swarming in *S. enterica*. Some mutants are defective in LPS synthesis while others are defective in chemotaxis or lack some portion of the two-component signaling system. Failure to identify specific swarming signals such as amino acids, pH changes, oxygen, iron starvation, increased viscosity,



**Fig. 7-39. Electron micrograph of a wild-type swarmer cell of *P. mirabilis*.** The cell was taken from the periphery of a swarming colony grown on an agar plate at 37 °C for 6 hours and negatively stained with uranyl acetate. Bar equals 5  $\mu$ m. (Source: From Belas, R. *J. Bacteriol.* 173:6279–6288, 1991.)

flagellar rotation, or autoinducers has led to the conclusion that the production of external slime is a major factor in the development of swarming motility.

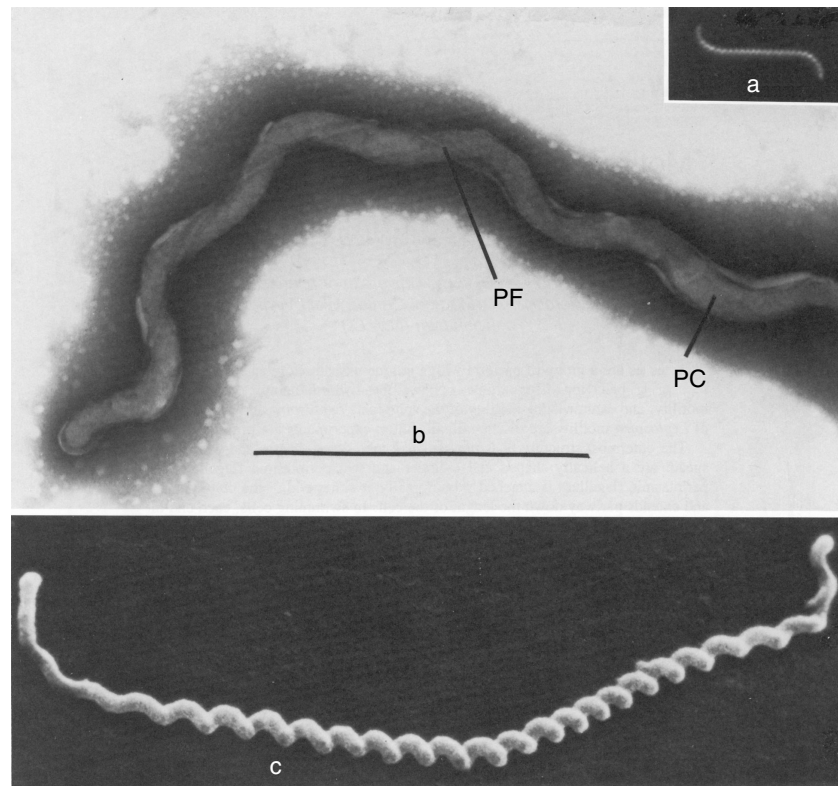
### Motility in Spirochetes

Spirochetes are helical in shape and are motile by means of flagella enclosed within the periplasmic space. Major species of spirochetes include *Brachyspira*, *Spirochaeta*, *Treponema*, *Borrelia*, and *Leptospira*. The handedness of the helices (right- or left-handed) is species specific, and the organisms vary in shape and size. A right-handed helix rotates clockwise (CW), and a left-handed helix rotates counterclockwise (CCW), going away from an observer. Attached near each end of the protoplasmic cell cylinder are from one to several hundred periplasmic flagella that extend toward the center of the cell where they may or may not overlap. The size of the spirochetes, the number of flagella attached at each end, and the degree of overlap of the flagella at the center vary from species to species.

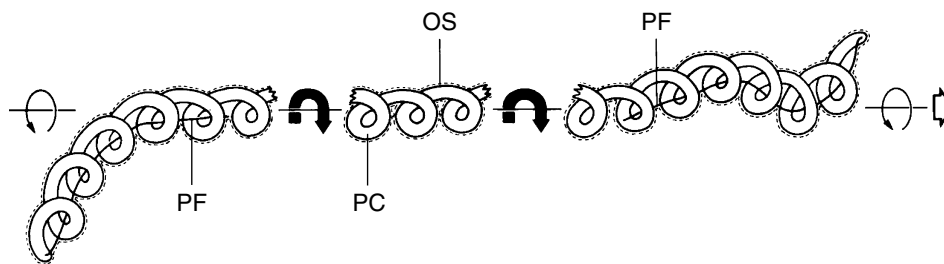
The flagella of spirochetes are comparable to those of other bacteria in that they contain a basal body, rod, flagellar hook, and filament. Beyond that, however, spirochetes differ from other bacteria in that their organelles for motility reside inside the cell and within the periplasmic space. Surrounding both the protoplasmic cylinder and the periplasmic flagella is an outer membrane sheath. This unique morphological structure allows them to bore through viscous gel-like media that inhibit the motility of most other bacteria. In fact, increasing the viscosity of the medium by the addition of a compound such as methylcellulose increases the speed of spirochete motility. Although they are structurally similar, the swimming motions of *Leptospira* and *Borrelia burgdorferi* appear to be quite different, and models have been developed to explain these differences.

The cell cylinder of a spirochete such as *Leptospira biflexa* forms a right-handed helix. Only one periplasmic flagellum extends from each end of the cell along the center axis of the helix, but flagella do not overlap. In a translating cell, the posterior end is hook shaped and the anterior end is spiral shaped (see Fig. 7-40). When the cells reverse direction, the end of the cell concomitantly changes shape. The periplasmic flagella rotate between the outer membrane sheath and the right-handed protoplasmic cylinder. They are more rigid than the cell cylinder, causing the end of the cell to conform to the shape of its flagellum. As viewed from the center of the cell toward one of the cell ends, CCW rotation of a periplasmic flagellum results in the flagellum at that end forming a left-handed helix whose helical pitch and helical diameter are larger than those of the cell body. This form is referred to as the spiral-shaped end. On the other hand, rotation of that periplasmic flagellum in the CW direction results in the flagellum and cell body at that end being hook shaped. When the periplasmic flagella are rotating in the same direction (both CCW or CW), the cell does not translate.

In the model for rotation of a *Leptospira*, forward thrust results from two rotations. The model can be conveniently viewed from behind a swimming cell (Fig. 7-41). First, both periplasmic flagella rotate CCW. Second, the torque of the rotating flagellum produces a CW counterrotation of the right-handed helical cell body. Two modes of thrust are proposed. CCW rotation of the anterior periplasmic flagellum causes that end of the cell to gyrate (i.e., to bend in a circular manner without necessarily rotating); it generates a backward-moving spiral wave. This wave is predicted to be left-handed and is sufficient for the cell to translate forward in a low-viscosity medium. Concomitantly,



**Fig. 7-40. *Leptospira biflexa* serovar patoc.** (a) Dark-field micrograph illustrating the hook-shaped ends and helical cell morphology. Cell length is 10  $\mu\text{m}$ . (b) Transmission electron micrograph of one cell end. Note the protoplasmic cell cylinder (PC) and periplasmic flagellum (PF). (c) Scanning electron micrograph illustrating the hook-shaped ends and right-handed helical PC. Cell length is 19  $\mu\text{m}$ . (Source: From S. F. Goldstein and N. W. Charon, *Cell Motility Cytoskel.* 9:101–110, 1988.)



**Fig. 7-41. Possible conformations and gyrations of periplasmic flagella and protoplasmic cylinder of *Leptospira*.** Motion is toward the right of the figure (open arrow). The anterior spiral end is left-handed. An observer behind the cell (left side of figure) sees the spiral and hook-shaped ends gyrate CCW (thin arrows) because of the rotation of the periplasmic flagella in that direction, while the protoplasmic cylinder rotates CW (wide filled arrows). (Source: From Goldstein, S. F. and N. W. Charon, *Cell Motility Cytoskel.* 9:101–110, 1988.)

CW rolling of the cell cylinder gives the organism the unique capacity to bore through gel-like media or connective tissue. Due to the roll of the cell cylinder, the torque is counterbalanced by the gyrations of the cell ends.

The model for *Leptospira* motility applies to *Treponema phagedenis* as well. This spirochete has a right-handed cell cylinder with four to eight short, nonoverlapping periplasmic flagella attached subterminally at each end. As with *Leptospira*, these structures cause the cell ends to be left-handed or irregular in shape. However, CCW gyration of the bent-shaped ends does not yield sufficient thrust to cause the cell to move in a low-viscosity medium. But because the cell is helical and right-handed, *T. phagedenis* can translate in a highly viscous gel-like medium such as 1% methylcellulose. The CW counterrotation of the cell cylinder is considered to cause cell movement.

*Borrelia burgdorferi* differs from other spirochetes in that it has a flat-wave morphology (nonhelical; planar wave). Cells swim with the flat waves being propagated from the anterior end of a translating cell to the posterior. These flat waves are reminiscent of eukaryotic cell motility and flagella. In *B. burgdorferi* there are between 7 and 11 periplasmic flagella attached near each end and they overlap in the center. Nonmotile mutants are rod shaped, indicating that the periplasmic flagella dictate the shape of the entire cell.

The flagellar filaments of most species of spirochetes contain three proteins (FlaB1, FlaB2, and FlaB3) and are attached subterminally at each end of the cell cylinder and within the periplasmic sheath that contains the sheath protein FlaA. A separate gene encodes each of these proteins. Periplasmic flagella consist of one to two FlaA proteins and three to four FlaB proteins. *B. burgdorferi* is an exception in that it has one FlaA protein and only one FlaB protein. Studies with mutants of *Brachyspira hyodysenteriae* indicate that the interaction of FlaA with the FlaB core impacts periplasmic flagella helical morphology.

Exclusive of chemotaxis genes, at least 36 motility genes have been identified in *B. burgdorferi* and *T. pallidum*. The motility genes of spirochetes appear to be very tightly clustered. Only 8 motility operons are present in *B. burgdorferi* and nine or 10 in *T. pallidum*. In *B. burgdorferi* there is a large motility gene cluster of 21 kb. The gene order is similar to that of a gene cluster found in *B. subtilis*. This cluster appears to be initiated by a  $\sigma^{70}$ -like promoter mapping upstream of the rod protein *figB*. The direct involvement of this housekeeping protein suggests the importance of motility in all phases of growth and survival of this spirochete in nature.

### Gliding Motility

Gliding bacteria display an unusual form of motility in that they glide on solid surfaces without the aid of flagella or other obvious external organs of locomotion. A diverse assortment of bacteria from many different phylogenetic groups displays this form of motility. Gliding motility has been observed in cyanobacteria (*Anabaena*, *Oscillatoria*, and many others), *Flavobacterium*, *Cytophaga*, *Flexibacter*, *Myxococcus*, *Beggiatoa*, *Leucothrix*, *Chloroflexus*, *Helicobacterium*, and *Mycoplasma*. As more studies have been conducted, it has become apparent that there are probably several different mechanisms of gliding motility.

*Myxococcus xanthus*, one of the most well-studied members of the gliding bacteria, travels by means of coordinated rafts or swarms. No obvious organs of locomotion

have been identified. Gliding motility by vegetative cells is controlled by at least two systems of genes. The S or social system is required for the rafts of cells to move out from the edge of a vegetative swarm. The A or adventurous system is required for single cells to move out from the edge of a vegetative swarm.

A number of genes affecting S motility have been identified. Type IV pili (see next section) are components of the S system. Mutants unable to synthesize type IV pili lack S motility. Cell interactions are required for S motility; individual cells are nonmotile. This form of motility may very likely be related to the twitching motility that requires type IV pili as described in the next section. The A system of motility is governed by a large number of genes but does not require pili. These include genes coding for the production of surface LPS. One class of mutants defective in A motility involves several *cgl* (for contact gliding) genes (*cglB* through *cglF*). Mutants in this class become transiently motile after contact with wild-type cells. The *cglB* gene codes for a protein that is essential for movement, but its function has not been elucidated. Another gene, *mglA*, is required for the proper functioning of both the A-motility and S-motility systems, suggesting that MglA (a GTPase) may play a regulatory role in both systems.

Motility defects caused by “frizzy” (*frz*) mutations affect the frequency of reversal in the direction of gliding. These mutants produce frizzy filaments of aggregating cells during development, which would be characteristic of cells unable to migrate to centers of aggregation. The *frz* genes show striking homology to the *che* genes that govern the chemotactic signal transduction systems in flagellated bacteria.

*Flavobacterium johnsoniae* (formerly known as *Cytophaga johnsonae*) cells move in groups, known as rafts, away from the point of inoculation to form thin, veil-like colonies with feathery edges. On solid media, colonies of motile cells spread much faster than can be accounted for by growth alone. Raft movement can be observed on the edge of the colony. Motility in this organism has been correlated with the ability to move polystyrene-latex beads over the cell surface. Mutants lacking this ability are all nongliding. Secretion of extracellular slime has been shown to mediate surface adhesion during gliding motility. Slime is often deposited on slime tracks on agar surfaces over which the cells move, and these tracks serve as preferred paths of movement for other cells. Extracellular material appears to be secreted ahead and to the sides of the leading organisms in a mass of cells, indicating that the cells are moving through the secreted slime. An alternative hypothesis involves cell surface macromolecules (proteins or glycoproteins) either in or on the outer membrane that promote adherence to the substratum. Some gliding motility mutants of *F. johnsoniae* are defective in the production of sulfonolipids present in the outer membranes. Motility in one of these mutants could be reestablished by the addition of cysteate, a sulfonolipid precursor that restores normal levels of cellular sulfonolipids.

## PILI OR FIMBRIAE

Bacteria from many different genera produce a multiplicity of nonflagellar, filamentous, proteinaceous appendages called **pili** (Latin, *pilus*, meaning “hair”) or **fimbriae** (Latin, *fimbria*, meaning “fringe”). Conjugative or **F pili** are involved in the mating process described in Chapter 3. A number of other types of pili, sometimes referred to as **common** or **generalized pili**, are also produced by numerous bacterial species. These



pili play a role in adherence of microorganisms to surfaces or to specific receptors on eukaryotic cells, aiding in the colonization of these ecological niches through the formation of biofilms. Type IV pili are involved in a form of motility known as **twitching**.

Members of the *Enterobacteriaceae* produce a wide assortment of pili or fimbriae (Table 7-7). In *E. coli*, as well as in most members of the *Enterobacteriaceae*, the most prevalent class of fimbriae are of type 1. These pili are composed of a short-tip fibrillar structure containing FimG and the FimH adhesin attached to a rod composed of FimA subunits. The FimH adhesin mediates binding to mannose oligosaccharides (mannose sensitivity). The chaperone-usher pathway, a donor strand complementation mechanism in which the chaperone donates a strand to complete pilin formation, mediates pilus synthesis. Expression of the type 1 fimbriae in *E. coli* exhibits recombination-mediated on-off phase variation. This phase variation correlates with the orientation of a short,

**TABLE 7-7. Pili or Fimbriae Produced by *Enterobacteriaceae* and Other Pathogenic Bacteria**

| Pilus Type | Properties                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | Encoded by genes ( <i>fimA–fimH</i> ) clustered at 98 min on <i>E. coli</i> genetic map; 17 kDa pilin subunits encoded by <i>fimA</i> ; agglutination of fowl or guinea pig erythrocytes inhibited by $\alpha$ -D-mannose (mannose sensitivity); produced by <i>E. coli</i> , <i>Klebsiella</i> , <i>Salmonella</i> , and several other species of <i>Enterobacteriaceae</i> .                                                                 |
| 2          | Nonhemagglutinating; observed in <i>Escherichia</i> and <i>Salmonella</i> strains; may be type 1 antigenic variants.                                                                                                                                                                                                                                                                                                                           |
| 3          | Mediate mannose-resistant agglutination of tannic acid–treated animal erythrocytes; expression in <i>K. pneumoniae</i> involves at least six genes ( <i>mrkA–mrkF</i> ). Produced by <i>Klebsiella</i> , <i>Salmonella</i> , <i>Yersinia</i> , <i>Proteus</i> , and <i>Providencia</i> .                                                                                                                                                       |
| IV         | Bundle-forming pili of enteropathogenic <i>E. coli</i> , other human and animal pathogens including <i>Neisseria gonorrhoeae</i> ; associated with “twitching” motility (nonflagellar movement). Encoded by <i>pilE</i> genes that result from nonreciprocal reactions with numerous silent loci, <i>pilS</i> .                                                                                                                                |
| P          | Produced by uropathogenic or pyelonephritis-associated (Pap) <i>E. coli</i> . Mediates binding Pap to digalactoside receptors (galactose sensitive) in urinary tract epithelium. Eleven genes ( <i>papA–pakK</i> ) are involved in biosynthesis and expression of P pili; Pap is an adhesin located at the tips of pili. Expression of Pap pili subject to on-off phase variation involving DNA methylation by deoxyadenosine methylase (Dam). |
| S          | Mediate mannose-resistant hemagglutination; facilitate binding to $\alpha$ -sialic acid-2,3- $\beta$ -D-Gal residues; encoded by <i>sfa</i> gene cluster in <i>E. coli</i> K1.                                                                                                                                                                                                                                                                 |
| F1C        | Nonhemagglutinating; encoded by 6 genes in <i>foc</i> gene cluster; <i>focA</i> encodes major fimbrial subunit; <i>focC</i> gene product required for fimbria formation; <i>focG</i> , <i>focH</i> encode minor fimbrial subunits; mediate adherence to collecting ducts and distal tubules of human kidney.                                                                                                                                   |
| K88        | Produced by porcine enterotoxigenic <i>E. coli</i> . Plasmid-borne structural genes ( <i>faeC–faeH</i> ) are involved.                                                                                                                                                                                                                                                                                                                         |
| K99        | Produced by bovine enterotoxigenic <i>E. coli</i> .                                                                                                                                                                                                                                                                                                                                                                                            |
| F17        | Mediate binding to <i>N</i> -acetylglucosamine receptors in calf intestinal mucosal cells; F17 gene cluster contains at least 4 genes.                                                                                                                                                                                                                                                                                                         |

invertible 314 bp DNA element (switch) located immediately upstream of *fimA*. The promoter for *fimA* is thought to reside within this invertible element and to direct the transcription of *fimA* when the element is in one orientation (on) but not when the element is in the alternate orientation (off).

Two genes, *fimB* and *fimE*, mapping immediately adjacent to the *fim* invertible element, are believed to encode site-specific recombinases. A model for the roles of *fimB* and *fimE* suggests that *fimB* promotes recombination of the *fim* invertible element in both directions, whereas *fimE* promotes recombination primarily from “on” to “off.” However, the on orientation of the *fim* invertible element is necessary but not totally sufficient for fimbrial expression, since strains in which the invertible element is locked in the on orientation continue to exhibit phase-variable expression of type 1 fimbriae. Other genes whose products affect the invertible element include *pilG*, which encodes the histone-like protein H1, and *himA* and *himD/hip*, which together encode IHF.

Immediately 3' to *fimA* are two genes (*fimC* and *fimD*) whose products are required for polymerization of the *fimA* gene product into pili. The *fimD* product is apparently incorporated into the outer membrane and serves as a scaffold or polymerization channel for pilus assembly. Genes encoding minor components of type 1 fimbriae (*fimF*, *fimG*, and *fimH*) are located distally to *fimA* and encode products similar to pilin structure. The *fimH* product appears to be the type 1 pilus adhesion component that interacts directly with the eukaryotic cell receptor and confers mannose sensitivity. It has been suggested that the *fimF* product aids in starting new pili and that the *fimG* product is an inhibitor of pilus polymerization.

An additional gene, *lrp*, is also required for normal activity of the *fim* switch. The *lrp* gene product, Lrp, is a site-specific DNA-binding protein that protects extended regions of DNA in vitro from nuclease digestion. It affects the transcription of many genes, collectively called the leucine-Lrp regulon. Lrp could influence the *fim* inversion either indirectly by altering the expression of other proteins, such as FimB and FimE or IHF, or directly by participating in the *fim* inversion as an auxiliary factor. Lrp is known to influence phase variation of Pap fimbriation by blocking the methylation of two *dam* sites in the promoter region of *papAB*. Lrp may be required either for the expression of an additional factor that promotes the *fim* inversion or as a direct participant in the recombination reaction.

Uropathogenic *E. coli* produce P pili encoded by genes *papA* through *papK* located in the *pap* (pilus associated with pyelonephritis) gene cluster. The P pili are composite fibers containing a rod joined to a thinner-tip fibrillum. Biogenesis of P pili occurs via the highly conserved chaperone-usher pathway. In this system, periplasmic chaperones escort pilus subunits to the usher, a large protein complex that facilitates the translocation and assembly of subunits across the outer membrane.

*Neisseria gonorrhoeae* and *N. meningitidis* produce type IV pili, which are essential determinants of virulence. These type IV pili, encoded by the *pilE* gene, mediate adhesion to host cells, preventing phagocytosis by polymorphonuclear leukocytes. Expression of *pilE* is controlled transcriptionally by the products of two linked chromosomal genes: *pilA* and *pilB*. In *N. gonorrhoeae*, antigenic variation occurs when DNA sequences from one of several silent pilin gene copies (*pilS*) are transferred unidirectionally to replace variable sequences within the *pilE* gene.

The *pilE* sequence changes alter the amino acid sequence of the pilin protein and the antigenicity of the pilus. Presumably, these antigenic changes aid in the avoidance

of the host immune response. DNA recombination is required for natural DNA transformation, DNA repair, and pilus antigenic variation. RecA mediates all these processes, as discussed in Chapter 3. A homologue of the recombination-dependent growth gene, *rdgC*, is involved in gonococcal pilin antigenic variation. There are two routes for generating pilin variation in *N. gonorrhoeae*: intragenomic recombination and recombination following transformation with exogenous DNA from lysed gonococci. The incoming donor sequence can be either a second *pilE* locus or any *pilS* locus. In either case, the antigenicity of the produced pilin differs from that originally present.

**Twitching** is a form of motility that is expressed as a jerky translocation over solid surfaces. In liquid medium, *P. aeruginosa* swims rapidly by means of flagella. However, once attached to a solid surface, large groups of cells are translocated by means of a flagellum-independent activity dependent on the presence of type IV pili. Other bacteria—for example, *N. gonorrhoeae*, and *E. coli*—also exhibit this unusual form of motility. Twitching is essential for biofilm formation and undoubtedly contributes to the ability of these organisms to form biofilms or to effect attachment to cells, thereby aiding in the invasion of human or animal hosts.

*Pseudomonas aeruginosa* is freely motile in an aqueous environment by means of polar flagella. However, it can display a swarming type of motility on agar surfaces through the development of alternate forms of flagella. *P. aeruginosa* also produces type IV pili that are involved in twitching motility and are presumed to mediate adherence to surfaces including those of eukaryotic cells. It has been shown that the initiation of biofilm formation in *P. aeruginosa* correlates with the emergence of hyperpiliated and highly adherent phenotypic variants that are deficient in swimming, swarming, and twitching motilities. These variants produced small, rough colonies and autoaggregated in liquid cultures, and rapidly initiated the formation of strongly adherent biofilms. In contrast, the large-colony variant (parent form) was poorly adherent, homogeneously dispersed in liquid cultures, and produced scant polar pili. It has been suggested that phase variation ensures the presence of different phenotypic forms that adapt readily to initiate biofilm formation under the appropriate environmental conditions.

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## CHAPTER 8

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# CENTRAL PATHWAYS OF CARBOHYDRATE METABOLISM

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The study of carbohydrate metabolism involves consideration of the following factors:

1. Central pathways of carbohydrate metabolism
2. Conversion of compounds to intermediates usable in central pathways
3. Mechanisms of energy (ATP) production
  - a. Substrate-level phosphorylation
  - b. Oxidative phosphorylation
  - c. Other mechanisms of energy transfer
4. Metabolic steps involved in the generation and use of reducing activity
  - a. Reduction of pyruvate or other substrates to fermentation end products
  - b. Biosynthetic reactions requiring reducing action
5. Oxygen involvement in energy-generating reactions
  - a. Aerobic metabolism
  - b. Anaerobic metabolism
  - c. Facultative metabolism
6. Metabolic intermediates serving as biosynthetic precursors
7. Reactions that replenish biosynthetic intermediates (anapleurotic reactions)
8. Metabolic and genetic regulatory systems

Carbohydrates are not the only compounds utilized as sources of energy by microorganisms. Fatty acids, lipids, amino acids, purines, pyrimidines, and a wide variety of other compounds can also serve as carbon and energy sources. Generally, utilization of an alternate substrate involves its conversion to an intermediate intrinsic to one of the central pathways of carbohydrate metabolism. Some of these alternate pathways are discussed in Chapter 10. The details of energy production and metabolite transport are discussed in Chapter 9.

## ALTERNATE PATHWAYS OF CARBOHYDRATE METABOLISM

### Fructose Bisphosphate Aldolase Pathway

One major pathway of glucose degradation is accomplished by the series of reactions shown in Figure 8-1. This pathway is often referred to as the Embden-Meyerhof-Parnas (EMP) pathway in recognition of some of the earliest scientists who contributed to its elucidation. Several important reactions occur in the EMP pathway:

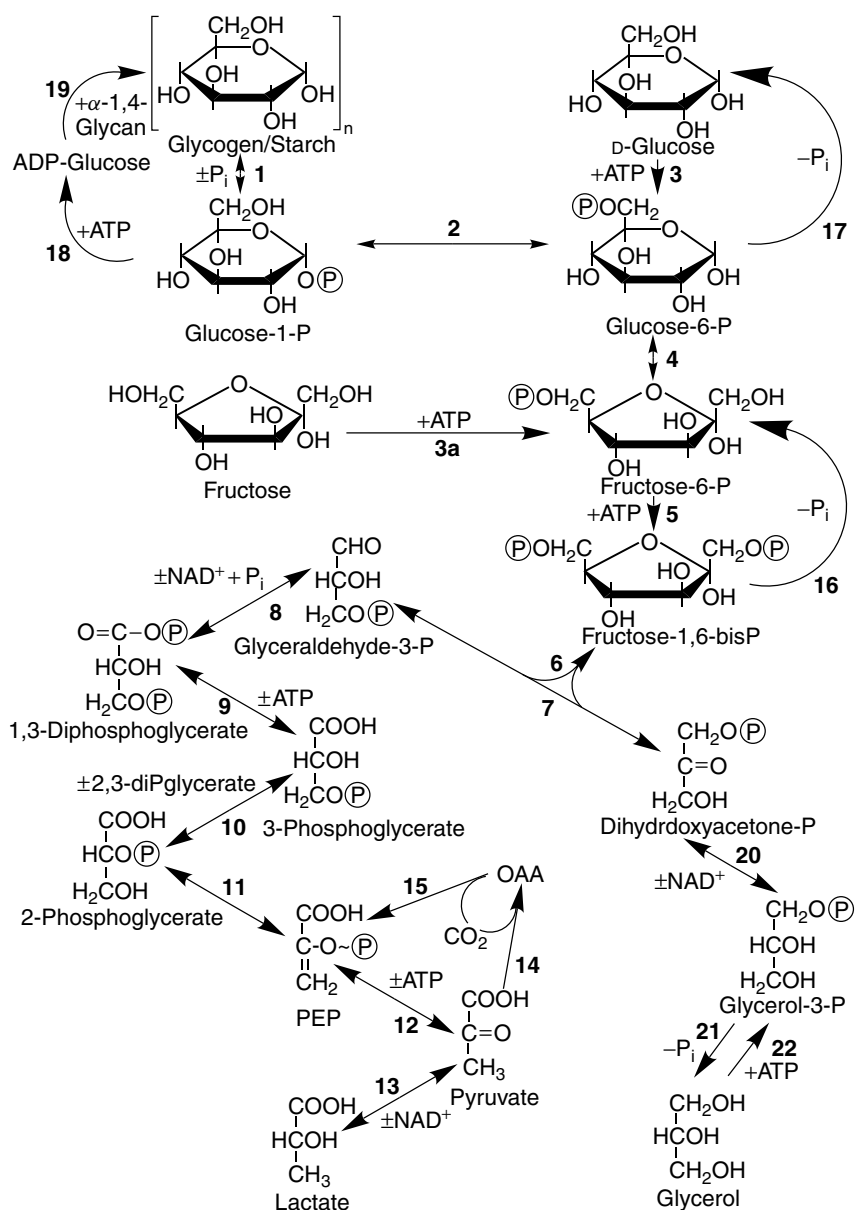


Fig. 8-1

1. Phosphorylation of glucose and fructose-6-phosphate by ATP
2. Cleavage of fructose-1,6-bisphosphate to trioses by a specific aldolase
3. Structural rearrangements
4. Oxidation–reduction and  $P_i$  (inorganic phosphate) assimilation



**Fig. 8-1 Fructose biphosphate (FBP) aldolase or Embden-Meyerhof-Parnas (EMP) pathway of glycolysis.** Enzymes involved in gluconeogenesis are also shown. The italicized three-letter designation indicates the structural gene for the enzyme in *E. coli*.

#### Glycolytic enzymes

1. Phosphorylase. Degradation of glycogen or starch to G-1-P.
2. Phosphoglucomutase. Isomerization of G-1-P to G-6-P.
3. Hexokinase. Phosphorylation of glucose to G-6-P, using ATP Hexokinase also phosphorylates fructose to F-6-P using ATP (reaction 3a).
4. Phosphoglucisomerase (*pgi*). Isomerization of G-6-P to F-6-P.
5. Phosphofructokinase (*pfkA*). Phosphorylation of F-6-P to FBP using ATP.
6. Fructose biphosphate (FBP) aldolase (*fbaA*). Cleaves FBP to GA-3-P and dihydroxyacetone phosphate.
7. Triose phosphate isomerase (*tpi*). Interconverts GA-3-P and dihydroxyacetone-P.
8. Glyceraldehyde-3-phosphate dehydrogenase (*gap*). Oxidizes GA-3-P to 1,3-diphosphoglycerate using nicotinamide adenine dinucleotide ( $NAD^+$ ) and inorganic phosphate ( $P_i$ ) to form  $NADP + H^+$ .
9. Phosphoglycerokinase (*pgk*). Generates ATP from ADP.
10. Phosphoglyceromutase (*pgm*). Uses 2,3-diphosphoglycerate to convert 3-phosphoglycerate to 2-phosphoglycerate.
11. Enolase (*eno*). Enolization of 2-phosphoglycerate forms high-energy phosphate bond ( $\sim P$ ; encircled P) in phosphoenolpyruvate (PEP).
12. Pyruvate kinase (*pykA*; *pykF*). Generates ATP from ADP.
13. Lactate dehydrogenase. Reduces pyruvate to lactate using  $NADP + H^+$ .

#### Gluconeogenesis enzymes

14. Pyruvate carboxylase. Converts pyruvate to oxaloacetic acid (OAA) via carbon dioxide ( $CO_2$ ) fixation using ATP.
15. PEP carboxykinase. Forms phosphoenolpyruvate from OAA using GTP.
16. Fructose-1,6-bisphosphatase. Removes  $P_i$  from F-1,6-bisP to form F-6-P.
17. Glucose-6-phosphatase. Removes  $P_i$  from G-6-P to form glucose.
18. ATP-glucose pyrophosphorylase. Forms ADP-glucose from G-1-P and ATP.
19. Glycogen synthase. Adds  $\alpha$ -1,4-glycan to ADP-glucose to form glycogen or starch.

#### Glycerol formation or utilization

20. Glycerophosphate dehydrogenase. Reduces dihydroxyacetone-P to glycerol-3-P.
21. Phosphatase. Removes  $P_i$  from glycerol-3-P to form glycerol.
22. Glycerol kinase. Phosphorylates glycerol using ATP.

5. Energy transfer by phosphoglycerokinase and pyruvate kinase
6. Metabolic and genetic regulation of the pathway

The overall equation for the ethanol fermentation in yeast can be shown as



The EMP pathway is common to a great many microorganisms as well as higher forms. The enzyme fructose biphosphate (FBP) aldolase is one of the most critical steps in the pathway. In the absence of this enzyme, glucose or other hexose sugars must be metabolized via one of several alternative pathways, as discussed later.

In general, glycolysis in muscle tissue, yeast, and many bacterial species appears to be identical in terms of the intermediates involved. Pyruvate is the last common intermediate. In yeast, pyruvate is cleaved to acetaldehyde and carbon dioxide. The acetaldehyde is then reduced to ethanol by alcohol dehydrogenase. In muscle tissue and in lactic acid bacteria (*Streptococcus*, *Lactococcus*, *Lactobacillus*), pyruvate is reduced to lactate. Other microorganisms that use the EMP pathway have the capacity to convert pyruvate to a wide variety of other fermentation end products. These fermentation pathways are discussed in more detail in Chapter 10.

The enzymes of the glycolytic pathway and the tricarboxylic acid (TCA) cycle are regulated either positively or negatively by specific metabolic intermediates (**feed-back control**). As shown in Table 8-1, AMP or FBP stimulates several enzymes. Some metabolites, such as AMP, phosphoenolpyruvate (PEP), or dihydroxyacetone phosphate (DHAP), or cofactors such as reduced nicotinamide adenine dinucleotide (NADH), may have an inhibitory effect on certain catabolic enzymes. Elevation of the AMP concentration may signal a low-energy state. In the case of DHAP or PEP, AMP may regulate the flow of these metabolites into biosynthetic pathways or transport functions. The FBP-activated lactic acid dehydrogenases are characteristically produced by a number of lactic acid-producing bacteria. In streptococci, 6-phosphogluconate inhibits the activity of phosphohexose isomerase, and the activity of 6-phosphogluconate dehydrogenase is inhibited by FBP (Table 8-1). Details of the regulation of prokaryotic gene expression as they apply to carbohydrate metabolism are discussed in Chapter 5.

The structural genes for the enzymes in the glycolytic pathway of *E. coli* have been identified as shown in Figure 8-1. The expression of *pgi*, *pfk*, *fbaA*, and *pykA*, genes that code for enzymes in both the upper and lower portions of the glycolytic pathway, are unaffected by glucose. However, the *gap* gene (coding for glyceraldehyde-3-phosphate dehydrogenase) and the phosphoglucokinase (*pgk*) operon are activated by glucose. This activation is dependent on catabolite-control protein A (CcpA), a global regulator that represses several catabolic operons involved in the degradation of secondary carbon sources. This regulator is also involved in glucose repression of genes encoding enzymes of the TCA cycle (see later discussion). The genes and operons required for the utilization of specific carbon sources are usually expressed only if the carbon source is present and glucose or other sugars that can be used via the glycolytic pathway are absent from the growth medium.

**TABLE 8-1. Metabolic Regulation of Enzymes of Glycolysis and Tricarboxylic Acid Cycle**

| Enzyme                            | Positive Effector | Negative Effector     |
|-----------------------------------|-------------------|-----------------------|
| 6-Phosphogluconate dehydrogenase  |                   | FBP                   |
| Phosphohexose isomerase           |                   | 6-PG                  |
| Fructose-1,6-bisphosphatase       |                   | AMP                   |
| Phosphofructokinase               | ADP               | PEP                   |
| Pyruvate kinase 1                 | FBP               |                       |
| Pyruvate kinase 2                 | AMP               |                       |
| Citrate synthase                  | AMP               | NADH, or $\alpha$ -KG |
| Malate dehydrogenase              |                   | NADH                  |
| Malate enzyme                     |                   | NADH                  |
| Phosphoenolpyruvate carboxykinase |                   | NADH                  |
| Phosphoenolpyruvate carboxylase   | FBP, acetyl-CoA   | Aspartate             |

FBP, fructose-1,6-bisphosphate; 6-PG, 6-phosphogluconate; AMP, adenosine monophosphate; ADP, adenosine diphosphate; PEP, phosphoenolpyruvate; NADH, reduced nicotinamide adenine dinucleotide.

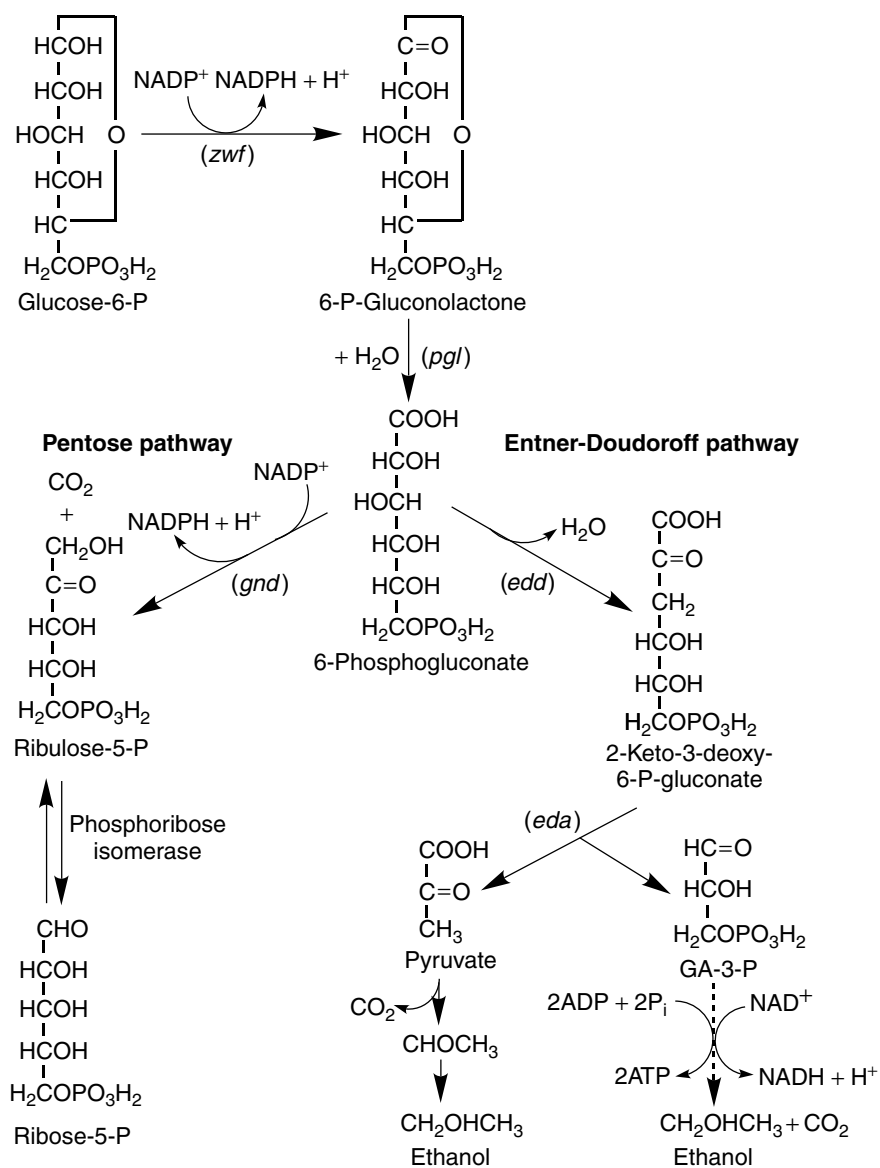
### Alternate Pathways of Glucose Utilization

Warburg and Christian provided the first evidence for the existence of an alternative pathway for the utilization of hexose sugars. They described the oxidation of glucose-6-phosphate to 6-phosphogluconate (6-P-G) via G-6-P dehydrogenase (*zwischenferment*). They also described the decarboxylation of 6-P-G to form a pentose sugar. For a long time, relatively little attention was paid to the real significance of this alternative pathway because Meyerhof and others strongly asserted that the FBP aldolase (EMP) pathway was the main route of glucose catabolism. Subsequently, ribulose-6-phosphate was shown to be the first product formed and was converted to ribose-5-phosphate via an isomerase reaction. This series of reactions is common to several alternate pathways of carbohydrate metabolism as shown in Figures 8-2 and 8-4 and discussed below.

### Entner-Doudoroff or Ketogluconate Pathway

The Entner-Doudoroff pathway branches from the ketogluconate pathway of hexose oxidation as shown in Figure 8-2. It consists of two enzymes: 6-phosphogluconate dehydratase (encoded by *edd*) and 2-keto-3-deoxy-6-phosphogluconate (KDPG) aldolase (*eda*). Thus, the Entner-Doudoroff pathway differs from the EMP pathway primarily in the form of the 6-carbon intermediate that undergoes C<sub>3</sub>-C<sub>3</sub> cleavage (aldol cleavage) to form three-carbon intermediates. Note that the formed glyceraldehyde-3-phosphate is metabolized to pyruvate via the triose phosphate pathway common to the EMP and phosphoketolase pathways. Originally described as a major pathway in *Pseudomonas*, the Entner-Doudoroff pathway is active in *Escherichia coli* and many other gram-negative species. It is also present in many other microorganisms including members of the archaeobacteria. The widespread occurrence of the Entner-Doudoroff pathway has led to its consideration as a much more significant pathway than previously recognized and has given rise to the suggestion that it may have predated the EMP pathway.

*E. coli* utilizes gluconate via the Entner-Doudoroff pathway. Gluconate dehydratase activity is virtually absent in cells grown on glucose and is induced only by gluconate.



**Fig. 8-2. Divergent pathways from 6-phosphogluconate.** The structural genes for the enzymes in *E. coli* are indicated by their three-letter designations. In the core pathway, glucose-6-phosphate dehydrogenase (*zwf*, for *zwischenferment*) oxidizes glucose-6-phosphate to 6-phosphogluconolactone. The lactone is dehydrated to 6-phosphogluconate via lactonase (*pgl*). In the pentose pathway, 6-phosphogluconate is oxidized to ribulose-5-phosphate and carbon dioxide by 6-phosphogluconate dehydrogenase (*gnd*). Phosphoribose isomerase maintains ribulose-5-phosphate and ribose-5-phosphate in equilibrium. In the Entner-Doudoroff pathway, 6-phosphogluconate is dehydrated by 6-phosphogluconate dehydratase (*edd*) to yield 2-keto-3-deoxy-6-phosphogluconate (KDPG). The enzyme KDPG aldolase (*eda*) cleaves KDPG to form pyruvate and GA-3-P (glyceraldehyde-3-phosphate). Pyruvate decarboxylase action yields ethanol and carbon dioxide. The GA-3-P is metabolized via the triose phosphate portion of the EMP pathway to yield ethanol and carbon dioxide. The net yield in the Entner-Doudoroff pathway is 2 ethanol + 2  $\text{CO}_2$ .

High basal levels of KDPG aldolase activity are present regardless of the carbon source. This enzyme plays a major role in the metabolism of pectin and aldohexuronate by *Erwinia* and other related organisms, as discussed in Chapter 10. In *E. coli* the *edd* and *eda* genes are closely linked to *zwf*, which codes for G-6-P dehydrogenase (Fig. 8-2). However, these genes are regulated under a separate set of regulatory controls. The *zwf* gene is subject to growth rate-dependent regulation at the level of transcription. On the other hand, the *edd-eda* operon is regulated by a gluconate-responsive promoter,  $P_1$ , located upstream of *edd*, which is responsible for induction of the Entner-Doudoroff pathway. High basal levels of KDPG aldolase are explained by constitutive transcription of *eda* from additional promoters ( $P_2$ ,  $P_3$ , and  $P_4$ ) within the *edd-eda* region but not from  $P_1$ .

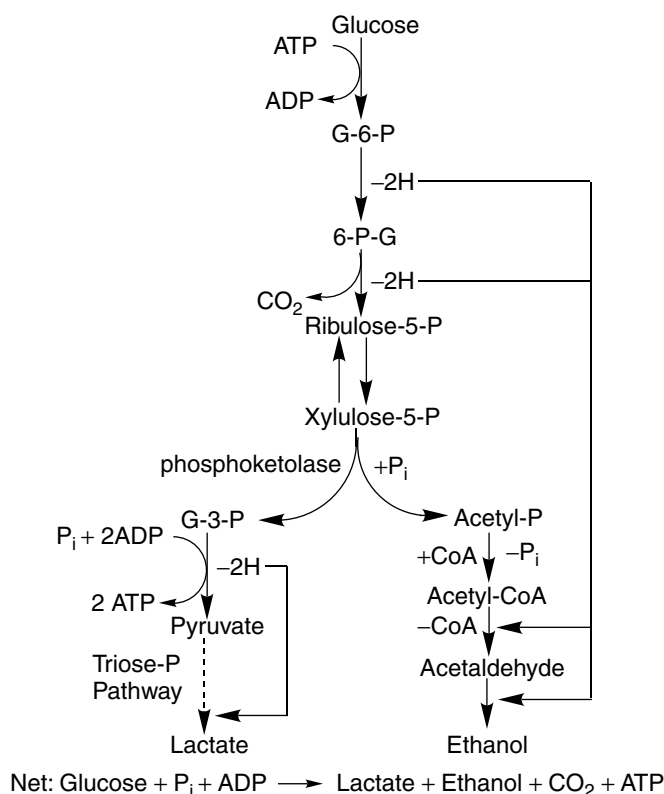
The Entner-Doudoroff pathway is active in several *Pseudomonas* species. It appears to be the sole pathway for the metabolism of glucose in *Zymomonas mobilis* and a major pathway in other members of the pseudomonad group of organisms as well as other gram-negative species. However, *Z. mobilis* is unique in that it is the only genus known to utilize the Entner-Doudoroff pathway anaerobically. This organism lacks an oxidative electron transport system and is, therefore, obligately fermentative. The pathway is inefficient in that it yields only 1 mol of ATP per mol of hexose fermented. It is of special interest for the industrial production of alcohol, since the yield of ethanol approaches the theoretical 2 mol/mol of substrate. Rapid production of ethanol by *Z. mobilis* as the sole product of sugar fermentation results from the presence of pyruvate decarboxylase, an enzyme not frequently observed in bacteria.

A cyclic version of the Entner-Doudoroff pathway is used by *P. aeruginosa* to metabolize carbohydrates. In this pathway, the catabolism of mannitol occurs via glucose-6-phosphate and requires the activity of glucose-6-phosphate dehydrogenase. Mannitol is converted to fructose by mannitol dehydrogenase. Fructose is phosphorylated to F-6-P by fructokinase. Phosphoglucisomerase converts F-6-P to G-6-P, which is then utilized through the Entner-Doudoroff pathway to form GA-3-P and pyruvate. The GA-3-P is then recycled to F-1,6-BP and F-6-P. The enzymes responsible for conversion of glucose to GA-3-P and pyruvate are coordinately regulated and induced by growth on glycerol, fructose, mannitol, glucose, and gluconate. The genes regulating this pathway, referred to as the *hex* regulon, are clustered in at least three operons near 39 minutes on the chromosome and are under the control of the *hexR* repressor.

### Phosphoketolase Pathway

One major fermentation pathway involves the conversion of ribulose-5-phosphate to xylulose-5-phosphate (X-5-P). The X-5-P is then cleaved to form a  $C_3$  compound (glyceraldehyde-3-phosphate) and a  $C_2$  compound (acetyl phosphate) by the action of the phosphoketolase enzyme (Fig. 8-3). Glyceraldehyde-3-phosphate is metabolized via the triose phosphate portion of the EMP pathway to form lactate. Acetyl phosphate is converted to acetyl CoA, which is then reduced to ethanol. The conversion of glucose to pentose sugars serves as a major source of the reduced NADP that drives a myriad of biosynthetic reactions. However, this pathway is not essential for the growth of *E. coli*. Mutants blocked in the pentose phosphate pathway still grow with glucose as the carbohydrate source without other nutritional supplements because other routes are available for the formation of pentoses and reduced NADP (e.g., the malate enzyme and isocitrate dehydrogenase, as discussed later).

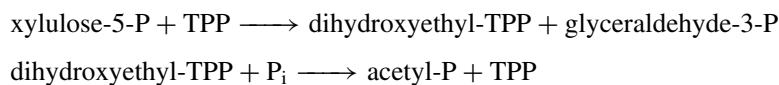


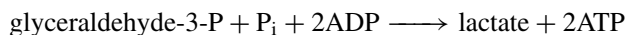
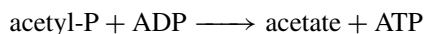


**Fig. 8-3. Phosphoketolase or hexose monophosphate pathway.** G-6-P, glucose-6-phosphate; 6-P-G, 6-phosphogluconate; G-3-P, glyceraldehyde-3-phosphate.

A number of microorganisms utilize the phosphoketolase pathway as the major route of glucose metabolism. *Leuconostoc mesenteroides*, a typical heterofermentative organism, utilizes this pathway, yielding lactate, ethanol, and carbon dioxide as shown in Figure 8-3. Within the genus *Lactobacillus* it has been possible to clearly differentiate **homofermentative** (*L. casei*, *L. pentosus*) from **heterofermentative** (*L. lysopersici*, *L. pentoaceticus*, *L. brevis*) types. Heterofermentative species are found in the genera *Streptococcus*, *Lactococcus*, *Pediococcus*, *Microbacterium*, some *Bacillus* species, and the mold *Rhizopus*. In *Lactobacillus pentoaceticus* and *Leuconostoc mesenteroides*, the basic pathway of glucose conversion leads to the formation of equimolar amounts of lactate, ethanol, and carbon dioxide. A commonly observed variation involves formation of considerable quantities of glycerol, as discussed in Chapter 11.

The pathway outlined in Figure 8-3 does not reveal some of the details of the reactions involved. In actuality, cleavage of the pentose molecule involves the cofactors thiamine pyrophosphate (TPP) and coenzyme A (CoA) as shown in the following series of reactions:

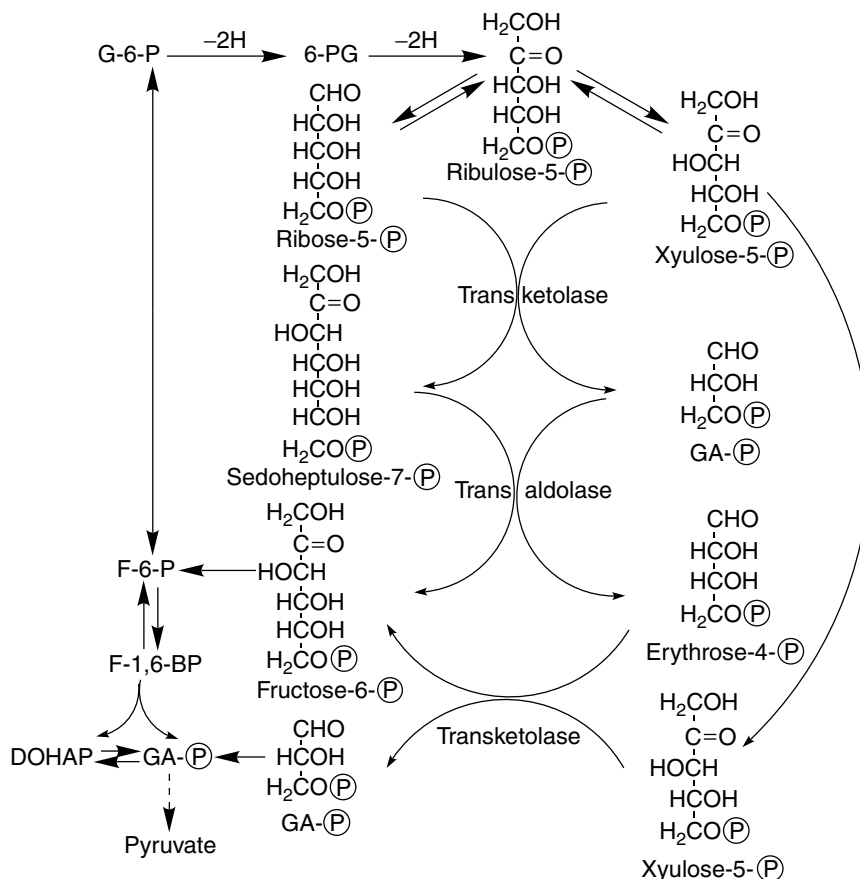




If acetyl phosphate kinase is present, an additional ATP will be generated. In this series of reactions, TPP or diphosphothiamine (DPT) functions as a  $\text{C}_2$  carrier in a similar manner to its function in the pyruvate dehydrogenase and  $\alpha$ -ketoglutarate dehydrogenase complex of reactions in the TCA cycle.

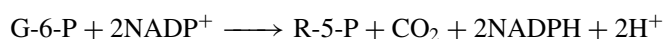
### Oxidative Pentose Phosphate Cycle

In some organisms a cyclic mechanism, the **oxidative pentose phosphate cycle**, accounts for the complete oxidation of carbohydrates (Fig. 8-4). In this cycle, G-6-P

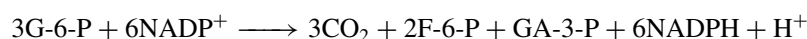


**Fig. 8-4. Oxidative pentose phosphate cycle.** Encircled P, phosphate group; G-6-P, glucose-6-phosphate; 6-PG, 6-phosphogluconate; F-6-P, fructose-6-phosphate; F-1,6-BP, fructose-1,6-bisphosphate; DOHAP, dihydroxyacetone phosphate; GA-P, glyceraldehyde-3-phosphate.

is converted to ribulose-5-phosphate and  $\text{CO}_2$ . Ribulose-5-phosphate is maintained in equilibrium with ribose-5-phosphate (R-5-P) and xylulose-5-phosphate (X-5-P) by the action of ribose phosphate isomerase and ribulose phosphate epimerase. Transketolase converts R-5-P and X-5-P to sedoheptulose-7-phosphate (SH-7-P) and glyceraldehyde-3-phosphate (GA-3-P). SH-7-P and GA-3-P are converted to F-6-P and erythrose-4-phosphate (E-4-P) via transaldolase. Transketolase also catalyzes the conversion of E-4-P and X-5-P to F-6-P and GA-3-P. By reversal of the FBP aldolase and G-6-P isomerase reactions, GA-3-P and F-6-P may be converted to G-6-P. The G-6-P can then reenter the oxidative pentose cycle. After one turn of the cycle, the net reaction is



Three turns of the cycle are required to produce one triose phosphate:



Repetitive action of the cycle could account for the complete oxidation of G-6-P:



However, the cycle does not appear to function in this manner under normal conditions. It is important to note that glycolysis generates NADH, which can be reoxidized by linkage to the electron transport system, or under anaerobic conditions it can be used to reduce an oxidized substrate, such as pyruvate, to lactate.

By contrast, the pentose phosphate pathway generates NADPH, which is used primarily for reducing power in biosynthetic reactions (e.g., the conversion of  $\alpha$ -ketoglutarate to glutamate or the incorporation of acetate into fatty acids) and is not linked to the terminal respiratory system. Operation of the oxidative pentose phosphate cycle provides for the formation of two very important biosynthetic precursors, SH-7-P and E-4-P, which serve as precursors to the aromatic amino acids as discussed in Chapter 15. The pentose phosphate cycle operates to yield a fermentative end product only under certain conditions and in a few unusual organisms. *Acetobacter xylinum* uses a variation of this pathway to produce acetic acid (see Chapter 11).

The oxidative pentose cycle (Fig. 8-4) assumes significance in the energy production of some organisms. Although the cycle is operative in *E. coli*, yeasts, streptomycetes, and fungi, it has been difficult to assess the relative importance of this pathway as compared to the combined operation of the EMP pathway and TCA cycle. The aerobic organism, *Gluconobacter suboxydans*, cannot ferment glucose and does not appear to contain the enzymes of the TCA cycle. Studies on isotope distribution into various products indicate that this organism uses a modification of the pentose phosphate cycle as a pathway of carbohydrate utilization.

The isotope distribution label from glucose labeled with  $^{14}\text{C}$  in the C-3 and C-4 positions is helpful in determining which major metabolic pathways are utilized by a given organism. As shown in Figure 11-2, utilization of the EMP pathway leads to distribution of  $^{14}\text{C}$  in the carbon dioxide formed. No other products are labeled. If the HMP pathway is used, there is no labeling of carbon dioxide, and the ethanol and lactate show an equal label distribution. If the Entner-Doudoroff pathway is utilized, half the label will be found in carbon dioxide and half in ethanol. Although this method

of determining the route of carbohydrate metabolism has some limitations, especially if more than one pathway is operative at the same time, it provides an important basis for later studies using a variety of other techniques.

## GLUCONEOGENESIS

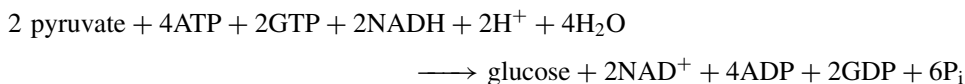
Growth of microorganisms on so-called poor carbon sources, such as L-malate, succinate, acetate, or glycerol, requires the ability to synthesize hexoses needed for the production of cell wall mucopeptides, storage glycogen, and other compounds derived from hexose, such as pentoses, for nucleic acid biosynthesis. Hexose synthesis involves a reversal of carbon flow from pyruvate (**gluconeogenesis**). This could potentially be achieved by the reversal of the enzymes in the EMP glycolytic pathway (Fig. 8-1). However, of the major enzymatic reactions involved in glycolysis, three are insufficiently reversible to allow carbon flow from pyruvate in the direction of hexose synthesis.

First, pyruvate kinase is not reversible because the free-energy requirement is too great. Instead, the formation of PEP is catalyzed by **PEP carboxykinase**, the first committed step in gluconeogenesis (reaction 15 in Fig. 8-1). This  $Mg^{++}$ -dependent enzyme requires GTP as the phosphate donor:



A second irreversible enzyme is phosphofructokinase (reaction 5 in Fig. 8-1). To overcome this deterrent to gluconeogenesis, fructose-1,6-bisphosphatase (fructose-1,6-bisphosphate 1-phosphohydrolase, reaction 16 in Fig. 8-1) dephosphorylates FBP to yield F-6-P and  $P_i$ . The relative rates of glycolytic phosphofructokinase and gluconeogenic fructose bisphosphatase determine the direction of net carbon flux in the EMP pathway. In *E. coli* and *S. enterica*, mutants lacking fructose-1,6-bisphosphatase (encoded by *fbp*) cannot grow on L-malate, succinate, glycerol, or acetate (so-called gluconeogenic substrates). Organisms grown on pentoses can make hexose phosphates from GA-3-P.

The third bypass reaction required for gluconeogenesis involves dephosphorylation of G-6-P (reaction 17 in Fig. 8-1). Glucose-6-phosphatase removes  $P_i$  from G-6-P to yield free glucose. Thus, the formation of glucose from pyruvate requires a considerable expenditure of energy:



## Regulation

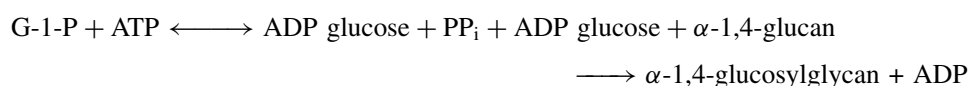
A major regulatory step in gluconeogenesis is PEP carboxykinase, encoded by *pckA* in *E. coli*. This enzyme is regulated by catabolite repression, a process in which gluconeogenesis is inhibited when glucose or other carbohydrate carbon sources are available. Maximum levels of PEP carboxylase are also induced at the onset of the stationary phase of growth, presumably to ensure the synthesis of adequate carbohydrate storage reserves or to provide metabolites from the upper

part of the EMP pathway as the organism converts proteins to gluconeogenic amino acids. The stationary phase induction of PEP carboxykinase requires cyclic AMP as well as a regulatory signal, the nature of which has not been fully elucidated.

In *B. subtilis*, the genes governing the reactions in the glycolytic cycle have been identified as shown in Figure 8-1.

### Glycogen Synthesis

Many organisms store glycogen as an energy reserve. Bacteria form glycogen using the enzyme ADP glucose pyrophosphorylase (reaction 18 in Fig. 8-1) and the branching enzyme glycogen synthase (1,4- $\alpha$ -D-glucan:1,4- $\alpha$ -D-glucan 6- $\alpha$ -D-glucanotransferase; reaction 19 in Figure 8-1):

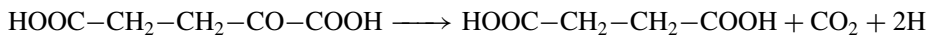
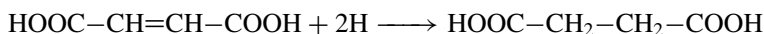


In *E. coli*, mutants defective either in glycogen synthase or in ADP glucose pyrophosphorylase are unable to accumulate glycogen. Glycogen synthesis in *E. coli* is regulated by both the *relA* gene, which mediates the stringent response to amino acid starvation when the cells are using glucose but not when the cells are using glycerol, and by cyclic AMP. These two regulatory controls are independent of each other in that each regulatory process can be expressed in the absence of the other.

Glycogen synthesis in *E. coli* is regulated at the level of ADP glucose pyrophosphorylase (encoded by *glgC*). Glucose ADP synthetase, the first unique step in glycogen synthesis in this organism, is activated by glycolytic intermediates with FBP as the activator and AMP, ADP, and  $\text{P}_i$  as inhibitors. The ADP glucose synthetases of *E. coli* and *S. enterica* show considerable similarity in that both consist of four identical subunits and have the same spectrum of activators and inhibitors. Genetically, the *glg* genes of both organisms are clustered at the same point (75 min) on their respective genetic maps. A number of genes encoding catabolic, biosynthetic, and amphibolic enzymes in enteric bacteria are transcriptionally regulated by a complex catabolite repression-activation mechanism that involves enzyme III of the phosphotransferase system as one of the regulatory components. Comparable systems have been described for the regulation of gluconeogenesis in a wide variety of microorganisms from yeast to symbiotic nitrogen-fixing bacteria.

### TRICARBOXYLIC ACID CYCLE

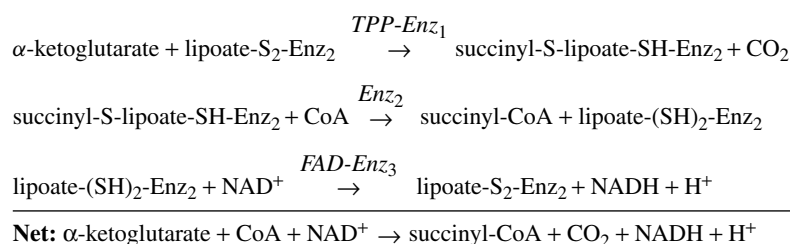
Sir Hans Krebs and co-workers demonstrated that  $\text{C}_4$  dicarboxylic acids, such as succinate and malate, the  $\text{C}_5$  compound  $\alpha$ -ketoglutarate, and the  $\text{C}_6$  compound citrate, were all oxidized by pigeon breast muscle. Citrate was synthesized from added oxaloacetate. The low yield of citrate was explained by the fact that both  $\alpha$ -ketoglutarate and citrate were formed. Demonstration of succinate formation from fumarate and oxaloacetate in the presence of malonate (an inhibitor of succinate dehydrogenase) led to the conclusion that succinate could be formed by either oxidative or reductive reactions:

*$\alpha$ -Ketoglutarate dehydrogenase**Fumarate reductase*

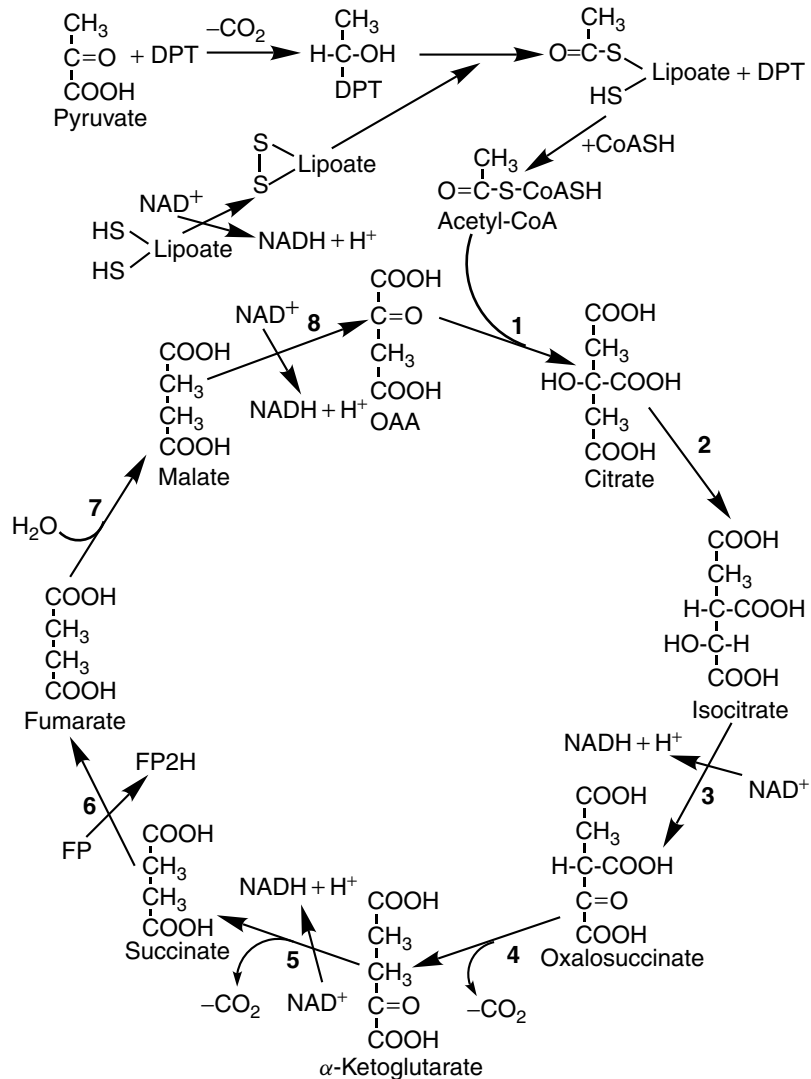
These observations led Krebs to propose the cyclic mechanism for the oxidation of pyruvate shown in Figure 8-5. He theorized that oxaloacetate condensed with a  $\text{C}_3$  compound (presumably pyruvate) derived from glycolysis to yield a  $\text{C}_7$  intermediate that was converted to citrate via decarboxylation. It was subsequently shown that a  $\text{C}_2$  intermediate (acetyl-CoA) condensed with oxaloacetate to form citrate. The presence of the condensing enzyme (citrate synthase) has been confirmed in mammalian systems as well as in yeast and a wide variety of bacteria.

Malate represents a pivotal point in the cycle. It participates in several alternative reactions. It may be oxidized to oxaloacetate via the NAD-linked malate dehydrogenase (Fig. 8-5) as observed in *E. coli* or via a direct cytochrome-linked dehydrogenase as seen in *Pseudomonas* and *Serratia*. In either case, energy is generated. Many organisms can decarboxylate malate to pyruvate (malic enzyme), and subsequently carboxylate pyruvate, to form oxaloacetate. These reactions provide a scavenger system for reclaiming carbon dioxide (anapleurotic reactions). A few organisms apparently utilize this system for completion of the TCA cycle when the malate dehydrogenase enzyme is absent. However, the malic enzyme is not a normal link in the cyclic operation of the TCA cycle of most organisms. The fact that it is linked to NADP rather than NAD is of major significance in that it provides an alternate route of generating reducing activity for biosynthetic reactions that require reduced NADP.

The  $\alpha$ -ketoglutarate dehydrogenase system is a complex containing three enzyme components: a thiamine pyrophosphate-dependent  $\alpha$ -ketoglutarate dehydrogenase ( $\text{Enz}_1$ ), dihydrolipoyl *trans*-succinylase ( $\text{Enz}_2$ ), and an FAD-dependent dihydrolipoyl dehydrogenase ( $\text{Enz}_3$ ):



This complex of enzymes is comparable to the pyruvate dehydrogenase complex that catalyzes the oxidative decarboxylation of pyruvate at the initial stage of the TCA cycle (Fig. 8-5). However, the individual components differ from each other in physicochemical properties and specificity except for the third enzyme in which the components are identical and functionally interchangeable. Studies with mutants deficient in various components of the  $\alpha$ -ketoglutarate dehydrogenase complex indicate that dihydrolipoyl components of both systems are encoded by *lpd*.



**Fig. 8-5. The Krebs tricarboxylic (citric) acid cycle.** OAA, oxaloacetate; DPT, diphosphothiamine; FP, flavoprotein. Structural genes for the enzymes in *Bacillus subtilis* are indicated by italicized three-letter designations: 1. Citrate synthase (*citZ*). 2. Aconitase (*citB*). 3 and 4. Isocitrate dehydrogenase (*citC*). 5.  $\alpha$ -ketoglutarate dehydrogenase (*odhA*, *pdhA*). 6. Succinate dehydrogenase (*sdhA*). 7. Fumarase (*citG*). 8. Malate dehydrogenase (*citH*). The conversion of pyruvate to  $\text{CO}_2$  and acetyl-CoA shown at the top of the diagram involves a group of enzymes known as the pyruvate dehydrogenase complex.

Under anaerobic conditions the TCA cycle no longer functions as such because the links to terminal respiration are required to maintain the activities and synthesis of succinate dehydrogenase and the  $\alpha$ -ketoglutarate dehydrogenase complex. However, synthesis of these intermediates is required for biosynthetic reactions. Fumarate reductase activity is increased, providing a mechanism for continued succinate

synthesis. Thus, the TCA cycle now functions as a branched biosynthetic pathway: one branch operating as a reductive pathway reversing the sequence from succinate to oxaloacetate and the other branch continuing to operate oxidatively to convert oxaloacetate to  $\alpha$ -ketoglutarate as shown in Figure 8-6.

In actuality, the activity levels of a large number of enzymes that serve primarily aerobic functions are markedly reduced when *E. coli* is grown under anaerobic conditions. In this organism, a two-component signal transduction system, consisting of a transmembrane sensor protein, ArcB, and a cytoplasmic regulatory protein, ArcA, controls the expression of genes encoding enzymes involved in aerobic respiration. When oxygen is excluded, the Arc (aerobic respiratory control) system represses the expression of the structural genes for several flavoprotein-linked dehydrogenases, the cytochrome *o* complex, enzymes of the TCA cycle, glyoxylate shunt, and fatty acid degradation. Conversely, the oxygen-scavenging cytochrome *d* oxidase (encoded by the *cydAB* operon) is activated by the Arc system. The Fnr protein functions as an anaerobic repressor of both the cytochrome *o* oxidase complex (encoded by *cyoABCDE*) and cytochrome *d* oxidase (encoded by *cydAB*). Operation of this regulatory system is discussed in some detail in Chapter 5.

In Chapter 1 it is stated that all of the components of the cell are synthesized from only 12 precursor metabolites. Thus, the TCA cycle provides both reducing equivalents

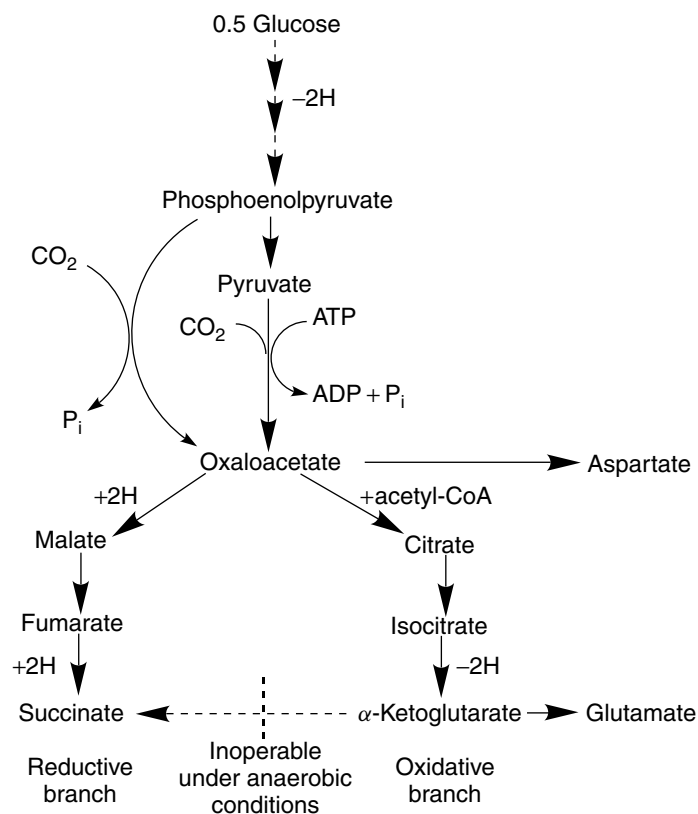


Fig. 8-6. Operation of the TCA cycle under anaerobic conditions.



to the terminal respiratory system and intermediates for the biosynthesis of amino acids and other vital cell constituents. Synthesis of these compounds at the expense of TCA cycle intermediates tends to diminish the activity of the cycle. However, these metabolites are replenished through carbon dioxide fixation by the following reactions:

*Pyruvate carboxylase*



*Phosphoenolpyruvate carboxykinase*



*Malic enzyme*

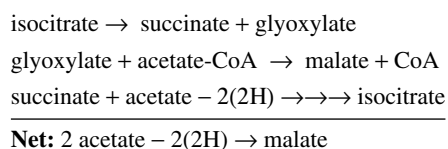


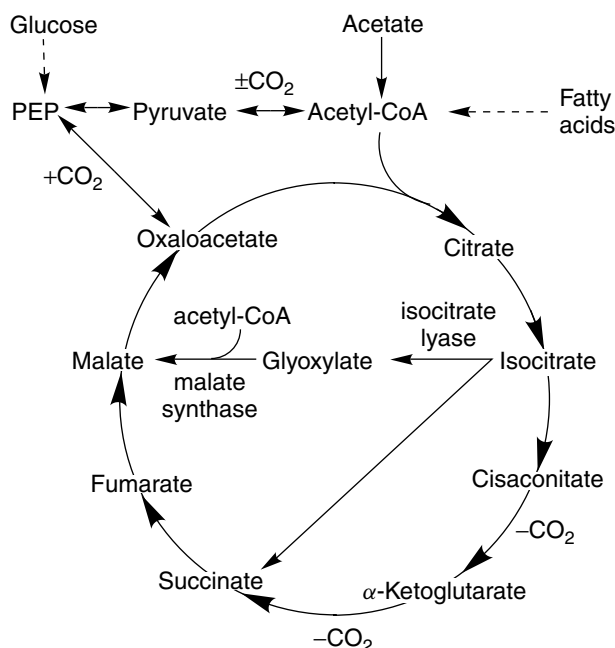
Heterotrophic organisms require at least one enzyme of this nature in order to grow aerobically on hexoses or glycolytic intermediates. The generation of oxaloacetate permits the continued flow of hexose carbon through the TCA cycle under conditions in which intermediates are being removed for biosynthesis. The same purpose is also served when oxaloacetate is used to initiate gluconeogenesis. Enzymes that serve in this capacity are termed **anapleurotic** (from the Greek, meaning “to fill up”).

Transaminase reactions may also serve to generate oxaloacetate (from aspartate) or  $\alpha$ -ketoglutarate (from glutamate). The aspartase reaction can provide oxaloacetate by producing fumarate that can be reduced to oxaloacetate. The glyoxylate cycle, discussed in the next section, can serve an anapleurotic role by allowing a bypass of carbon dioxide-releasing reactions. However, the glyoxylate cycle can function in this capacity only in organisms capable of utilizing acetate following induction on this substrate. Certain organisms deficient in pyruvate carboxylase activity, such as *Arthrobacter pyridinolis*, exhibit a nutritional requirement for malate in order to grow on glucose.

## GLYOXYLATE CYCLE

Activation of acetate with CoA to form acetyl-CoA and the combined activities of the enzymes isocitritase (isocitrate lyase) and malate synthase provide for the operation of a C<sub>4</sub> cycle called the glyoxylate cycle as shown in Figure 8-7. Isocitrate lyase converts isocitrate to succinate and glyoxylate. Malate synthase couples acetyl-CoA and glyoxylate to form malate. Operation of these enzymes results in the net formation of malate from 2 mol of acetate:





**Fig. 8-7. The glyoxylate cycle or glyoxylate shunt.** Note that the two  $\text{CO}_2$ -evolving steps of the TCA cycle are bypassed. Either glucose or fatty acids can serve as sources of acetate.

Activity of the glyoxylate bypass explains the ability of bacteria, yeast, and other microorganisms to utilize acetate as a sole source of carbon for growth.

The enzymes of the glyoxylate cycle are repressed by the presence of glucose or another more rapidly utilized substrate (see Chapter 5). As discussed previously, the Arc (anaerobic respiratory control) system and other regulatory factors repress the TCA cycle and the glyoxylate cycle under anaerobic conditions. Because of the very low redox potential of ferredoxin (see Fig. 9-6), many anaerobes and certain photosynthetic bacteria can form pyruvate, and in some cases,  $\alpha$ -ketoglutarate, via reductive decarboxylation reactions. By this means, these organisms can circumvent the irreversibility of the oxidative decarboxylation of pyruvate to acetyl-CoA and  $\text{CO}_2$  and of  $\alpha$ -ketoglutarate to succinyl-CoA and  $\text{CO}_2$  observed in other organisms. Thus, the utilization of acetate can occur in organisms that utilize ferredoxin in reductive metabolism.

The yeast *Saccharomyces cerevisiae* utilizes a reaction analogous to the conversion of isocitrate to glyoxylate and succinate in the metabolism of propionyl-CoA via a 2-methylcitrate cycle. This cycle is initiated by the formation of 2-methylcitrate from propionyl-CoA and oxaloacetate. 2-Methylcitrate is then converted into 2-methylisocitrate, which is subsequently cleaved to form pyruvate and succinate by a mitochondrial 2-methylisocitrate lyase encoded by *ICL2*. This reaction is very similar to the conversion of isocitrate to succinate and glyoxylate by the *ICL1*-encoded isocitrate lyase. The 2-methylcitrate cycle may be involved in the degradation of carbon skeletons of certain amino acids. For example, oxidative decarboxylation of 2-ketoisobutyrate, an intermediate in threonine catabolism, yields propionyl-CoA.

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## CHAPTER 9

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# ENERGY PRODUCTION AND METABOLITE TRANSPORT

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### ENERGY PRODUCTION

The second law of thermodynamics states that systems will *spontaneously* change from states of higher order to lesser order. The cell, however, must do the opposite to survive. It must make order out of chaos. To accomplish this feat, cells first unleash and then harness the energy trapped within molecules encountered in the environment and use that energy to increase order (e.g., synthesize macromolecules or move chemicals into or out of the cell against a gradient). In this chapter, we discuss how cells steal energy from the environment.

Any discussion of energy must begin with a definition of **entropy**. Entropy (S), simply put, is the *measure* of disorder; the larger the value of S, the greater the disorder. Chemical reactions that cause a large *increase* in S [where the difference between product and precursor entropies ( $\Delta S$ ) is greater than zero] are favored and will occur spontaneously with the *release* of energy. Reactions that cause a *decrease* in S [ $\Delta S < 0$ ] increase order. These reactions require the input of energy and thus cannot occur spontaneously. An accounting device used to deduce the entropy change that occurs following a reaction is called the **Gibbs free energy (G)**. G combines all the physical properties of a system including heat, pressure, volume, and energy. The free-energy *change* ( $\Delta G$ ) during a reaction (i.e., the G of the products minus the G of the starting materials) is a measure of the *disorder* created when the reaction occurs. Energetically favored reactions are those in which a large amount of free energy is released (i.e., they have a large *negative*  $\Delta G$ ), creating much disorder. In contrast, reactions with a large *positive*  $\Delta G$  (such as would occur during peptide bond formation) create order and thus cannot occur spontaneously.

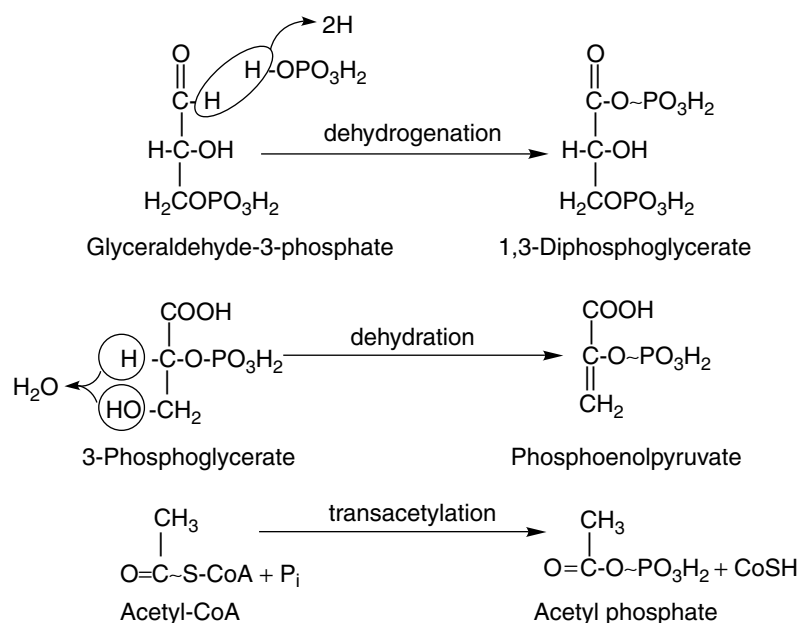
The goal of the cell is to harness the  $-\Delta G$  that results from converting glucose (a highly ordered compound) into  $\text{CO}_2$  and  $\text{H}_2\text{O}$  and to use that energy to drive biosynthetic reactions that have a positive  $\Delta G$ . Since ATP is the fuel used for most anabolic reactions, a major objective of the cell is to synthesize this high-energy

nucleotide. The  $\Delta G$  at equilibrium ( $\Delta G^0$ ) for  $\text{ATP} \rightleftharpoons \text{ADP} + \text{P}_i$  is  $-7.3$  Kcal/mol. However, at normal cellular concentrations where there are 10 ATP molecules per ADP [ $(\text{ATP}) = 10(\text{ADP})$ ],  $\Delta G^0$  is even higher at  $-11$  to  $-13$  Kcal/mol. The cell must be able to *couple* the release of this much energy from glucose to the synthesis of ATP. This is accomplished by substrate-level phosphorylation and oxidative phosphorylation as discussed below.

The energy captured within ATP can then be harnessed to create order in the form of biosynthetic reactions. In a hypothetical enzyme reaction that converts substrates  $\text{A-H}$  and  $\text{B-OH}$  to  $\text{A-B}$  and  $\text{H}_2\text{O}$ , the energy from ATP hydrolysis is first used to convert  $\text{B-OH}$  to a higher-energy intermediate,  $\text{B-O-PO}_4$ . This compound is only transiently formed, with the energy released during its decay used by the enzyme to form  $\text{A-B}$ . Thus, the energy released from the ATP hydrolysis reaction (large  $-\Delta G$ ) is coupled to the synthesis reaction (large  $+\Delta G$ ). In this way, the cell can progressively create order.

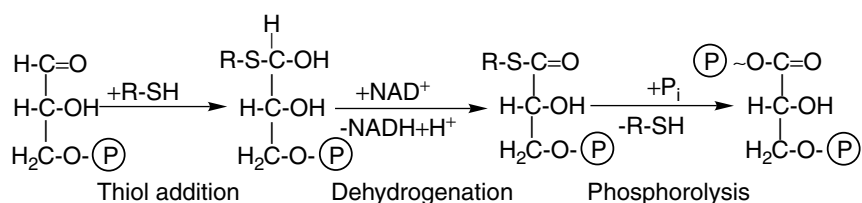
### Substrate-Level Phosphorylation

During glycolysis, energy in the form of ATP is generated by substrate-level phosphorylation. The oxidative steps in glycolysis or other comparable metabolic systems give rise to this energy. In the EMP pathway (Fig. 8-1) oxidation of glyceraldehyde-3-phosphate and the incorporation of  $\text{P}_i$  result in the formation of 1,3-diphosphoglycerate. One of the phosphate bonds achieves a high-energy state. The energy inherent in this bond is used to transfer the  $\text{P}_i$  to ADP to form ATP. Subsequent conversion of the 3-phosphoglycerate from this reaction to phosphoenolpyruvate again provides for the transfer of energy by the formation of ATP from ADP (Fig. 9-1).



**Fig. 9-1.** Reactions involved in substrate-level formation of high-energy phosphate bonds.

The special physicochemical properties of certain compounds such as ATP enable them to transfer energy into an energy-requiring system. In this case, the energy transfer is associated with the transfer of a chemical unit—the phosphoryl group. In other systems, the energy transfer is associated with the transfer of electrons and protons in oxidation–reduction reactions. Although the energy of high-energy compounds is often said to reside in the P–O bond, it is more correct to refer to the **phosphoryl transfer potential** associated with phosphorylated compounds. The nucleoside triphosphates (ATP, GTP, etc.) are the most universal compounds with energy transfer potential in biological systems and serve as the immediate energy sources for biosynthesis as well as physicochemical activities, such as flagellar motor rotation or active transport. In the oxidation of triose phosphate, as shown in Figure 9-1, some of the details are not shown. In this reaction, the combined events of thiol addition, hydrogen transfer, and phosphorolysis by glyceraldehyde-3-phosphate dehydrogenase lead to formation of a phosphate bond that may be transferred to ADP:

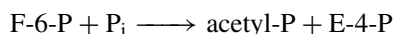


This reaction is initiated by condensation of the thiol (-SH) group at a specific cysteine residue at the catalytic site of the enzyme with the aldehyde carbonyl. Since thiol esters yield considerably more energy on hydrolysis than ordinary oxygen esters, this facilitates the addition of the phosphate group to the carbonyl group. Generation of high-energy phosphate bonds by reactions of this type are referred to as **substrate-level phosphorylations** to distinguish them from those generated by electron transport.

The number of moles of ATP generated will depend on the cleavage pathway utilized during the degradation of carbohydrates (Table 9-1). For example, pentose fermentation proceeds via C<sub>2</sub>-C<sub>3</sub> cleavage to yield lactate and acetate as products:



As shown in the last part of Figure 8-3, pentose fermentation will yield 2 mol of ATP per mol of pentose used if acetyl phosphate is formed as an intermediate. Phosphoketolase splits the pentose into acetyl phosphate and glyceraldehyde-3-phosphate. Acetyl phosphate can be converted to acetate by acetate kinase, yielding 1 mol of ATP. The glyceraldehyde-3-phosphate is utilized via the triose portion of the EMP pathway to yield 2 additional molecules of ATP. The phosphoketolase reaction has been shown to occur in *Acetobacter xylinum*, *Lactobacillus* (heterofermentative species), and *Leuconostoc*. In the genus *Bifidobacterium*, hexose phosphoketolase cleavage of F-6-P yields acetyl phosphate and erythrose-4-phosphate (E-4-P):



Pentose phosphate is formed by the actions of transaldolase and transketolase. The pentose thus formed is then split by the phosphoketolase to form acetyl phosphate

TABLE 9-1. Energy Yields from Various Fermentation Pathways

| Substrate | Cleavage Type                | Products                                | Moles Triose-P Formed | Net ATP Yield (mol) |
|-----------|------------------------------|-----------------------------------------|-----------------------|---------------------|
| Hexose    | FBP-aldolase <sup>a</sup>    | 2 lactate                               | 2                     | 2                   |
| Hexose    | Phosphoketolase              | 1 lactate, 2 acetate, 1 CO <sub>2</sub> | 1                     | 1                   |
| Pentose   | Phosphoketolase              | 2 lactate, 1 acetate                    | 1                     | 2                   |
| Hexose    | KDPG-aldolase <sup>b</sup>   | 2 ethanol, 2 CO <sub>2</sub>            | 1                     | 1                   |
| Aldonic   | FBP-aldolase + KDPG-aldolase | 1.83 lactate, 0.5                       | 1                     | 1.33 <sup>c</sup>   |

<sup>a</sup>FBP-aldolase, fructose-1,6-bisphosphate aldolase.

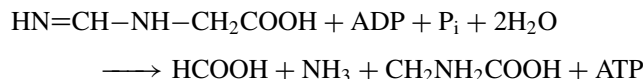
<sup>b</sup>KDPG-aldolase, 2-keto-3-deoxy-6-phosphogluconate aldolase.

<sup>c</sup>Resting cells produce 1.5 lactate + 0.5 C (actually missing, but calculated as acetate via phosphoketolase. 1.83 lactate formed in the presence of arsenite.

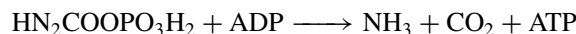
Source: From Gunsalus, I. C., and C. W. Shuster, 1961. Energy-yielding metabolism in bacteria. In *The Bacteria*, Vol. II, Metabolism. I. C. Gunsalus and R. Y. Stanier (eds.). Academic Press, New York.

and glyceraldehyde-3-phosphate as shown previously. The hexose phosphoketolase reaction appears to be unique in the bifidobacteria and has not been found in other heterofermentative organisms. Anaerobic glycolytic organisms do not appear to contain these phosphoketolases.

Other energy-yielding reactions at the substrate level have been described. In *Clostridium cylindrosporum* the fermentation of purines yields ATP via the reaction:



Reactions that lead to the formation of carbamoyl phosphate via energy coupling may eventually yield energy through the transfer of the phosphate group to ADP:



The contribution of inorganic pyrophosphate (PP<sub>i</sub>) and polyphosphate to the overall energy yield may be considerable. Although earlier workers had suspected that PP<sub>i</sub> could serve as an energy source, most biochemists considered that hydrolysis of PP<sub>i</sub> served to render irreversible reactions in which PP<sub>i</sub> was formed. As discussed later in relation to the high-energy yields observed in propionic acid fermentations, PP<sub>i</sub> can be used in place of ATP as a source of energy in certain fermentation reactions. Such reactions may be used more widely than is currently appreciated.

### Oxidative Phosphorylation

When a carbohydrate is oxidized via a respiratory mechanism, energy is generated by passing electrons through a series of electron acceptors and donors until they ultimately reach a final e<sup>-</sup> acceptor such as O<sub>2</sub> or nitrate. The energy inherent in the carbohydrate is gradually released during this series of coupled oxidation-reduction reactions and used to pump protons out of the cell via the membrane-bound cytochrome systems. Since membranes are impermeable to protons, this establishes an electrochemical

gradient ( $\Delta\mu\text{H}^+$ ) or proton motive force across the cell membrane. Proton motive force ( $\Delta p$ ) is composed of electrical (charge) and chemical (pH) components according to the following relationship:

$$\Delta p = \frac{\Delta\mu\text{H}^+}{F} = \Delta\Psi - \frac{2.3RT}{F}\Delta\text{pH}$$

where  $\Delta\Psi$  represents the transmembrane electrical potential and  $\Delta\text{pH}$  is the pH difference across the membrane.  $R$ ,  $T$ , and  $F$  are the gas constant, the absolute temperature, and the Faraday constant, respectively. Thus,  $(2.3RT/F)$  at  $36^\circ\text{C}$  equals 60 mV, and the total driving force across the membrane generated by proton extrusion in millivolts (mV) is  $\Delta\mu\text{H}^+ = \Delta\Psi - 60\text{ mV } (\Delta\text{pH})$ . The cell uses this energy to synthesize ATP. According to chemiosmotic theory, the two components of  $\Delta p$  are thermodynamically equivalent. A change of  $\Delta\text{pH}$  by 1 unit results in the same change in the ATP/ADP equilibrium as a shift of  $\Delta\Psi$  by 60 mV. PMF is also used directly to power flagellar movement and some transport systems.

### Measurement of PMF

Electrical potential ( $\Delta\Psi$ ) across a cell membrane can be measured using lipophilic cations that will freely pass through the membrane. Since the interior side of the membrane is negatively charged, lipophilic cations such as tetraphenyl phosphonium ion ( $\text{TPP}^+$ ) will accumulate inside a cell depending on the extent of the negative charge. In a similar manner,  $\Delta\text{pH}$  can be measured experimentally using radiolabeled weak acids such as benzoic acid. Weak acids disassociate based on their disassociation constants ( $\text{pK}_a$ ) as follows:



The protonated form (HA) passes freely back and forth across the membrane, whereas the deprotonated form ( $\text{A}^-$ ) does not. At a pH value equal to the  $\text{pK}_a$ , 50% of the acid will be disassociated. At 1 pH unit above the  $\text{pK}_a$ , 90% will be disassociated, whereas at 1 pH unit below the  $\text{pK}_a$ , only 10% will be disassociated.

So, 10% of a weak acid with a  $\text{pK}_a$  of 5.5 will be protonated at pH 6.5. If cells are also present, the HA will pass into the cells, equilibrate with external HA, and disassociate intracellularly according to the intracellular pH. The disassociated form will become trapped and thus cannot equilibrate with the outside. This disassociation concomitantly lowers internal HA, which will reequilibrate with external HA, bringing more HA into the cell. This HA also disassociates according to the pH of the cell, continuing the cycle. Consequently, by measuring the accumulation of a radiolabeled weak acid, internal pH can be calculated according to the following formula:

$$\text{pH}_i = \log \left[ \frac{\text{total acid concentration in}}{\text{total acid concentration out}} (10^{\text{pK}} + 10^{\text{pH out}}) - 10^{\text{pK}} \right]$$

Under aerobic conditions at pH 6, the PMF generated is approximately 190 mV with  $\Delta\Psi$  and  $\Delta\text{pH}$  contributing equally. However, what happens when  $\Delta\text{pH} = 0$  — that is,  $\text{pH}_i = \text{pH}_o$ ? In this situation, all of PMF is derived from  $\Delta\Psi$ , the electrical component. In fact,  $\Delta\Psi$  increases to compensate for the loss of  $\Delta\text{pH}$ . This illustrates that the two components of PMF can be exchanged according to the needs of the cell. Ion circulations involving the exchange of protons for other cations such as  $\text{Na}^{2+}$  or  $\text{K}^+$  are



responsible for the interconversion between  $\Delta\Psi$  and  $\Delta\text{pH}$  (see chemiosmotic-driven transport in this chapter).

### Electron Transport Systems

The cytoplasmic membranes of bacteria contain the electron transfer systems (ETS) that generate PMF by coupling the oxidation of NADH and other substrates to the expulsion of protons. The electron transport systems are composed of heme-containing components (cytochromes), iron-sulfur cluster enzymes, flavoproteins (containing FMN), and quinolones. A typical electron transport chain is depicted in Figure 9-2. Remember that different organisms, and even *E. coli* under different conditions, will use an ETS that employs a variety of substrates and terminal electron acceptors. Thus, the respiratory systems can be visualized as being branched at the dehydrogenase and terminal oxidase sites (Fig. 9-3). The electron transfer reactions are initiated by specific modular units. These are NADH, lactate, succinate, formate, or glycerol-3-phosphate dehydrogenases. Figure 9-4 illustrates how NAD-dependent dehydrogenases stereospecifically transfer a substrate-bound H (as deuterium) with two  $\text{e}^-$  to  $\text{NAD}^+$ , forming NADH. Some enzymes, such as succinate dehydrogenase, contain flavoproteins and are linked directly to the next step in the system, bypassing the need for  $\text{NAD}^+$ . FAD functions in flavoprotein-linked reactions as shown in Figure 9-5. Each oxidation–reduction system has a definite oxidation–reduction potential ( $E'_0$ ) at which it functions. By determining the electrode potential for each system, the order in which each functions can be deduced (Fig. 9-6).

The quinone pool acting as a universal adapter collects the electrons from the different donor modules. The quinone pool passes the electrons on to the appropriate acceptor module depending on what terminal electron acceptor is being used (e.g.,  $\text{O}_2$ , nitrate, fumarate). Quinones are mobile hydrogen carriers that shuttle between the large dehydrogenase and oxidase complexes. The predominant quinone, ubiquinone-8, functions in aerobic respiration while anaerobic respiratory chains may require menaquinone-8. The ETS system in Figure 9-2 begins with NADH dehydrogenase,

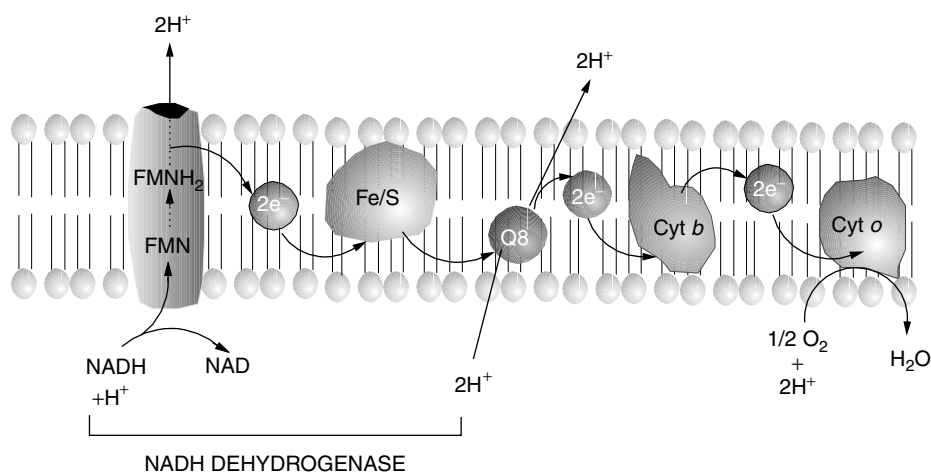
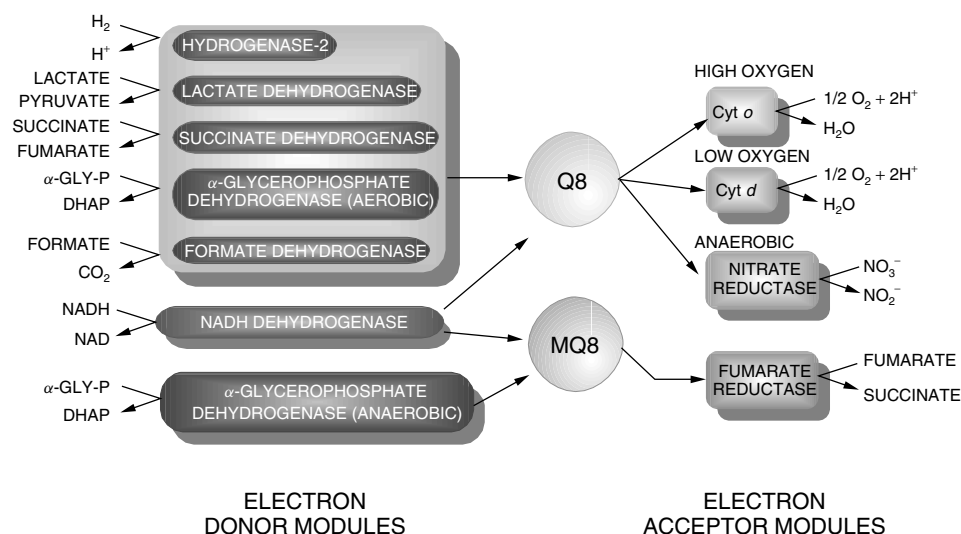
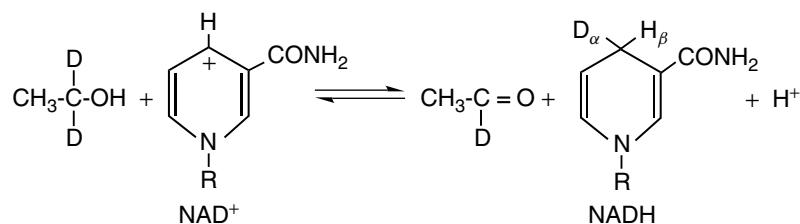


Fig. 9-2. An electron transport chain of *E. coli*.



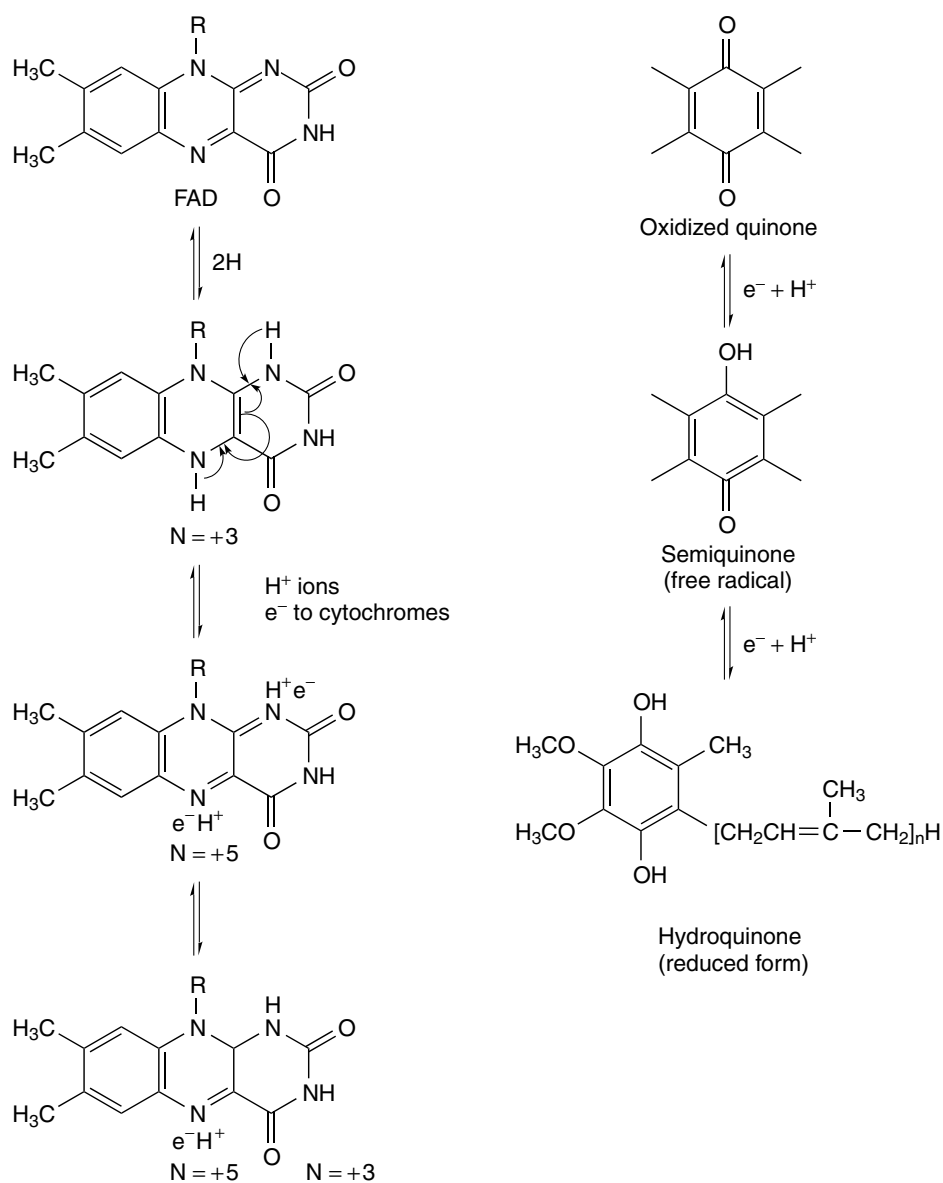
**Fig. 9-3. Menu of electron donor and acceptor modules for *E. coli*.** Selection (induction) of module combinations depends on the substrate available (left side of figure) and the redox potential of the final electron acceptor (right side of figure). Q8, ubiquinone; MQ8, menaquinone.



**Fig. 9-4. Deuterium labeling of  $\text{NAD}^+$  during reduction by alcohol dehydrogenase.** In the presence of alcohol dehydrogenase, deuterium (D) from deuterium-labeled alcohol is added stereospecifically to  $\text{NAD}^+$ . Under comparable circumstances  $\text{NADP}^+$ , which contains an additional phosphate group on the pentose of the adenylic acid moiety, reacts similarly.

a flavoprotein complex containing FMN and nonheme iron-sulfur clusters. NADH transfers two electrons and protons to FMN, forming  $\text{FMNH}_2$ . Transfer of the electrons to the Fe/S group causes the release of the 2 protons ( $\text{H}^+$ ) to the outside, generating PMF. The electrons in the Fe/S group are then transferred to ubiquinone. Ubiquinone picks up the  $2\text{e}^-$ , as well as  $2\text{H}^+$  from the cytosol, to form hydroquinone.

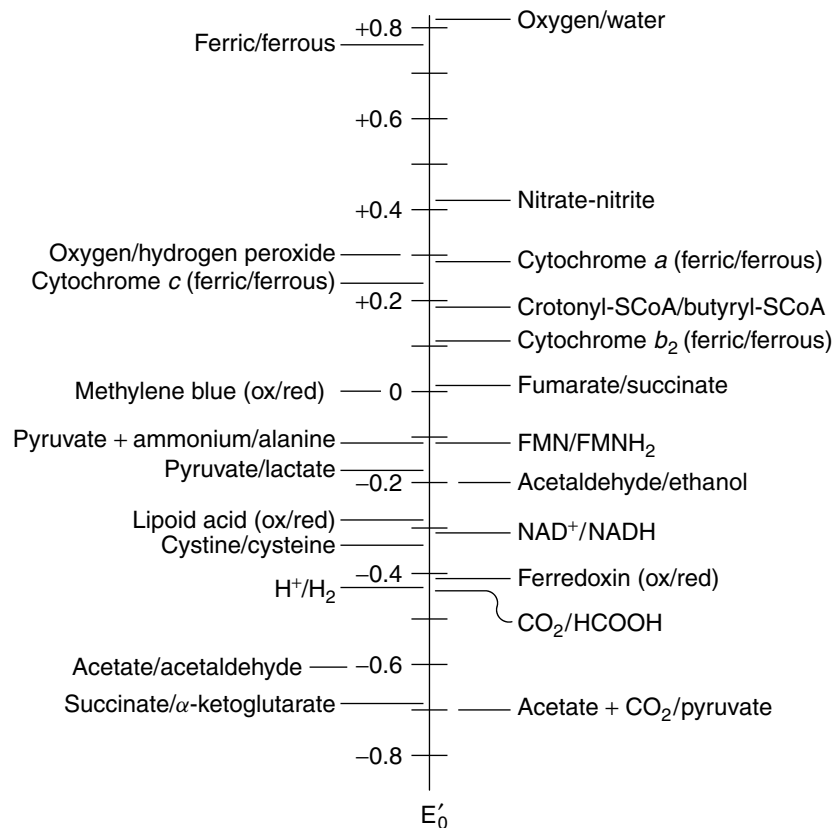
Under conditions of high oxygenation, hydroquinone will transfer its electrons to the cytochrome-*o* complex and release  $2\text{H}^+$  to the cell exterior, adding to PMF. The cytochrome-*o* complex is composed of four polypeptides that make up two spectral signals, termed cytochromes  $b_{555}$  and  $b_{562}$ . The subscript numbers correspond to maximum wavelength absorbance values. It may be difficult to visualize how proton translocation across the membrane can occur during the transfer of  $\text{e}^-$  from hydroquinone to cytochrome *o*. It seems the ubiquinol oxidase site of the Cyt *o* complex faces the periplasm while the dioxygen reductase site faces the cytoplasm. Thus, the



**Fig. 9-5. Functions of flavin adenine dinucleotide (FAD) and ubiquinone in hydrogen ( $H^+$ ) and electron ( $e^-$ ) transfer reactions.**

release of two protons to the periplasm is coupled to the consumption of two protons in the cytoplasm to convert  $\frac{1}{2}O_2 + 2H^+ + 2e^-$  to  $H_2O$ .

When *E. coli* grows under low oxygen conditions, it requires a terminal oxidase with a higher affinity for oxygen. Thus, the high-affinity cytochrome *d* complex is induced and the lower-affinity Cyt *o* complex is repressed. A combination of regulatory proteins including Arc (see Chapter 18) are involved in this control. The Cyt *d* complex utilizes cytochrome *b*<sub>558</sub> and cytochrome *d*.

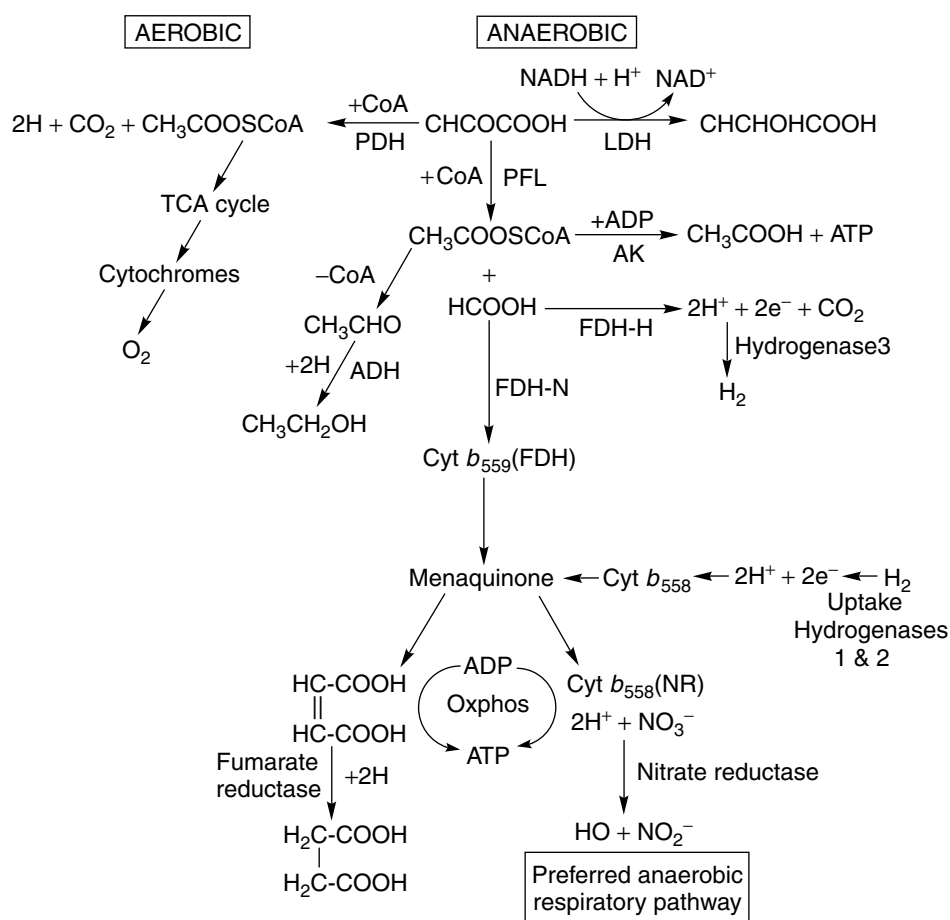


**Fig. 9-6. Electrode potentials of some important biological systems.**  $E'_0$ , oxidation–reduction potential in volts.

### Anaerobic Respiration

When placed under anaerobic conditions, enteric bacteria (such as *Escherichia coli*) cleave pyruvate to acetyl CoA and formate. If there are no appropriate alternative electron acceptors available, acetyl CoA and formate will be converted to fermentative end products (Fig. 9-7). But as noted above, many organisms such as *E. coli* can respire even under anaerobic conditions if alternative electron acceptors are present. Each system for accepting electrons can be viewed as a modular unit. Some of these acceptor modules are shown in Figure 9-3. They are complex membrane-associated enzymes. For example, fumarate reductase has an FAD-containing subunit, an iron-sulfur protein, and two membrane anchor proteins. Nitrate reductase is even more complex. A given modular system is induced when cells are grown in medium containing a specific acceptor molecule. There is, in fact, a hierarchy of transcriptional control corresponding to the potential energy that can be produced and the electrical potential of each acceptor.

The Fnr, NarL, and Arc systems described in Chapter 18 control the expression of the respiratory genes. Anaerobic growth on specific acceptor molecules also results in the synthesis of cytochromes specific for each acceptor module. Thus, the cytochrome *b* used for fumarate reductase is different from that used for nitrate reductase. As



**Fig. 9-7. Alternate pathways of pyruvate utilization under aerobic and anaerobic conditions.** PDH, pyruvate dehydrogenase; PFL, pyruvate formate lyase; TCA, tricarboxylic acid cycle; FDH-N, formate dehydrogenase-N; FDH-H, formate dehydrogenase-H; Oxphos, oxidative phosphorylation; AK, acetate kinase; LDH, lactate dehydrogenase; ADH, alcohol dehydrogenase.

illustrated when oxygen was the terminal  $\text{e}^-$  acceptor, the transmission of electrons to nitrate or fumarate also translocates protons to the cell exterior, generating PMF. Figure 9-7 illustrates the alternative pathways for pyruvate that are possible depending on the type of terminal electron acceptor present.

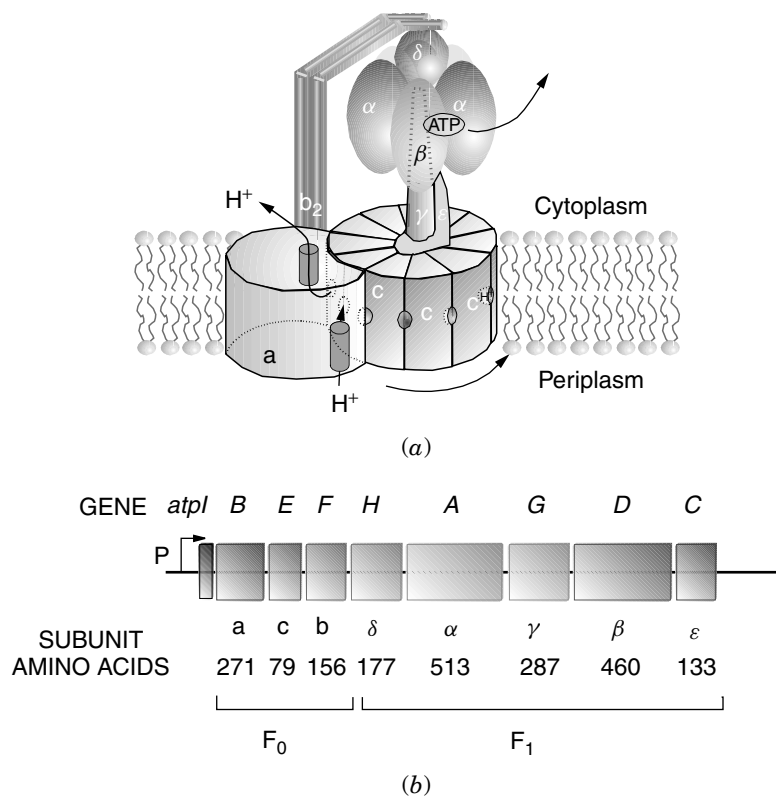
### Conversion of PMF to Energy

How PMF is converted into usable energy by the cell is certainly one of the keys to understanding life. There are several ways the cell can use PMF. The cell can directly generate ATP from PMF by reversing the action of the major  $\text{H}^+$ -translocating ATPase. These are the so-called  **$\text{F}_1\text{F}_0$ -type ATP synthases** that are widely distributed in the bacterial kingdom. This type of ATPase is composed of two structurally and

functionally distinct entities. The  $F_1$  moiety lies extrinsic to the membrane in the cytoplasm and catalyzes the hydrolysis of ATP to ADP.  $F_1$  is attached to the membrane by a pore complex termed  $F_0$ .  $F_0$  serves as the conductor of  $H^+$  across the membrane. Protons passing into the cell (i.e., down the electrochemical gradient) through  $F_1F_0$  will drive  $F_1$  to synthesize ATP (see Fig. 1-13 and Fig. 9-8). It takes three to four  $H^+$  to generate one ATP.

PMF can also be used to drive the transport of some metabolites into the cell. Lactose, for instance, is transported into the cell by the integral membrane permease LacY. The energy required to pump lactose into the cell against an increasing lactose gradient comes from the symport of  $H^+$  with lactose and the partial dissipation of the PMF. Additionally, the flagellar motor is driven by PMF. Each flagellar rotation requires the influx of  $256H^+$ !

Another consideration is how a facultative organism such as *E. coli* generates PMF under anaerobic conditions in the absence of any electron acceptor molecules—that is, under fermentative conditions. Even under these conditions the cell requires a PMF to



**Fig. 9-8. ATP synthase rotor. The structure of *E. coli*  $F_1F_0$  complex and genetic organization of the *atp* operon.** (a) The ATP synthase shown with the cytoplasmic  $F_1$  portion synthesizing ATP as  $H^+$  is transferred through the  $F_0$  pore and into the cell. (b) The *atp* operon is transcribed as a single polycistronic message from the promoter (P) region. The generic designations are shown above the operon; the corresponding protein products and the number of amino acids in each are shown below the operon.

transport carbohydrates and use its flagella. But how can an electrochemical gradient be formed if there are no active electron transport chains to pump  $H^+$  out of the cell? Under these conditions, the  $F_1F_0$  ATPase will use an ATP to pump  $H^+$  out of the cell, thereby generating a PMF. Loss of the  $F_1F_0$  ATPase, therefore, will prevent anaerobic growth in instances where the ATPase is required to generate PMF. In addition, these mutants will not grow aerobically on TCA cycle intermediates, since substrate-level phosphorylation is not possible. The  $H^+$ -translocating ATPase is the only way to synthesize ATP when growing on TCA cycle intermediates.

### Structure of $F_1F_0$ and the *atp* Operon

As noted above, the ATP synthase consists of two functionally distinct entities,  $F_1$  and  $F_0$ , which function in a rotary-like manner. The  $F_1$  moiety retains ATPase function when stripped from the membrane. It consists of five subunits in an unusual stoichiometric ratio of  $\alpha_3\beta_3\gamma\delta\epsilon$  (aggregate molecular mass ca. 380 kDa). The  $F_0$  pore consists of three subunits also in an unusual stoichiometry of  $a_1b_2c_{10\pm1}$  (aggregate molecular mass ca. 150 kDa).

Figure 9-8a illustrates that  $F_1$  is built as a hexamer of alternating  $\alpha$  and  $\beta$  subunits that surround an asymmetric core consisting of the  $\gamma$  subunit. The interfaces between  $\alpha$  and  $\beta$  subunits form three nucleotide-binding catalytic sites where ADP is converted to ATP (or vice versa).  $F_1$  binds to  $F_0$  through two distinct linkages: (1) a central stalk (or rotor) comprised of  $\gamma\epsilon$  anchored to the  $c$ -ring of the  $F_0$  subunit and (2) a peripheral stalk (or stator) comprised of  $b_2\delta$  that anchors the  $\alpha_3\beta_3$  hexamer to the  $F_0a$  subunit. The  $a/c$  interface forms two unaligned partial channels for proton transport that open to opposite sides of the membrane. During ATP synthesis, each proton enters through the periplasmic channel in the  $a$  subunit and binds to a deprotonated  $c$  subunit. The  $c$ -ring then rotates one step to the right relative to the  $a$  subunit. This rotation moves a previously protonated  $c$  subunit into alignment with the cytoplasmic channel where the proton is lost and released to the cytoplasm.

The key to ATP synthesis is this  $c$ -ring rotation. Because the  $c$ -ring is anchored to the  $\gamma\epsilon$  pin, rotation of the  $c$ -ring rotor forces  $\gamma$  to rotate. The  $\gamma$  pin is forced to rotate within the catalytic  $\alpha_3\beta_3$  hexamer because the stator moors the hexamer to the  $a$  subunit. Rotation of the  $\gamma$  pin within the hexamer drives the binding changes required to achieve net synthesis of ATP. One of the elegant proofs for subunit rotation involved attaching an actin filament to the  $c$ -ring and observing its rotation by fluorescence microscopy.

As noted earlier, when needed, the ATPase can work in reverse to pump protons out of the cell, generating the PMF needed to drive flagellar movement as well as a variety of transport systems. Thus, the cytoplasmic  $\alpha_3\beta_3$  hexamer will also catalyze ATP hydrolysis, the energy from which rotates the  $\gamma$  pin in the opposite direction. This, in turn, reverses rotation of the  $c$ -ring, so cytoplasmic protons enter the cytoplasmic channel of the  $a$  subunit and are released on the periplasmic side, generating PMF.

It is worth discussing at this point how *E. coli* synthesizes the proper ratio of ATPase subunits. All of the genes encoding  $F_1F_0$  reside in a single operon (Fig. 9-8b). RNA transcription of the entire operon occurs from a promoter region upstream of the first gene such that all of the genes are cotranscribed. With this in mind, it seems odd that with the unusual stoichiometries of the ATPase subunits, there is little if any excess  $a$ ,  $b$ ,  $\gamma$ ,  $\delta$ , or  $\epsilon$  subunits in the cytoplasm. The proteins are all synthesized in the ratios

as they occur in the assembled complex, but there is no differential transcriptional control that would explain why, for example, more *c* protein is produced than *b* or *a*. The answer appears to lie with the translational efficiencies of the various **translational initiation regions (TIR)** located at the beginning of each gene within the polycistronic message. Genes for subunits required at high stoichiometric ratios contain more efficient TIRs than do those required in smaller amounts. TIR efficiency can be correlated to the Shine-Dalgarno (SD) consensus sequence as well as to surrounding levels of secondary structure that could block ribosome access to the SD sites.

### Energy Yield

The amount of energy derived from the oxidation of a given carbohydrate during respiration varies depending on several factors. These factors include the type of cytochrome system used to pump protons and generate PMF, the type of terminal electron acceptor and the efficiency of the ATP synthetase. For example, both *E. coli* and yeast growing aerobically will produce  $\text{NADH} + \text{H}^+$ . Transfer of  $\text{e}^-$  from NADH to the cytochrome system of *E. coli* will pump  $4\text{H}^+$  out of the cell while the yeast cytochrome system will pump  $6\text{H}^+$ . Furthermore, the number of protons required to generate ATP via the *E. coli* ATPase is perhaps two or three while the value for yeast mitochondrial ATP synthetase is closer to  $2\text{H}^+$ . So, 1  $\text{NADH} + \text{H}^+$  in *E. coli* may yield between 1 and 2 ATP molecules, whereas yeast can produce 3 ATP molecules. Figure 9-9 illustrates the energy yield from glucose using yeast as the model system.

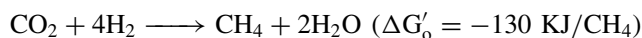
### Generating ATP in Alkalophiles

Bacteria that live in very alkaline environments will have a difficult time generating a PMF, since there are usually more protons in the cell than outside (internal pH is more acidic than the external environment, e.g.,  $\text{pH}_i$  8 vs.  $\text{pH}_o$  10). In these situations, the solution for generating ATP is to rely on **sodium motive force**. Again, an  $\text{F}_1\text{F}_0$  ATPase is involved, but it couples ATP synthesis to movement of  $\text{Na}^+$  rather than  $\text{H}^+$ .

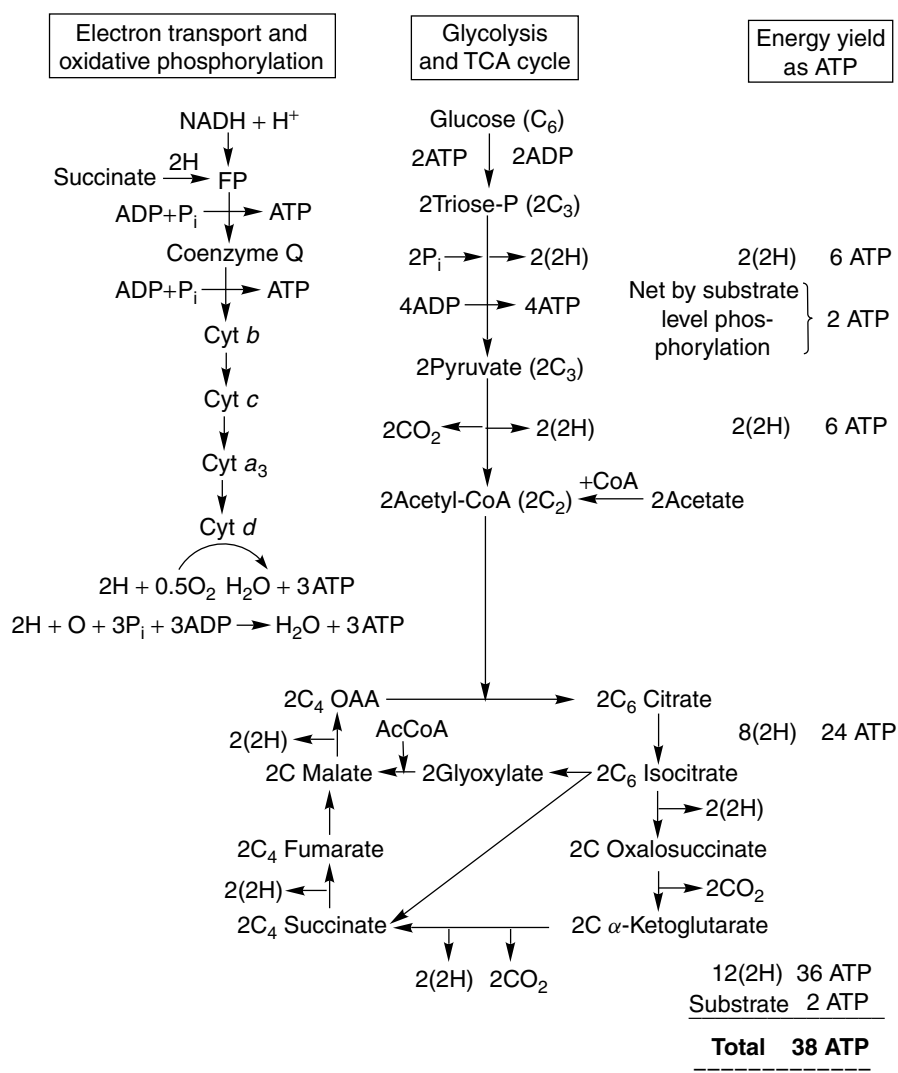
### Energetics of Chemolithotrophs

The chemolithotrophic bacteria derive energy from the oxidation of inorganic substrates. Best known are those oxidizing hydrogen to water, ammonia to nitrite and nitrite to nitrate, sulfur to sulfate and iron(II) to iron(III). Most chemolithotrophs contain all the components of a “normal” electron transport chain and therefore generate proton motive force in much the same manner as chemoorganotrophs (e.g., *Escherichia coli*). The inorganic compounds serve as sources of energy in the form of electrons that can be used to pump  $\text{H}^+$  (or other cations) out of the cell.

**Hydrogen Oxidation.** Aerobic hydrogen oxidation is well known in eubacteria such as *Escherichia coli*. Hydrogen, via hydrogenase 1 or 2 (hydrogenase 3 in *E. coli* is involved with the production of  $\text{H}_2$  as part of the formate-hydrogen lyase system), is introduced into the electron transport chain by transfer to  $\text{NAD}^+$ , forming NADH. Subsequent electron transfers are equivalent to what have already been discussed. Anaerobic hydrogen oxidation is known best in the methanogens in which  $\text{CO}_2$  serves as the hydrogen acceptor:

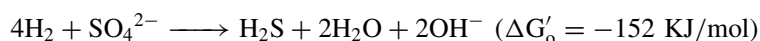




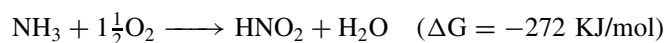


**Fig. 9-9. Theoretical energy yield as ATP from glycolysis and the TCA cycle.** The calculations shown here assume that each pair of hydrogen atoms (2H) released from the substrate yields 3 ATP. The reaction shown as ADP + P<sub>i</sub> → ATP represents the action of ATP synthase. Two turns of the TCA cycle are required to completely oxidize the 2 acetyl-CoA derived from glucose. Each 2H generated by the system yields 1 molecule of water (2H + 0.5O<sub>2</sub> → H<sub>2</sub>O). Overall reaction: C<sub>6</sub>H<sub>12</sub>O<sub>6</sub> + 38P<sub>i</sub> + 38ADP + 6O<sub>2</sub> → 6CO<sub>2</sub> + 6H<sub>2</sub>O + 38ATP. Total ATP from TCA cycle: 12(2H) + 6O<sub>2</sub> + 36P<sub>i</sub> → 6H<sub>2</sub>O + 36ATP.

Hydrogen can also be oxidized by some sulfate-reducing bacteria as:



**Inorganic Nitrogen Compounds.** Nitrifying bacteria use either ammonia or nitrite as their energy source:



**Iron Oxidation.** Organisms that can derive metabolic energy from  $\text{Fe}^{2+}$  include *Thiobacillus ferrooxidans* and *Leptospirillum ferrooxidans*. In all cases, oxidation takes place at acidic pH values, physiological optimum pH 2. This allows an increased solubility of the ferric ion product. These organisms, then, are acidophilic, which presents some interesting problems in terms of maintaining internal pH (see “pH Homeostasis”).



These and other forms of chemolithotrophy are discussed in further detail below.

### pH Homeostasis

Most enzymes and proteins have a rather narrow pH optima for activity or function. Nevertheless, bacteria have an amazing capacity to grow over a wide range of pH values. The bacteria can be divided into three groups depending on their preferred range of growth pH. Organisms that prefer acidic, neutral, or alkaline environments are called acidophiles, neutralophiles, and alkalophiles, respectively. *E. coli* is a neutralophile, but even it can grow over a 1,000-fold range of pH, from pH 5 to pH 8, and it can survive over a 10 million-fold range, from pH 2.0 to pH 9.0! Over external pH values from 4 to 9, the internal pH ( $\text{pH}_i$ ) does not vary more than 10-fold and usually much less than that. Over growth pH values from 5.5 to 8, for example,  $\text{pH}_i$  only wavers between 7.2 and 7.7. How this extraordinary degree of homeostasis is accomplished remains one of the fundamental questions of bacterial physiology.

There are a number of factors that can influence  $\text{pH}_i$ . Among these are the buffer capacity of the cell, the outward transport of protons associated with respiration or ATP hydrolysis, and electroneutral transport systems that exchange protons for certain cations, particularly  $\text{Na}^+$  and  $\text{K}^+$ . For example, as environmental pH becomes more acidic, any decrease in  $\text{pH}_i$  could trigger a  $\text{K}^+/\text{H}^+$  antiport system that would bring  $\text{K}^+$  into the cell while extruding a proton. This will maintain internal pH near 7.5 and create a larger  $\Delta\text{pH}$ . Likewise, an increase in external pH will cause alkalization of the cytoplasm, which could trigger  $\text{Na}^+/\text{H}^+$  antiport systems that will extrude  $\text{Na}^+$  and import protons to acidify the cytoplasm (see chemiosmotic-driven transport).

Aside from pH homeostasis, bacteria can induce acid protection systems called the acid tolerance response (ATR) and acid resistance, both of which increase survival during exposure to extremely low pH. The specifics of ATR systems differ among bacteria but are generally believed to include mechanisms designed to protect or repair macromolecules exposed to low pH. In organisms such as *Salmonella* and *E. coli*, they protect cells to about pH 3 but not below. *Shigella* and *E. coli* have additional acid resistance systems that are remarkable in their ability to protect cells under acid conditions as low as pH 2. For technical reasons, we do not know what happens to  $\text{pH}_i$  between pH 2 and 3, but details of the acid resistance systems have been exposed.

*E. coli* possess three acid resistance systems that are induced under different conditions. Two of these systems depend on the presence of particular amino acids at the acidic pH to protect the cell against acid stress: a glutamate-dependent system

and an arginine-dependent system. Specific amino acid decarboxylases are involved with each system as well as membrane-bound antiporters that exchange substrate (e.g., glutamate) in the media for the decarboxylation product (e.g., gamma-amino butyric acid). How the systems function to protect the cell against protons is not clear, but mounting evidence suggests they may not directly enhance pH homeostasis, as originally proposed. They may be somehow linked to energy production at low pH. The importance of these AR systems is that they enable very low numbers of *Shigella* or *E. coli* to survive the gastric acidity barrier and cause disease.

## METABOLITE TRANSPORT

The cytoplasmic membrane forms a hydrophobic barrier impermeable to most hydrophilic molecules. Consequently, mechanisms must exist that enable the cell to transport growth solutes into the cell and waste solutes out (useful transport websites include: <http://www.biology.ucsd.edu/~msaier/transport/http://www.biology.ucsd.edu/~ipaulsen/transport/index.html>). The outer membranes of gram-negative cells (e.g., *E. coli*) contain narrow channels that permit passive diffusion of hydrophilic compounds such as sugars, amino acids, and certain ions to the periplasm. Only compounds smaller than 600 to 700 daltons can pass. Hydrophobic compounds of any size are excluded. These pores are formed by porin proteins (e.g., OmpF or OmpC of *E. coli*). Chapter 7 discusses the structure of pores in detail and Chapter 18 provides details about their regulation.

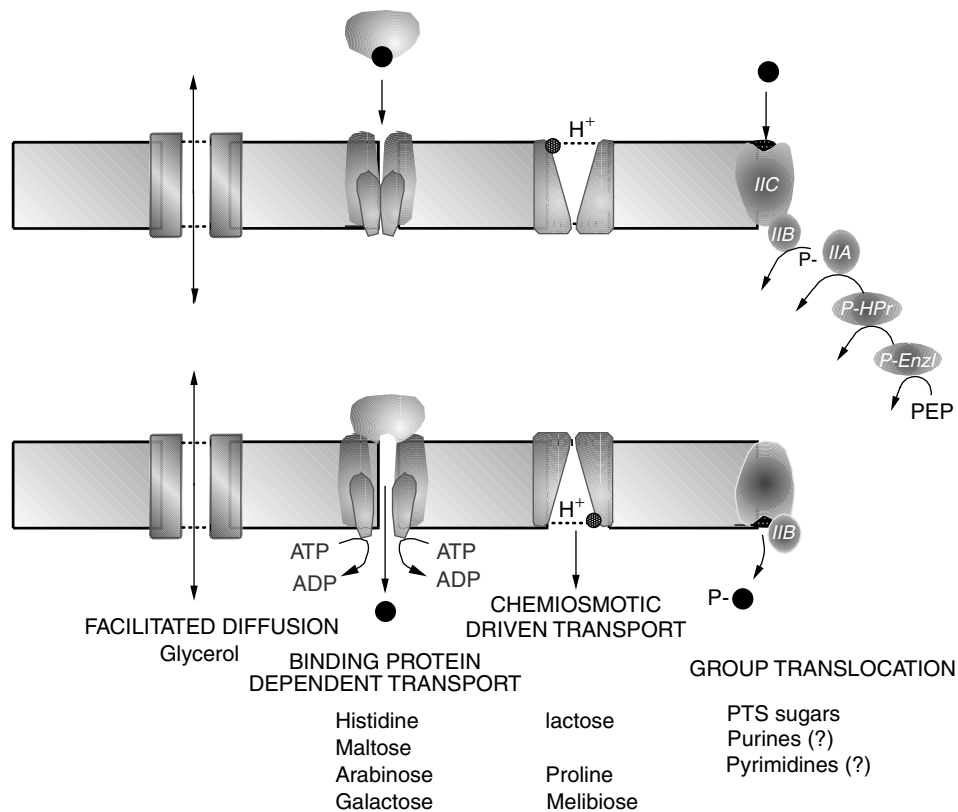
A few compounds essential to bacterial growth, such as vitamin B<sub>12</sub>, oligosaccharides, and iron chelates that are too large to cross the outer membrane via normal pores, have specific transport proteins or specific pores in the outer membrane that mediate their passage. For example, maltose traverses the outer membrane via a specific pore composed of maltoporin.

Oxygen, carbon dioxide, ammonia, and water all diffuse through both the outer and inner membranes without hindrance. But while most essential nutrients can diffuse through the outer membrane, all require one or more transport systems to cross the inner membrane and gain access to the cytoplasm. The goal of most of these transport systems is to concentrate nutrients inside the cell that may be in short supply externally. Consequently they work against a concentration gradient and require energy in some form to accomplish their tasks. There are several general transport mechanisms involved in membrane transport. These are diagrammed in Figure 9-10 and are divided into four groups: facilitated diffusion, chemiosmotic-driven transport, binding protein-dependent transport, and group translocation.

### Facilitated Diffusion

Facilitated diffusion is the only transport system that does not require energy. As a result, it only allows for passage of a substrate *down* a concentration gradient. Consequently, the substrate will never achieve a concentration inside the cell greater than what exists outside. Inner membrane **channel proteins** are responsible for this type of transport. The glycerol transporter (GlpF), a mechanosensitive ion channel (MscL), and a voltage-sensitive ion channel are examples in *E. coli*.

**Glycerol** is one of the few compounds that enters prokaryotic cells by facilitated diffusion. The glycerol facilitator (GlpF) is a pore-type channel protein present in



**Fig. 9-10. Possible models for cytoplasmic membrane transport systems.** Each of the systems probably has one or more membrane-spanning proteins that forms a specific channel in the cytoplasmic membrane. Facilitated diffusion and ion-driven transport systems require only one gene product, whereas binding protein-dependent transport and group translocation systems require several gene products. Facilitated diffusion allows entry and exit of substrate through a specific pore. Binding of substrate to specific sites accessible to the periplasmic side of the active transport systems coupled with an appropriate energy source allows a conformational change in the carrier proteins and substrate release inside the cell.

the inner membrane of *E. coli*. The *glpF* gene occurs in an operon with *glpK* (encoding glycerol kinase). This porin-type channel also allows passage of polyhydric alcohols and other unrelated small molecules such as urea and glycine but excludes charged molecules such as glyceraldehyde-3-phosphate (G3P) and dihydroxyacetone phosphate (DOHAP).

**Aquaporin-Z (AqpZ).** The entry and release of water from cells is a necessary feature of many physiological functions. However, the diffusion of water through a lipid bilayer is too low to explain the high osmotic permeability exhibited by cell membranes. The Aqp channel will mediate inwardly and outwardly directed osmotic fluxes of water triggered by abrupt changes in the extracellular osmolality. However, since *aqpZ* mutants remain viable, it is believed that other water channels are present that will compensate for the loss.

### Mechanosensitive Channels

Turgor pressure (see Chapter 18) is defined as the outwardly directed pressure used by the cell to maintain shape and to allow for expansion of the cell membrane as the cell grows. The concentration of intracellular solutes (glutamate, potassium) determines the extent of turgor pressure (usually around 4 atm for *E. coli*). Upon transfer to medium that is *lower* in osmolarity, water will rapidly enter the cell. In the absence of mechanisms to reduce turgor, the pressure inside could rise to 11 atm, threatening the integrity of the cell. It has been proposed that catastrophic damage to the cell membrane is avoided by rapid activation of mechanosensitive channels (MscL, MscS). In contrast to aquaporin, mechanosensitive channels will permit inward or outward flow of ions. Transient opening of these channels allows  $K^+$ , glutamate, other compatible solutes, and ATP to exit and sodium and  $H^+$  to enter. The result is a lowering of intracellular-compatible solute concentration and, thus, turgor.

### ATP-Binding Cassette Transporter Family

Periplasmic proteins of gram-negative bacteria can be released into the surrounding medium if cells are subjected to osmotic shock (cells suspended in buffered 20% glucose containing EDTA are centrifuged and rapidly resuspended into cold  $MgCl_2$ ). Among these periplasmic proteins are **binding proteins** for specific nutrients such as  $SO_4^{2-}$ , amino acids, sugars, and other compounds. In gram-positive bacteria, where no outer membrane is present to compartmentalize proteins, the binding proteins are present either as lipoproteins, tethered to the external surface of the cytoplasmic membrane, or as cell surface-associated proteins, bound to the external membrane surface via electrostatic interactions. The binding proteins function by transferring the bound substrate to a compatible membrane-bound complex of four proteins of which two are integral membrane proteins and two are peripheral, membrane-associated cytoplasmic proteins (Fig. 9-10). It is remarkable that these transport systems, sometimes called **traffic ATPases**, bear striking subsequence homologies to each other across phylogenetic domains. The cytoplasmic members of these systems contain two specific ATP-binding motifs: the Walker A box [GX(S/T)GXGK(S/T)(S/T)], which forms a so-called P-loop that interacts with the  $\gamma$  phosphate of ATP, and the Walker B box (ILLLD). As a result, these proteins are called ATP-binding cassette, or ABC proteins.

The transport process is energized by ATP or other high-energy phosphate compounds such as acetyl phosphate. Approximately 40% of the substrates transported by *E. coli* involve periplasmic-binding protein and ABC transporter mechanisms. It appears with these systems that transfer of the substrate from the periplasmic-binding protein to the membrane complex triggers ATP hydrolysis, which in turn leads to the opening of a pore that allows unidirectional diffusion of the substrate to the cytoplasm. The histidine permease of *E. coli* is one of the best-studied examples of a traffic ATPase. More information on ABC transporters can be found at <http://ir2lcb.cnrs-mrs.fr/ABCdb/>.

### Chemiosmotic-Driven Transport

These systems accomplish movement of a molecule across the membrane at the expense of a previously established ion gradient such as a proton motive or a sodium motive

force. About 40% of the substrates that enter *E. coli* involve ion-driven transport. There are three basic types: symport, antiport, and uniport. **Symport** involves the simultaneous transport of two substrates in the same direction by a single carrier. For example, a proton gradient can allow symport of an oppositely charged ion or a neutral molecule. Transport of lactose by the LacY permease is such an example. **Antiport** is the simultaneous transport of two like-charged compounds in opposite directions by a common carrier. The  $\text{Na}^+/\text{H}^+$  antiporters of *E. coli* (NhaA and NhaB) are examples that are believed to be important for generating sodium motive force and maintaining neutral internal pH under alkaline growth conditions. **Uniport** occurs when movement of a substrate is independent of any coupled ion. Transport of glycerol is an example of uniport and was described above as facilitated diffusion. Table 9-2 lists examples of chemiosmotic transporters and their coupling ions.

**Group translocation** couples transport of a substrate to its chemical modification (e.g., by attaching a phosphate or coenzyme A group to the substrate). This traps the substrate within the cell in a form different from the exogenous substrate, so the concentration gradient of unmodified substrate never equilibrates. The phosphotransferase system (*pts*) involved in the transport of many carbohydrates utilizes this approach (see below).

TABLE 9-2. Representative Chemiosmotic Porters

| Uniport               | Symport                                               | Antiport                                                     |
|-----------------------|-------------------------------------------------------|--------------------------------------------------------------|
| Glycerol <sup>a</sup> | $\text{H}^+$ /amino acid                              | $\text{H}^+$ : cation                                        |
|                       | $\text{H}^+$ /glycine <sup>a</sup>                    | $\text{H}^+$ : $\text{Ca}^{2+}$ <sup>a</sup>                 |
|                       | $\text{H}^+$ /histidine <sup>a</sup>                  | $\text{H}^+$ : $\text{CaHPO}_4^a$                            |
|                       | $\text{H}^+$ /lysine <sup>a</sup>                     | $\text{H}^+$ : $\text{K}^{+a}$                               |
|                       | $\text{H}^+$ /phenylalanine <sup>a</sup>              | $\text{H}^+$ : $\text{Na}^{+a}$                              |
|                       | $\text{H}^+$ /organic acid                            | $\text{K}^+$ : cation                                        |
|                       | $\text{H}^+$ /DL-lactate <sup>a</sup>                 | $\text{K}^+$ : $\text{CH}_3\text{NH}_3^{+a}$                 |
|                       | $\text{H}^+$ /pyruvate <sup>a</sup>                   | $\text{H}_2\text{PO}_4^-$ : organic anion                    |
|                       | $\text{H}^+$ /succinate <sup>a</sup>                  | $\text{H}_2\text{PO}_4^-$ :hexose 6-phosphate <sup>a</sup>   |
|                       | $\text{H}^+$ /gluconate <sup>a</sup>                  | $\text{H}_2\text{PO}_4^-$ :glycerol 3-phosphate <sup>a</sup> |
|                       | $\text{H}^+$ /sugar                                   | $\text{H}_2\text{PO}_4^-$ :phosphoenolpyruvate <sup>b</sup>  |
|                       | $\text{H}^+$ /arabinose <sup>a</sup>                  |                                                              |
|                       | $\text{H}^+$ /galactose <sup>a</sup>                  |                                                              |
|                       | $\text{H}^+$ /lactose <sup>a</sup>                    |                                                              |
|                       | $\text{H}^+$ /inorganic anion                         |                                                              |
|                       | $\text{H}^+$ /phosphate <sup>a</sup>                  |                                                              |
|                       | $\text{Na}^+$ /amino acid                             |                                                              |
|                       | $\text{Na}^+$ , $\text{H}^+$ /glutamate <sup>a</sup>  |                                                              |
|                       | $\text{Na}^+$ ( $\text{H}^+$ )/proline <sup>a</sup>   |                                                              |
|                       | $\text{Na}^+$ /sugar                                  |                                                              |
|                       | $\text{Na}^+$ ( $\text{H}^+$ )/melibiose <sup>a</sup> |                                                              |
|                       | $\text{Mg}^{2+}$ /organic acid                        |                                                              |
|                       | $\text{Mg}^{2+}$ , $\text{H}^+$ /citrate <sup>c</sup> |                                                              |

<sup>a</sup>*E. coli*.

<sup>b</sup>*Salmonella typhimurium*.

<sup>c</sup>*Bacillus subtilis*.

### Establishing Ion Gradients

The establishment of ion gradients is of supreme importance to microorganisms. Proton and sodium gradients are important in various organisms for energy production (see above) and in transport. Some consideration should be given at this point to how the gradients are established and maintained. The cell membrane is impermeable, for the most part, to charged ions, so the cell has an opportunity to control the flow of ions across the membrane through ion-specific transport systems. As already detailed, proton gradients (known as the proton motive force) are established principally by the electron transport systems that pump protons out of the cell as electrons are transferred down the system. There are also specific membrane-bound ATPases that can couple the hydrolysis of ATP with the export of  $H^+$ ,  $Na^+$ , and  $K^+$ . In addition, as shown in Table 9-2, there are a variety of antiport systems that can exchange ions such that one gradient can help build another (e.g.,  $Na^+/H^+$ ). So the picture that develops is of an interrelated series of ion circulations whose purpose is to provide energy to the cell.

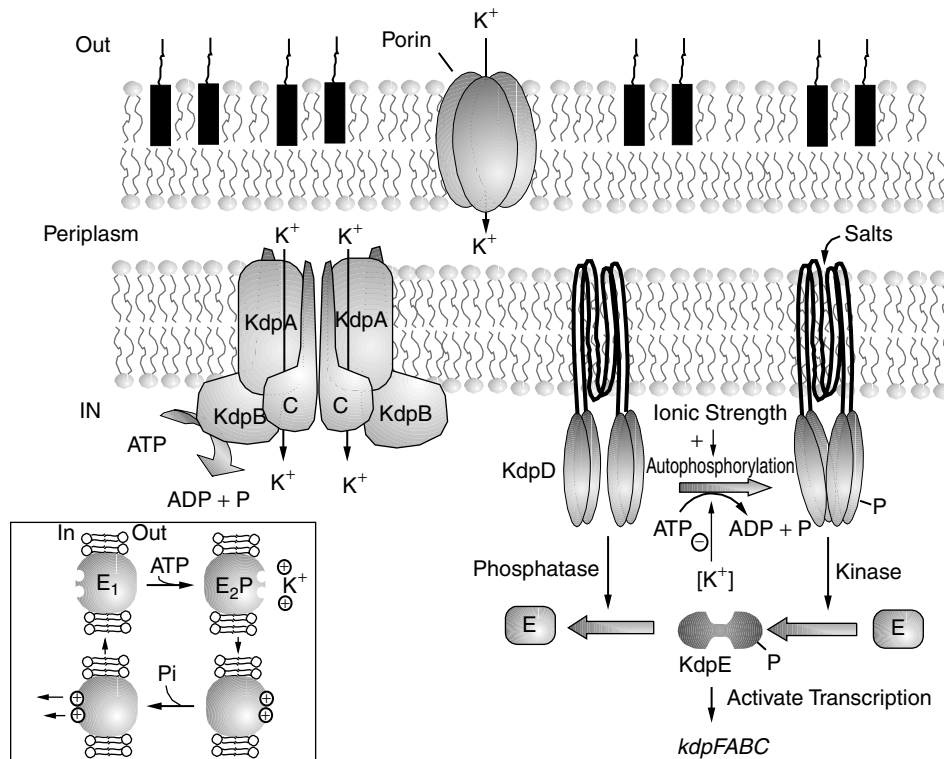
## SPECIFIC TRANSPORT SYSTEMS

### ATP-Linked Ion Motive Pumps

The two major classes of these ion pumps are the F-type ( $F_1F_0$ ) and P-type ( $E_1E_2$ ) ATPases. The F-type ATPase is involved with pumping  $H^+$  out of the cell or in coupling  $H^+$  movement into the cell with the generation of ATP. The P-type ATPases are remarkably similar to each other despite the wide range of ions they transport. They include potassium, magnesium, calcium, cadmium (resistance), and arsenate (resistance). All members have the following properties: (1) a phosphorylated intermediate (aspartyl-phosphate in all ATPases where the phosphorylated amino acid has been determined); (2) two conformational forms of the phosphorylated intermediate, referred to as  $E_1$  and  $E_2$ , which differ in reactivity to substrates and proteases; (3) a large (ca. 100 kDa) membrane-bound subunit with six to eight membrane-spanning regions and several regions of amino acid sequence homology. Most P-type ATPases only have this single large subunit. An exception is the Kdp ATPase.

The **Kdp ATPase** of *Escherichia coli* was first identified from the analysis of mutants that could not grow on low concentrations of potassium. It is a three-subunit enzyme whose subunit ratio appears to be  $A_2B_2C_2$ , although the exact stoichiometry is unknown (Fig. 9-11). The KdpB protein spans the membrane seven times, is homologous to other P-type ATPases (e.g., contains a conserved DKTGT motif), and is the subunit that is phosphorylated by ATP (putative phosphorylation site Asp307), which couples energy to transport. The KdpA subunit, which traverses the membrane 10 times, binds  $K^+$  and probably forms most, if not all, of the membrane channel for  $K^+$ .

All other P-type ATPases couple the export of one ion from the cytoplasm to the import of a different ion from outside the cell (e.g.,  $Na^+K^+$  ATPase). The Kdp ATPase is different from all other P-type ATPases in that it transports ions exclusively to the intracellular compartment. The inset in Figure 9-11 illustrates the proposed kinetic model for  $K^+$  transport via the Kdp system. The low-energy state of the system ( $E_1$ ) changes to high energy ( $E_2P$ ) following phosphorylation. In this state, sites on the



**Fig. 9-11. The Kdp potassium transport ATPase of *E. coli* consists of three inner membrane proteins: KdpA, KdpB, and KdpC.** KdpB is homologous to other E<sub>1</sub>-E<sub>2</sub> ATPases. The outer membrane porin protein is not specified, and there is no periplasmic  $K^+$ -binding protein. The regulatory proteins KdpD and KdpE form a two-component signal transduction system that undergoes a cycle of auto- and *trans*-phosphorylation. Autophosphorylation is activated by intracellular ionic strength and an extracellular rise in osmolarity, and repressed by potassium. The phosphorylated form of KdpE activates transcription of the *kdpFABC*.

exterior of KdpA are exposed and  $K^+$  can enter. Opening of the KdpA membrane channel and dephosphorylation accompany movement of  $K^+$  to the inner surface and a return of the complex to E<sub>1</sub>. The number of ions transported per ATP hydrolyzed is not known.

Transcriptional control of the *kdpFABC* operon is unusual in that the system is expressed under two seemingly contradictory conditions: when cells are unable to maintain the desired pool of  $K^+$  and when experiencing osmotic upshift. Another, constitutively expressed, non-P-type system called Trk is involved with “housekeeping” potassium transport during normal growth conditions. The *kdp* operon is regulated by a typical two-component system involving KdpD, the sensor kinase, and KdpE, the response regulator (Fig. 9-11). However, the environmental signal recognized by KdpD has been unclear. It now appears that the sensing of  $K^+$  and osmotic upshock by KdpD are mechanistically discriminated. Increased ionic strength triggers autophosphorylation of KdpD while high internal  $K^+$  concentration inhibits autophosphorylation. Although this makes intuitive sense, osmotic upshift by salts can override the downregulating



effect of high internal  $K^+$  concentration and stimulate KdpD phosphorylation. Under any of these scenarios, phosphorylated KdpD transphosphorylates KdpE, which in turn activates transcription of the *kdpFABC* operon. The higher level of Kdp proteins increases  $K^+$  transport. The reasons for the dual environmental control of *kdp* make perfect sense. Obviously, the cell needs to increase  $K^+$  uptake when internal levels fall too low, but since  $K^+$  is a compatible solute, the cell will also want to increase  $K^+$  transport following osmotic shock (see Chapter 18).

### The Histidine Permease

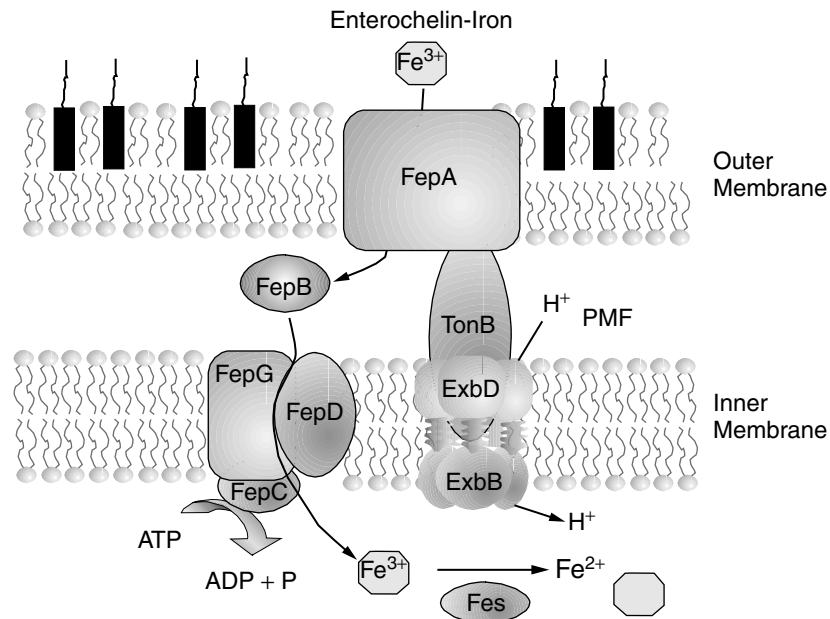
A member of the traffic ATPases, the histidine permease comprises the histidine-binding protein, HisJ, and three proteins, HisQ, HisM, and HisP, which form the membrane-bound complex. The membrane-bound complex consists of one copy each of HisQ and HisM and two copies of the ATP-binding subunit HisP. The receptor HisJ binds histidine and changes conformation, allowing it to interact with the membrane-bound complex. Translocation then occurs through a series of conformational changes concomitant with ATP hydrolysis and is hypothesized to take place through a pore formed by HisQ and HisM.

### Iron

Iron is an essential nutrient for bacteria, serving as a cofactor for many enzymes and as a redox center in cytochromes and iron-sulfur proteins. However, iron is initially inaccessible to bacteria in the natural environment. Within animal fluids and tissues, proteins such as transferrin, lactoferrin, or ferritin sequester iron. Outside hosts, FeII rapidly oxidizes to FeIII and forms insoluble ferric hydroxide polymers. Bacteria have evolved elaborately clever mechanisms to wrest iron from the environment. A common strategy, used also by our reference organism *E. coli*, involves the synthesis and secretion of small, high-affinity, iron-binding chelates (called **siderophores**) that bind environmental iron. The iron chelate is then transported back into the cell by a specialized transport system and the iron is released.

A major siderophore of *E. coli* is **enterochelin** (Fig. 15-21). Its chemical synthesis and structure are described in Chapter 15. The mechanism by which desferrienterochelin is secreted remains a mystery. Once in the environment, it will bind to  $Fe^{3+}$ , after which the products of the *fep* genes enable movement of ferrienterochelin back into the cell (Fig. 9-12). The outer membrane FepA protein (79 kDa) functions as a monomer and enables transport of the complex across the outer membrane. It contains as many as 29  $\beta$ -sheet transmembrane spanning regions and an amino acid sequence called a **Ton box**. This region is predicted to contact the TonB inner membrane protein in a way that will “transduce” potential energy from the cytoplasmic membrane to FepA and drive transport of the siderophore across the outer membrane (Fig. 9-12).

Two other inner membrane proteins, ExbB and ExbD, form a heterohexamer with a central channel that accommodates the transmembrane segment of TonB. PMF is harnessed by the passage of protons through the ExbDB hexamer. This in some way energizes TonB to interact with FepA (and other TonB-dependent outer membrane proteins), triggering a conformational change in FepA that facilitates ferrienterochelin transport to the periplasm. Next, a periplasmic ferrienterochelin-binding protein (FepB) delivers the complex to the inner membrane transport system comprised of FepG, FepD,



**Fig. 9-12. The Ent enterochelin iron system transports the trimeric catechol enterochelin (hexagon).** Enterochelin passes through the FepA protein of the outer membrane in a process requiring energy coupling through TonB. The periplasmic FepB protein passes the  $\text{Fe}^{3+}$  enterochelin to the inner membrane proteins FepG and FepD. Passage across the inner membrane requires ATP energy through FepC. In the cytoplasm, the Fes enterochelin esterase cleaves enterochelin, allowing the release and subsequent reduction of iron.

and FepC. FepC is a membrane-associated ATPase that provides the energy required to effect transport of the complex through FepG and FepD and into the cytoplasm. Once inside the cell, the enterochelin backbone is cleaved by Fes esterase, reducing the affinity for iron from  $10^{-52}$  to  $10^{-8}$ .

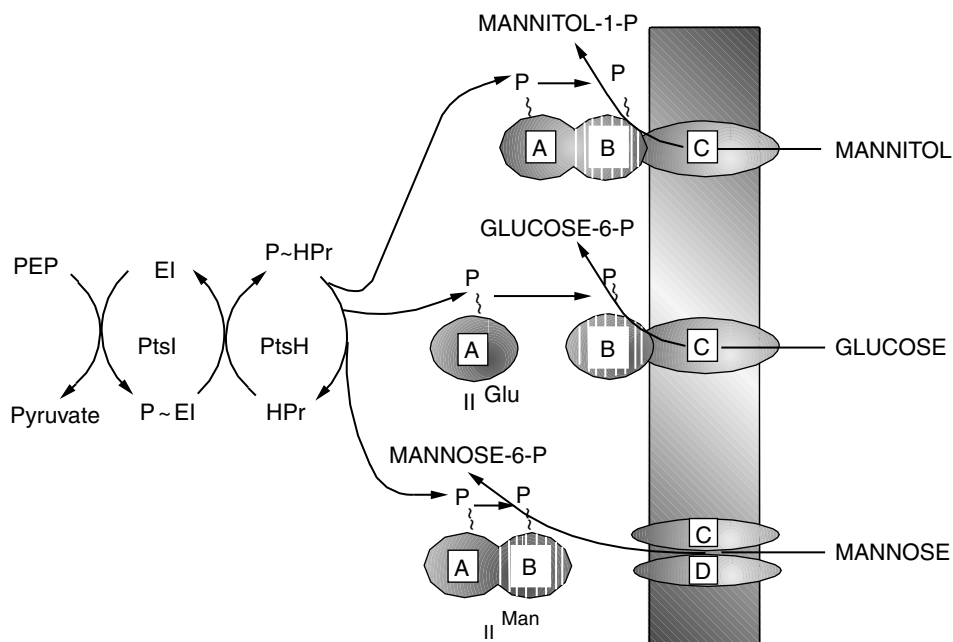
Regulation of this system is also interesting. The  $\text{Fe}^{3+}$  released in the cell is reduced by an unknown mechanism to  $\text{Fe}^{2+}$ . As ferrous iron accumulates in the cell, it is thought to bind to the regulatory protein **Fur** (ferric uptake regulator). Fur is actually a  $\text{Mn}^{2+}$ -binding protein with two binding sites for this metal. How and where (and even if) iron binds to Fur is as yet unclear. Nevertheless, as iron levels increase in the cell, metallated Fur binds to a 17 bp consensus sequence (the IRON or FUR box) located in front of the seven transcriptional units in *E. coli* that contain the *ent*, *fep*, and *fes* genes and repress their expression. Fur, however, has a role in the cell beyond that of controlling iron uptake. At least 50 *E. coli* or *S. enterica* genes are regulated by Fur, some in an iron-independent fashion.

### Phosphotransferase System

The phosphotransferase system (PTS) is involved in both the transport and phosphorylation of a large number of carbohydrates, in movement toward these carbon sources (see "Chemotaxis" in Chapter 7) and in the regulation of several other metabolic

pathways (see “catabolite control” in Chapter 5). In this group translocation transport system, carbohydrate phosphorylation is coupled to carbohydrate transport, the energy for which is provided by the EMP intermediate phosphoenolpyruvate (PEP). Figure 9-13 illustrates that there are two proteins common to all of the PTS carbohydrates. They are enzyme I (*ptsI* in *E. coli*) and the histidine protein HPr (*ptsH*). They are soluble, cytoplasmic proteins that participate in the phosphorylation of all PTS carbohydrates in a given organism and therefore are called the **general PTS proteins**.

In contrast, the enzyme IIs (EIIs) are carbohydrate specific. EIIs consist of three domains (A, B, and C) that may be combined in a single membrane-bound protein or split into two or more proteins (depending on the system) called IIA, IIB, and IIC. The B and C domains may form one protein called IICB. Likewise, the A and B subunits may form one protein called IIAB. IIC is an integral membrane protein (or domain). IIA (formerly called enzyme III) is soluble. In any scenario, the phospho group from PEP is transferred to the incoming carbohydrate via phospho intermediates of EI, HPr, EIIA, and EIIB. EIIC forms the translocation channel and at least part of the specific carbohydrate-binding site. On the basis of sequence homologies, EIIs may be grouped into four classes: the **mannitol class**, the **glucose class**, the **mannose class**,



**Fig. 9-13. Organization of PTS systems.** EI and HPr are the general proteins for all PTSs. Of the many EIIs, only three are shown: those specific for mannitol (Mtl), glucose (Glc), and mannose (Man). Each contains two hydrophilic domains: IIA (formerly EIII or III) containing the first phosphorylation site (P-His), and IIB containing the second phosphorylation site (either a P-Cys or a P-His residue). The membrane-bound, hydrophobic domain IIC may be split into two domains (IIC and IID).  $\text{II}^{\text{Mtl}}$ ,  $\text{II}^{\text{Glc}}$ , and  $\text{II}^{\text{Man}}$  are specific for mannitol, glucose, and mannose, respectively. P indicates the phosphorylated forms of the various proteins. (Source: Derived from Postma, et al. *Microbiol Rev.* 57:543–594, 1993.)

and in a few gram-positive organisms, galactose and the disaccharide lactose are PTS carbohydrates forming the **lactose class**.

A discussion of the PTS role in catabolite repression of gene expression can be found in Chapter 4. The PTS system is also involved in chemotaxis, a basic discussion of which is found in Chapter 7. The PTS allows chemotaxis toward PTS carbohydrates. Stimulation corresponds to uptake and phosphorylation of a substrate through EII. No MCP is involved in this process. CheB and CheR also are not involved. While the exact mechanism is not clear, it is thought that dephosphorylated EI or HPr that forms during carbohydrate transport could directly or indirectly decrease the level of P-CheA or P-CheY, resulting in increased smooth swimming and positive chemotaxis.

## SUMMARY

The remarkable diversity of compounds tapped by the microbial world for energy underscores the basic evolutionary concept of “finding your own niche.” While one energy source in a given environment (e.g., glucose) may be scarce, there will surely be an organism that has evolved to unleash energy intrinsic to a different compound plentiful to that environment (e.g.,  $\text{Fe}^{2+}$ ). So, too, the multitude of mechanisms used by various species to generate PMF belies the force of necessity to avoid competition with other species, if at all possible.

Microbiology students should stand in awe of the complexity of biochemical processes interwoven to make a living cell. Nowhere is this more evident than when considering all the ramifications involved in transporting an ion from one cell compartment to another. Moving just one  $\text{Na}^+$  ion will have a ripple effect on many other ion movements—all so that the cell can maintain an overall homeostasis. In fact, the volume and diversity of ion traffic that occurs back and forth across the bacterial membrane makes the Los Angeles freeway look underutilized.

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## CHAPTER 10

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# METABOLISM OF SUBSTRATES OTHER THAN GLUCOSE

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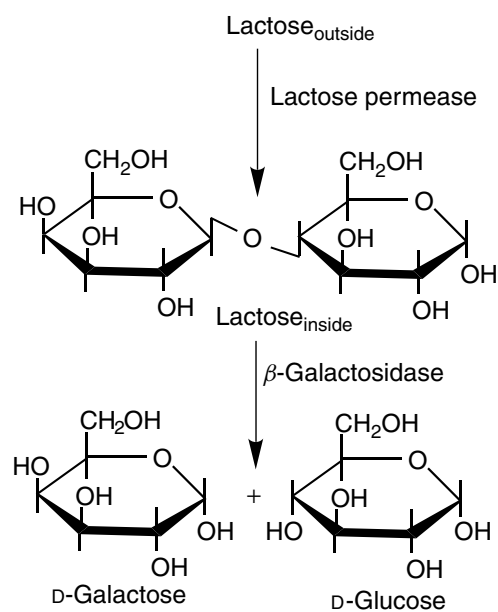
### UTILIZATION OF SUGARS OTHER THAN GLUCOSE

Glucose is the preferred carbon source for most of the common heterotrophic bacteria. Nevertheless, microorganisms possess remarkable versatility in their ability to use a wide range of other compounds as sources of carbon and energy. Most (if not all) of these pathways invariably lead to the production of intermediates that can enter one of the central pathways described in Chapter 8 (i.e., EMP, Entner-Doudoroff, pentose shunt, TCA cycle, or glyoxylate cycle). Many of these alternate carbon sources are utilized only after a period of **induction** of the enzymes required for their transport and metabolism. The presence of glucose in the growth medium generally inhibits the expression of catabolic enzymes for these substrates.

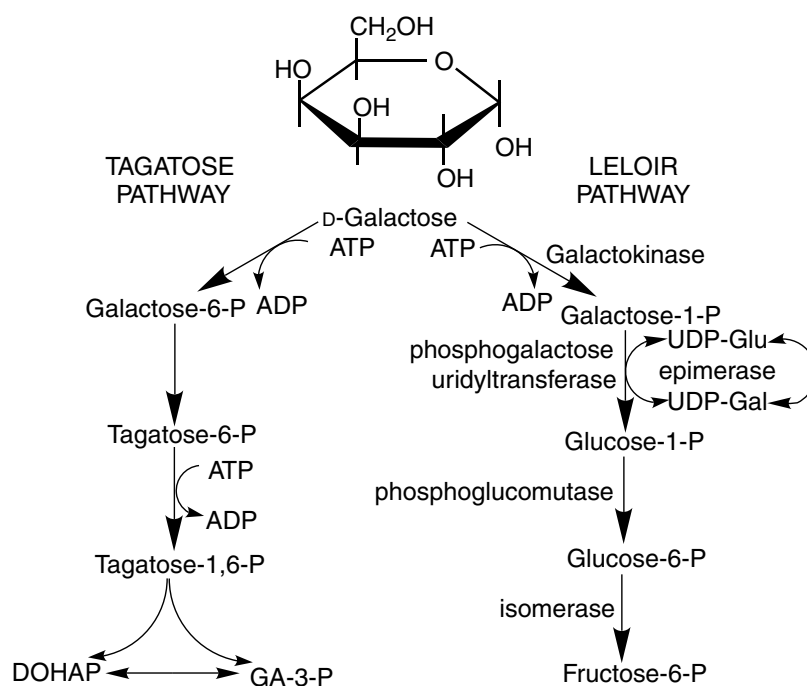
Monod first described this glucose effect in 1947 for the inhibition of  $\beta$ -galactosidase synthesis in *Escherichia coli*, and in 1961 Magasanik coined the term **catabolite repression** for this phenomenon. In media containing glucose and another carbohydrate, bacteria exhibit two complete growth cycles (diauxic growth) separated by a lag period. Glucose prevents entry of the second substrate by a process known as **inducer exclusion** and represses induction of the genes coding for the enzymes required for utilization of the second substrate. The details of the genetic regulation of carbohydrate metabolism are in Chapter 5.

### Lactose

In *E. coli*, lactose utilization requires the induction, under the control of the *lac* operon, of a specific permease for its transport, and  $\beta$ -galactosidase, the enzyme that cleaves it to form D-galactose and D-glucose (Fig. 10-1). The D-galactose is phosphorylated to galactose-1-phosphate and metabolized via the Leloir pathway to yield fructose-6-phosphate (Fig. 10-2). Fructose-6-phosphate is ultimately utilized via the EMP pathway.



**Fig. 10-1. Lactose utilization in *E. coli*.**



**Fig. 10-2. The tagatose and Leloir pathways of galactose utilization.**

Some organisms, such as *Lactobacillus casei*, do not contain the *lac* operon system but possess a PEP phosphotransferase system (PTS) that phosphorylates lactose via a specific enzyme II (see Chapter 5). The intracellular lactose is split by phospho- $\beta$ -galactosidase to yield glucose and galactose-6-phosphate. Galactose-6-phosphate is metabolized via the tagatose-6-phosphate pathway to yield glyceraldehyde-3-phosphate and dihydroxyacetone phosphate (Fig. 10-2). In both cases, the D-glucose is metabolized via the EMP pathway. In *Staphylococcus aureus*, the tagatose-6-phosphate pathway is required for the utilization of galactose as well as lactose. Strains of *Klebsiella* lacking both *lac* and *gal* operons ( $\text{Lac}^- \text{Gal}^-$ ) give rise to lactose-utilizing mutants that transport lactose via the PEP-PTS and metabolize lactose via phospho- $\beta$ -galactosidase. The resultant galactose-6-phosphate is metabolized via the tagatose phosphate pathway.

### Galactose

The inducible enzyme system required for galactose utilization in *S. cerevisiae* is under the control of a complex system of structural and regulatory genes that includes a galactose permease (encoded by *GAL2*) and three enzymes of the Leloir pathway: galactokinase, galactose-1-phosphate uridylyltransferase, and uridine diphosphoglucose-4-epimerase encoded by the structural genes *GAL1*, *GAL7*, and *GAL10*. The enzyme phosphoglucomutase converts glucose-1-phosphate to glucose-6-phosphate, which then enters the glycolytic pathway. The galactose pathway enzymes are coordinately controlled by a positive factor required for the expression of structural genes and a negative factor that interacts with the inducer (galactose) to modulate the function of the positive factor. Expression of the structural genes is controlled by carbon catabolite repression.

### Maltose

Utilization of maltose, a disaccharide of glucose, by *E. coli* requires the expression of genes concerned with maltose uptake (*malEGF*, *malK*, and *lamB*) and induction of amylomaltase and maltodextrin phosphorylase (encoded by the *malPG* operon). Expression of these genes is induced by maltose and is mediated indirectly by the *malT* activator. The cAMP receptor protein (CRP) binds to cAMP and positively regulates the *malT* gene and the *malEGF* operon. The MalT protein mediates the action of cAMP-CRP on *malPQ* genes. Amylomaltase hydrolyzes maltose with the production of D-glucose and the polysaccharide maltodextrin (Fig. 10-3). Maltodextrin is converted to G-1-P by maltodextrin phosphorylase. With the isomerization of G-1-P to G-6-P by phosphoglucomutase, both products are utilized via the EMP pathway. Similar systems for the utilization of maltose are active in a number of other organisms.

### Mannitol

Catabolism of mannitol and other hexitols (D-glucitol or sorbitol, and D-galactitol) by *E. coli* involves transport by specific enzyme IIs of the PEP-PTS. The phosphorylated derivative is then oxidized to the corresponding sugar phosphate by a specific hexitol phosphate dehydrogenase (Fig. 10-4). The catabolic enzymes for mannitol, glucitol, and galactitol are encoded within three operons (*mtl*, *gut*, and *gat*) in *E. coli*. All



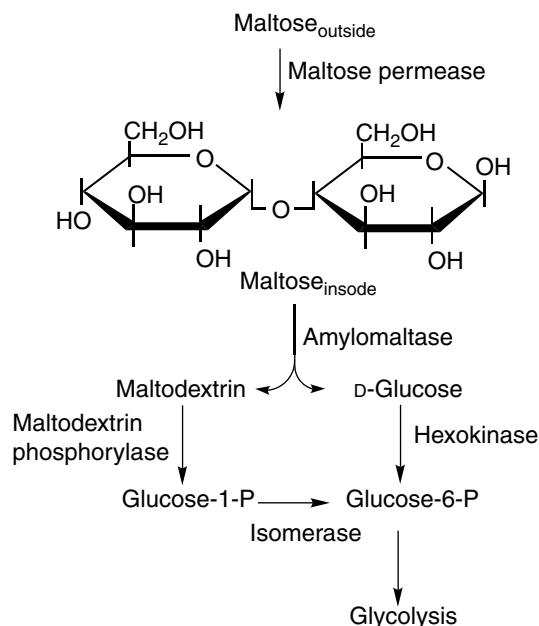


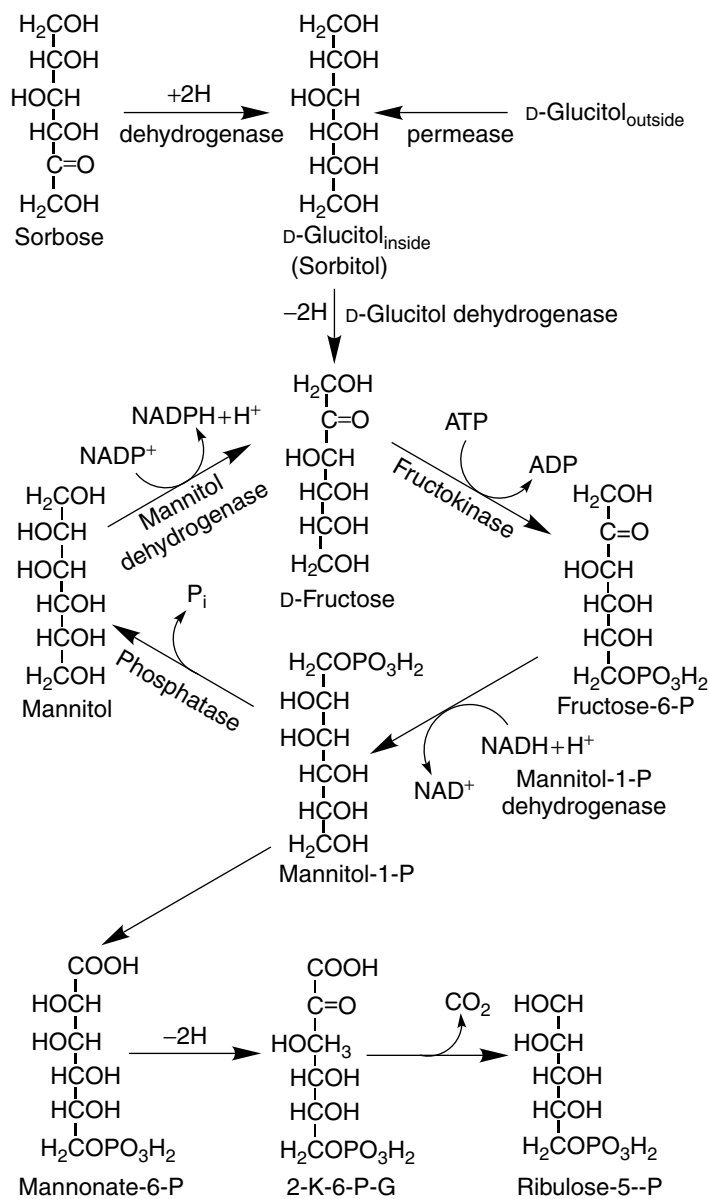
Fig. 10-3. Maltose utilization pathways in *Escherichia coli*.

three operons have the same gene order: a regulatory gene (C), the hexitol-specific enzyme II of the PEP-PTS (A), and the hexitol phosphate dehydrogenase (D). In the case of mannitol, mannitol-1-phosphate is oxidized by a specific NAD-dependent dehydrogenase to yield F-6-P, which enters the EMP pathway. *Bacillus subtilis* transports D-glucitol (sorbitol) through an inducible permease. The free hexitol is oxidized to D-fructose by an inducible glucitol dehydrogenase. D-Fructose then enters the EMP pathway via F-6-P.

Mannitol is also produced as an end product of glucose metabolism in *E. coli*, *Staphylococcus aureus*, *Lactobacillus brevis*, *Leuconostoc mesenteroides*, *Absidia glauca*, and *Rhodococcus erythropolis*. Many organisms utilize a mannitol cycle as a means of regenerating NADP through the NADP-dependent mannitol dehydrogenase that oxidizes mannitol to fructose (Fig. 10-4). The fruiting body of *Agaricus bisporus* accumulates large amounts of mannitol, whereas only low amounts are found in the mycelium. *E. coli* can also convert D-mannitol to D-ribose utilizing a pathway involving the synthesis of ribulose-5-phosphate (Fig. 10-4).

### Fucose and Rhamnose

*E. coli* uses interesting parallel pathways for L-fucose and L-rhamnose metabolism. These pathways involve the induction of specific permeases, kinases, isomerases, and aldolases (Fig. 10-5). A cluster of genes at 60.2 minutes on the *E. coli* genetic map encodes the enzymes involved in fucose utilization. The *fuc* genes are organized into four operons under the influence of *fucR*, a positive regulatory protein. Fuculose-1-phosphate is a true inducer. The genes for the rhamnose system constitute a well-defined operon, whereas the fucose system maintains the gene for aldolase (*ald*) under separate



**Fig. 10-4. Pathways of metabolism of sorbose and mannitol.** In many organisms the mannitol cycle functions as a means of regulating reduced NADP levels.

control. The pathways converge with the formation of dihydroxyacetone phosphate and lactaldehyde. Dihydroxyacetone phosphate enters the central metabolic pathway at the triose level.

Under aerobic conditions an NAD-linked dehydrogenase oxidizes L-lactaldehyde to L-lactate and an FAD-linked dehydrogenase oxidizes L-lactate to pyruvate, which then enters the central pathway. Anaerobically, a single NAD-linked oxidoreductase

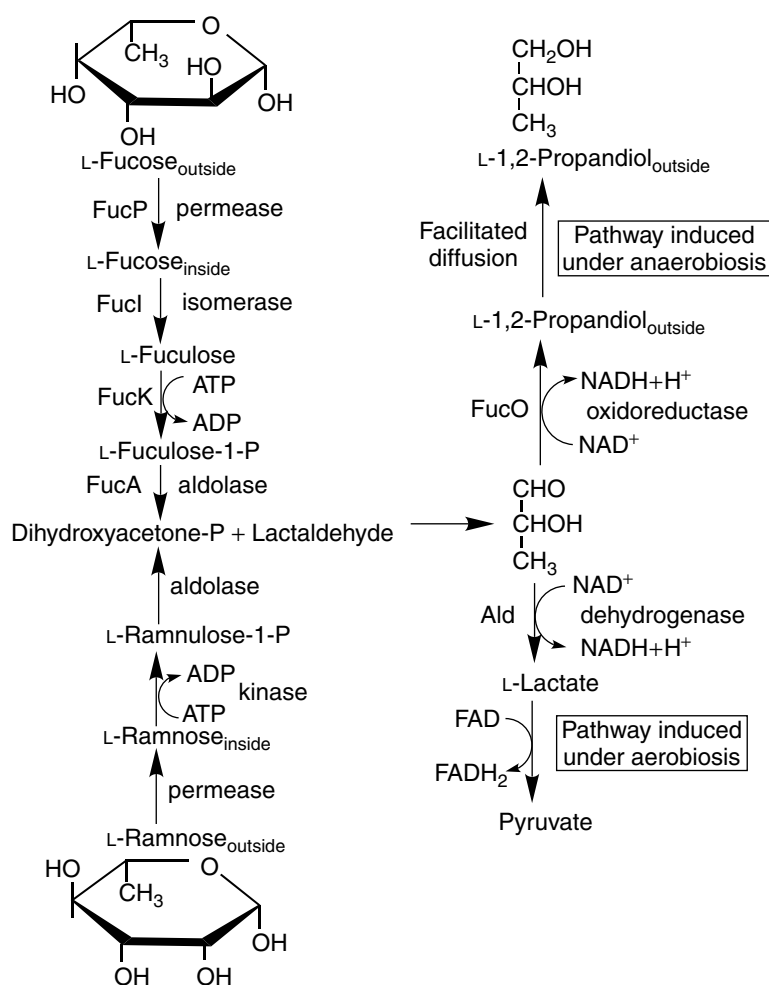


Fig. 10-5. Pathways for the utilization of L-fucose and L-rhamnose.

is induced by either L-fucose or L-rhamnose and serves both pathways. In either case, L-1,2-propanediol is formed and excreted. However, each methyl pentose exerts its influence at a different level. L-fucose is a transcriptional activator of *fucO*, whereas L-rhamnose functions posttranscriptionally. The L-fucose pathway can be recruited for the utilization of 6-deoxy-L-talitol and D-arabinose. Identical systems for the dissimilation of L-fucose and L-rhamnose are present in *S. enterica* and *K. pneumoniae*.

### Mellibiose, Raffinose, Stachyose, and Guar Gum

These compounds all contain  $\alpha$ -1,6-linked galactose residues, as shown in Figure 10-6. Many species of enteric bacteria can ferment mellibiose and raffinose, but relatively few can utilize galactomannans such as guar gum. *Bacteroides ovatus*, an obligately anaerobic resident of the intestinal tract of humans, develops  $\alpha$ -galactosidase activity when grown on mellibiose, raffinose, or galactomannan. However,  $\beta$ -D-mannanases,

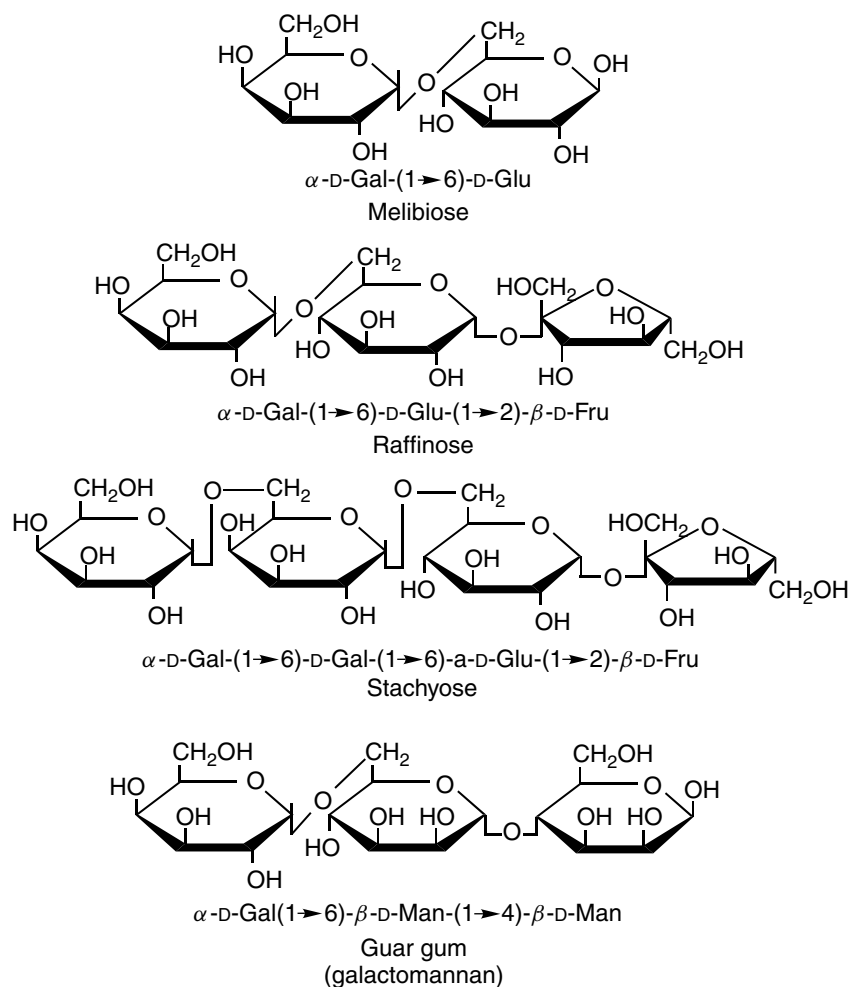
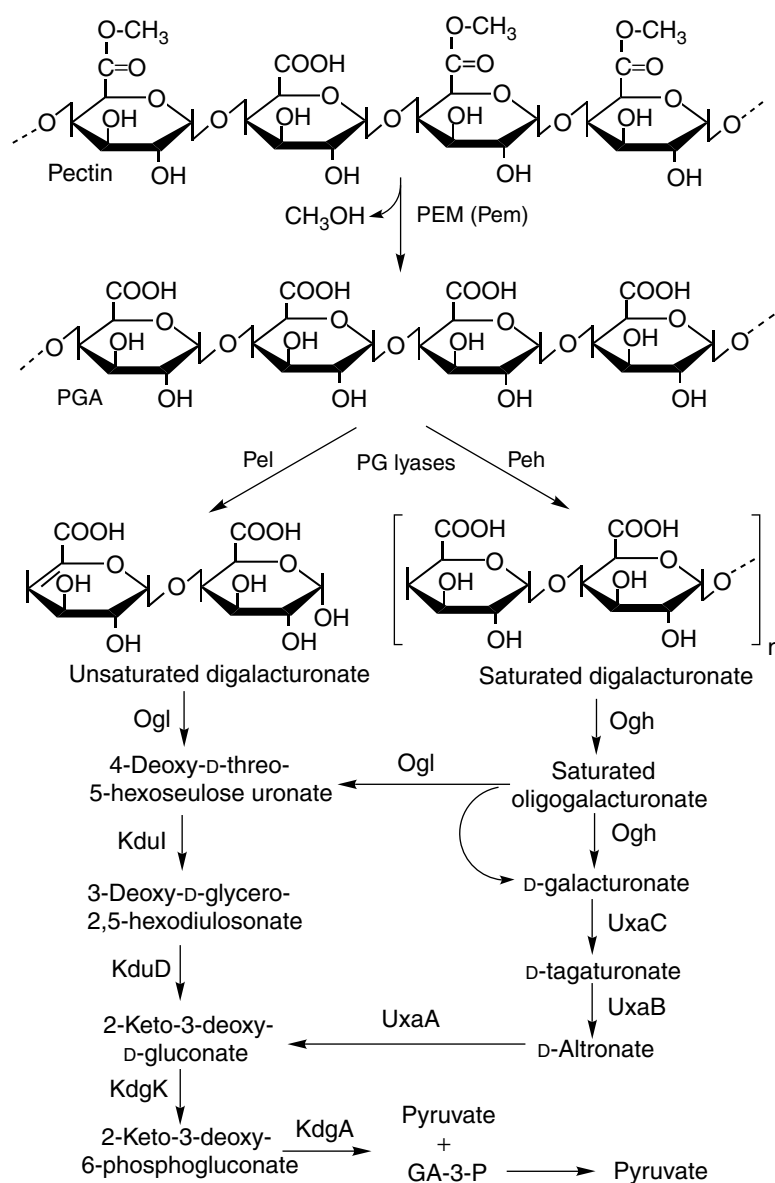


Fig. 10-6. Structures of melibiose, raffinose, stachyose, and guar gum (galactomannan).

enzymes that degrade the mannose backbone of guar gum (galactomannan), are produced only during growth on this substrate. The  $\alpha$ -galactosidase I differs from the  $\alpha$ -galactosidase II produced during growth on melibiose, raffinose, or stachyose.

## PECTIN AND ALDOHEXURONATE PATHWAYS

Pectin, a highly methylated form of poly- $\beta$ -1,4-D-galacturonic acid, is a major constituent of plant cell walls. The ability to degrade pectin contributes to the virulence of bacterial and fungal phytopathogens such as *Erwinia chrysanthemi* and *E. carotovora*, *Moloninia fructigena*, *Cladosporium cucumerinum*, and *Botrytis cinerea*. The rumen bacteria, *Butyrivibrio fibrisolvens* and *Lachnospira multiparus*, provide nutrients to the ruminant animal because of their ability to degrade pectin and cellulose.



**Fig. 10-7. Pathways of pectin catabolism in bacteria.** The three-letter designation for the structural enzymes are: Pem, pectin methylesterase (PME); polygalacturonate (PGA) lyases (PelA to PelE) are extracellular enzymes. Further degradation of the breakdown products are conducted by: Ogh, oligogalacturonate hydrolase; Ogl, oligogalacturonate lyase; KduI, 4-deoxy-L-threo-5-hexoseulose uronate isomerase; KduD, keto-deoxyuronate dehydrogenase; KdgK, 2-keto-3-deoxygluconate kinase; KdgA, 2-keto-3-deoxy-6-phosphogluconate aldolase; UxaA, altronate hydrolase; UxaB, altronate oxidoreductase; UxaC, uronate isomerase.

*Bacteroides thetaiotaomicron*, a gram-negative anaerobe found in the human intestinal flora, can also degrade pectin and other plant polysaccharides.

*E. chrysanthemi* and *E. carotovora* initiate the degradation of pectin by means of a group of enzymes acting sequentially as shown in Figure 10-7. Pectin methylesterase (Pem) removes methoxyl groups linked to C-6 to yield polygalacturonate, which is degraded by pectate lyases (encoded by *pelABCDE*) to form unsaturated digalacturonate. Alternatively, an exopolygalacturonase produces digalacturonate. The polygalacturonate residues are then degraded via either of the two alternative pathways that lead to the formation of 2-keto-3-deoxy-6-phosphogluconate (KDPG) as shown in Figure 10-7. This metabolite then follows the Entner-Doudoroff pathway to pyruvate and triose phosphate.

## CELLULOSE DEGRADATION

Cellulose is a  $\beta$ -1,4-linked glucose polymer that occurs in crystalline or amorphous forms and is usually found along with other oligosaccharides in the walls of plants and fungi. Cellulose-digesting microorganisms in the rumen of herbivorous animals are responsible for the ability of ruminants to use cellulose as a source of energy and building blocks for biosynthesis. A major obstacle to economical animal protein production has been the inefficient utilization of cellulose-containing materials by the rumen microflora. The ubiquitous distribution of cellulose in municipal, agricultural, and forestry waste emphasizes the potential use of cellulose for conversion to useful products such as single-cell protein or fermentation products such as methane or alcohol. As a consequence, the degradation of cellulose has been a continuing subject of intense study.

Cellulose degradation requires the combined activities of three basic types of enzymes (Fig. 10-8). Initially, an *endo*- $\beta$ -1,4-glucanase cleaves cellulose to smaller oligosaccharides with free-chain ends. Then *exo*- $\beta$ -1,4-glucanases remove disaccharide cellobiose units from either the reducing or nonreducing ends of the oligosaccharide chains. Cellobiose is then hydrolyzed to glucose by  $\beta$ -glucosidases.

These cellulolytic enzymes may be produced as extracellular proteins by organisms such as *Trichoderma* or *Phanerochete* (filamentous fungi), or by *Cellulomonas*, *Microbispora*, or *Thermomonaspora* (Actinomycetes). Rumen bacteria such as *Ruminococcus flavofaciens* and *Fibrobacter succinogenes*, or gram-positive anaerobes such as *Clostridium thermocellum*, *C. cellulovorans*, or *C. cellulolyticum*, produce a cell-bound multienzyme complex called the **cellulosome**. With the aid of the electron microscope, cellulosomes can be seen as protuberances on the cell surface (Fig. 10-9). The cellulosome of *C. cellulovorans* contains three major subunits: a scaffolding protein, CipA; an exoglucanase, ExgS; and an endoglucanase, EngE. Also present are endoglucanases EngB, EngL, and EngY, and a mannanase, ManA. The scaffolding protein serves as a cellulose-binding factor. Another component, present in duplicate and referred to as dockerin, mediates the association of cellulose fibers with the scaffolding protein. Various models have been proposed to conceptualize the complex interaction of the cellulosome with cellulose fibers during the digestion process.

A wide diversity of actively cellulolytic organisms is important in industrial applications, in the rumen of animals, and in the digestive systems of arthropods that degrade wood. Termites and other arthropods that degrade wood owe their ability to digest cellulose to the presence of specific microbial symbionts in their digestive tract.

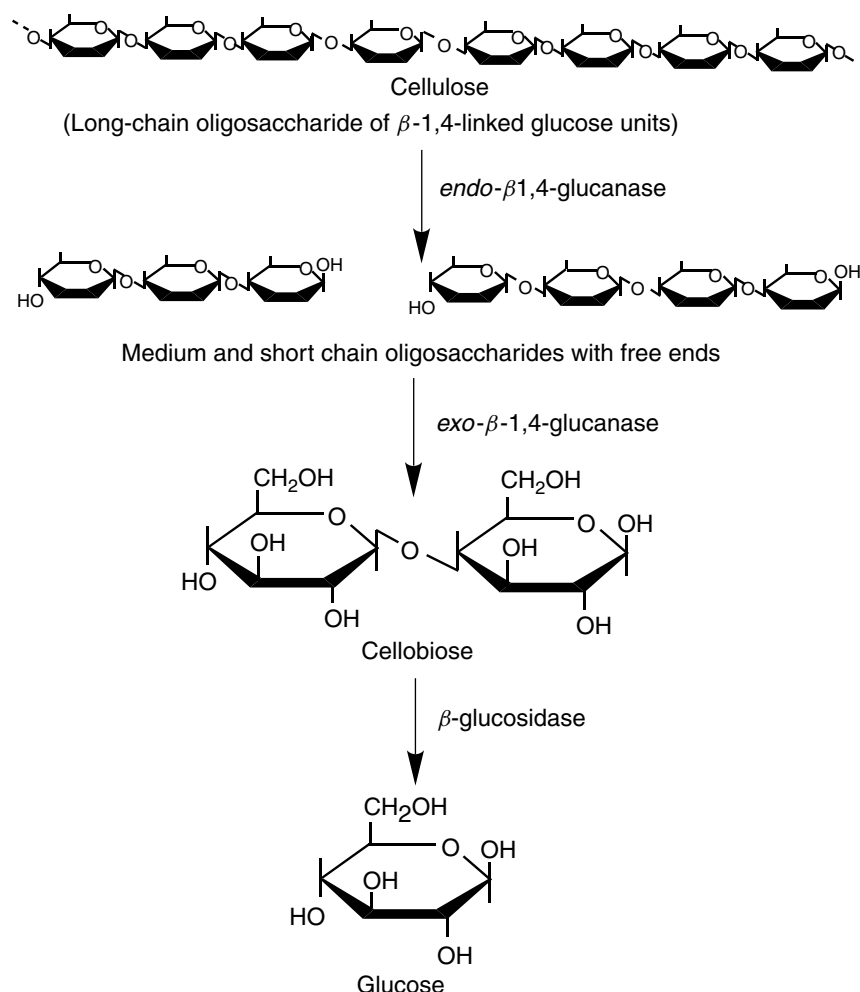
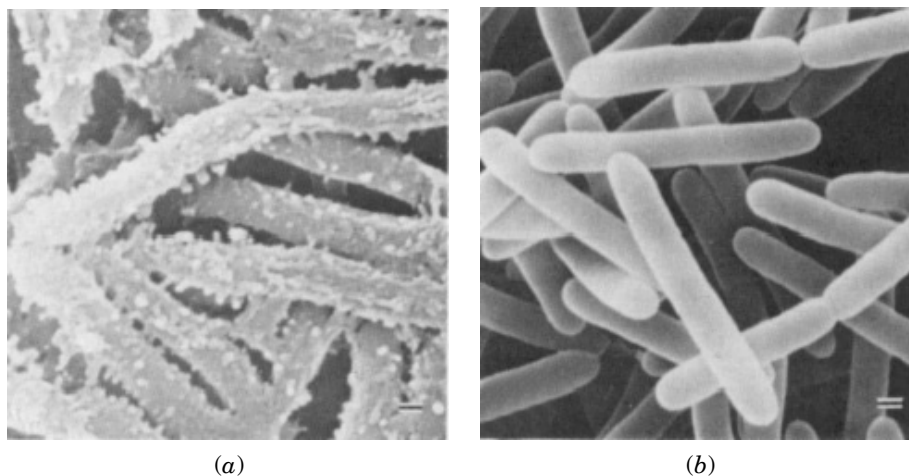


Fig. 10-8. Enzymatic degradation of cellulose.

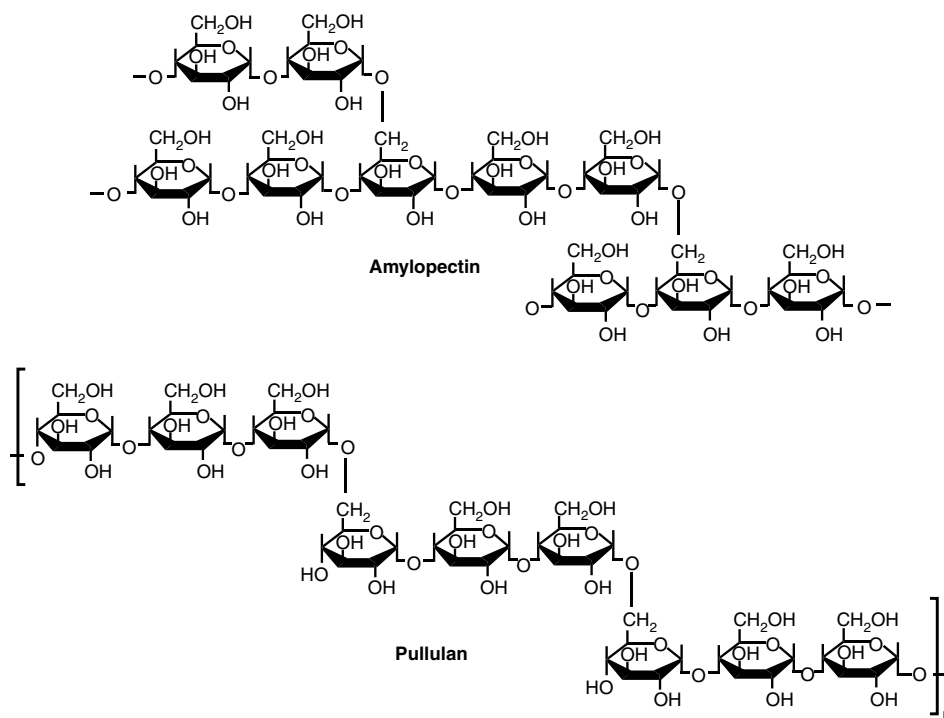
## STARCH, GLYCOGEN, AND RELATED COMPOUNDS

Starch, one of the most common storage compounds in plants, is a mixture of 25% amylose and 75% amylopectin. Amylose consists of long, unbranched chains of glucose in  $\alpha$ -(1,4) linkage. Although not truly soluble in water, amylose forms hydrated micelles in which the polysaccharide chain forms a helical coil. By comparison, amylopectin is a highly branched form of starch in which the backbone consists of glucose chains in  $\alpha$ -(1,4) linkage with  $\alpha$ -(1,6) linkages at the branch points (Fig. 10-10). Glycogen, the main storage compound in animal cells, is comparable to amylopectin in that the main backbone also consists of glucose units in  $\alpha$ -(1,4) linkage but with more frequent  $\alpha$ -(1,6) branches. Pullulan, a starch-like polysaccharide, is a 1,4:1,6- $\alpha$ -D-glycan composed of maltotriose units in 1,6- $\alpha$ -linkage.

Amylose can be hydrolyzed by  $\alpha$ -amylase, which cleaves the  $\alpha$ -(1,4) linkages to yield a mixture of  $\alpha$ -glucose and  $\alpha$ -maltose. Amylose is also hydrolyzed by  $\beta$ -amylase,



**Fig. 10-9. Cellulosomes of *Clostridium thermocellum*.** Scanning electron micrographs of ferritin-labeled cellobiose-grown cells. Prior to processing for electron microscopy, the cells were treated with cationized ferritin. Wild-type cellulose-digesting cells (a) are easily distinguishable from mutant cells (b) by the protuberances that cover the entire cell surface of the wild type. Bars = 200 nm. (source From Bayer, E. A. and R. Lamed, 1986. *J. Bacteriol.* **167**:828–836.)



**Fig. 10-10. structure of amylopectin and pullulan.** Amylopectin contains chains of glucose molecules in  $\alpha$ -(1,4) linkage with occasional  $\alpha$ -(1,6)-linked branches. The structure of glycogen is similar except that there are more frequent  $\alpha$ -(1,6)-linked branches. Pullulan is composed of maltotriose units of three glucose molecules in  $\alpha$ -(1,4) linkage connected by  $\alpha$ -(1,6) linkages.



liberating  $\beta$ -maltose in stepwise fashion from the nonreducing end of the molecule. Thus  $\beta$ -amylases invert the configuration at the C-1 position during cleavage of the  $\alpha$ -(1,4)-glucosidic bond, whereas  $\alpha$ -amylases retain the  $\alpha$  configuration at the C-1 position. These enzymes also hydrolyze amylopectin or glycogen to yield glucose, maltose, and a highly branched core, limit dextrin. A debranching enzyme,  $\alpha$ -(1,6)-glucosidase, is capable of hydrolyzing the  $\alpha$ -(1,6) linkages in limit dextrin. The combined action of  $\alpha$ -(1,6)-glucosidase and  $\alpha$ -amylase is required to completely degrade amylopectin or glycogen to glucose and maltose. Thus,  $\alpha$ -amylases are endoenzymes that can bypass the  $\alpha$ -(1,6) branch points of amylopectin, whereas  $\beta$ -amylases are exoenzymes that cannot hydrolyze amylopectin internally to the  $\alpha$ -(1,6) branch points. Both  $\alpha$ - and  $\beta$ -amylases and debranching enzymes (also called pullulanases) are produced by a number of bacteria and fungi. *Aspergillus* species produce glucoamylases, enzymes that degrade starch to glucose. Extracellular  $\beta$ -amylases have been observed in *B. polymyxa*, *B. megaterium*, and other *Bacillus* species.

The amylases secreted by a variety of *Bacillus* species have been studied intensively because of their industrial application. The  $\alpha$ -amylases from *B. subtilis* are saccharifying enzymes that produce mostly glucose and maltose from starch. On the other hand, the  $\alpha$ -amylases produced by *B. licheniformis* and *B. amyloliquefaciens* are liquefying enzymes that yield mostly maltosaccharides. Under the conditions of nutrient deprivation, *B. subtilis* activates the structural gene for  $\alpha$ -amylase, *amyE*. The synthesis of  $\alpha$ -amylase is not inducible in the classic sense in that no compound can be shown to trigger the action of *amyE*. However, synthesis of the enzyme is repressed by glucose and other readily metabolizable carbon sources. Several genes appear to increase the formation of  $\alpha$ -amylase and its secretion.

For ecological as well as industrial reasons, the thermostability of  $\alpha$ -amylases has been of considerable interest. Thermophilic species, such as *B. acidocaldarius* and *B. stearothermophilus*, produce amylases that are stable at temperatures ranging from 58°C to 80°C. However, mesophilic species, such as *B. licheniformis* and *B. amyloliquefaciens*, also produce amylases that are active at temperatures in excess of 75°C. Comparative studies of the enzymes and the structural genes of mesophilic and thermophilic species reveal considerable amino acid homology between the enzymes from the two groups, indicating that these proteins are related on an evolutionary basis.

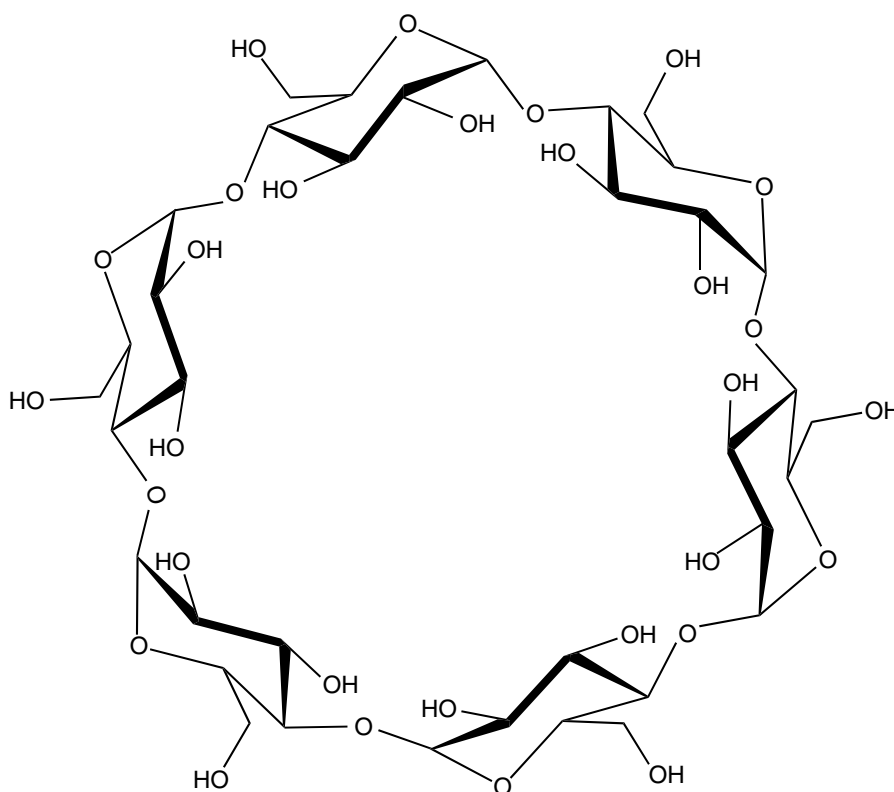
Thermophilic anaerobes produce highly active thermostable starch-degrading enzymes. *Clostridium thermosulfurogenes* produces an extracellular, thermoactive, and thermostable  $\beta$ -amylase and a cell-bound glucoamylase. It does not produce a debranching enzyme. *Clostridium thermosulfuricum* produces a cell-bound glucoamylase and a debranching enzyme, pullulanase, which are thermoactive and thermostable. Mixed cultures of these organisms exhibit enhanced production of  $\beta$ -amylase, glucoamylase, and pullulanase, with concomitant increase in ethanol production from starch. The  $\beta$ -amylase of *C. thermosulfurogenes* is expressed at high levels only when the organism is grown on maltose or other carbohydrates containing maltose units. Glucose represses  $\beta$ -amylase synthesis, but cAMP does not eliminate the repressive effect.

*Bacteroides thetaiotaomicron*, a gram-negative anaerobe found in high numbers in the human colon, can ferment a variety of polysaccharides, including amylose, amylopectin, and pullulan. In this organism the degradative enzymes are not extracellular but are cell associated, indicating that the first step in starch utilization involves binding of starch to the bacterial surface. This action is followed by translocation

through the outer membrane into the periplasm where the starch-degrading enzymes are located. *B. thetaiotaomicron* produces four maltose-inducible outer membrane proteins and one enzyme located in the cytoplasmic membrane.

A 115 kDa outer membrane protein is essential for maltoheptaose utilization, whereas the other outer membrane proteins are involved in starch utilization. Two constitutively produced outer membrane proteins are also involved in starch utilization. Various bacteria produce four types of pullulan-hydrolyzing enzymes. A glucoamylase hydrolyzes pullulan from the nonreducing ends to produce glucose. A pullulanase (type I) found in *K. pneumoniae* and *B. thetaiotaomicron* breaks  $\alpha$ -(1,6) glucosidic linkages to produce maltotriose. An isopullulanase found in *Aspergillus niger* hydrolyzes  $\alpha$ -(1,4) glucosidic linkages in pullulan to produce panose [a branched trisaccharide with two  $\alpha$ -(1,4) linkages and one  $\alpha$ -(1,6) branch]. An  $\alpha$ -glucosidase is found in *B. thetaiotaomicron* in addition to pullulanases I and II.

Cyclodextrins are cyclic oligosaccharides containing from 6 to 12 glucopyranose units bonded through  $\alpha$ -(1,4) linkages, as shown in Figure 10-11. These compounds are widely used in the food and pharmaceutical industry as solubilizing and stabilizing agents. They have also found application as the bonded phase for high-performance liquid chromatography (HPLC) used in the separation and identification of organic compounds. Certain organisms produce cyclodextrins in the process



**Fig. 10-11. Cyclodextrin structure.** Cyclodextrins are cyclic oligosaccharides containing from 6 to 12 glucopyranose units bonded through  $\alpha$ -(1,4) linkage.

of degrading starch. *Bacillus stearothermophilus* degrades starch by means of the enzyme cyclodextrin glucanotransferase. Cyclodextrins are resistant to hydrolysis by many starch-splitting enzymes. *Bacillus macerans*, *B. coagulans*, *B. sphaericus*, and *C. thermohydrosulfuricum* produce a cyclodextrinase that can hydrolyze linear maltodextrins as well as cyclodextrin.

## METABOLISM OF AROMATIC COMPOUNDS

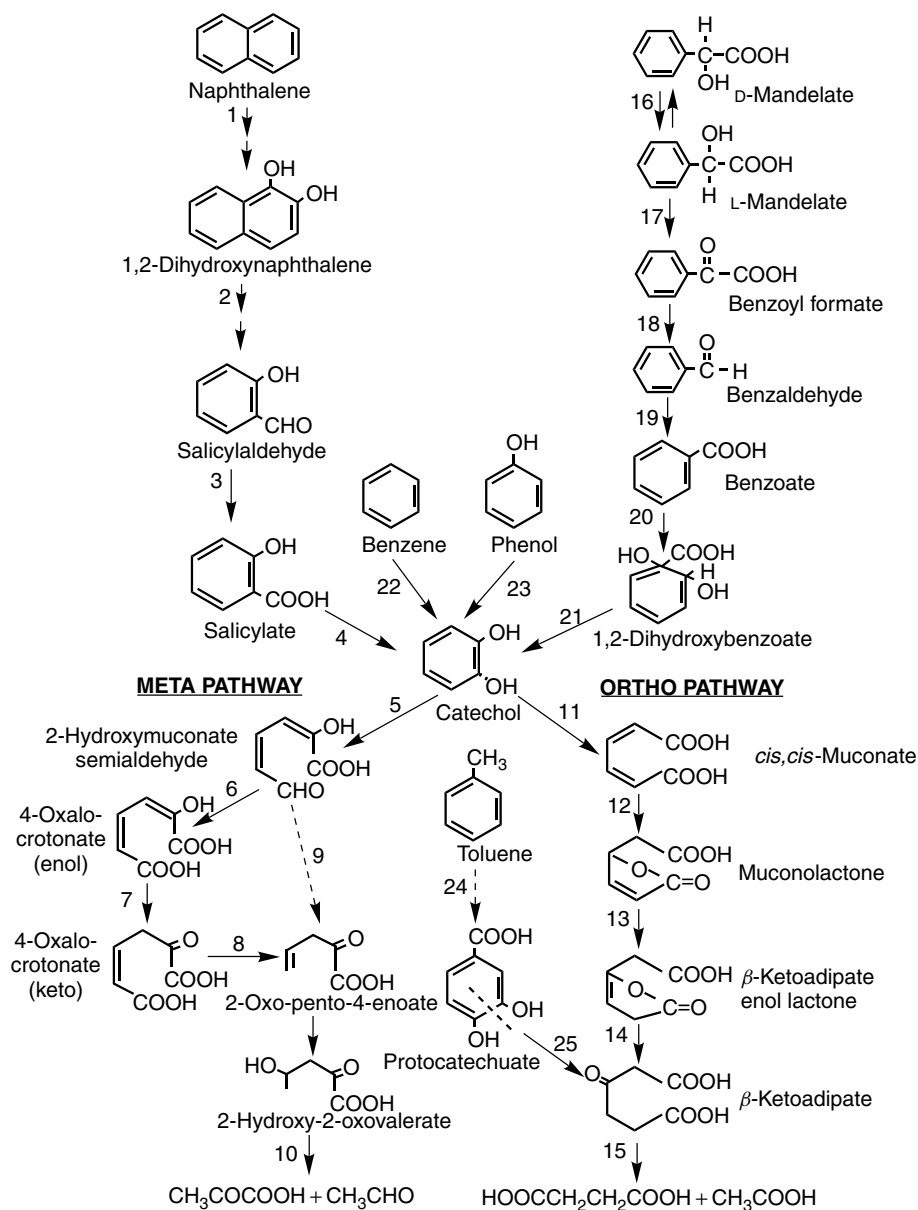
Many microorganisms can utilize aromatic compounds as their sole source of carbon. The aromatic ring structure is degraded and the products are converted to compounds that can enter central metabolic pathways to provide carbon sources and energy. Degradation of aromatic compounds is of major importance because of the widespread occurrence of these substances in natural as well as synthetic materials. Hydrocarbon-degrading microorganisms facilitate reentry of the carbon from these compounds into the natural biological cycles, preventing their accumulation in the environment. Accumulation of toxic aromatic hydrocarbons, particularly herbicides, pesticides, and other industrial waste products, can lead to serious contamination of ground water and cause serious ecological damage.

A fundamental aspect of the metabolism of aromatic ring compounds involves the conversion of the ring structure to a form that can be cleaved to yield an open chain, which can then be metabolized to intermediates in the central metabolic pathways. Under aerobic conditions, monooxygenases or dioxygenases introduce hydroxyl groups that facilitate ring cleavage. As shown in Figure 10-12, several pathways lead to the formation of catechol or similar compounds that can undergo either ortho- or meta-cleavage to yield intermediates leading into the general metabolic schemes. The meta-cleavage pathway results in the formation of pyruvate and acetaldehyde, whereas the ortho-pathway yields  $\beta$ -ketoadipate, a compound that is cleaved to form succinate and acetate.

Several ring oxidation mechanisms are known. *Pseudomonas putida* mt-2 oxidizes the methyl group of toluene to form benzyl alcohol. Subsequent reactions yield catechol, the substrate for the meta-cleavage pathway that leads to keto acid formation. *P. putida* PpF1 initiates the oxidation of toluene by incorporating oxygen into the aromatic nucleus to form *cis*-toluene dihydrodiol, which is converted to 3-methylcatechol. This compound is then degraded via the meta-cleavage pathway to yield an  $\alpha$ -keto acid. *P. mendocina* oxidizes toluene to *p*-cresol, which undergoes ortho-cleavage to form  $\beta$ -ketoadipate. This compound is split by succinyl-CoA transferase to yield succinate and acetate as intermediates that can be used in the TCA cycle. The keto acid intermediates formed via the meta-cleavage route ultimately lead to acetaldehyde and pyruvate, compounds that can enter the central metabolic pathways.

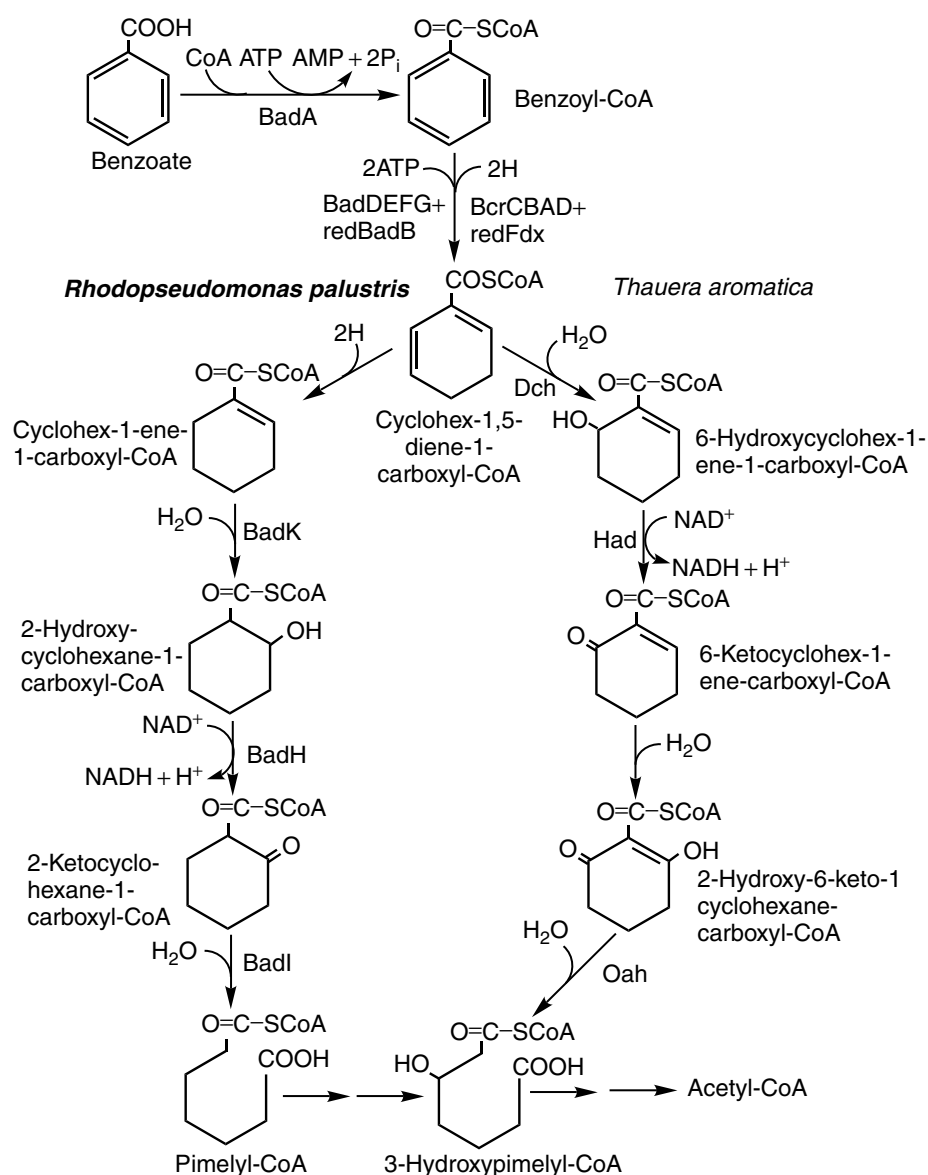
The demonstration by Stanier and his colleagues in 1947 of the mechanism of the oxidation of mandelate to intermediates of the TCA cycle by *P. putida* represents the first example of sequential induction of enzymes of a catabolic pathway. **Sequential induction** refers to the fact that each of the enzymes in the pathway is inducible by its specific substrate.

Compounds with more complex ring structures, such as naphthalene, anthracene, or phenanthrene, undergo sequential ring fission to form ring structures that can



**Fig. 10-12. Metabolism of aromatic hydrocarbons.** The enzymes are: (1) naphthalene oxygenase, (2) 1,2-dihydroxynaphthalene oxygenase, (3) salicylaldehyde dehydrogenase, (4) salicylate hydroxylase, (5) catechol 2,3-oxygenase, (6) 2-hydroxymuconic semialdehyde dehydrogenase, (7) 4-oxalocrotonate tautomerase, (8) 4-oxalocrotonate decarboxylase, (9) 2-hydroxymuconic semialdehyde hydrolase, (10) 4-hydroxy-2-oxovalerate aldolase, (11) catechol 1,2-oxygenase, (12) *cis, cis*-muconate lactonizing enzyme, (13) (+)-muconolactone isomerase, (14)  $\beta$ -keto adipate enol-lactone hydrolase, (15) succinyl CoA- $\beta$ -keto adipate thiolase, (16) mandelate racemase, (17) L-(+)-mandelate dehydrogenase, (18) benzoylformate decarboxylase, (19) NAD- and NADP-linked benzaldehyde dehydrogenases, (20, 21) benzoic acid oxidase, (22) benzene hydroxylase, (23) phenol hydroxylase, (24) toluene hydroxylase, (25) protocatechuate ortho-cleavage enzyme.

then be degraded via the ortho- or meta-cleavage pathways (Fig. 10-12). Chlorinated compounds, such as chlorobenzoate or polychlorinated biphenyls, often require dehalogenation before the aromatic ring can be hydroxylated to facilitate ring fission. In general, the more complex the ring structures, the more resistant polyaromatic hydrocarbons are to degradation. So far in this discussion, we have mentioned primarily pseudomonads that are well known for their ability to degrade aromatic hydrocarbons. Under natural conditions, a consortium of microorganisms may be



**Fig. 10-13.** Comparison of anaerobic benzoate degradation by *R. palustris* and *T. aromatica*. (Source: Redrawn from C. S. Harwood, et al., FEMS Microbiol. Rev. 22, 439–458 1999.)

required to rapidly degrade a compound with four or more fused rings. For example, benzo[a]pyrene, a potent carcinogen that commonly occurs in complex mixtures such as diesel fuel, contains five fused rings. Rapid mineralization of this compound was best accomplished by a combination of organisms phylogenetically related to the bacterial genera *Sphingobacterium*, *Pseudomonas*, *Aquabacterium*, *Burkholderia*, *Ralstonia*, *Sphingomonas*, *Mycobacterium*, and *Alcaligenes*.

Even more complex organic materials such as lignin are resistant to microbial attack. However, many fungi and a few bacteria are capable of degrading lignin or its breakdown products. These organisms are essential for the return of the constituents of these heteropolymers to the natural carbon and nitrogen cycles. The wood-rotting fungus *Phanerochaete chrysosporium* is capable of degrading the pesticide DDT and other complex organic compounds because of their structural similarity to components of the lignin complex.

Studies on the degradation of aromatic compounds cited thus far have all involved the incorporation of molecular oxygen into the aromatic nucleus to form dihydroxylated intermediates. Under anaerobic conditions, the aromatic ring structures are attacked reductively. A comparison of anaerobic benzoate degradation by *Rhodopseudomonas palustris* and *Thauera aromatica* is shown in Figure 10-13. Subsequent metabolism involves the formation of intermediates leading to the formation of acetyl-CoA. Acetyl-CoA can then enter the central metabolic pathway. A wide assortment of aromatic compounds are degraded to benzoyl-CoA via several novel reactions including carboxylation of phenolic compounds, reductive elimination of ring substituents such as hydroxyl or amino groups, oxidation of methyl substituents, *O*-demethylation reactions, and shortening of aliphatic side chains.

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## CHAPTER 11

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# FERMENTATION PATHWAYS

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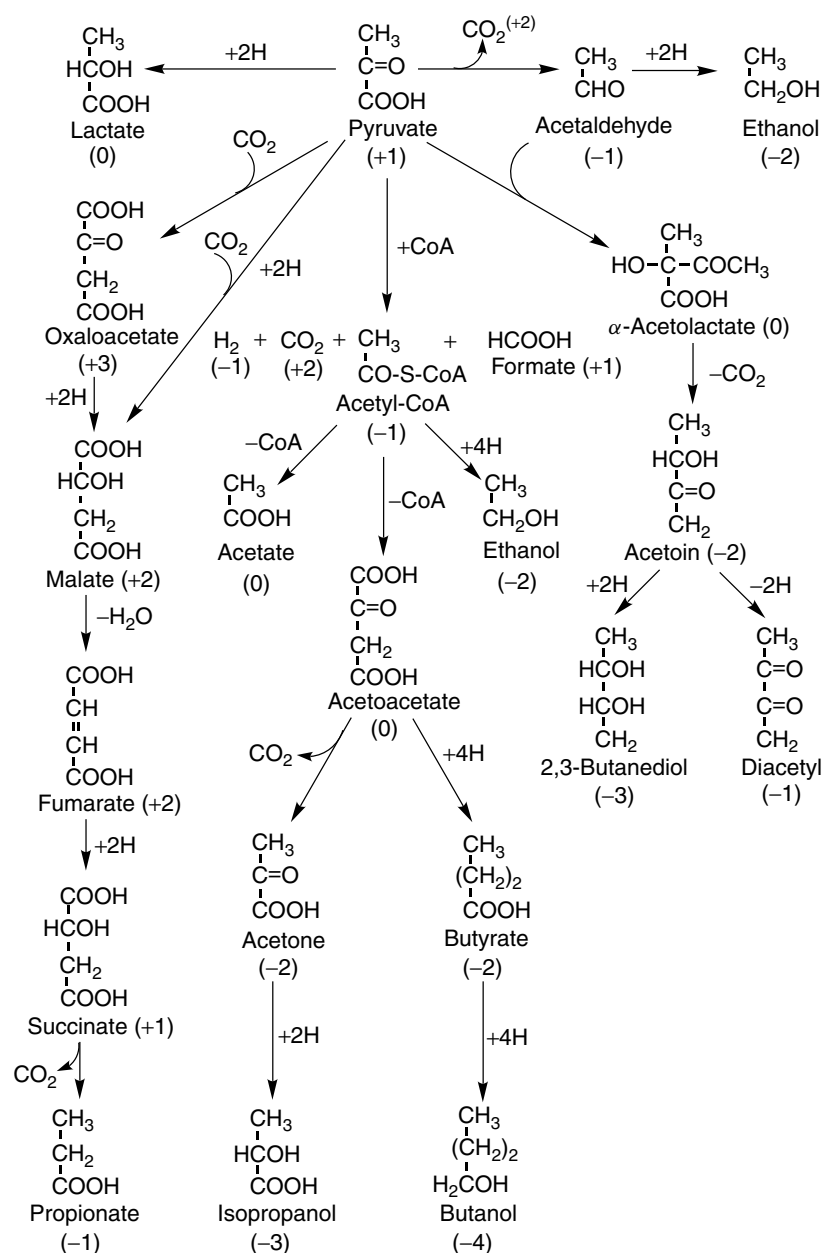
The major pathways used by microorganisms for the dissimilation of carbohydrates and the generation of biologically useful energy (as PMF or ATP) and reducing power (as NADH or NADPH) are delineated in previous chapters. The introductory reactions whereby a variety of alternate sugars and other substrates are metabolized to intermediates that can enter these central metabolic routes are also described. Individual organisms may use the fructose biphosphate aldolase, phosphoketolase, or ketogluconate aldolase pathways or variations of them. Additional fermentative products may be derived from pyruvate or other intermediates. The fermentative pathways occurring in some of the major groups of microorganisms are outlined in Figure 11-1 and are considered in detail in this chapter.

## FERMENTATION BALANCES

A thorough evaluation of the pathways of carbohydrate fermentation requires qualitative identification of and quantitative accounting for the amount of products recovered. To assess the accuracy of the analytical determinations, a **carbon balance** or **carbon recovery** can be calculated, as shown in the sample fermentation in Table 11-1. Under a number of circumstances, the balance may not be ideal because some portion of the carbohydrate may be assimilated into cellular constituents or compounds other than the major substrate added may be utilized to form some portion of the recovered products.

Oxidation–reduction reactions play a major role in the fermentative metabolism of carbohydrates. It is, therefore, advisable to perform an **oxidation–reduction** or **O–R balance** on the products formed. The O–R balance provides an indication as to whether the formed products balance with regard to their oxidized or reduced states. It may not be possible to balance the hydrogen and oxygen of the substrate directly because hydrations or dehydrations may occur as intermediary steps in the fermentation pathway. However, a simple way to achieve the same end is to calculate an **oxidation**





**Fig. 11-1. Major pathways of fermentation product formation from pyruvate.** Numbers in parentheses are the oxidation values calculated on the basis of the number of oxygen atoms less one-half the number of hydrogen atoms as shown in Table 11-1.

**value** for each compound and compare the total amount of oxidized and reduced products as shown in Table 11-1. If the ratio of oxidized products to reduced products is close to the theoretical value of 1.0, this provides further indication that the products are in balance.

TABLE 11-1. Fermentation Balance for *Lactobacillus pentoaceticus*

| Compound        | Mmol | Mmol Carbon <sup>a</sup> | Oxidation Value <sup>b</sup> | Oxidized Products | Reduced Products | C <sub>1</sub> Observed | C <sub>1</sub> Calculated <sup>c</sup> |
|-----------------|------|--------------------------|------------------------------|-------------------|------------------|-------------------------|----------------------------------------|
| Glucose         | 100  | 600                      | 0                            |                   |                  |                         |                                        |
| Lactate         | 96   | 288                      | 0                            |                   |                  |                         |                                        |
| Glycerol        | 7    | 21                       | -1                           |                   | 7                |                         |                                        |
| Ethanol         | 86   | 172                      | -2                           |                   | 172              |                         | 86                                     |
| Acetate         | 7    | 14                       | 0                            |                   |                  |                         | 7                                      |
| CO <sub>2</sub> | 89   | 89                       | +2                           | 178               |                  | 89                      |                                        |
| Totals          |      | 584                      |                              | 178               | 179              | 89                      | 93                                     |

Carbon Recovery

$$\frac{\text{mmol product}}{\text{mmol substrate}} \times 100$$

$$\frac{584}{600} \times 100 = 97.4\%$$

Oxidation-Reduction

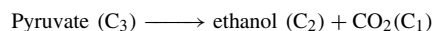
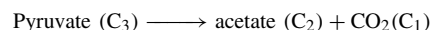
$$\frac{\text{Oxidized products}}{\text{Reduced products}}$$

$$\frac{178}{179} = 1.0$$

C<sub>1</sub> Balance

$$\frac{\text{Observed C}_1}{\text{Calculated C}_1}$$

$$\frac{89}{93} = 0.96$$

<sup>a</sup>mmol C = mm compound × number of C atoms.<sup>b</sup>Oxidation value = (number of O atoms) – (number of H atoms/2).<sup>c</sup>Calculated C<sub>1</sub> = expected amount for each C<sub>2</sub> observed. In the example above, for each mmole of pyruvate converted to ethanol, an equal amount of CO<sub>2</sub> is expected:Similarly, for each mmol of pyruvate converted to acetate, an equal amount of CO<sub>2</sub> is expected:

The **oxidation value** of a compound is determined from the number of oxygen atoms less one-half the number of hydrogen atoms. For glucose, which has 6 oxygen atoms and 12 hydrogen atoms, the oxidation value is 0 (Table 11-1). Thus, glucose is referred to as a **neutral compound**, and to be equally balanced the fermentation products should contain equivalent amounts of oxidized and reduced products. The carbon in the end products in the example shown in Table 11-1 is very close to the amount of fermented glucose (97.4%). In addition, the amount of oxidized product (carbon dioxide) is approximately equal to the amount of reduced products (glycerol and ethanol), so the O–R balance is very close to the theoretical value of 1.0. Figure 11-1 shows the general routes of formation and the oxidation values for a number of common fermentation products.

Another aspect of fermentation balances is the **C<sub>1</sub> balance**. The amount of expected C<sub>1</sub> product is calculated from the amounts of those products for which CO<sub>2</sub> or formate is expected as an accompanying product. For example, if a C<sub>2</sub> compound such as ethanol or acetate is among the final products, an equal amount of CO<sub>2</sub> will be expected, since ethanol is derived from pyruvate by decarboxylation. Note that in Table 11-1 the C<sub>1</sub> balance (0.96) is very close to the theoretical value of 1.0.

## YEAST FERMENTATION

Although well over 800 species of yeasts are known, the paradigm for studies of physiology, intermediary metabolism, and genetics is *Saccharomyces cerevisiae*. This yeast species uses the EMP pathway of glucose metabolism under the conditions of neutral or slightly acid pH and an anaerobic environment. The major products formed under these conditions are carbon dioxide and ethanol. The sequence of reactions involved in this pathway is presented in Figure 8-1. However, certain facets of the alcoholic fermentation of yeast bear additional consideration.

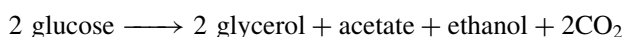
When yeast cells are grown in high concentrations of glucose, the cells function under a regulatory control system called carbon catabolite repression or glucose repression. By using  $^{13}\text{C}$ -labeled substrates, it is possible to make reliable estimates as to the metabolic fluxes occurring in the cell. In batch culture, 16.2 of every 100 molecules of glucose consumed by the cells enters the pentose phosphate pathway, whereas the same relative flux is 44.2 per 100 molecules in a chemostat. The TCA cycle does not operate as a cycle in batch cultures of growing yeast cells. In contrast, chemostat cultures of yeast use the TCA cycle in its intact form.

The ethanolic fermentation in yeast may be altered as a result of a number of changes in the culture medium. In the presence of sodium sulfite, acetaldehyde is trapped as a bisulfite addition complex, and glycerol is formed as a major product:

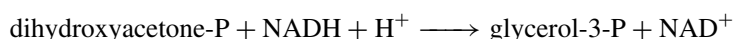
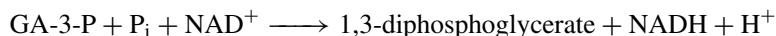
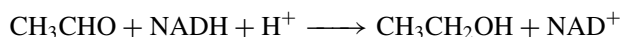
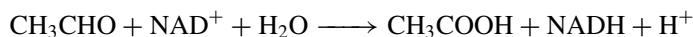


Under these conditions, acetaldehyde is unable to serve as a hydrogen acceptor. Dihydroxyacetone phosphate becomes the preferred hydrogen acceptor, yielding glycerol-3-phosphate, which is then hydrolyzed to glycerol and  $\text{P}_i$ . The fermentation is not shifted entirely in the direction of glycerol formation as it is not possible to add sufficient bisulfite to bind all of the acetaldehyde without incurring additional toxic effects. Thus, some ethanol will be found among the products.

Under alkaline conditions still another type of fermentation pattern is observed:



Under these conditions a dismutation reaction occurs in which 1 mol of acetaldehyde is oxidized to acetate and another is reduced to ethanol. This sequence is catalyzed by two NAD-linked dehydrogenases and requires a balance between the following reactions:



This series of reactions is sometimes referred to as the glycerol-3-phosphate shuttle. The oxidation of acetaldehyde to acetate yields no ATP, since the reaction proceeds directly without the intermediary formation of acetyl-CoA.

The fermentation products formed by yeast can also be altered drastically through metabolic engineering. For example, introduction of a lactate dehydrogenase gene from bovine muscle (*LDH-A*) into *S. cerevisiae* engenders the production of lactic acid at levels rivaling those achieved by lactic acid bacteria.

In cell-free yeast extracts the addition of  $P_i$  results in a marked increase in the fermentation rate. The rate eventually subsides to that of the control without added  $P_i$ . Addition of a second quantity of  $P_i$  will again increase the fermentation rate. This phenomenon (sometimes referred to as the Harden-Young effect) results from incorporation of  $P_i$  into organic phosphate esters, particularly fructose-1,6-bisphosphate, F-6-P, G-6-P, and G-1-P. Inorganic phosphate is assimilated during the formation of 1,3-diphosphoglycerate. In subsequent steps, ADP is phosphorylated to ATP, which in turn permits the phosphorylation of additional glucose. Accumulation of phosphate esters limits the amount of  $P_i$  available for the metabolism of additional glucose. This effect can be counteracted by the addition of arsenate ion, which results in the formation of phosphoglyceryl arsenate. This compound hydrolyzes nonenzymatically, preventing the accumulation of phosphorylated intermediates. During the usual procedure of preparing cell-free extracts of yeast, the enzyme ATPase may be inactivated. Preparations in which ATPase is active, or to which exogenous ATPase is added, do not exhaust the  $P_i$ , and phosphate esters do not accumulate. Under these conditions, fermentation continues at a constant rate.

In a resting yeast cell suspension that is fermenting glucose, the introduction of oxygen results in the cessation of ethanol formation. This phenomenon, termed the **Pasteur effect**, has been studied extensively in attempts to elucidate the mechanism involved. Growing yeast cells do not show a noticeable Pasteur effect. Growing cells respire only 3 to 20% of the catabolized sugar and the rest is fermented. The mechanism involved in the cessation of fermentation in resting cells (i.e., under conditions of nitrogen starvation) is a progressive inactivation of the sugar transport systems. As a consequence, the contribution of respiration to glucose catabolism, which is small in growing cells, becomes quite significant under the conditions of nitrogen starvation.

Some of the difficulty in studying alcoholic fermentation in yeast results from the fact that several isozymes of alcohol dehydrogenase (ADH) are present, and the activities of these and other enzymes concerned with carbohydrate metabolism are partitioned between the cytoplasm, the mitochondrial membrane, and the mitochondrial cytosol. The key enzyme of fermentative alcohol production (ADH1) reduces acetaldehyde to ethanol in the presence of NADH and serves to maintain the redox balance of glucose fermentation occurring in the cytoplasm of yeast cells. During ethanol oxidation, ADH2 catalyzes the formation of acetaldehyde.

Oxidative utilization of ethanol in the mitochondrion involves ADH3. The inner mitochondrial membrane is impermeable to  $NAD^+$  or NADH. Yeast mutants lacking ADH1 may regenerate cytoplasmic  $NAD^+$  by reducing DHAP to glycerol-3-phosphate. Glycerol-3-phosphate can enter the mitochondrial membrane, where it is reoxidized to DHAP by an FAD-linked glycerol-3-phosphate dehydrogenase. The electrons generated in this oxidation are transferred to the respiratory chain. For this reason, ADH1-deficient cells are unable to grow anaerobically. As a result of the release of  $P_i$  from glycerol-3-phosphate, glycerol appears as an end product. However, yeast deficient in at least four of the known ADH isozymes still produces up to one-third of the theoretical maximum yield of ethanol from glucose. This may result from the presence of additional forms of

ADH. The published genome for *S. cerevisiae* identified at least seven genes that could potentially code for members of the ADH family. Glycerol is a major fermentation product, but acetaldehyde and acetate are also produced. Ethanol production in multiply ADH-deficient cells is dependent on mitochondrial electron transport associated with the inner mitochondrial membrane.

## LACTIC ACID-PRODUCING FERMENTATIONS

In lactic acid-producing bacteria two major types of lactic acid fermentation occur. Those species that ferment glucose primarily to lactic acid are **homofermentative** and those that produce a mixture of other products along with lactic acid are **heterofermentative**. Some homofermentative species can form a wider variety of products if the conditions of the fermentation are altered. At alkaline pH the amounts of formate, acetate, and ethanol are increased at the expense of lactate (Table 11-2). As discussed earlier, isotope-labeling studies are helpful in determining the major pathway used in the formation of end products. *Lactobacillus casei*, *L. pentosus*, and *S. faecalis* were shown to produce  $^{14}\text{C}$ -methyl-labeled lactate from glucose-1- $^{14}\text{C}$  with 50% dilution and  $^{14}\text{C}$ -carboxyl-labeled lactate from glucose-3,4- $^{14}\text{C}$  without dilution of the specific activity. These labeling patterns fit the distribution of carbon atoms expected from utilization of the EMP pathway, as shown in Figure 11-2. In *L. lactis* an operon has been shown to code for phosphofructokinase, pyruvate kinase, and lactate dehydrogenase, three enzymes crucial to the operation of the EMP pathway. Distribution of  $^{14}\text{C}$ -labeled substrates indicates that more than 90% of the carbon in fermented sugars is converted into metabolic end products, usually lactic acid (see Fig. 11-2). As little as 5% of the glucose carbon consumed is converted into biomass.

It has been proposed that the term **homolactic** be used to refer to lactic acid bacteria that contain aldolase but not transketolase. True homolactic acid bacteria metabolize glucose via the EMP pathway while those that are truly heterofermentative presumably follow a hexose monophosphate pathway. However, some data suggest that a combination of both pathways may be operative in certain organisms.

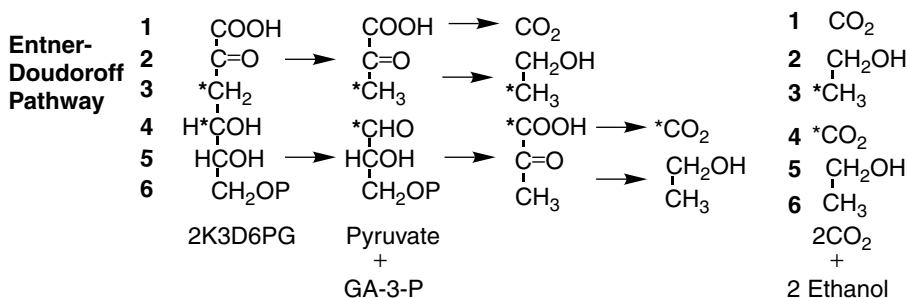
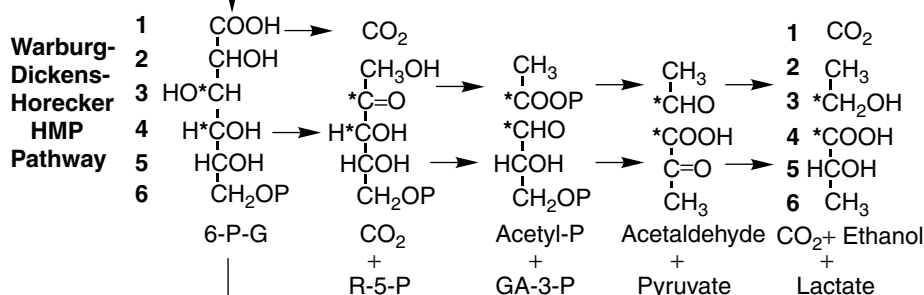
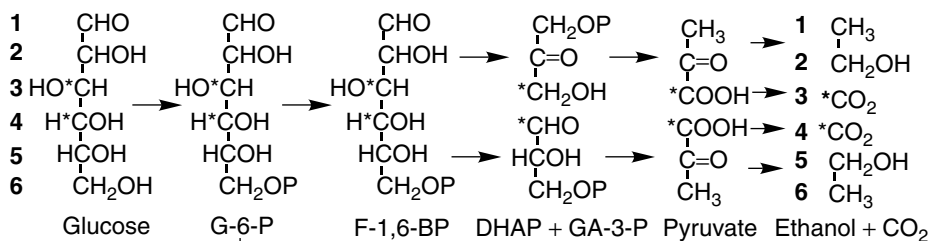
TABLE 11-2. Effect of pH on Glucose Fermentation by *S. faecalis*

| Products Formed <sup>a</sup> | pH    |       |       |
|------------------------------|-------|-------|-------|
|                              | 5.0   | 7.0   | 9.0   |
| Lactic acid                  | 174.0 | 146.0 | 122.0 |
| Acetic acid                  | 12.2  | 18.8  | 31.2  |
| Formic acid                  | 15.4  | 33.6  | 52.8  |
| Ethanol                      | 7.0   | 14.6  | 22.4  |
| Carbon recovery %            | 95.0  | 90.0  | 88.0  |
| O-R index <sup>b</sup>       | 1.02  | 1.18  | 1.18  |

<sup>a</sup>Amounts of products shown as  $\mu\text{mol}/100 \mu\text{mol}$  of fermented glucose.

<sup>b</sup>See Table 7-5 for definition of O-R (oxidation-reduction) index and its calculation.

Source: From Wood, 1961. Fermentation of carbohydrates and related compounds. In *The Bacteria*, Vol. II, Metabolism. I. C. Gunsalus and R. Y. Stanier (Eds.). Academic Press, New York, p. 59.

**Embden-Meyerhof-Parnas Pathway**

**Fig. 11-2. Isotope-labeling patterns in central pathways of carbohydrate metabolism in microorganisms.** \*C indicates <sup>14</sup>C-labeled carbon atoms.

One of the major factors controlling the type of product formed is the amount of reducing equivalents available. In the oxidative step in the triose phosphate pathway, NAD<sup>+</sup> is reduced. The electrode potential of this system (NAD<sup>+</sup>/NADH<sub>2</sub>) is -0.28 V. The electrode potential of the lactate dehydrogenase reaction is -0.18 V. Since this is at a higher potential, reduced NAD will donate its hydrogen atoms to pyruvate, reducing it to lactate. However, if some other hydrogen acceptor is available, reduced NAD will donate its hydrogen atoms to this acceptor rather than to pyruvate if the electrode potential of the alternative reaction is higher (see Fig. 9-6). For example, the potential for the reduction of acetaldehyde to ethanol is -0.07 V. If acetaldehyde is formed, then NADH will donate its hydrogen atoms to this system. Under the usual conditions of lactic acid fermentation, acetaldehyde does not appear to be formed in any appreciable quantity. However, by alteration of the pH, the fermentation may be diverted to a variety of products (Table 11-3). Since the streptococci appear to utilize the EMP pathway, the diversion of fermentation must occur at a step after pyruvate